# Online Supplement

*Online Supplement accompanying the final manuscript.*

The primary analysis concerns purchased/eligible Freddie Mac mortgages and 90+ day delinquency during the first 24 scheduled-payment months. Risky/non-risky labels are analytical classifications; the primary 5:1 FN:FP loss ratio is illustrative, not measured lender economics. Appendix A numbers relocated displays separately from Tables S1–S10 and Figures S1/S3. All existing scientific quantities, table rows and figure assets are retained.

# Appendix A. Supporting Main-Analysis Displays

## Appendix Table A1. Cohort design and eligibility.

Official Freddie Mac Standard annual samples contain purchased/eligible mortgages, not applications or all U.S. originations. The unit is one mortgage at origination and the outcome is 90+ day delinquency during the first 24 scheduled-payment months. Month 1 is First Payment Date and month 24 ends 23 calendar months later, no later than the role-specific cutoff. The primary observed-delinquency rule retains observed numeric 03–99 events despite earlier gaps; non-events require complete valid reporting through month 24 or valid Code 01 payoff without an earlier event. Ambiguous exits and unresolved histories are excluded; payoff does not create a counterfactual post-payoff history. Gross, calendar-mature and eligible counts have distinct denominators. Nineteen origination predictors are retained, and predictor missingness alone does not exclude a mortgage.

A. Final temporal roles

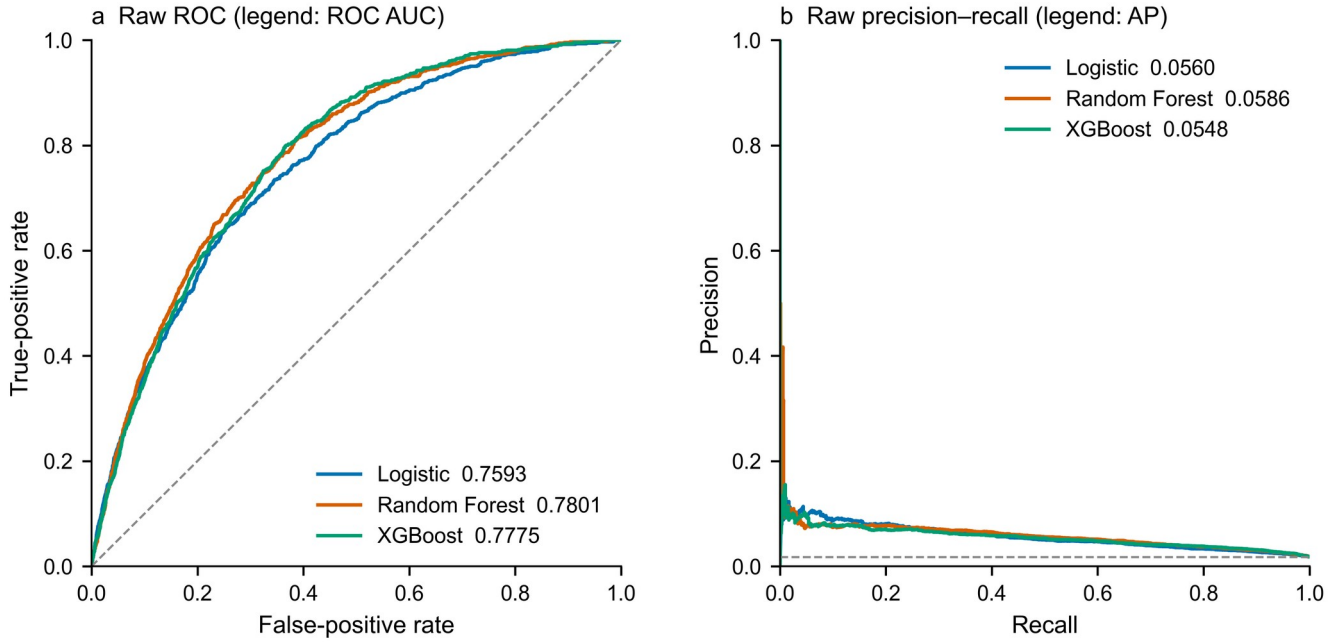
| Role | Vintages | Endpoint cutoff | Eligible N | Events | Event (%) |
|---|---|---|---|---|---|
| Development | 2015–2019 | 2022-12 | 239,435 | 4,550 | 1.900 |
| Validation | 2020 | 2022-12 | 45,364 | 771 | 1.700 |
| Frozen test | 2023 | 2026-03 | 49,338 | 886 | 1.796 |

B. Annual source-to-cohort flow

| Vintage | Raw N | Calendar- mature N | Eligible N | Events | Event (%) | Excluded N |
|---|---|---|---|---|---|---|
| 2015 | 50,000 | 50,000 | 47,221 | 252 | 0.534 | 2,779 |
| 2016 | 50,000 | 50,000 | 48,031 | 330 | 0.687 | 1,969 |
| 2017 | 50,000 | 50,000 | 48,211 | 397 | 0.823 | 1,789 |
| 2018 | 50,000 | 49,999 | 47,393 | 1,279 | 2.699 | 2,607 |
| 2019 | 50,000 | 49,996 | 48,579 | 2,292 | 4.718 | 1,421 |
| 2020 | 50,000 | 45,745 | 45,364 | 771 | 1.700 | 4,636 |
| 2023 | 50,000 | 49,964 | 49,338 | 886 | 1.796 | 662 |

## Appendix Figure A1. Raw-score discrimination in the primary 2023 cohort.

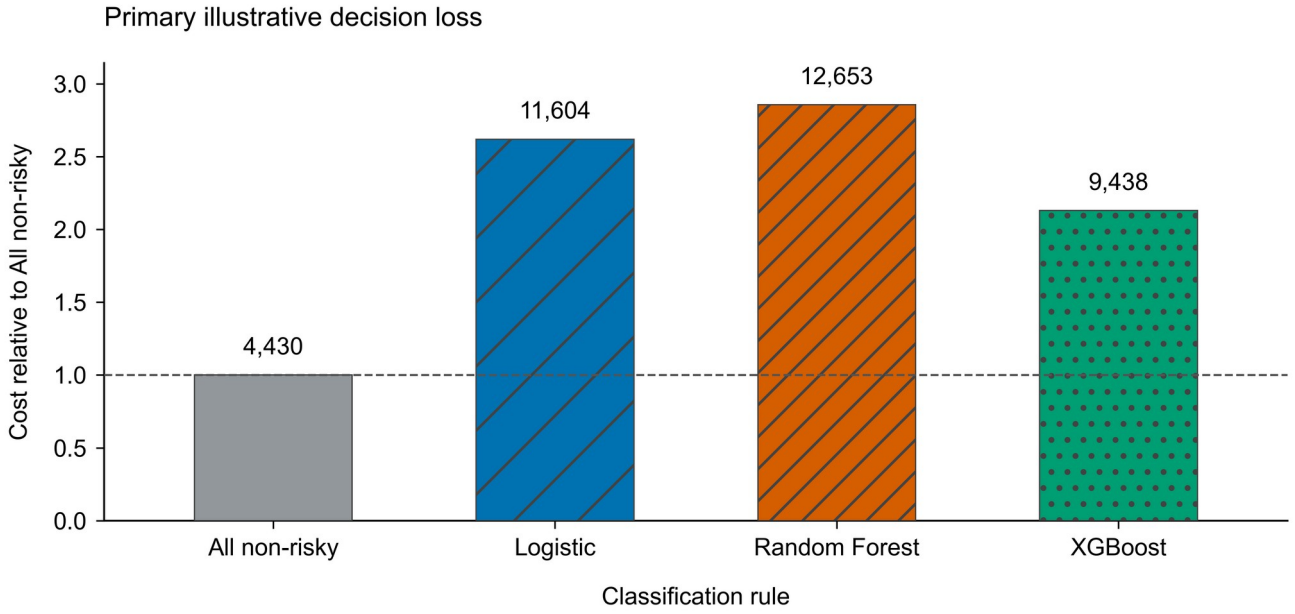
ROC and precision–recall curves compare Logistic Regression, Random Forest and XGBoost for 90+ day delinquency during the first 24 scheduled-payment months among 49,338 purchased/eligible mortgages, including 886 events. Only raw model scores are used for discrimination. The ROC diagonal is a no-discrimination reference; the horizontal precision reference is the observed prevalence of 1.796%. Legend AP values are average precision, not trapezoidal precision–recall area. Axes retain the full unit range. These curves describe ranking performance, not institutional decision value or statistical superiority; no uncertainty bands are shown.




Frozen 2023 purchased/eligible mortgages (N = 49,338; 886 events). Outcome: 90+ day delinquency during the first 24 scheduled-payment months. Raw scores only. AP denotes average precision, not trapezoidal PR area. Dashed lines show the ROC diagonal and PR prevalence baseline (1.796%). No confidence intervals or statistical-superiority claim.

## Appendix Figure A2. Primary illustrative decision loss relative to all non-risky.

Frozen historical calibrated probabilities classify 49,338 purchased/eligible mortgages as risky at p ≥ 1/6 for 90+ day delinquency during the first 24 scheduled-payment months (886 events). The illustrative 5:1 FN:FP loss is FP + 5FN. The benchmark classifies all mortgages as non-risky and costs 4,430. Bar heights show cost relative to this same-cohort benchmark, absolute-cost annotations reproduce the reported costs, and the dashed reference is one. XGBoost has the lowest cost among the three model-based rules, but still exceeds the all-non-risky benchmark. Illustrative loss is not measured lender economics; these are analytical risky/non-risky classifications. No raw-probability or validation-optimal rule is included.




## Appendix Table A2. Compact robustness summary.

Each row uses its own specification's purchased/eligible cohort and all-non-risky benchmark. Discrimination uses raw ROC AUC. The lowest three-model calibrated common 5:1 cost is taken from the same model row as its benchmark ratio; this is a retrospective descriptive minimum, not test-based model selection. The primary outcome is 90+ day delinquency during the first 24 scheduled-payment months; R2 changes it to the distress composite, R3 conditions on remaining on book, and R7 adds First Payment Date as a timing proxy rather than exact origination month, extrapolating beyond development support. R3 and R7 materially change interpretation. All three primary model-based calibrated common 5:1 rules remain above their own all-non-risky benchmark in the primary analysis and R1–R7, while rankings and model-induced reclassification magnitude are more specification-sensitive. Event and disagreement rates are percentages in the reader table; the 5:1 FN:FP loss ratio is illustrative, not measured lender economics.

Cohort and raw discrimination

| Specification | Scientific change | Test N | Event rate | Logistic raw ROC AUC | Random Forest raw ROC AUC | XGBoost raw ROC AUC |
|---|---|---|---|---|---|---|
| S10 | Candidate A / T2 reference | 49,338 | 1.796% | 0.7593 | 0.7801 | 0.7775 |

| Specification | Scientific change | Test N | Event rate | Logistic raw ROC AUC | Random Forest raw ROC AUC | XGBoost raw ROC AUC |
|---|---|---|---|---|---|---|
| R1 | Candidate C complete-history 90+ outcome | 49,328 | 1.776% | 0.7614 | 0.7812 | 0.7801 |
| R2 | Candidate B distress-composite outcome | 49,338 | 1.796% | 0.7594 | 0.7816 | 0.7779 |
| R3 | Conditional-on-remaining-on-book cohort | 40,514 | 2.187% | 0.7321 | 0.7354 | 0.7024 |
| R4 | Exclude all within-window Code 96 loans | 49,308 | 1.736% | 0.7583 | 0.7834 | 0.7724 |
| R5 | Exclude unusual First Payment Dates | 49,321 | 1.796% | 0.7594 | 0.7740 | 0.7752 |
| R6 | T1 temporal design | 37,022 | 1.723% | 0.7735 | 0.7960 | 0.7903 |
| R7 | Three time-feature sensitivity | 49,338 | 1.796% | 0.7687 | 0.7463 | 0.7762 |

Illustrative common 5:1 loss and model-induced reclassification

| Specification | Lowest model cost | Model attaining minimum | Cost / own All non-risky | Logistic–XGBoost disagreement N | Disagreement rate | Interpretation class |
|---|---|---|---|---|---|---|
| S10 | 9,438 | XGBoost | 2.130 | 5,950 | 12.06% | reference |
| R1 | 8,401 | XGBoost | 1.918 | 6,966 | 14.12% | quantitatively changed but same conclusion |
| R2 | 7,440 | XGBoost | 1.679 | 6,690 | 13.56% | quantitatively changed but same conclusion |
| R3 | 23,145 | XGBoost | 5.225 | 11,983 | 29.58% | materially changes interpretation |
| R4 | 8,693 | XGBoost | 2.031 | 5,752 | 11.67% | quantitatively changed but same conclusion |
| R5 | 7,685 | XGBoost | 1.735 | 6,539 | 13.26% | quantitatively changed but same conclusion |
| R6 | 6,852 | XGBoost | 2.148 | 4,734 | 12.79% | quantitatively changed but same conclusion |
| R7 | 6,619 | XGBoost | 1.494 | 41,218 | 83.54% | materially changes interpretation |

Interpretation notes

• S10: Frozen Candidate A/T2 reference, not a sensitivity.

• R1: Complete-history event eligibility changes quantitative results; the common calibrated 5:1 loss conclusion remains.

• R2: The distress-composite outcome changes the event definition; the common calibrated 5:1 loss conclusion remains.

• R3: Restriction to remaining on book changes the estimand; raw Logistic discrimination exceeds XGBoost and disagreement rises to 29.58%. This is not post-payoff counterfactual risk.

• R4: Excluding all within-window effective-date Code 96 exits changes the cohort without recoding Code 96; the common calibrated 5:1 loss conclusion remains.

• R5: Excluding unusual First Payment Dates changes the cohort; the common calibrated 5:1 loss conclusion remains.

• R6: T1 changes validation maturity and restricts test to 2023Q2–Q4; discrimination differs, while the common calibrated 5:1 loss conclusion remains.

• R7: First Payment Date timing-proxy features alter rankings and yield 83.54% disagreement. Out-of-support timing is consistent with extrapolation sensitivity, not proof of a causal mechanism.

# S.A Reproducible features and training

## Table S1. Predictors, preprocessing and frozen grids

The primary models share exactly 19 origination predictors (seven numeric and twelve categorical). Predictor definitions, exclusions, full hyperparameter grids and selected settings are reported below. Medians, missingness indicators and category handling are learned only within the earlier-year fit subset; Logistic Regression additionally standardizes numeric inputs. Four chronological folds select models by mean log loss. No class weighting or resampling is used. Historical isotonic calibration uses training-period forward out-of-fold predictions; FT-Transformer remains exploratory, not a fourth primary benchmark.

### Primary predictor definitions

*Column panel 1 of 2 — Primary predictor definitions*

| feature_name | source_field | conceptual_group | clean_type | missing_rule | allowed_values_or_range | primary_predictor |
|---|---|---|---|---|---|---|
| classic_fico | Classic FICO | borrower | integer_nullable | 9999 or blank -> null; out-of-range -> null plus diagnostic | 300-850 | yes |
| original_dti | Original Debt-to-Income Ratio | borrower | integer_nullable | 999 or blank -> null; out-of-range -> null plus diagnostic | 0-65 | yes |
| first_time_homebuyer | First Time Homebuyer Indicator | borrower | categorical | 9 or blank -> unknown | yes; no; unknown | yes |
| borrower_structure | Number of Borrowers | borrower | categorical | 99 or blank -> unknown; invalid -> unknown plus diagnostic | one; multiple; unknown | yes |
| original_upb | Original UPB | loan | integer_nullable | blank/nonpositive/invalid -> null plus diagnostic | >0; rounded to nearest $1,000 | yes |
| original_ltv | Original Loan-to-Value | loan | integer_nullable | 999 or blank -> null; outside vintage-specific range -> null plus diagnostic | 6-105 through 2018Q1; 1-998 from 2018Q2 | yes |
| original_cltv | Original Combined Loan-to-Value | loan | integer_nullable | 999 or blank -> null; outside vintage-specific range -> null plus diagnostic | 6-200 through 2018Q1; 1-998 from 2018Q2 | yes |
| original_interest_rate | Original Interest Rate | loan | float_nullable | blank, nonnumeric, or negative -> null plus diagnostic | >=0; no finite maximum documented | yes |
| original_loan_term | Original Loan Term | loan | integer_nullable | blank, nonnumeric, or nonpositive -> null plus diagnostic | >0 months; calculated from maturity and first payment dates | yes |
| loan_purpose | Loan Purpose | loan | categorical | 9 or blank -> unknown; other invalid -> unknown plus diagnostic | purchase; cash_out_refinance; no_cash_out_refinance; refinance_not_specified; unknown | yes |
| channel | Channel | loan | categorical | 9 or blank -> unknown; other invalid -> unknown plus diagnostic | retail; broker; correspondent; third_party_not_specified; unknown | yes |
| super_conforming_flag | Super Conforming Flag | loan | categorical | blank -> unknown; invalid -> unknown plus diagnostic | yes; no; unknown | yes |
| harp_indicator | HARP Indicator | loan | categorical | blank -> unknown; invalid -> unknown plus diagnostic | yes; no; unknown | yes |
| special_eligibility_program | Special Eligibility Program | loan | categorical | blank -> not_available_or_not_applicable; invalid nonblank -> unknown | home_possible; hfa_advantage; refi_possible; not_available_or_not_applicable; unknown | yes |
| occupancy_status | Occupancy Status | property | categorical | 9 or blank -> unknown; invalid -> unknown plus diagnostic | primary_residence; investment_property; second_home; unknown | yes |
| property_type | Property Type | property | categorical | 99 or blank -> unknown; invalid -> unknown plus diagnostic | cooperative; condominium; planned_unit_development; single_family; manufactured_housing; unknown | yes |
| number_of_units | Number of Units | property | categorical | 99 or blank -> unknown; invalid -> unknown plus diagnostic | one; two; three; four; unknown | yes |
| property_state | Property State | geography | categorical | blank/invalid -> unknown plus diagnostic | two-letter state or territory code; unknown | yes |
| msa_or_division | MSA or Metropolitan Division | geography | categorical | blank -> nonmetropolitan_or_unknown; invalid nonblank -> unknown | five-digit MSA/division; nonmetropolitan_or_unknown; unknown | yes |

*Column panel 2 of 2 — Primary predictor definitions*

| feature_name | notes | feature_kind | record_type |
|---|---|---|---|
| classic_fico | Release 47 General User Guide p.4 | numeric | predictor |
| original_dti | all HARP loans officially 999; Guide pp.5-6 | numeric | predictor |
| first_time_homebuyer | 9 is officially Not Available or Not Applicable; Guide p.4 | categorical | predictor |
| borrower_structure | through 2018Q1 code 2 means >1; from 2018Q2 exact 1-10; Guide p.8 | categorical | predictor |
| original_upb | no numeric missing sentinel documented; Guide p.6 | numeric | predictor |
| original_ltv | HARP official range is 1-998; harmonized by identifier vintage rule; Guide p.6 | numeric | predictor |
| original_cltv | HARP official range is 1-998; harmonized by identifier vintage rule; Guide p.5 | numeric | predictor |
| original_interest_rate | note rate at origination; Guide p.6 | numeric | predictor |
| original_loan_term | Guide p.8 | numeric | predictor |
| loan_purpose | Guide pp.7-8 | categorical | predictor |
| channel | Guide pp.6-7 | categorical | predictor |
| super_conforming_flag | Guide p.9 | categorical | predictor |
| harp_indicator | Relief Refinance indicator; Guide p.9 | categorical | predictor |
| special_eligibility_program | enumerations disclosed for loans funded from 2015-03-01; Guide p.9 | categorical | predictor |
| occupancy_status | Guide p.5 | categorical | predictor |
| property_type | Guide p.7 | categorical | predictor |
| number_of_units | Guide p.5 | categorical | predictor |
| property_state | Guide p.7 | categorical | predictor |
| msa_or_division | official null conflates non-MSA with unknown; Guide p.4 | categorical | predictor |

## Preprocessing and information exclusions

| record_type | parameter | setting |
|---|---|---|
| preprocessing | numeric_cast | Decode frozen sentinels; numeric blanks/unparseable values become missing. |
| preprocessing | numeric_imputation | Median fitted in each earlier-year fit subset; no validation/test information. |
| preprocessing | numeric_missing_indicators | SimpleImputer(add_indicator=True); indicators for columns missing during fitting. |
| preprocessing | categorical_missing | Blank/null becomes __missing__; documented explicit unknown/unavailable levels retained. |
| preprocessing | categorical_protected_levels | unknown, nonmetropolitan_or_unknown, not_available_or_not_applicable and __missing__ are protected from rare pooling when observed during fitting. |
| preprocessing | categorical_rare_unseen | Fit-subset category counts; below min_frequency and later unseen values map to __rare_or_unseen__; one-hot, no ordinal assumptions. |
| preprocessing | numeric_scaling | Logistic standardizes the imputed numeric block and indicators using fit-subset parameters; RF/XGBoost do not scale. |
| preprocessing | fold_preprocessing | Preprocessing is refitted independently within every forward fold. |

| record_type | parameter | setting |
|---|---|---|
| preprocessing | weighting_resampling | No class weights, SMOTE, oversampling or undersampling. |
| preprocessing | selection | Unweighted four-forward-fold mean log loss; fixed 0.0001 effective-tie tolerance and frozen family-specific simplicity preference. |
| preprocessing | probability_calibration | Historical isotonic mapping fitted to pooled 2016–2019 forward OOF scores only; mappings locked before 2020/2023. |
| preprocessing | excluded_information | Loan ID, outcome, vintage, First Payment Date and performance/audit fields excluded; remaining exclusions follow frozen feature definitions. |

Complete prespecified grids and fixed settings

| record_type | parameter | setting | model | setting_role |
|---|---|---|---|---|
| grid | C | [0.01, 0.1, 1.0, 10.0] | logistic | tuned_grid |
| grid | min_frequency | [25, 50, 100, 200] | logistic | tuned_grid |
| grid | penalty | "l2" | logistic | fixed_setting |
| grid | solver | "liblinear" | logistic | fixed_setting |
| grid | max_iter | 2000 | logistic | fixed_setting |
| grid | class_weight | null | logistic | fixed_setting |
| grid | resampling | false | logistic | fixed_setting |
| grid | n_estimators | [400] | rf | tuned_grid |
| grid | max_depth | [12, null] | rf | tuned_grid |
| grid | min_samples_leaf | [5, 50] | rf | tuned_grid |
| grid | max_features | ["sqrt", 0.5] | rf | tuned_grid |
| grid | min_frequency | [50, 200] | rf | tuned_grid |
| grid | criterion | "gini" | rf | fixed_setting |
| grid | bootstrap | true | rf | fixed_setting |
| grid | random_state | 20260925 | rf | fixed_setting |
| grid | n_jobs | -1 | rf | fixed_setting |
| grid | class_weight | null | rf | fixed_setting |
| grid | resampling | false | rf | fixed_setting |
| grid | learning_rate | [0.03, 0.1] | xgb | tuned_grid |
| grid | max_depth | [3, 6] | xgb | tuned_grid |
| grid | min_child_weight | [1, 10] | xgb | tuned_grid |
| grid | min_frequency | [50, 200] | xgb | tuned_grid |
| grid | n_estimators | 600 | xgb | fixed_setting |
| grid | subsample | 0.8 | xgb | fixed_setting |
| grid | colsample_bytree | 0.8 | xgb | fixed_setting |
| grid | reg_lambda | 1.0 | xgb | fixed_setting |
| grid | reg_alpha | 0.0 | xgb | fixed_setting |
| grid | objective | "binary:logistic" | xgb | fixed_setting |
| grid | eval_metric | "logloss" | xgb | fixed_setting |
| grid | tree_method | "hist" | xgb | fixed_setting |
| grid | random_state | 20260925 | xgb | fixed_setting |
| grid | n_jobs | -1 | xgb | fixed_setting |
| grid | scale_pos_weight | null | xgb | fixed_setting |
| grid | class_weight | null | xgb | fixed_setting |

| record_type | parameter | setting | model | setting_role |
|---|---|---|---|---|
| grid | resampling | false | xgb | fixed_setting |

Frozen selected settings

| record_type | parameter | setting | model | source_parameter |
|---|---|---|---|---|
| selected_setting | C | 0.01 | logistic | C |
| selected_setting | min_frequency | 200 | logistic | categorical_min_frequency |
| selected_setting | penalty | "l2" | logistic | penalty |
| selected_setting | solver | "liblinear" | logistic | solver |
| selected_setting | max_iter | 2000 | logistic | max_iter |
| selected_setting | class_weight | null | logistic | class_weight |
| selected_setting | resampling | false | logistic | resampling |
| selected_setting | n_estimators | 400 | rf | n_estimators |
| selected_setting | max_depth | null | rf | max_depth |
| selected_setting | min_samples_leaf | 50 | rf | min_samples_leaf |
| selected_setting | max_features | "sqrt" | rf | max_features |
| selected_setting | min_frequency | 200 | rf | categorical_min_frequency |
| selected_setting | criterion | "gini" | rf | criterion |
| selected_setting | bootstrap | true | rf | bootstrap |
| selected_setting | random_state | 20260925 | rf | random_state |
| selected_setting | n_jobs | -1 | rf | n_jobs |
| selected_setting | class_weight | null | rf | class_weight |
| selected_setting | resampling | false | rf | resampling |
| selected_setting | learning_rate | 0.03 | xgb | learning_rate |
| selected_setting | max_depth | 3 | xgb | max_depth |
| selected_setting | min_child_weight | 10 | xgb | min_child_weight |
| selected_setting | min_frequency | 200 | xgb | categorical_min_frequency |
| selected_setting | n_estimators | 600 | xgb | n_estimators |
| selected_setting | subsample | 0.8 | xgb | subsample |
| selected_setting | colsample_bytree | 0.8 | xgb | colsample_bytree |
| selected_setting | reg_lambda | 1.0 | xgb | reg_lambda |
| selected_setting | reg_alpha | 0.0 | xgb | reg_alpha |
| selected_setting | objective | "binary:logistic" | xgb | objective |
| selected_setting | eval_metric | "logloss" | xgb | eval_metric |
| selected_setting | tree_method | "hist" | xgb | tree_method |
| selected_setting | random_state | 20260925 | xgb | random_state |
| selected_setting | n_jobs | -1 | xgb | n_jobs |
| selected_setting | scale_pos_weight | null | xgb | scale_pos_weight |
| selected_setting | class_weight | null | xgb | class_weight |
| selected_setting | resampling | false | xgb | resampling |

## Table S2. Complete forward-fold development results

Complete configuration-by-fold development evidence is retained, with a selected-configuration overview followed by all model/configuration sections. Each fold fits preprocessing on earlier years and evaluates the next year. AP is average precision. These training-only metrics are not independent final-test evidence: the same development

history supports selection and out-of-fold construction, rather than nested independent evaluation. All rows and section breaks are retained; the table is not forced into a one-page image.

## Selected configurations

*Column panel 1 of 5 — Selected configurations*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n |
|---|---|---|---|---|---|---|---|---|---|
| logistic | NA | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.01 | 200 | 47221 | 252 |
| logistic | NA | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.01 | 200 | 95252 | 582 |
| logistic | NA | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.01 | 200 | 143463 | 979 |
| logistic | NA | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.01 | 200 | 190856 | 2258 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | NA | 200 | 47221 | 252 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | NA | 200 | 95252 | 582 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | NA | 200 | 143463 | 979 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | NA | 200 | 190856 | 2258 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | NA | 200 | 47221 | 252 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | NA | 200 | 95252 | 582 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | NA | 200 | 143463 | 979 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | NA | 200 | 190856 | 2258 |

*Column panel 2 of 5 — Selected configurations*

| model | config_id | fold_id | validation_n | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|---|
| logistic | NA | 1 | 48031 | 330 | 0.8172752238851483 | 0.02912578236766845 | 0.006756960937007503 | 0.037008550033971126 | 0.006870562761549832 |
| logistic | NA | 2 | 48211 | 397 | 0.8259605151321573 | 0.04472243105857647 | 0.008016667659020975 | 0.041856144592576854 | 0.008234635249216984 |
| logistic | NA | 3 | 47393 | 1279 | 0.7427851831184389 | 0.07816561303869263 | 0.026016701374618 | 0.12223765428074762 | 0.026987107800730065 |
| logistic | NA | 4 | 48579 | 2292 | 0.6968496425914784 | 0.10035160140766261 | 0.04529900889404784 | 0.21061654759735857 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 48031 | 330 | 0.8407280070997811 | 0.03639981049866016 | 0.006738904876528 75 | 0.03604234186228811 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 48211 | 397 | 0.8369533643118976 | 0.05276286832662946 4 | 0.008019730834344142 | 0.04138148891503908 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 47393 | 1279 | 0.7125919844497284 | 0.06704730264720829 | 0.026207042173596318 | 0.12492488089923727 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 48579 | 2292 | 0.6845073820666121 | 0.09519682804863408 | 0.04551238304266364 | 0.20984763703907838 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 48031 | 330 | 0.8332574185281676 | 0.04030083074232704 | 0.006713737733662128 4 | 0.03586766496300697 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 48211 | 397 | 0.837522899135072 | 0.04617902453629771 4 | 0.008006484247744083 | 0.04099542647600174 | 0.008234635249216984 |

| model | config_id | fold_id | validation_n | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 47393 | 1279 | 0.7236771938517396 | 0.07083513974854984 | 0.026086924597620964 | 0.1286497563123703 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 48579 | 2292 | 0.6879870378495562 | 0.10004652405116389 | 0.04531946778297424 | 0.21859955787658691 | 0.04718088062743161 |

*Column panel 3 of 5 — Selected configurations*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | convergence_iterations | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | NA | 1 | 0.006290040898461558 | 137 | 8.0 | True | 0.0 | 0.209754 | 0.031202 | 0.240956 |
| logistic | NA | 2 | 0.008552337572713588 | 200 | 9.0 | True | 0.0 | 0.314339 | 0.078969 | 0.393308 |
| logistic | NA | 3 | 0.011201601935241922 | 242 | 9.0 | True | 0.0 | 0.439707 | 0.129814 | 0.569521 |
| logistic | NA | 4 | 0.013609486355580139 | 268 | 8.0 | True | 0.0 | 0.561377 | 0.174124 | 0.735501 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.00483057846055267 | 137 | NA | NA | NA | 0.21007 | 0.909332 | 1.119401 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007441555457873739 | 200 | NA | NA | NA | 0.321589 | 1.87277 | 2.194359 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010829348891769851 | 242 | NA | NA | NA | 0.434741 | 3.206962 | 3.641703 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.012989613938273166 | 268 | NA | NA | NA | 0.569217 | 5.481556 | 6.050773 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.0046556126326322556 | 137 | NA | NA | NA | 0.136994 | 0.849856 | 1.007459 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.007318955380469561 | 200 | NA | NA | NA | 0.217374 | 0.937583 | 1.179093 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.010314136743545532 | 242 | NA | NA | NA | 0.300201 | 1.028204 | 1.362918 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012954357080161572 | 268 | NA | NA | NA | 0.370155 | 1.136328 | 1.537205 |

*Column panel 4 of 5 — Selected configurations*

| model | config_id | fold_id | configuration | selected | average_precision | n_estimators | max_depth | min_samples_leaf | max_features |
|---|---|---|---|---|---|---|---|---|---|
| logistic | NA | 1 | {"C":0.01,"min_frequency":200} | True | 0.02912578236766845 | NA | NA | NA | NA |
| logistic | NA | 2 | {"C":0.01,"min_frequency":200} | True | 0.04472243105857647 | NA | NA | NA | NA |
| logistic | NA | 3 | {"C":0.01,"min_frequency":200} | True | 0.07816561303869263 | NA | NA | NA | NA |
| logistic | NA | 4 | {"C":0.01,"min_frequency":200} | True | 0.10035160140766261 | NA | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True | 0.03639981049866016 | 400.0 | NA | 50.0 | sqrt |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True | 0.052762868326629464 | 400.0 | NA | 50.0 | sqrt |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True | 0.06704730264720829 | 400.0 | NA | 50.0 | sqrt |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True | 0.09519682804863408 | 400.0 | NA | 50.0 | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True | 0.04030083074232704 | NA | 3.0 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True | 0.046179024536297714 | NA | 3.0 | NA | NA |

| model | config_id | fold_id | configuration | selected | average_precision | n_estimators | max_depth | min_samples_leaf | max_features |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True | 0.07083513974854984 | NA | 3.0 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True | 0.10004652405116389 | NA | 3.0 | NA | NA |

*Column panel 5 of 5 — Selected configurations*

| model | config_id | fold_id | tree_count | fit_key | config_index | status | completed_at | learning_rate | min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | NA | 1 | NA | NA | NA | NA | NA | NA | NA |
| logistic | NA | 2 | NA | NA | NA | NA | NA | NA | NA |
| logistic | NA | 3 | NA | NA | NA | NA | NA | NA | NA |
| logistic | NA | 4 | NA | NA | NA | NA | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 400.0 | NA | NA | NA | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 400.0 | NA | NA | NA | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 400.0 | NA | NA | NA | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 400.0 | NA | NA | NA | NA | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | xgb_lr_0p03_depth_3_child_10_freq_200::fold_1 | 4.0 | complete | 2026-09-25T17:36:37+01:00 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | xgb_lr_0p03_depth_3_child_10_freq_200::fold_2 | 4.0 | complete | 2026-09-25T17:36:38+01:00 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | xgb_lr_0p03_depth_3_child_10_freq_200::fold_3 | 4.0 | complete | 2026-09-25T17:36:39+01:00 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | xgb_lr_0p03_depth_3_child_10_freq_200::fold_4 | 4.0 | complete | 2026-09-25T17:36:41+01:00 | 0.03 | 10.0 |

## Logistic — {"C":0.01,"min_frequency":100}

*Column panel 1 of 3 — Logistic — {"C":0.01,"min_frequency":100}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.01 | 100 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.01 | 100 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.01 | 100 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.01 | 100 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.01,"min_frequency":100}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8173273795797431 | 0.02902023609495871 | 0.006757135976808478 | 0.03701435990680961 | 0.006870562761549832 | 0.00629822514127087 | 199 | 8.0 |
| logistic | 2 | 0.825735461689867 | 0.04476735950372751 | 0.008016856645739269 | 0.04186062721990482 | 0.008234635249216984 | 0.00856684650209891 | 263 | 9.0 |

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 3 | 0.7426950505737505 | 0.07819751475921309 | 0.02601572736439286 | 0.12222472584926361 | 0.026987107800730065 | 0.011212410976211552 | 311 | 9.0 |
| logistic | 4 | 0.6968064433411527 | 0.10034978041646667 | 0.04529862835854605 | 0.2106130484987741 | 0.04718088062743161 | 0.013613864854782916 | 349 | 8.0 |

*Column panel 3 of 3 — Logistic — {"C":0.01,"min_frequency":100}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209138 | 0.033008 | 0.242146 | {"C":0.01,"min_frequency":100} | False | 0.02902023609495871 |
| logistic | 2 | True | 0.0 | 0.33049 | 0.078424 | 0.408914 | {"C":0.01,"min_frequency":100} | False | 0.04476735950372751 |
| logistic | 3 | True | 0.0 | 0.448368 | 0.127984 | 0.576352 | {"C":0.01,"min_frequency":100} | False | 0.07819751475921309 |
| logistic | 4 | True | 0.0 | 0.55163 | 0.171911 | 0.723541 | {"C":0.01,"min_frequency":100} | False | 0.10034978041646667 |

## Logistic — {"C":0.01,"min_frequency":200}

*Column panel 1 of 3 — Logistic — {"C":0.01,"min_frequency":200}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.01 | 200 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.01 | 200 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.01 | 200 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.01 | 200 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.01,"min_frequency":200}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8172752238851483 | 0.02912578236766845 | 0.006756960937007503 | 0.037008550033971126 | 0.006870562761549832 | 0.006290040898461558 | 137 | 8.0 |
| logistic | 2 | 0.8259605151321573 | 0.04472243105857647 | 0.008016667659020975 | 0.041856144592576854 | 0.008234635249216984 | 0.008552337572713588 | 200 | 9.0 |
| logistic | 3 | 0.7427851831184389 | 0.07816561303869263 | 0.02601670137461816 | 0.12223765428074762 | 0.026987107800730065 | 0.011201601935241922 | 242 | 9.0 |
| logistic | 4 | 0.6968496425914784 | 0.10035160140766261 | 0.04529900889404784 | 0.21061654759735857 | 0.04718088062743161 | 0.013609486355580139 | 268 | 8.0 |

*Column panel 3 of 3 — Logistic — {"C":0.01,"min_frequency":200}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209754 | 0.031202 | 0.240956 | {"C":0.01,"min_frequency":200} | True | 0.02912578236766845 |
| logistic | 2 | True | 0.0 | 0.314339 | 0.078969 | 0.393308 | {"C":0.01,"min_frequency":200} | True | 0.04472243105857647 |
| logistic | 3 | True | 0.0 | 0.439707 | 0.129814 | 0.569521 | {"C":0.01,"min_frequency":200} | True | 0.07816561303869263 |
| logistic | 4 | True | 0.0 | 0.561377 | 0.174124 | 0.735501 | {"C":0.01,"min_frequency":200} | True | 0.10035160140766261 |

## Logistic — {"C":0.01,"min_frequency":25}

*Column panel 1 of 3 — Logistic — {"C":0.01,"min_frequency":25}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.01 | 25 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.01 | 25 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.01 | 25 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.01 | 25 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.01,"min_frequency":25}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8161895468807274 | 0.028713360275695658 | 0.006758334708575664 | 0.0370444224799937 | 0.006870562761549832 | 0.006302309029516584 | 341 | 8.0 |
| logistic | 2 | 0.825911574437427 | 0.045189995703829935 | 0.008015706332544099 | 0.041842813262378875 | 0.008234635249216984 | 0.008577076389753188 | 443 | 9.0 |
| logistic | 3 | 0.7426923716907445 | 0.07819829471657118 | 0.02601542476968013 | 0.1222204448996887 | 0.026987107800730065 | 0.011215276282459533 | 489 | 9.0 |
| logistic | 4 | 0.6969005051607032 | 0.100399186433299 | 0.04529733225109746 | 0.2105831068263065 | 0.04718088062743161 | 0.013619457631517758 | 512 | 9.0 |

*Column panel 3 of 3 — Logistic — {"C":0.01,"min_frequency":25}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.216906 | 0.031869 | 0.248774 | {"C":0.01,"min_frequency":25} | False | 0.028713360275695658 |
| logistic | 2 | True | 0.0 | 0.314825 | 0.078818 | 0.393643 | {"C":0.01,"min_frequency":25} | False | 0.045189995703829935 |
| logistic | 3 | True | 0.0 | 0.448542 | 0.13033 | 0.578872 | {"C":0.01,"min_frequency":25} | False | 0.07819829471657118 |
| logistic | 4 | True | 0.0 | 0.568876 | 0.214164 | 0.78304 | {"C":0.01,"min_frequency":25} | False | 0.100399186433299 |

## Logistic — {"C":0.01,"min_frequency":50}

*Column panel 1 of 3 — Logistic — {"C":0.01,"min_frequency":50}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.01 | 50 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.01 | 50 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.01 | 50 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.01 | 50 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.01,"min_frequency":50}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8164148772689475 | 0.028756710910821447 | 0.006758219954888032 | 0.03703999506745104 | 0.006870562761549832 | 0.006301414770143428 | 266 | 8.0 |
| logistic | 2 | 0.8259198453621553 | 0.04514961690664095 | 0.00801582768813987 | 0.04184393549005804 | 0.008234635249216984 | 0.008576198681017591 | 347 | 9.0 |
| logistic | 3 | 0.7426781295279269 | 0.07817422788873749 | 0.0260157308899594 | 0.12222427923745642 | 0.026987107800730065 | 0.011214588676928518 | 411 | 9.0 |
| logistic | 4 | 0.6968809179815244 | 0.10038946348572834 | 0.04529761182286585 | 0.210586654238848 | 0.04718088062743161 | 0.01361951483472072 | 449 | 8.0 |

*Column panel 3 of 3 — Logistic — {"C":0.01,"min_frequency":50}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.207412 | 0.032051 | 0.239463 | {"C":0.01,"min_frequency":50} | False | 0.028756710910821447 |
| logistic | 2 | True | 0.0 | 0.32633 | 0.079303 | 0.405633 | {"C":0.01,"min_frequency":50} | False | 0.04514961690664095 |
| logistic | 3 | True | 0.0 | 0.441721 | 0.132316 | 0.574036 | {"C":0.01,"min_frequency":50} | False | 0.07817422788873749 |
| logistic | 4 | True | 0.0 | 0.557483 | 0.174652 | 0.732135 | {"C":0.01,"min_frequency":50} | False | 0.10038946348572834 |

## Logistic — {"C":0.1,"min_frequency":100}

*Column panel 1 of 3 — Logistic — {"C":0.1,"min_frequency":100}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.1 | 100 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.1 | 100 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.1 | 100 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.1 | 100 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.1,"min_frequency":100}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8457886976513419 | 0.04203899703778429 | 0.00672507123081339 | 0.03557448319982211 | 0.006870562761549832 | 0.004880370553159456 | 199 | 10.0 |
| logistic | 2 | 0.8430279107359657 | 0.05409386119940244 | 0.00797978752203351 | 0.0406666931514154 | 0.008234635249216984 | 0.007784975765996633 | 263 | 10.0 |
| logistic | 3 | 0.7328389652553282 | 0.07397602014706171 | 0.02604905739306756 | 0.1256436447060777 | 0.026987107800730065 | 0.01089165352342222 | 311 | 11.0 |
| logistic | 4 | 0.6961657597180593 | 0.10003299228388197 | 0.04535005641031875 | 0.21436988692912087 | 0.04718088062743161 | 0.013240387440168349 | 349 | 11.0 |

*Column panel 3 of 3 — Logistic — {"C":0.1,"min_frequency":100}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209138 | 0.05051 | 0.259648 | {"C":0.1,"min_frequency":100} | False | 0.04203899703778429 |
| logistic | 2 | True | 0.0 | 0.33049 | 0.114013 | 0.444503 | {"C":0.1,"min_frequency":100} | False | 0.05409386119940244 |

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 3 | True | 0.0 | 0.448368 | 0.228131 | 0.676499 | {"C":0.1,"min_frequency":100} | False | 0.07397602014706171 |
| logistic | 4 | True | 0.0 | 0.55163 | 0.344621 | 0.896251 | {"C":0.1,"min_frequency":100} | False | 0.10003299228388197 |

## Logistic — {"C":0.1,"min_frequency":200}

*Column panel 1 of 3 — Logistic — {"C":0.1,"min_frequency":200}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.1 | 200 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.1 | 200 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.1 | 200 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.1 | 200 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.1,"min_frequency":200}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8454533384409069 | 0.04147947840077325 | 0.006727523184398829 | 0.035603947565793236 | 0.006870562761549832 | 0.00487746644916857 | 137 | 10.0 |
| logistic | 2 | 0.8434584202702349 | 0.05437141208562124 | 0.007978293318019941 | 0.04065152422005386 | 0.008234635249216984 | 0.007762275080910802 | 200 | 10.0 |
| logistic | 3 | 0.7328286396872855 | 0.07436169919222316 | 0.026047008294436680 | 0.12563178512710788 | 0.026987107800730065 | 0.010897211272516048 | 242 | 10.0 |
| logistic | 4 | 0.6962107216259916 | 0.10007210837067738 | 0.045349754970030146 | 0.21435173176883374 | 0.04718088062743161 | 0.013241290921581252 | 268 | 10.0 |

*Column panel 3 of 3 — Logistic — {"C":0.1,"min_frequency":200}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209754 | 0.053367 | 0.263121 | {"C":0.1,"min_frequency":200} | False | 0.04147947840077325 |
| logistic | 2 | True | 0.0 | 0.314339 | 0.112327 | 0.426666 | {"C":0.1,"min_frequency":200} | False | 0.05437141208562124 |
| logistic | 3 | True | 0.0 | 0.439707 | 0.191449 | 0.631156 | {"C":0.1,"min_frequency":200} | False | 0.07436169919222316 |
| logistic | 4 | True | 0.0 | 0.561377 | 0.290339 | 0.851716 | {"C":0.1,"min_frequency":200} | False | 0.10007210837067738 |

## Logistic — {"C":0.1,"min_frequency":25}

*Column panel 1 of 3 — Logistic — {"C":0.1,"min_frequency":25}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.1 | 25 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.1 | 25 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.1 | 25 | 143463 | 979 | 47393 | 1279 |

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.1 | 25 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.1,"min_frequency":25}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8452919162485 | 0.041803870353111616 | 0.006726822813738364 | 0.03559485055573405 | 0.006870562761549832 | 0.004880048077180475 | 341 | 10.0 |
| logistic | 2 | 0.8429517339387861 | 0.05453055984817312 | 0.007977567389172158 | 0.04065052543980341 | 0.008234635249216984 | 0.007794868894396074 5 | 443 | 10.0 |
| logistic | 3 | 0.7330862363297703 | 0.07424730322603718 | 0.026044536251404828 | 0.12559513124827795 | 0.026987107800730065 | 0.010894021890686225 | 489 | 11.0 |
| logistic | 4 | 0.6965724623263514 | 0.10030960951176995 | 0.04534431013697259 | 0.2142664417790633 | 0.04718088062743161 | 0.013251235069464611 | 512 | 11.0 |

*Column panel 3 of 3 — Logistic — {"C":0.1,"min_frequency":25}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.216906 | 0.050644 | 0.26755 | {"C":0.1,"min_frequency":25} | False | 0.041803870353111616 |
| logistic | 2 | True | 0.0 | 0.314825 | 0.115399 | 0.430224 | {"C":0.1,"min_frequency":25} | False | 0.05453055984817312 |
| logistic | 3 | True | 0.0 | 0.448542 | 0.230877 | 0.679419 | {"C":0.1,"min_frequency":25} | False | 0.07424730322603718 |
| logistic | 4 | True | 0.0 | 0.568876 | 0.346424 | 0.9153 | {"C":0.1,"min_frequency":25} | False | 0.10030960951176995 |

## Logistic — {"C":0.1,"min_frequency":50}

*Column panel 1 of 3 — Logistic — {"C":0.1,"min_frequency":50}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 0.1 | 50 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 0.1 | 50 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 0.1 | 50 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 0.1 | 50 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":0.1,"min_frequency":50}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8453367663342297 | 0.041896189511325295 | 0.006726411855162318 | 0.03559119183787445 | 0.006870562761549832 | 0.004880711841631607 5 | 266 | 10.0 |
| logistic | 2 | 0.8428104433647639 | 0.05471877531975221 5 | 0.00797678055563866 | 0.04064663949831509 | 0.008234635249216984 | 0.007792979866155341 | 347 | 10.0 |
| logistic | 3 | 0.7330219261826667 | 0.074068392292334 | 0.02604720860332879 | 0.12562010892972714 | 0.026987107800730065 | 0.010889707553511567 | 411 | 11.0 |
| logistic | 4 | 0.6963809641876613 | 0.10017117128758565 | 0.045346236184542685 | 0.21429789069462693 | 0.04718088062743161 | 0.013250033455990408 | 449 | 11.0 |

*Column panel 3 of 3 — Logistic — {"C":0.1,"min_frequency":50}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.207412 | 0.050468 | 0.25788 | {"C":0.1,"min_frequency":50} | False | 0.041896189511325295 |
| logistic | 2 | True | 0.0 | 0.32633 | 0.116789 | 0.44312 | {"C":0.1,"min_frequency":50} | False | 0.054718775319752215 |
| logistic | 3 | True | 0.0 | 0.441721 | 0.22522 | 0.666941 | {"C":0.1,"min_frequency":50} | False | 0.074068392292334 |
| logistic | 4 | True | 0.0 | 0.557483 | 0.350506 | 0.907989 | {"C":0.1,"min_frequency":50} | False | 0.10017117128758565 |

## Logistic — {"C":1,"min_frequency":100}

*Column panel 1 of 3 — Logistic — {"C":1,"min_frequency":100}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 1.0 | 100 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 1.0 | 100 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 1.0 | 100 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 1.0 | 100 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":1,"min_frequency":100}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8354062839671107 | 0.04282494817763704 | 0.006719580320685379 | 0.03580500425329501 | 0.006870562761549832 | 0.004660476217910834 5 | 199 | 13.0 |
| logistic | 2 | 0.8344957406844891 | 0.05148429643459613 | 0.007998522586853019 | 0.04103138531752048 | 0.008234635249216984 | 0.007640709622885178 | 263 | 14.0 |
| logistic | 3 | 0.7239263723587018 | 0.07098133779924334 | 0.026094775031887313 | 0.1279852508734955 | 0.026987107800730065 | 0.010797229475580592 | 311 | 14.0 |
| logistic | 4 | 0.6922468911338548 | 0.09863061450258737 | 0.045393520257984626 | 0.21652909561590966 | 0.04718088062743161 | 0.013202559480406498 | 349 | 14.0 |

*Column panel 3 of 3 — Logistic — {"C":1,"min_frequency":100}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209138 | 0.084285 | 0.293422 | {"C":1,"min_frequency":100} | False | 0.04282494817763704 |
| logistic | 2 | True | 0.0 | 0.33049 | 0.230412 | 0.560902 | {"C":1,"min_frequency":100} | False | 0.05148429643459613 |
| logistic | 3 | True | 0.0 | 0.448368 | 0.361351 | 0.809719 | {"C":1,"min_frequency":100} | False | 0.07098133779924334 |
| logistic | 4 | True | 0.0 | 0.55163 | 0.517237 | 1.068867 | {"C":1,"min_frequency":100} | False | 0.09863061450258737 |

## Logistic — {"C":1,"min_frequency":200}

*Column panel 1 of 3 — Logistic — {"C":1,"min_frequency":200}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 1.0 | 200 | 47221 | 252 | 48031 | 330 |

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 1.0 | 200 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 1.0 | 200 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 1.0 | 200 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":1,"min_frequency":200}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8373984282141345 | 0.04327307831611364 | 0.006718968354205443 | 0.03573985428251407 | 0.006870562761549832 | 0.004635006926717437 | 137 | 13.0 |
| logistic | 2 | 0.8370137894753588 | 0.05454025833990894 | 0.007988955242154669 | 0.04086820150355946 | 0.008234635249216984 | 0.007593841713031645 | 200 | 13.0 |
| logistic | 3 | 0.7243603853156112 | 0.07138112829070936 | 0.026089752382987805 | 0.12784930671436007 | 0.026987107800730065 | 0.010817350380228024 | 242 | 14.0 |
| logistic | 4 | 0.6927682136164565 | 0.09876143871002399 | 0.0453883020849127 | 0.21628081962790713 | 0.047180880627431 61 | 0.013191345472972 | 268 | 13.0 |

*Column panel 3 of 3 — Logistic — {"C":1,"min_frequency":200}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209754 | 0.088818 | 0.298572 | {"C":1,"min_frequency":200} | False | 0.04327307831611364 |
| logistic | 2 | True | 0.0 | 0.314339 | 0.19292 | 0.507258 | {"C":1,"min_frequency":200} | False | 0.05454025833990894 |
| logistic | 3 | True | 0.0 | 0.439707 | 0.398503 | 0.83821 | {"C":1,"min_frequency":200} | False | 0.07138112829070936 |
| logistic | 4 | True | 0.0 | 0.561377 | 0.482833 | 1.044211 | {"C":1,"min_frequency":200} | False | 0.09876143871002399 |

## Logistic — {"C":1,"min_frequency":25}

*Column panel 1 of 3 — Logistic — {"C":1,"min_frequency":25}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 1.0 | 25 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 1.0 | 25 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 1.0 | 25 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 1.0 | 25 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":1,"min_frequency":25}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8345622637985481 | 0.04255649534004288 | 0.006722207378160638 | 0.03584276163484855 6 | 0.006870562761549832 | 0.004651888690661089 | 341 | 14.0 |
| logistic | 2 | 0.832928479470037 | 0.052006362945168 | 0.00799816528542895 | 0.04110372439790961 | 0.008234635249216984 | 0.00766699901735325 | 443 | 14.0 |
| logistic | 3 | 0.7241493130716639 | 0.07136297606701987 | 0.026087607880168538 | 0.1279802294657569 | 0.026987107800730065 | 0.010831698822653607 | 489 | 15.0 |

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 4 | 0.6933543962433939 | 0.0993426312362547 | 0.04537967797896961 | 0.2163424412412417 | 0.04718088062743161 | 0.013242259880725757 | 512 | 16.0 |

*Column panel 3 of 3 — Logistic — {"C":1,"min_frequency":25}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.216906 | 0.098527 | 0.315433 | {"C":1,"min_frequency":25} | False | 0.04255649534004288 |
| logistic | 2 | True | 0.0 | 0.314825 | 0.203168 | 0.517993 | {"C":1,"min_frequency":25} | False | 0.052006362945168 |
| logistic | 3 | True | 0.0 | 0.448542 | 0.415915 | 0.864457 | {"C":1,"min_frequency":25} | False | 0.07136297606701987 |
| logistic | 4 | True | 0.0 | 0.568876 | 0.663874 | 1.23275 | {"C":1,"min_frequency":25} | False | 0.0993426312362547 |

## Logistic — {"C":1,"min_frequency":50}

*Column panel 1 of 3 — Logistic — {"C":1,"min_frequency":50}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 1.0 | 50 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 1.0 | 50 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 1.0 | 50 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 1.0 | 50 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":1,"min_frequency":50}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.8350956367727504 | 0.04303284286278763 | 0.006721155027931703 | 0.03581005998494382 | 0.00687056276154983 | 0.004656415966934532 | 266 | 14.0 |
| logistic | 2 | 0.8332092167813585 | 0.05198178489843357 | 0.007994726480610427 | 0.04105672340017022 | 0.00823463524921698 | 0.007642434208693352 | 347 | 14.0 |
| logistic | 3 | 0.724136308620615 | 0.07113647298336762 | 0.026091870250022815 | 0.12798800620427322 | 0.02698710780073006 | 0.01081267449200683 | 411 | 15.0 |
| logistic | 4 | 0.6926957749870101 | 0.09911332728796443 | 0.045381833108473095 | 0.21642357030847956 | 0.04718088062743161 | 0.013227016098984726 | 449 | 16.0 |

*Column panel 3 of 3 — Logistic — {"C":1,"min_frequency":50}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.207412 | 0.099212 | 0.306624 | {"C":1,"min_frequency":50} | False | 0.043032842862787635 |
| logistic | 2 | True | 0.0 | 0.32633 | 0.210598 | 0.536928 | {"C":1,"min_frequency":50} | False | 0.05198178489843357 |
| logistic | 3 | True | 0.0 | 0.441721 | 0.519805 | 0.961525 | {"C":1,"min_frequency":50} | False | 0.07113647298336762 |
| logistic | 4 | True | 0.0 | 0.557483 | 0.677467 | 1.23495 | {"C":1,"min_frequency":50} | False | 0.09911332728796443 |

## Logistic — {"C":10,"min_frequency":100}

*Column panel 1 of 3 — Logistic — {"C":10,"min_frequency":100}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 10.0 | 100 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 10.0 | 100 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 10.0 | 100 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 10.0 | 100 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":10,"min_frequency":100}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.7982099352468947 | 0.038385758461090984 | 0.006743239370951493 | 0.037194317697413465 | 0.006870562761549832 | 0.004693948203311509 | 199 | 18.0 |
| logistic | 2 | 0.8121143022832283 | 0.049588920767196044 | 0.008020871066917045 | 0.04200866110778667 | 0.008234635249216984 | 0.00762885492934542 | 263 | 18.0 |
| logistic | 3 | 0.708322794415431 | 0.06831649357625347 | 0.026128407822780315 | 0.1308749622534821 | 0.026987107800730065 | 0.010720291645965111 | 311 | 18.0 |
| logistic | 4 | 0.6888306156169353 | 0.09774189134912249 | 0.04541435443696143 | 0.2180077191460469 | 0.04718088062743161 | 0.013214949610501393 | 349 | 18.0 |

*Column panel 3 of 3 — Logistic — {"C":10,"min_frequency":100}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209138 | 0.149928 | 0.359066 | {"C":10,"min_frequency":100} | False | 0.038385758461090984 |
| logistic | 2 | True | 0.0 | 0.33049 | 0.368527 | 0.699017 | {"C":10,"min_frequency":100} | False | 0.049588920767196044 |
| logistic | 3 | True | 0.0 | 0.448368 | 0.631729 | 1.080097 | {"C":10,"min_frequency":100} | False | 0.06831649357625347 |
| logistic | 4 | True | 0.0 | 0.55163 | 0.928539 | 1.48017 | {"C":10,"min_frequency":100} | False | 0.09774189134912249 |

## Logistic — {"C":10,"min_frequency":200}

*Column panel 1 of 3 — Logistic — {"C":10,"min_frequency":200}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 10.0 | 200 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 10.0 | 200 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 10.0 | 200 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 10.0 | 200 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":10,"min_frequency":200}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.811564588252708 | 0.039825368873288254 | 0.006729338997882537 | 0.03665685673510947 | 0.006870562761549832 | 0.0046204091726492 | 137 | 23.0 |
| logistic | 2 | 0.8228246756770226 | 0.05207781885273945 | 0.008000100421871392 | 0.04140125262319715 | 0.008234635249216984 | 0.007564854465424989 5 | 200 | 16.0 |
| logistic | 3 | 0.7135090271405775 | 0.0696390838621968 | 0.026113423485278 26 | 0.12991809327731518 | 0.026987107800730065 | 0.010753771846139 57 | 242 | 18.0 |
| logistic | 4 | 0.6909607637695324 | 0.09818416769316052 | 0.04540100703586779 | 0.2171157796034479 | 0.04718088062743161 | 0.013191108254085442 | 268 | 16.0 |

*Column panel 3 of 3 — Logistic — {"C":10,"min_frequency":200}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.209754 | 0.161446 | 0.3712 | {"C":10,"min_frequency":200} | False | 0.039825368873288254 |
| logistic | 2 | True | 0.0 | 0.314339 | 0.34493 | 0.659269 | {"C":10,"min_frequency":200} | False | 0.05207781885273945 |
| logistic | 3 | True | 0.0 | 0.439707 | 0.642069 | 1.081776 | {"C":10,"min_frequency":200} | False | 0.0696390838621968 |
| logistic | 4 | True | 0.0 | 0.561377 | 0.935069 | 1.496446 | {"C":10,"min_frequency":200} | False | 0.09818416769316052 |

## Logistic — {"C":10,"min_frequency":25}

*Column panel 1 of 3 — Logistic — {"C":10,"min_frequency":25}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 10.0 | 25 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 10.0 | 25 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 10.0 | 25 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 10.0 | 25 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":10,"min_frequency":25}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.7919174555136066 | 0.03754138743547223 | 0.006757916407711635 | 0.03762743952225827 | 0.006870562761549832 | 0.004658906641928515 5 | 341 | 32.0 |
| logistic | 2 | 0.8020890986156578 | 0.044955407637677504 | 0.008059981730703514 | 0.04269208870295223 | 0.008234635249216984 | 0.007723261566648074 | 443 | 20.0 |
| logistic | 3 | 0.7064715167086172 | 0.06777865913235956 | 0.026137610809132225 | 0.13158481006251285 | 0.026987107800730065 | 0.010822842390979 64 | 489 | 20.0 |
| logistic | 4 | 0.6884901116416429 | 0.09819976621592727 | 0.04541195815155373 | 0.21858411615127094 | 0.04718088062743161 | 0.013269576118371026 | 512 | 21.0 |

*Column panel 3 of 3 — Logistic — {"C":10,"min_frequency":25}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.216906 | 0.24324 | 0.460146 | {"C":10,"min_frequency":25} | False | 0.03754138743547223 |
| logistic | 2 | True | 0.0 | 0.314825 | 0.418393 | 0.733219 | {"C":10,"min_frequency":25} | False | 0.044955407637677504 |

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 3 | True | 0.0 | 0.448542 | 0.682289 | 1.130831 | {"C":10,"min_frequency":25} | False | 0.06777865913235956 |
| logistic | 4 | True | 0.0 | 0.568876 | 1.151711 | 1.720587 | {"C":10,"min_frequency":25} | False | 0.09819976621592727 |

## Logistic — {"C":10,"min_frequency":50}

*Column panel 1 of 3 — Logistic — {"C":10,"min_frequency":50}*

| model | fold_id | notice | train_vintages | validation_vintage | C | min_frequency | train_n | train_positive_n | validation_n | validation_positive_n |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 10.0 | 50 | 47221 | 252 | 48031 | 330 |
| logistic | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 10.0 | 50 | 95252 | 582 | 48211 | 397 |
| logistic | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 10.0 | 50 | 143463 | 979 | 47393 | 1279 |
| logistic | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 10.0 | 50 | 190856 | 2258 | 48579 | 2292 |

*Column panel 2 of 3 — Logistic — {"C":10,"min_frequency":50}*

| model | fold_id | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence | mean_predicted_probability | transformed_feature_count | convergence_iterations |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | 0.7970101636901076 | 0.03877469238388885 | 0.006751545132921436 | 0.03733768340006385 | 0.006870562761549832 | 0.004679526180182927 | 266 | 30.0 |
| logistic | 2 | 0.8055300140268561 | 0.04894972621305807 | 0.008026904202178869 | 0.04233925293915237 | 0.008234635249216984 | 0.007629199053116128 | 347 | 20.0 |
| logistic | 3 | 0.7069883715792487 | 0.068086703922116 | 0.026130240352032565 | 0.13130515732252074 | 0.026987107800730065 | 0.010783941297651698 | 411 | 19.0 |
| logistic | 4 | 0.6880764338107364 | 0.09807878005898535 | 0.045407357073427 | 0.21843628272086463 | 0.04718088062743161 | 0.013243976808651444 | 449 | 20.0 |

*Column panel 3 of 3 — Logistic — {"C":10,"min_frequency":50}*

| model | fold_id | converged | convergence_warning_n | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected | average_precision |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 1 | True | 0.0 | 0.207412 | 0.213602 | 0.421014 | {"C":10,"min_frequency":50} | False | 0.03877469238388885 |
| logistic | 2 | True | 0.0 | 0.32633 | 0.388114 | 0.714444 | {"C":10,"min_frequency":50} | False | 0.04894972621305807 |
| logistic | 3 | True | 0.0 | 0.441721 | 0.685249 | 1.12697 | {"C":10,"min_frequency":50} | False | 0.068086703922116 |
| logistic | 4 | True | 0.0 | 0.557483 | 1.098467 | 1.65595 | {"C":10,"min_frequency":50} | False | 0.09807878005898535 |

## Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 330 | 0.8264332175235511 | 0.033957877533911524 | 0.00674304799859196 | 0.03647235941879935 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 397 | 0.8320333757626505 | 0.04950607756949375 | 0.008016709290731746 | 0.04166044899997651 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 1279 | 0.7060205318410169 | 0.06510356885790625 | 0.02622283401143734 | 0.1260124252230204 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2292 | 0.6805490469187784 | 0.09483804665151033 | 0.04559591410880351 | 0.20943918119742488 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.004869370020362729 | 137 | 0.21007 | 0.838712 | 1.048781 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007372596038643426 5 | 200 | 0.321589 | 1.288576 | 1.610165 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.010559800746017391 | 242 | 0.434741 | 2.002462 | 2.437203 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012835566570718658 | 268 | 0.569217 | 3.021938 | 3.591155 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.033957877533911524 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.04950607756949375 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.06510356885790625 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.09483804665151033 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |

## Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 330 | 0.8395403056793804 | 0.03646083735848914 | 0.006743476927915912 | 0.036272883785083976 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 397 | 0.8377407352736184 | 0.051809740201821976 | 0.008032119240318828 | 0.041781999348822596 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 1279 | 0.7063873692633035 | 0.0666952605308615 | 0.026276700873730928 | 0.12670881392623784 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2292 | 0.6802369622626505 | 0.09462802343071144 | 0.04562238116710027 | 0.20923072811753643 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004813301013531823 | 137 | 0.21007 | 0.749091 | 0.95916 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.007210774651812034 6 | 200 | 0.321589 | 1.317176 | 1.638765 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.010038123862833821 | 242 | 0.434741 | 1.919569 | 2.354309 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.0128356175806574 | 268 | 0.569217 | 2.852741 | 3.421957 | {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.03646083735848914 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.051809740201821976 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.0666952605308615 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.09462802343071144 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |

Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 330 | 0.8182135816986238 | 0.03411758036962895 | 0.006753514996010268 | 0.036951870131287035 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 397 | 0.8287550867504105 | 0.045886402006589715 | 0.00804058585267298 | 0.042244684175166955 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 1279 | 0.6986508907811598 | 0.06347426479718779 | 0.026292075364201854 | 0.127663181716197 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2292 | 0.678925818356682 | 0.09427792686767167 | 0.04565352786340941 | 0.2093287286642748 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.004869786407794138 | 266 | 0.20236 | 0.651676 | 0.854037 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007232909925381738 | 347 | 0.334008 | 1.210391 | 1.544399 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.009859348938208752 | 411 | 0.436965 | 1.765066 | 2.202031 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012781920756560315 | 449 | 0.574631 | 2.420721 | 2.995352 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.03411758036962895 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.045886402006589715 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.06347426479718779 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.09427792686767167 | 400.0 | 12.0 | 5.0 | sqrt | 400.0 |

## Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 330 | 0.8372324320753075 | 0.03721177897439956 | 0.006753910108969777 | 0.03671172241810044 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 397 | 0.8310992880788369 | 0.050636334587852175 | 0.008051722870390158 | 0.04239686274434906 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 1279 | 0.7016678047398122 | 0.0647286376288503 | 0.026329143421651165 | 0.12780192487448597 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2292 | 0.6794528718329992 | 0.09303359096889356 | 0.045685519395295845 | 0.2093828961626628 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004869468603723089 | 266 | 0.20236 | 0.647266 | 0.849627 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.007086196727865568 | 347 | 0.334008 | 1.084061 | 1.418069 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009698919558492383 | 411 | 0.436965 | 1.758332 | 2.195297 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.012738888086127472 | 449 | 0.574631 | 2.280669 | 2.8553 | {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.03721177897439956 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.050636334587852175 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.0647286376288503 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.09303359096889356 | 400.0 | 12.0 | 50.0 | sqrt | 400.0 |

## Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 330 | 0.8220645587126373 | 0.02994287783281559 | 0.00675177577108881 | 0.03666922904084889 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 397 | 0.8270755095390102 | 0.04363829762729724 | 0.008026815838825228 | 0.041556959194166715 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 1279 | 0.7092103185283452 | 0.06579435989637236 | 0.02614123063697413 | 0.13057387952098568 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2292 | 0.669081677255243 | 0.09192658940333451 | 0.045459457441712085 | 0.22246088234997274 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.00505094272844287 | 137 | 0.21007 | 2.116949 | 2.327018 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.007821382213659236 | 200 | 0.321589 | 5.206055 | 5.527644 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011404911913950048 | 242 | 0.434741 | 9.671614 | 10.106355 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.01294557592037249 | 268 | 0.569217 | 17.931452 | 18.500669 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.02994287783281559 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.04363829762729724 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.06579435989637236 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.09192658940333451 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |

Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 330 | 0.8397111298727618 | 0.03552224528774166 | 0.006730801716680959 | 0.03579628858950473 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 397 | 0.8361900159086233 | 0.05032636077159142 | 0.007982870240909195 | 0.040852651024121736 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 1279 | 0.7184356964483741 | 0.06895192159599234 | 0.026103398878162352 | 0.12866326356478033 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2292 | 0.6743092766954306 | 0.0937089012419573 | 0.04540690652352954 | 0.2221389084864037 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004883758232971338 | 137 | 0.21007 | 1.926968 | 2.137037 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007473811142344571 | 200 | 0.321589 | 5.27127 | 5.592859 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010397204484506673 | 242 | 0.434741 | 9.727546 | 10.162287 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.012943746833412 52 | 268 | 0.569217 | 17.93138 | 18.500597 | {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.03552224528774166 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.05032636077159142 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.06895192159599234 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.0937089012419573 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |

Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 330 | 0.8217385697396599 | 0.02850874810486173 | 0.006761612374386131 | 0.03676871316408859 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 397 | 0.8250054867312768 | 0.04258689315491025 | 0.00803102554195993 | 0.04167159301083173 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 1279 | 0.7082410545738316 | 0.06613228316315009 | 0.026159547816396195 | 0.1317089124391802 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2292 | 0.6667458872862089 | 0.09227081980556244 | 0.04546274878368069 | 0.2222147819949983 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.004956927988468189 | 266 | 0.20236 | 2.133238 | 2.335599 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.007664992296825264 | 347 | 0.334008 | 5.629285 | 5.963293 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.01090370172987723 | 411 | 0.436965 | 9.517876 | 9.954841 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012860893849419287 | 449 | 0.574631 | 16.857599 | 17.43223 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.02850874810486173 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.04258689315491025 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.06613228316315009 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.09227081980556244 | 400.0 | 12.0 | 5.0 | 0.5 | 400.0 |

Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 330 | 0.8393253301976389 | 0.03389874062609588 | 0.006731297703241873 | 0.03579793122426348 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 397 | 0.8346255994708294 | 0.04974666207330165 | 0.007984616709625986 | 0.04091541304747387 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 1279 | 0.7195199658676396 | 0.06903321120337483 | 0.026099961451835827 | 0.1286075023897751 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2292 | 0.6754137843444409 | 0.09400567841743172 | 0.04540389338379716 | 0.22187315003447333 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.004896315437906641 | 266 | 0.20236 | 1.995831 | 2.198192 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007471742662748061 | 347 | 0.334008 | 5.784798 | 6.118806 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.010406043294216641 | 411 | 0.436965 | 10.63196 | 11.068925 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012959757668006737 | 449 | 0.574631 | 17.753113 | 18.327744 | {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":12,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | max_depth | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.03389874062609588 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.04974666207330165 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.06903321120337483 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.09400567841743172 | 400.0 | 12.0 | 50.0 | 0.5 | 400.0 |

## Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 330 | 0.8164744338629581 | 0.03164620819665059 | 0.006738140609874207 | 0.03653747326257436 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 397 | 0.8214685074268163 | 0.044904625484053516 | 0.00800815324912326 | 0.04149415577173598 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 1279 | 0.7093206189928803 | 0.06889328223302982 | 0.02604505905964805 | 0.12446043453523174 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2292 | 0.678161381088045 | 0.09531019014448162 | 0.045389477206279706 | 0.21494450352213762 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.00496772379950651 | 137 | 0.21007 | 0.966483 | 1.176553 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.007893536665995628 | 200 | 0.321589 | 2.580287 | 2.901877 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.01242939093866895 | 242 | 0.434741 | 5.041728 | 5.476469 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.013283893181886242 | 268 | 0.569217 | 10.435363 | 11.00458 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.03164620819665059 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.044904625484053516 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.06889328223302982 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.09531019014448162 | 400.0 | 5.0 | sqrt | 400.0 |

## Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 330 | 0.8407280070997811 | 0.03639981049866016 | 0.00673890487652875 | 0.03604234186228811 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 397 | 0.8369533643118976 | 0.052762868326629464 | 0.008019730834344142 | 0.04138148891503908 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 1279 | 0.7125919844497284 | 0.06704730264720829 | 0.026207042173596318 | 0.12492488089923727 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2292 | 0.6845073820666121 | 0.09519682804863408 | 0.04551238304266364 | 0.20984763703907838 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.00483057846055267 | 137 | 0.21007 | 0.909332 | 1.119401 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007441555457873739 | 200 | 0.321589 | 1.87277 | 2.194359 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010829348891769851 | 242 | 0.434741 | 3.206962 | 3.641703 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.012989613938273166 | 268 | 0.569217 | 5.481556 | 6.050773 | {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | True |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.03639981049866016 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.052762868326629464 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.06704730264720829 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.09519682804863408 | 400.0 | 50.0 | sqrt | 400.0 |

Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 330 | 0.8162208339447811 | 0.03215471981731033 | 0.006739231728404201 | 0.03656749061417764 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 397 | 0.8219945277033307 | 0.045602144339792774 | 0.008011730012445742 | 0.041495655428180594 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 1279 | 0.7100054550874582 | 0.0686582950263595 | 0.026068721098404773 | 0.12374518110906547 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2292 | 0.6779941925427632 | 0.09526529020864796 | 0.04541365044800027 | 0.21336224043755808 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.004884203915025282 | 266 | 0.20236 | 1.154884 | 1.357245 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.007787591123225894 | 347 | 0.334008 | 2.671986 | 3.005994 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.012191144296709264 | 411 | 0.436965 | 5.256216 | 5.693181 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.013275805932464536 | 449 | 0.574631 | 9.103905 | 9.678535 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.03215471981731033 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.045602144339792774 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.0686582950263595 | 400.0 | 5.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.09526529020864796 | 400.0 | 5.0 | sqrt | 400.0 |

## Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 330 | 0.8377629463329973 | 0.03677017564967788 | 0.006747159158126422 | 0.03635517091205998 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 397 | 0.8361842209932084 | 0.05077410995829808 | 0.0080327335502495 | 0.04168527403689397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 1279 | 0.7097578449139015 | 0.06674338032214995 | 0.026243802377149048 | 0.12544017169690447 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2292 | 0.684268094227038 | 0.09494034012373534 | 0.04555650109200142 | 0.20905407283179628 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004808235924459073 | 266 | 0.20236 | 0.753124 | 0.955485 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.007356954481049943 | 347 | 0.334008 | 1.611038 | 1.945046 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010622399921085043 | 411 | 0.436965 | 2.931424 | 3.368389 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012962301813836641 | 449 | 0.574631 | 4.149641 | 4.724272 | {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":"sqrt","min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.03677017564967788 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.05077410995829808 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.06674338032214995 | 400.0 | 50.0 | sqrt | 400.0 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.09494034012373534 | 400.0 | 50.0 | sqrt | 400.0 |

Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 330 | 0.7884484665526992 | 0.027325214700250674 | 0.006773235578321084 | 0.05341876978492354 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 397 | 0.7995407845620082 | 0.03864814769632911 | 0.00810367576939308 | 0.05442014892976078 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 1279 | 0.6994192300327335 | 0.06626668017073269 | 0.026056078114020276 | 0.1715179117376807 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2292 | 0.6570001392405249 | 0.09033935393278951 | 0.04539623244480453 | 0.38806925431900624 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.0053879276324808635 | 137 | 0.21007 | 3.050003 | 3.260072 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.008567593198189477 | 200 | 0.321589 | 10.009474 | 10.331063 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.014047380358366026 | 242 | 0.434741 | 21.294606 | 21.729346 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.013864895522341678 | 268 | 0.569217 | 48.086102 | 48.655318 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.027325214700250674 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.03864814769632911 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.06626668017073269 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.09033935393278951 | 400.0 | 5.0 | 0.5 | 400.0 |

## Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 330 | 0.8356975871797363 | 0.03544773709454352 | 0.006730109193053086 | 0.036032649265242606 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 397 | 0.8313473894801634 | 0.05022745276762364 | 0.007985048994619626 | 0.04112217987144976 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 1279 | 0.7165769161736477 | 0.06992349126523086 | 0.026074346999320184 | 0.12892183026667878 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2292 | 0.673180233229576 | 0.09375945514380349 | 0.045385817900779946 | 0.23799991294002756 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.004905369477546337 | 137 | 0.21007 | 2.319942 | 2.530012 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007552111412888347 | 200 | 0.321589 | 7.320992 | 7.642581 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.010826607040696535 | 242 | 0.434741 | 15.089287 | 15.524027 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.013036216313765335 | 268 | 0.569217 | 29.205352 | 29.774569 | {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":200,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.03544773709454352 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.05022745276762364 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.06992349126523086 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.09375945514380349 | 400.0 | 50.0 | 0.5 | 400.0 |

## Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 330 | 0.7860221785579744 | 0.0250521570405602 | 0.006792747185564058 | 0.05636092214403732 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 397 | 0.7917280269187519 | 0.03583092029263312 | 0.00812320111361961 | 0.05900161395562077 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 1279 | 0.7000703681527878 | 0.06649201244226516 | 0.0260719142353332 | 0.1803146644229054 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2292 | 0.6543975611454611 | 0.08953186270072319 | 0.04541118514415433 | 0.39942308077516275 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.00527069139919047 2 | 266 | 0.20236 | 3.883657 | 4.086017 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.00851874247225599 4 | 347 | 0.334008 | 11.311221 | 11.645229 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.01342200328642529 8 | 411 | 0.436965 | 25.296859 | 25.733824 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.01383608726658748 7 | 449 | 0.574631 | 50.077471 | 50.652102 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":5,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.0250521570405602 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.03583092029263312 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.06649201244226516 | 400.0 | 5.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.08953186270072319 | 400.0 | 5.0 | 0.5 | 400.0 |

## Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}

*Column panel 1 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 330 | 0.8338463141297463 | 0.0333784907835279 | 0.006731779916207558 | 0.03666905544388343 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 397 | 0.8300060246047893 | 0.05046025028795234 | 0.007985220704522986 | 0.04112502706497617 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 1279 | 0.7169822159130195 | 0.06987043228473347 | 0.02606819417230029 | 0.12960010950354175 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2292 | 0.6731729658016901 | 0.09368649663972416 | 0.04538680553159157 | 0.24029371402791275 | 0.04718088062743161 |

*Column panel 3 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004903949945948333 | 266 | 0.20236 | 2.64344 | 2.845801 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.007562386031294974 | 347 | 0.334008 | 7.62302 | 7.957028 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.010867405965686814 | 411 | 0.436965 | 15.95885 | 16.395815 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.013032327214514881 | 449 | 0.574631 | 28.25499 | 28.829621 | {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400} | False |

*Column panel 4 of 4 — Random Forest — {"max_depth":null,"max_features":0.5,"min_frequency":50,"min_samples_leaf":50,"n_estimators":400}*

| model | config_id | fold_id | average_precision | n_estimators | min_samples_leaf | max_features | tree_count |
|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.0333784907835279 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.05046025028795234 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.06987043228473347 | 400.0 | 50.0 | 0.5 | 400.0 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.09368649663972416 | 400.0 | 50.0 | 0.5 | 400.0 |

XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 330 | 0.837003417119138 | 0.040513087056998826 | 0.006717824377119541 | 0.03590932860970497 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 397 | 0.8374946620926874 | 0.047031218164936155 | 0.008000323548913002 | 0.04093410074710846 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 1279 | 0.7221474499254881 | 0.07145356557684987 | 0.026098906993865967 | 0.12932616472244263 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2292 | 0.6871092626394144 | 0.09981883982278371 | 0.04532662034034729 | 0.21850892901420593 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.00463639199733734 1 | 137 | 0.135983 | 0.850686 | 1.006669 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.00732619315385818 5 | 200 | 0.212856 | 0.933148 | 1.170647 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.0099892383441329 | 242 | 0.286644 | 1.023756 | 1.337827 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.01294004265218973 2 | 268 | 0.360717 | 1.121801 | 1.512598 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.040513087056998826 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_200::fold_1 | 2.0 | complete | 2026-09-25T17:36:26+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.047031218164936155 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_200::fold_2 | 2.0 | complete | 2026-09-25T17:36:27+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.07145356557684987 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_200::fold_3 | 2.0 | complete | 2026-09-25T17:36:29+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.09981883982278371 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_200::fold_4 | 2.0 | complete | 2026-09-25T17:36:30+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.03 | 1.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 330 | 0.8368806193631668 | 0.04126833008795586 | 0.006717883516103029 | 0.03590044379234314 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 397 | 0.8373987825830973 | 0.050055240458039184 | 0.007993252016603947 | 0.04091592878103256 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 1279 | 0.7227992747212495 | 0.07042842911597287 | 0.026105279102921486 | 0.12932512164115906 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2292 | 0.6882878773157126 | 0.10049215516139583 | 0.045312151312828064 | 0.21829888224601746 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004652868490666151 | 266 | 0.140903 | 0.859989 | 1.021417 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.00730919698253274 | 347 | 0.213836 | 0.940614 | 1.178762 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.010019875131547451 | 411 | 0.287511 | 1.025026 | 1.339839 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.012949401512742043 | 449 | 0.361403 | 1.131984 | 1.523283 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.04126833008795586 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_50::fold_1 | 1.0 | complete | 2026-09-25T17:36:21+01:00 |

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.05005524045803918 4 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_50::fold_2 | 1.0 | complete | 2026-09-25T17:36:22+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.07042842911597287 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_50::fold_3 | 1.0 | complete | 2026-09-25T17:36:23+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.10049215516139583 | 3.0 | xgb_lr_0p03_depth_3_child_1_freq_50::fold_4 | 1.0 | complete | 2026-09-25T17:36:25+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.03 | 1.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 330 | 0.8332574185281676 | 0.04030083074232704 | 0.006713737733662128 4 | 0.03586766496300697 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 397 | 0.837522899135072 | 0.046179024536297714 | 0.008006484247744083 | 0.04099542647600174 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 1279 | 0.7236771938517396 | 0.07083513974854984 | 0.026086924597620964 | 0.1286497563123703 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2292 | 0.6879870378495562 | 0.10004652405116389 | 0.04531946778297424 | 0.21859955787658691 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.0046556126326322556 | 137 | 0.136994 | 0.849856 | 1.007459 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.0073189553804695 61 | 200 | 0.217374 | 0.937583 | 1.179093 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.0103141367435455 32 | 242 | 0.300201 | 1.028204 | 1.362918 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True |

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012954357080161572 | 268 | 0.370155 | 1.136328 | 1.537205 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200} | True |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.04030083074232704 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_200::fold_1 | 4.0 | complete | 2026-09-25T17:36:37+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.046179024536297714 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_200::fold_2 | 4.0 | complete | 2026-09-25T17:36:38+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.07083513974854984 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_200::fold_3 | 4.0 | complete | 2026-09-25T17:36:39+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.10004652405116389 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_200::fold_4 | 4.0 | complete | 2026-09-25T17:36:41+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.03 | 10.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 330 | 0.8349788423214557 | 0.03980194167578516 | 0.006715235766023397 | 0.03585707023739815 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 397 | 0.8370358628349843 | 0.04828976318055842 | 0.00799594447016716 | 0.04095526039600372 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 1279 | 0.7231142638888978 | 0.07081027602142484 | 0.02609255351126194 | 0.12891259789466858 | 0.026987107800730065 |

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2292 | 0.6888437931320902 | 0.10040856992679877 | 0.04531169310212135 | 0.21846966445446014 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.004625238943845034 | 266 | 0.133989 | 0.846066 | 1.000524 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.007349071558564901 | 347 | 0.20877 | 0.947652 | 1.180987 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.010261072777211666 | 411 | 0.287209 | 1.054194 | 1.368266 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.012942951172590256 | 449 | 0.497896 | 1.260887 | 1.795168 | {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.03980194167578516 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_50::fold_1 | 3.0 | complete | 2026-09-25T17:36:31+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.04828976318055842 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_50::fold_2 | 3.0 | complete | 2026-09-25T17:36:32+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.07081027602142484 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_50::fold_3 | 3.0 | complete | 2026-09-25T17:36:34+01:00 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.10040856992679877 | 3.0 | xgb_lr_0p03_depth_3_child_10_freq_50::fold_4 | 3.0 | complete | 2026-09-25T17:36:36+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.03 | 10.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 330 | 0.8100482614874346 | 0.03315854452444982 | 0.006769730243831873 | 0.03832545503973961 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 397 | 0.8211621407850467 | 0.03703475273468293 | 0.00809402298182249 | 0.04248432815074921 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 1279 | 0.7151232745662133 | 0.0662678453874736 | 0.026181930676102638 | 0.13242517411708832 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2292 | 0.6848190991096562 | 0.09892614803849645 | 0.04534348100423813 | 0.22292259335517883 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.00392463710159063 | 137 | 0.138587 | 1.454503 | 1.613784 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.00677322596311569 | 200 | 0.222259 | 1.774316 | 2.02538 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.01019413582980632 | 242 | 0.341699 | 1.733793 | 2.109583 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.01290087588131427 | 268 | 0.369724 | 1.828358 | 2.229236 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.03315854452444982 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_200::fold_1 | 6.0 | complete | 2026-09-25T17:36:50+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.03703475273468293 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_200::fold_2 | 6.0 | complete | 2026-09-25T17:36:52+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.0662678453874736 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_200::fold_3 | 6.0 | complete | 2026-09-25T17:36:54+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.09892614803849645 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_200::fold_4 | 6.0 | complete | 2026-09-25T17:36:57+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.03 | 1.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 330 | 0.8121216568104475 | 0.03650364072812492 | 0.006751677021384239 | 0.037977416068315506 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 397 | 0.8203456108625795 | 0.03754644164580798 | 0.008093895390629768 | 0.04246950149536133 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 1279 | 0.7161201411886636 | 0.06826282067936575 | 0.02615066058933735 | 0.13241852819919586 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2292 | 0.686136959023885 | 0.09921855200905794 | 0.04533864185214043 | 0.22232882678508759 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.004025229252874851 | 266 | 0.136458 | 1.471665 | 1.629681 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.006838262081146246 | 347 | 0.209999 | 1.586949 | 1.820993 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.010179199278354645 | 411 | 0.289416 | 1.696881 | 2.014138 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.012887703254818916 | 449 | 0.371149 | 1.828531 | 2.230485 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.03650364072812492 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_50::fold_1 | 5.0 | complete | 2026-09-25T17:36:42+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.03754644164580798 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_50::fold_2 | 5.0 | complete | 2026-09-25T17:36:44+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.06826282067936575 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_50::fold_3 | 5.0 | complete | 2026-09-25T17:36:46+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.09921855200905794 | 6.0 | xgb_lr_0p03_depth_6_child_1_freq_50::fold_4 | 5.0 | complete | 2026-09-25T17:36:49+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.03 | 1.0 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.03 | 1.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 330 | 0.8189303572188626 | 0.0372632509400017 | 0.006740559358149767 | 0.03675711899995804 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 397 | 0.8279858380696232 | 0.03955292269164933 | 0.008061904460191727 | 0.04187261313199997 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 1279 | 0.714530419106499 | 0.06821220928934008 | 0.026112161576747894 | 0.13074688613414764 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2292 | 0.6860383727356119 | 0.09951480473546617 | 0.045311328023672104 | 0.2219030112028122 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.00461178692057728 | 137 | 0.140479 | 1.443147 | 1.603852 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.00716021843254566 2 | 200 | 0.217366 | 1.56951 | 1.812232 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.01081143040210008 6 | 242 | 0.29212 | 1.889835 | 2.209809 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.01312554534524679 2 | 268 | 0.373534 | 1.847035 | 2.252447 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.0372632509400017 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_200::fold_1 | 8.0 | complete | 2026-09-25T17:37:06+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.03955292269164933 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_200::fold_2 | 8.0 | complete | 2026-09-25T17:37:08+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.06821220928934008 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_200::fold_3 | 8.0 | complete | 2026-09-25T17:37:10+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.09951480473546617 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_200::fold_4 | 8.0 | complete | 2026-09-25T17:37:12+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.03 | 10.0 |

## XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 330 | 0.8206393932405965 | 0.03531623479199687 | 0.00674152048304677 | 0.03671201691031456 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 397 | 0.8270640250702791 | 0.04037889543368633 | 0.008059804327785969 | 0.041913144290447235 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 1279 | 0.7165074432425228 | 0.06962250867137613 | 0.026094552129507065 | 0.13016191124916077 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2292 | 0.6855360106047514 | 0.09942289053493136 | 0.04531338810920715 | 0.22200247645378113 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.00460507255047559 7 | 266 | 0.137822 | 1.43273 | 1.591326 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.00717154424637556 1 | 347 | 0.214518 | 1.589653 | 1.829325 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.01092441473156213 8 | 411 | 0.292897 | 1.723924 | 2.044399 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.01312687713652849 2 | 449 | 0.371542 | 1.849748 | 2.251788 | {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.03531623479199687 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_50::fold_1 | 7.0 | complete | 2026-09-25T17:36:58+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.04037889543368633 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_50::fold_2 | 7.0 | complete | 2026-09-25T17:37:00+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.06962250867137613 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_50::fold_3 | 7.0 | complete | 2026-09-25T17:37:02+01:00 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.09942289053493136 | 6.0 | xgb_lr_0p03_depth_6_child_10_freq_50::fold_4 | 7.0 | complete | 2026-09-25T17:37:04+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.03,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.03 | 10.0 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.03 | 10.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 330 | 0.8026997401109054 | 0.03562876057810091 | 0.006766169797629118 | 0.03836836665868759 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 397 | 0.8191389777705991 | 0.03628506515946588 | 0.00813412107527256 | 0.04276391118764877 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 1279 | 0.7083683439040135 | 0.06581267234164852 | 0.026216965168714523 | 0.13374321162700653 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2292 | 0.6835262934409795 | 0.09960924612907143 | 0.04530627280473709 | 0.22204966843128204 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004332290962338448 | 137 | 0.139202 | 0.836909 | 0.996599 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.00696936110034585 | 200 | 0.216917 | 0.91698 | 1.15885 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.01029791496694088 | 242 | 0.29363 | 1.011391 | 1.333327 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.01314939558506012 | 268 | 0.370428 | 1.102078 | 1.503599 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.03562876057810091 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_200::fold_1 | 10.0 | complete | 2026-09-25T17:37:19+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.03628506515946588 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_200::fold_2 | 10.0 | complete | 2026-09-25T17:37:20+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.06581267234164852 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_200::fold_3 | 10.0 | complete | 2026-09-25T17:37:21+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.09960924612907143 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_200::fold_4 | 10.0 | complete | 2026-09-25T17:37:23+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.1 | 1.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 330 | 0.8054408998477257 | 0.035872470596354046 | 0.006781482137739658 | 0.03837093338370323 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 397 | 0.8183062220849706 | 0.038040549821818856 | 0.008136788383126259 | 0.04272044822573662 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 1279 | 0.7106979378670727 | 0.06685521836852976 | 0.026205597445368767 | 0.1329224556684494 | 0.026987107800730065 |

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2292 | 0.6873337045659921 | 0.10066496623995196 | 0.04529505968093872 | 0.22117622196674347 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.0043994891457259655 | 266 | 0.137477 | 0.834387 | 0.992518 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.0071669882163405442 | 347 | 0.210641 | 0.923677 | 1.159947 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.0105922687798738448 | 411 | 0.289619 | 1.009377 | 1.326405 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.0131913730874657633 | 449 | 0.368544 | 1.108387 | 1.507868 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.035872470596354046 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_50::fold_1 | 9.0 | complete | 2026-09-25T17:37:13+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.038040549821818856 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_50::fold_2 | 9.0 | complete | 2026-09-25T17:37:15+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.06685521836852976 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_50::fold_3 | 9.0 | complete | 2026-09-25T17:37:16+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.10066496623995196 | 3.0 | xgb_lr_0p1_depth_3_child_1_freq_50::fold_4 | 9.0 | complete | 2026-09-25T17:37:18+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.1 | 1.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 330 | 0.8048100764039632 | 0.032479287578531926 | 0.006766162347048521 | 0.03766518458724022 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 397 | 0.8214707463714084 | 0.036533918167438235 | 0.008101494051516056 | 0.04242817685008049 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 1279 | 0.7151162976019284 | 0.06894925117979625 | 0.026112791150808334 | 0.13040569424629211 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2292 | 0.6864404189115101 | 0.09959524093942089 | 0.045294877141714096 | 0.22151458263397217 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004617859143763781 | 137 | 0.142203 | 0.842128 | 1.00512 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.007082318887114525 | 200 | 0.219052 | 0.933187 | 1.176926 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.011180009692907333 | 242 | 0.29059 | 1.031637 | 1.349142 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.01328342780470848 | 268 | 0.366727 | 1.106665 | 1.504381 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.032479287578531926 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_200::fold_1 | 12.0 | complete | 2026-09-25T17:37:29+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.036533918167438235 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_200::fold_2 | 12.0 | complete | 2026-09-25T17:37:30+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.06894925117979625 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_200::fold_3 | 12.0 | complete | 2026-09-25T17:37:32+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.09959524093942089 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_200::fold_4 | 12.0 | complete | 2026-09-25T17:37:33+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.1 | 10.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 330 | 0.8079080357250626 | 0.0331383278263891 | 0.006766742095351219 | 0.037527747452259064 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 397 | 0.8210967899434827 | 0.03766180098580184 | 0.008095023222267628 | 0.042383864521980286 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 1279 | 0.714227569687157 | 0.06898668348099925 | 0.02611512877047062 | 0.13028950989246368 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2292 | 0.6862981008052385 | 0.09967763115001373 | 0.04529774188995361 | 0.22160448133945465 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.0045889681205153465 | 266 | 0.137693 | 0.845187 | 1.004668 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.0071636694483459 | 347 | 0.212912 | 1.107452 | 1.345605 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.011022808030247688 | 411 | 0.297623 | 1.029733 | 1.356235 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.013249117881059647 | 449 | 0.366533 | 1.129373 | 1.527832 | {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.0331383278263891 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_50::fold_1 | 11.0 | complete | 2026-09-25T17:37:24+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.03766180098580184 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_50::fold_2 | 11.0 | complete | 2026-09-25T17:37:25+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.06898668348099925 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_50::fold_3 | 11.0 | complete | 2026-09-25T17:37:26+01:00 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.09967763115001373 | 3.0 | xgb_lr_0p1_depth_3_child_10_freq_50::fold_4 | 11.0 | complete | 2026-09-25T17:37:28+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":3,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.1 | 10.0 |

XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 330 | 0.7577217744625135 | 0.024663627020614105 | 0.006871653255075216 | 0.049226436764001846 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 397 | 0.7761710233367567 | 0.026262250123569735 | 0.008332904428243637 | 0.0491173192858696 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 1279 | 0.6871996662722153 | 0.05791562190939881 | 0.026599351316690445 | 0.15011189877986908 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2292 | 0.6761611700215789 | 0.09368568137539884 | 0.04558457434177399 | 0.23552341759204865 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.0017635166877880692 | 137 | 0.150641 | 1.45441 | 1.627686 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.004947231151163578 | 200 | 0.218845 | 1.571407 | 1.816133 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008800492621958256 | 242 | 0.297203 | 1.684158 | 2.010733 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.012501650489866734 | 268 | 0.372367 | 1.789273 | 2.195657 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.024663627020614105 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_200::fold_1 | 14.0 | complete | 2026-09-25T17:37:43+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.026262250123569735 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_200::fold_2 | 14.0 | complete | 2026-09-25T17:37:45+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.05791562190939881 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_200::fold_3 | 14.0 | complete | 2026-09-25T17:37:47+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.09368568137539884 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_200::fold_4 | 14.0 | complete | 2026-09-25T17:37:49+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.1 | 1.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 330 | 0.7652901629023724 | 0.0280408191269328 | 0.006840030662715435 | 0.04822254925966263 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 397 | 0.7712429219059287 | 0.02549498857692945 | 0.008334985002875328 | 0.049287013709545135 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 1279 | 0.6861871841355328 | 0.05976989257673336 | 0.026585545390844345 | 0.1491805762052536 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2292 | 0.6775430134643288 | 0.09614299433852325 | 0.04549076780676842 | 0.23455092310905457 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.0019044623477384448 | 266 | 0.136043 | 1.461635 | 1.618481 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.004912102594971657 | 347 | 0.212313 | 1.58539 | 1.822591 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.009027849882841 11 | 411 | 0.289826 | 1.696892 | 2.014842 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.012507797218859196 | 449 | 0.372859 | 2.012829 | 2.418943 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.0280408191269328 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_50::fold_1 | 13.0 | complete | 2026-09-25T17:37:35+01:00 |

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.02549498857692945 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_50::fold_2 | 13.0 | complete | 2026-09-25T17:37:37+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.05976989257673336 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_50::fold_3 | 13.0 | complete | 2026-09-25T17:37:39+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.09614299433852325 | 6.0 | xgb_lr_0p1_depth_6_child_1_freq_50::fold_4 | 13.0 | complete | 2026-09-25T17:37:41+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":1,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.1 | 1.0 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.1 | 1.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 200 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 200 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 200 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 |2018 | 2019 | 200 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 330 | 0.7661906586038156 | 0.02578469325236084 | 0.006869308650493622 | 0.04089401289820671 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 397 | 0.7954113278374356 | 0.02942963049416516 | 0.008260298520326614 | 0.045391712337732315 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 1279 | 0.6968421021934186 | 0.0617462214972945 | 0.026369446888566017 | 0.14018099009990692 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2292 | 0.6769291184664645 | 0.09264692367982039 | 0.045522239059209824 | 0.2308150827884674 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.00425694882869720 5 | 137 | 0.152127 | 1.436121 | 1.611667 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.00659902207553386 7 | 200 | 0.236359 | 1.576469 | 1.840172 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.01036971434950828 6 | 242 | 0.309941 | 1.707 | 2.046921 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.013425095938146114 | 268 | 0.378866 | 1.828774 | 2.241894 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.02578469325236084 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_200::fold_1 | 16.0 | complete | 2026-09-25T17:37:59+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.02942963049416516 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_200::fold_2 | 16.0 | complete | 2026-09-25T17:38:00+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.0617462214972945 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_200::fold_3 | 16.0 | complete | 2026-09-25T17:38:03+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.09264692367982039 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_200::fold_4 | 16.0 | complete | 2026-09-25T17:38:05+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":200}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.1 | 10.0 |

## XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}

*Column panel 1 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | notice | train_vintages | validation_vintage | min_frequency | train_n | train_positive_n | validation_n |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015 | 2016 | 50 | 47221 | 252 | 48031 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016 | 2017 | 50 | 95252 | 582 | 48211 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017 | 2018 | 50 | 143463 | 979 | 47393 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | DEVELOPMENT / QC — NOT FINAL PAPER RESULTS | 2015|2016|2017|2018 | 2019 | 50 | 190856 | 2258 | 48579 |

*Column panel 2 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | validation_positive_n | roc_auc | pr_auc | brier_score | log_loss | observed_prevalence |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 330 | 0.7616477451397055 | 0.024248508544919296 | 0.006875363178551197 | 0.041021473705768585 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 397 | 0.7937853272530975 | 0.03076832883427942 | 0.008252519182860851 | 0.045363783836364746 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 1279 | 0.6932892505614549 | 0.060901821758469914 | 0.026385711506009102 | 0.13987714052200317 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2292 | 0.675472536456001 | 0.09386755374125236 | 0.04549688845872879 | 0.23164059221744537 | 0.04718088062743161 |

*Column panel 3 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | mean_predicted_probability | transformed_feature_count | preprocessing_runtime_seconds | model_runtime_seconds | runtime_seconds | configuration | selected |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.00427178619429469 1 | 266 | 0.148652 | 1.434641 | 1.605692 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.00667774258181452 75 | 347 | 0.218689 | 1.589758 | 1.834644 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.01057146862149238 6 | 411 | 0.302999 | 1.855225 | 2.19521 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.01326012890785932 5 | 449 | 0.417487 | 1.876265 | 2.336637 | {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50} | False |

*Column panel 4 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | average_precision | max_depth | fit_key | config_index | status | completed_at |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.02424850854491929 6 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_50::fold_1 | 15.0 | complete | 2026-09-25T17:37:50+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.03076832883427942 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_50::fold_2 | 15.0 | complete | 2026-09-25T17:37:52+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.06090182175846991 4 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_50::fold_3 | 15.0 | complete | 2026-09-25T17:37:55+01:00 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.09386755374125236 | 6.0 | xgb_lr_0p1_depth_6_child_10_freq_50::fold_4 | 15.0 | complete | 2026-09-25T17:37:57+01:00 |

*Column panel 5 of 5 — XGBoost — {"learning_rate":0.1,"max_depth":6,"min_child_weight":10,"min_frequency":50}*

| model | config_id | fold_id | learning_rate | min_child_weight |
|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.1 | 10.0 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.1 | 10.0 |

# S.B Complete prediction and calibration evidence

## Table S3. Complete prediction and calibration evidence

Complete probability-quality and discrimination evidence is reported separately for 2020 validation (45,364 mortgages; 771 events) and 2023 test (49,338 mortgages; 886 events). The outcome is 90+ day delinquency during the first 24 scheduled-payment months in purchased/eligible mortgages. Raw ROC AUC and AP (average precision) are the basis for discrimination comparisons; calibrated ranking measures are archived but can be affected by isotonic-induced ties. Historical mappings are applied unchanged. ECE uses ten fixed equal-width bins, last including one; intercept and slope are jointly estimated from probabilities clipped at $10^{-6}$ before the logit. FT-Transformer is explicitly exploratory/appendix-only. Years provide temporal-transport evidence, not IID comparisons.

### 2020 Logistic

*Column panel 1 of 4 — 2020 Logistic*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| logistic | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | raw | 45364 | 771 | 0.01699585574464333 | 0.00928662939284793 | 0.7754174861187957 | 0.06309510462751056 | 0.01644945338653597 |

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| logistic | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | calibrated | 45364 | 771 | 0.01699585574464333 | 0.021595779056357737 | 0.7750634118300049 | 0.06053227297895611 | 0.01633315043506912 |

*Column panel 2 of 4 — 2020 Logistic*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept | calibration_slope |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.08004728640539988 | 2020 | validation | primary_benchmark | not_applicable | 0.06309510462751056 | ranking_basis_raw | 0.8056929077120383 | 1.0430304349628383 |
| logistic | 0.07789155824037552 | 2020 | validation | primary_benchmark | not_applicable | 0.06053227297895611 | archival_only_isotonic_ties_no_ranking | 0.11073096785176038 | 1.1077835617166332 |

*Column panel 3 of 4 — 2020 Logistic*

| model | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before | delta_roc_auc_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| logistic | 0.007709226351795399 | 0.0077092263517954 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | -0.003109303040080993 | -0.00011630295146684844 | -0.002155728165024351 | -0.0003540742887907511 |
| logistic | 0.004621713844893278 | 0.004599923311714407 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | -0.003109303040080993 | -0.00011630295146684844 | -0.002155728165024351 | -0.0003540742887907511 |

*Column panel 4 of 4 — 2020 Logistic*

| model | delta_pr_auc_after_minus_before |
|---|---|
| logistic | -0.002562831648554448 |
| logistic | -0.002562831648554448 |

## 2020 Random Forest

*Column panel 1 of 4 — 2020 Random Forest*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| rf | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | raw | 45364 | 771 | 0.01699585574464333 | 0.014784540645620991 | 0.7354624269546357 | 0.04965069398349603 | 0.01647803442125307 |
| rf | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | calibrated | 45364 | 771 | 0.01699585574464333 | 0.03703736352387975 | 0.7346328166585678 | 0.04696065228857252 | 0.01703894568104938 |

*Column panel 2 of 4 — 2020 Random Forest*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept | calibration_slope |
|---|---|---|---|---|---|---|---|---|---|
| rf | 0.0802259718639594 | 2020 | validation | primary_benchmark | not_applicable | 0.04965069398349603 | ranking_basis_raw | 0.9291755922683321 | 1.2016777315191016 |

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept | calibration_slope |
|---|---|---|---|---|---|---|---|---|---|
| rf | 0.08729272876695134 | 2020 | validation | primary_benchmark | not_applicable | 0.04696065228857252 | archival_only_isotonic_ties_no_ranking | -0.4606347020279198 | 1.120551548537699 |

*Column panel 3 of 4 — 2020 Random Forest*

| model | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before | delta_roc_auc_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| rf | 0.002211315099022339 | 0.002211315099022339 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.01783019268021408 | 0.0005609112597963102 | 0.007066756902991939 | -0.00082961029606787 31 |
| rf | 0.020041507779236425 | 0.020041507779236425 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.01783019268021408 | 0.0005609112597963102 | 0.007066756902991939 | -0.00082961029606787 31 |

*Column panel 4 of 4 — 2020 Random Forest*

| model | delta_pr_auc_after_minus_before |
|---|---|
| rf | -0.0026900416949235118 |
| rf | -0.0026900416949235118 |

## 2020 XGBoost

*Column panel 1 of 4 — 2020 XGBoost*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| xgb | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | raw | 45364 | 771 | 0.01699585574464333 | 0.011386558454047157 | 0.7647251464702965 | 0.05299260214716412 | 0.016457431455713115 |
| xgb | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | calibrated | 45364 | 771 | 0.01699585574464333 | 0.029318767298621633 | 0.7641619753677613 | 0.05134053728276112 6 | 0.016585547740555737 |

*Column panel 2 of 4 — 2020 XGBoost*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept | calibration_slope |
|---|---|---|---|---|---|---|---|---|---|
| xgb | 0.07988968029553474 | 2020 | validation | primary_benchmark | not_applicable | 0.05299260214716412 | ranking_basis_raw | -0.16087844627665215 | 0.8543806778448414 |
| xgb | 0.0820187876343607 | 2020 | validation | primary_benchmark | not_applicable | 0.05134053728276112 6 | archival_only_isotonic_ties_no_ranking | 0.3713156606834705 | 1.298597823104236 |

*Column panel 3 of 4 — 2020 XGBoost*

| model | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before | delta_roc_auc_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| xgb | 0.005852352775749001 | 0.005609297290596173 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.00671361426338213 | 0.00012811628484262275 | 0.002129107338825955 | -0.00056317110253523 15 |

| model | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before | delta_roc_auc_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| xgb | 0.012322911553978307 | 0.012322911553978303 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.00671361426338213 | 0.00012811628484262275 | 0.00212910733882595 85 | -0.00056317110253523 15 |

*Column panel 4 of 4 — 2020 XGBoost*

| model | delta_pr_auc_after_minus_before |
|---|---|
| xgb | -0.0016520648644029953 |
| xgb | -0.0016520648644029953 |

## 2020 FT-Transformer (exploratory)

*Column panel 1 of 4 — 2020 FT-Transformer (exploratory)*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| ft | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | raw | 45364 | 771 | 0.01699585574464333 | 0.011174369832982367 | 0.7503338815689492 | 0.0554647596005568 | 0.016446921271209565 |
| ft | OUT-OF-TIME 2020 VALIDATION ASSESSMENT — NOT 2023 TEST RESULTS | calibrated | 45364 | 771 | 0.01699585574464333 | 0.01374264326689487 | 0.7477854250766036 | 0.05016584376860103 | 0.0164405454639796 |

*Column panel 2 of 4 — 2020 FT-Transformer (exploratory)*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept |
|---|---|---|---|---|---|---|---|---|
| ft | 0.08054595238157519 | 2020 | validation | exploratory_appendix_only | B_EXPLORATORY_APPENDIX_ONLY | 0.0554647596005568 | ranking_basis_raw | -0.1639186366217448 |
| ft | 0.07997381564375075 | 2020 | validation | exploratory_appendix_only | B_EXPLORATORY_APPENDIX_ONLY | 0.05016584376860103 | archival_only_isotonic_ties_no_ranking | -0.1860274220459509 |

*Column panel 3 of 4 — 2020 FT-Transformer (exploratory)*

| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| ft | 0.853341108401046 | 0.005821485911660962 | 0.005821485911660963 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | -0.002568273433912502 5 | -6.375807229966335e-06 | -0.00057213673782444 13 |
| ft | 0.8926159863587914 | 0.003517628388599718 5 | 0.003253212477748460 7 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | -0.002568273433912502 5 | -6.375807229966335e-06 | -0.00057213673782444 13 |

*Column panel 4 of 4 — 2020 FT-Transformer (exploratory)*

| model | delta_roc_auc_after_minus_before | delta_pr_auc_after_minus_before |
|---|---|---|
| ft | -0.0025484564923455855 | -0.005298915831955769 |
| ft | -0.0025484564923455855 | -0.005298915831955769 |

## 2023 Logistic

*Column panel 1 of 4 — 2023 Logistic*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| logistic | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | raw | 49338 | 886 | 0.017957760752361263 | 0.11082463683451976 | 0.7593236022936014 | 0.055997885695977155 | 0.03220168934018628 |
| logistic | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | calibrated | 49338 | 886 | 0.017957760752361263 | 0.13866772425857274 | 0.7563671495225825 | 0.048042173183384364 | 0.03560760074025593 6 |

*Column panel 2 of 4 — 2023 Logistic*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept |
|---|---|---|---|---|---|---|---|---|
| logistic | 0.1519606249221206 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.055997885695977155 | ranking_basis_raw | -2.0763091818182358 |
| logistic | 0.17876073344417556 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.048042173183384364 | archival_only_isotonic_ties_no_ranking | -1.5596454482342204 |

*Column panel 3 of 4 — 2023 Logistic*

| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| logistic | 0.9713027726214917 | 0.09286687608215853 | 0.0928668760821585 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.027843087424052976 | 0.0034059114000696555 | 0.026800108522054972 |
| logistic | 1.4519158182765555 | 0.12070996350621147 | 0.12070996350621148 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.027843087424052976 | 0.0034059114000696555 | 0.026800108522054972 |

*Column panel 4 of 4 — 2023 Logistic*

| model | delta_roc_auc_after_minus_before | delta_pr_auc_after_minus_before |
|---|---|---|
| logistic | -0.002956452771018858 | -0.007955712512592791 |
| logistic | -0.002956452771018858 | -0.007955712512592791 |

## 2023 Random Forest

*Column panel 1 of 4 — 2023 Random Forest*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| rf | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | raw | 49338 | 886 | 0.017957760752361263 | 0.046628117624305375 | 0.7800783358885917 | 0.058578153735247926 | 0.01811078687483048 |
| rf | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | calibrated | 49338 | 886 | 0.017957760752361263 | 0.12298008395947001 | 0.7737787289517317 | 0.04930538487640733 | 0.03003743855911028 |

*Column panel 2 of 4 — 2023 Random Forest*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept |
|---|---|---|---|---|---|---|---|---|
| rf | 0.09657820224946449 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.058578153735247926 | ranking_basis_raw | 4.277634369156922 |
| rf | 0.16235874275243167 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.04930538487640733 | archival_only_isotonic_ties_no_ranking | -1.1918722168573517 |

*Column panel 3 of 4 — 2023 Random Forest*

| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| rf | 2.86337994774849 | 0.028670356871944115 | 0.02867035687194411 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.07635196633516464 | 0.0119266516842798 | 0.06578054050296718 |
| rf | 1.5155360873673926 | 0.10502232320710876 | 0.10502232320710875 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.07635196633516464 | 0.0119266516842798 | 0.06578054050296718 |

*Column panel 4 of 4 — 2023 Random Forest*

| model | delta_roc_auc_after_minus_before | delta_pr_auc_after_minus_before |
|---|---|---|
| rf | -0.006299606936859936 | -0.009272768858840597 |
| rf | -0.006299606936859936 | -0.009272768858840597 |

## 2023 XGBoost

*Column panel 1 of 4 — 2023 XGBoost*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| xgb | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | raw | 49338 | 886 | 0.017957760752361263 | 0.07610699351178274 | 0.7774608539525935 | 0.05476970280237421 | 0.02349981381071771 |
| xgb | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | calibrated | 49338 | 886 | 0.017957760752361263 | 0.1042173408773576 | 0.7752814146285011 | 0.05082131692448367 | 0.03181630698408148 |

*Column panel 2 of 4 — 2023 XGBoost*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept |
|---|---|---|---|---|---|---|---|---|
| xgb | 0.11731174884793184 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.05476970280237421 | ranking_basis_raw | -1.5035500166333253 |
| xgb | 0.4005867846033775 | 2023 | frozen_test | primary_benchmark | not_applicable | 0.05082131692448367 | archival_only_isotonic_ties_no_ranking | -3.7166026978030544 |

*Column panel 3 of 4 — 2023 XGBoost*

| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| xgb | 1.0366400907521787 | 0.05814923275942148 | 0.05814923275942148 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.028110347365574856 | 0.008316493173363769 | 0.28327503575544566 |

| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| xgb | 0.1517906263078187 | 0.08625958012499634 | 0.08625958012499634 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.028110347365574856 | 0.008316493173363769 | 0.28327503575544566 |

*Column panel 4 of 4 — 2023 XGBoost*

| model | delta_roc_auc_after_minus_before | delta_pr_auc_after_minus_before |
|---|---|---|
| xgb | -0.0021794393240923693 | -0.003948385877890544 |
| xgb | -0.0021794393240923693 | -0.003948385877890544 |

## 2023 FT-Transformer (exploratory)

*Column panel 1 of 4 — 2023 FT-Transformer (exploratory)*

| model | artifact_label | probability_type | n | positive_n | prevalence | mean_probability | roc_auc | pr_auc | brier_score |
|---|---|---|---|---|---|---|---|---|---|
| ft | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | raw | 49338 | 886 | 0.017957760752361263 | 0.07797436418984843 | 0.7716738205822933 | 0.05828443239710509 | 0.02190514992826842 |
| ft | FROZEN 2023 FINAL TEST — SINGLE AUTHORIZED EVALUATION | calibrated | 49338 | 886 | 0.017957760752361263 | 0.08167877668858894 | 0.7472849837282818 | 0.03830594851749437 | 0.022083699556165036 |

*Column panel 2 of 4 — 2023 FT-Transformer (exploratory)*

| model | log_loss | year | cohort_role | model_role | ft_retention_decision | average_precision | discrimination_use | calibration_intercept |
|---|---|---|---|---|---|---|---|---|
| ft | 0.11865418656671077 | 2023 | frozen_test | exploratory_appendix_only | B_EXPLORATORY_APPENDIX_ONLY | 0.05828443239710509 | ranking_basis_raw | -0.4280378754311555 |
| ft | 0.1234960743077028 | 2023 | frozen_test | exploratory_appendix_only | B_EXPLORATORY_APPENDIX_ONLY | 0.03830594851749437 | archival_only_isotonic_ties_no_ranking | 0.7185058411585997 |

*Column panel 3 of 4 — 2023 FT-Transformer (exploratory)*

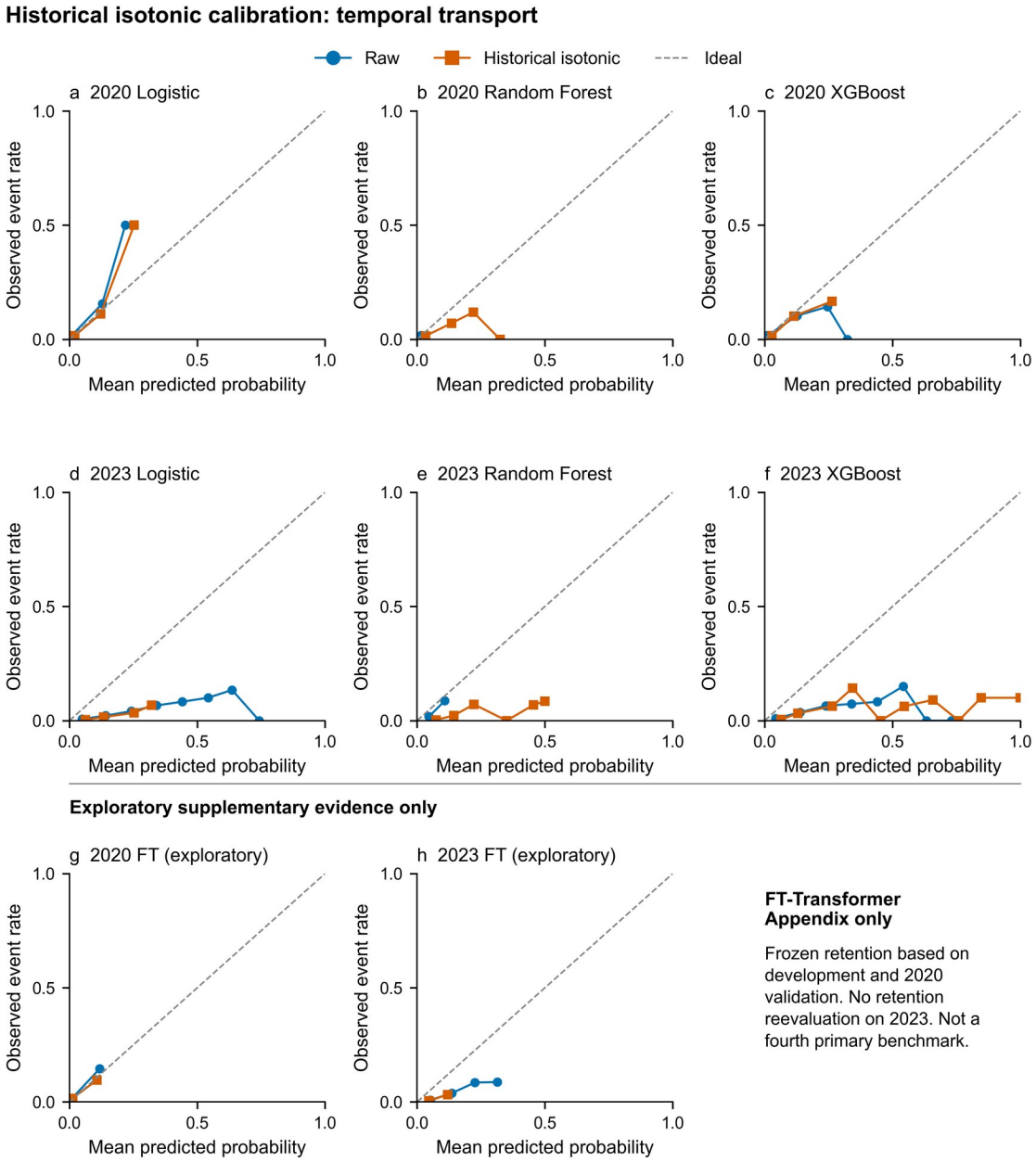
| model | calibration_slope | ece | mean_probability_error | ece_bin_rule | probability_clip_for_intercept_slope | delta_mean_probability_error_after_minus_before | delta_brier_after_minus_before | delta_log_loss_after_minus_before |
|---|---|---|---|---|---|---|---|---|
| ft | 1.5392685367031216 | 0.060016603437487166 | 0.06001660343748717 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.0037044124987405114 | 0.00017854962789661671 | 0.004841887740992035 |
| ft | 2.0745618341059004 | 0.06372101593622767 | 0.06372101593622768 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] | 1e-06 | 0.0037044124987405114 | 0.00017854962789661671 | 0.004841887740992035 |

*Column panel 4 of 4 — 2023 FT-Transformer (exploratory)*

| model | delta_roc_auc_after_minus_before | delta_pr_auc_after_minus_before |
|---|---|---|
| ft | -0.024388836854011497 | -0.019978483879610716 |
| ft | -0.024388836854011497 | -0.019978483879610716 |

Figure S1. Expanded calibration evidence across evaluation periods.

Full-range reliability diagrams compare raw and historically calibrated probabilities for 90+ day delinquency during the first 24 scheduled-payment months in the 2020 validation and 2023 purchased/eligible cohorts. The three primary models and separately labelled exploratory FT-Transformer evidence use the same ten equal-width bins, with the last including one. Empty bins are omitted and occupied-bin points are connected without smoothing. The diagonal indicates ideal reliability. Any labelled inset crops both axes to 0–0.20 and is accompanied by full-range context and an outside-inset count; cropping does not improve calibration. Historical isotonic mappings are applied unchanged. These periods assess temporal transport, not IID generalization, and FT-Transformer's appendix-only status is unchanged.




Outcome: 90+ day delinquency during the first 24 scheduled-payment months, among purchased/eligible mortgages. 2020 validation N = 45,364 (771 events); 2023 test N = 49,338 (886 events). Points use the same ten fixed equal-width bins; empty bins are omitted. Lines connect occupied bins in order without smoothing or refitting. All panels retain the full [0,1] probability range, including extreme points. The historical mapping improved Logistic probability quality in 2020 but worsened Brier and log loss for all three primary models in 2023. This is temporal transport, not a general claim about isotonic calibration.

# S.C Decision sensitivities

## Table S4. Common calibrated thresholds and illustrative loss

Common calibrated thresholds are 1/2, 1/3, 1/6 and 1/11 for illustrative FN:FP ratios 1:1, 2:1, 5:1 and 10:1. The primary rule remains 5:1. Classification counts, recall, precision and FP + kFN loss use the same purchased/eligible 2023 cohort and the ratio-specific all-non-risky benchmark. The outcome is 90+ day delinquency during the first 24 scheduled-payment months. Undefined precision remains NA. Exploratory FT-Transformer rows are separate and do not enter the primary three-model comparison. No rule is optimized on 2023; illustrative loss is not measured lender economics.

Primary benchmarks and all-non-risky reference

Classifications

*Column panel 1 of 2 — Classifications*

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:1 | All non-risky | NA | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 1:1 | Logistic | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 1:1 | Random Forest | 0.5 | 49338 | 886 | 401 | 0.008127609550447931 | 34 | 367 | 48085 | 852 | 0.03837471783295711 |
| 1:1 | XGBoost | 0.5 | 49338 | 886 | 454 | 0.009201832259110625 | 44 | 410 | 48042 | 842 | 0.04966139954853273 |
| 2:1 | All non-risky | NA | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 2:1 | Logistic | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 2:1 | Random Forest | 0.3333333333333333 | 49338 | 886 | 439 | 0.00889780696420609 | 36 | 403 | 48049 | 850 | 0.040632054176072234 |
| 2:1 | XGBoost | 0.3333333333333333 | 49338 | 886 | 465 | 0.009424784142040617 | 44 | 421 | 48031 | 842 | 0.04966139954853273 |
| 5:1 | All non-risky | NA | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 5:1 | Logistic | 0.16666666666666666 | 49338 | 886 | 10072 | 0.20414285135189914 | 483 | 9589 | 38863 | 403 | 0.5451467268623025 |
| 5:1 | Random Forest | 0.16666666666666666 | 49338 | 886 | 11625 | 0.23561960355101544 | 567 | 11058 | 37394 | 319 | 0.6399548532731377 |
| 5:1 | XGBoost | 0.16666666666666666 | 49338 | 886 | 7462 | 0.15124245003850986 | 409 | 7053 | 41399 | 477 | 0.4616252821670429 |
| 10:1 | All non-risky | NA | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 10:1 | Logistic | 0.09090909090909091 | 49338 | 886 | 35022 | 0.7098382585431108 | 840 | 34182 | 14270 | 46 | 0.9480812641083521 |
| 10:1 | Random Forest | 0.09090909090909091 | 49338 | 886 | 30242 | 0.612955531233532 | 825 | 29417 | 19035 | 61 | 0.9311512415349887 |
| 10:1 | XGBoost | 0.09090909090909091 | 49338 | 886 | 23575 | 0.4778264218249625 | 781 | 22794 | 25658 | 105 | 0.881489841986456 |

*Column panel 2 of 2 — Classifications*

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 1:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 1:1 | 0.9924254932716916 | 0.08478802992518704 | 0.9825898604328014 | 0.007574506728308429 | 0.9616252821670429 |
| 1:1 | 0.9915380170065219 | 0.09691629955947137 | 0.9827755502823009 | 0.008461982993478081 | 0.9503386004514672 |
| 2:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 2:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 2:1 | 0.9916824898868983 | 0.08200455580865604 | 0.9826172314362257 | 0.008317510113101627 | 0.9593679458239278 |
| 2:1 | 0.9913109881945018 | 0.09462365591397849 | 0.9827716735211671 | 0.008689011805498225 | 0.9503386004514672 |
| 5:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 5:1 | 0.8020927928671675 | 0.04795472597299444 | 0.9897366678551418 | 0.1979072071328325 | 0.4548532731376975 |
| 5:1 | 0.7717741269710229 | 0.0487741935483871 | 0.9915413783045635 | 0.22822587302897712 | 0.3600451467268623 |

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 5:1 | 0.854433253529266 | 0.054811042615920666 | 0.9886092272423345 | 0.14556674647073392 | 0.5383747178329571 |
| 10:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 10:1 | 0.29451828613885905 | 0.023984923762206612 | 0.9967868119586477 | 0.7054817138611409 | 0.05191873589164785 |
| 10:1 | 0.3928630397094031 | 0.027279941802790822 | 0.9968056137410977 | 0.6071369602905968 | 0.06884875846501129 |
| 10:1 | 0.5295550235284405 | 0.03312831389183457 | 0.9959243876877693 | 0.4704449764715595 | 0.11851015801354402 |

Illustrative loss

| FN:FP ratio | Model / benchmark | FP loss | FN loss | Total cost | Cost per loan | All non-risky cost | Relative cost versus all non-risky |
|---|---|---|---|---|---|---|---|
| 1:1 | All non-risky | 1 | 1 | 886 | 0.017957760752361263 | 886 | 1.0 |
| 1:1 | Logistic | 1 | 1 | 886 | 0.017957760752361263 | 886 | 1.0 |
| 1:1 | Random Forest | 1 | 1 | 1219 | 0.024707122299241963 | 886 | 1.3758465011286682 |
| 1:1 | XGBoost | 1 | 1 | 1252 | 0.025375977948031942 | 886 | 1.4130925507900678 |
| 2:1 | All non-risky | 1 | 2 | 1772 | 0.035915521504722527 | 1772 | 1.0 |
| 2:1 | Logistic | 1 | 2 | 1772 | 0.035915521504722527 | 1772 | 1.0 |
| 2:1 | Random Forest | 1 | 2 | 2103 | 0.04262434634561595 | 1772 | 1.1867945823927766 |
| 2:1 | XGBoost | 1 | 2 | 2105 | 0.04266488305160323 | 1772 | 1.187923250564334 |
| 5:1 | All non-risky | 1 | 5 | 4430 | 0.08978880376180631 | 4430 | 1.0 |
| 5:1 | Logistic | 1 | 5 | 11604 | 0.2351939681381491 | 4430 | 2.61941309255079 |
| 5:1 | Random Forest | 1 | 5 | 12653 | 0.256455470428473 | 4430 | 2.8562076749435668 |
| 5:1 | XGBoost | 1 | 5 | 9438 | 0.1912927155539341 | 4430 | 2.130474040632054 |
| 10:1 | All non-risky | 1 | 10 | 8860 | 0.17957760752361263 | 8860 | 1.0 |
| 10:1 | Logistic | 1 | 10 | 34642 | 0.7021362844055292 | 8860 | 3.9099322799097065 |
| 10:1 | Random Forest | 1 | 10 | 30027 | 0.6085978353399003 | 8860 | 3.3890519187358916 |
| 10:1 | XGBoost | 1 | 10 | 23844 | 0.48327860878025053 | 8860 | 2.691196388261851 |

## Exploratory FT-Transformer appendix block

### Classifications

*Column panel 1 of 2 — Classifications*

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:1 | FT-Transformer | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 2:1 | FT-Transformer | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 5:1 | FT-Transformer | 0.16666666666666666 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 10:1 | FT-Transformer | 0.09090909090909091 | 49338 | 886 | 23840 | 0.483197535368276 | 770 | 23070 | 25382 | 116 | 0.8690744920993227 |

*Column panel 2 of 2 — Classifications*

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 2:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 5:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 10:1 | 0.523858664245026 | 0.03229865771812081 | 0.9954506235783198 | 0.47614133575497397 | 0.1309255079006772 |

Illustrative loss

| FN:FP ratio | Model / benchmark | FP loss | FN loss | Total cost | Cost per loan | All non-risky cost | Relative cost versus all non-risky |
|---|---|---|---|---|---|---|---|
| 1:1 | FT-Transformer | 1 | 1 | 886 | 0.017957760752361263 | 886 | 1.0 |
| 2:1 | FT-Transformer | 1 | 2 | 1772 | 0.035915521504722527 | 1772 | 1.0 |
| 5:1 | FT-Transformer | 1 | 5 | 4430 | 0.08978880376180631 | 4430 | 1.0 |
| 10:1 | FT-Transformer | 1 | 10 | 24230 | 0.4911021930357939 | 8860 | 2.734762979683973 |

The FN:FP ratios are illustrative: 5:1 remains primary; 1:1, 2:1 and 10:1 are secondary sensitivity analyses. Frozen calibrated common thresholds are 1/2, 1/3, 1/6 and 1/11, respectively. These are analytical classifications as risky/non-risky, not approval decisions. The loss is not measured lender economics. Cost = FP + k × FN; the all-non-risky benchmark costs k × 886 events. Relative cost divides model cost by that benchmark. Benchmark threshold is not applicable; undefined precision remains NA. FT-Transformer is exploratory appendix evidence only. The cohort contains 49,338 purchased/eligible mortgages; outcome: 90+ day delinquency during the first 24 scheduled-payment months.

## Table S5. Threshold and probability-regime robustness

Decision evidence distinguishes calibrated common thresholds, raw-probability common thresholds and 2020 validation-optimal thresholds applied unchanged to 2023. The latter two are secondary sensitivities, not the primary rule or test-optimal choices. Counts concern analytical risky/non-risky classifications in the purchased/eligible 2023 cohort for 90+ day delinquency during the first 24 scheduled-payment months. Loss is FP + kFN under the stated illustrative ratio; each comparison uses its matching all-non-risky benchmark. Undefined quantities remain NA.

### A. Calibrated common reference

Classifications

*Column panel 1 of 2 — Classifications*

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 1:1 | Random Forest | 0.5 | 49338 | 886 | 401 | 0.008127609550447931 | 34 | 367 | 48085 | 852 | 0.03837471783295711 |
| 1:1 | XGBoost | 0.5 | 49338 | 886 | 454 | 0.009201832259110625 | 44 | 410 | 48042 | 842 | 0.04966139954853273 |
| 2:1 | Logistic | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 2:1 | Random Forest | 0.3333333333333333 | 49338 | 886 | 439 | 0.00889780696420609 | 36 | 403 | 48049 | 850 | 0.040632054176072234 |
| 2:1 | XGBoost | 0.3333333333333333 | 49338 | 886 | 465 | 0.009424784142040617 | 44 | 421 | 48031 | 842 | 0.04966139954853273 |
| 5:1 | Logistic | 0.16666666666666666 | 49338 | 886 | 10072 | 0.20414285135189914 | 483 | 9589 | 38863 | 403 | 0.5451467268623025 |
| 5:1 | Random Forest | 0.16666666666666666 | 49338 | 886 | 11625 | 0.23561960355101544 | 567 | 11058 | 37394 | 319 | 0.6399548532731377 |
| 5:1 | XGBoost | 0.16666666666666666 | 49338 | 886 | 7462 | 0.15124245003850986 | 409 | 7053 | 41399 | 477 | 0.4616252821670429 |
| 10:1 | Logistic | 0.09090909090909091 | 49338 | 886 | 35022 | 0.7098382585431108 | 840 | 34182 | 14270 | 46 | 0.9480812641083521 |
| 10:1 | Random Forest | 0.09090909090909091 | 49338 | 886 | 30242 | 0.612955531233532 | 825 | 29417 | 19035 | 61 | 0.9311512415349887 |
| 10:1 | XGBoost | 0.09090909090909091 | 49338 | 886 | 23575 | 0.4778264218249625 | 781 | 22794 | 25658 | 105 | 0.881489841986456 |

*Column panel 2 of 2 — Classifications*

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 1:1 | 0.9924254932716916 | 0.08478802992518704 | 0.9825898604328014 | 0.007574506728308429 | 0.9616252821670429 |

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 0.9915380170065219 | 0.09691629955947137 | 0.9827755502823009 | 0.008461982993478081 | 0.9503386004514672 |
| 2:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 2:1 | 0.9916824898868983 | 0.08200455580865604 | 0.9826172314362257 | 0.008317510113101627 | 0.9593679458239278 |
| 2:1 | 0.9913109881945018 | 0.09462365591397849 | 0.9827716735211671 | 0.008689011805498225 | 0.9503386004514672 |
| 5:1 | 0.8020927928671675 | 0.04795472597299444 | 0.9897366678551418 | 0.1979072071328325 | 0.4548532731376975 |
| 5:1 | 0.7717741269710229 | 0.0487741935483871 | 0.9915413783045635 | 0.22822587302897712 | 0.3600451467268623 |
| 5:1 | 0.854433253529266 | 0.054811042615920666 | 0.9886092272423345 | 0.14556674647073392 | 0.5383747178329571 |
| 10:1 | 0.29451828613885905 | 0.023984923762206612 | 0.9967868119586477 | 0.7054817138611409 | 0.05191873589164785 |
| 10:1 | 0.3928630397094031 | 0.027279941802790822 | 0.9968056137410977 | 0.6071369602905968 | 0.06884875846501129 |
| 10:1 | 0.5295550235284405 | 0.03312831389183457 | 0.9959243876877693 | 0.4704449764715595 | 0.11851015801354402 |

Illustrative loss

| FN:FP ratio | Model / benchmark | FP loss | FN loss | Total cost | Cost per loan | All non-risky cost | Relative cost versus all non-risky |
|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 1 | 1 | 886 | 0.017957760752361263 | 886 | 1.0 |
| 1:1 | Random Forest | 1 | 1 | 1219 | 0.024707122299241963 | 886 | 1.3758465011286682 |
| 1:1 | XGBoost | 1 | 1 | 1252 | 0.025375977948031942 | 886 | 1.4130925507900678 |
| 2:1 | Logistic | 1 | 2 | 1772 | 0.035915521504722527 | 1772 | 1.0 |
| 2:1 | Random Forest | 1 | 2 | 2103 | 0.04262434634561595 | 1772 | 1.1867945823927766 |
| 2:1 | XGBoost | 1 | 2 | 2105 | 0.04266488305160323 | 1772 | 1.187923250564334 |
| 5:1 | Logistic | 1 | 5 | 11604 | 0.2351939681381491 | 4430 | 2.61941309255079 |
| 5:1 | Random Forest | 1 | 5 | 12653 | 0.256455470428473 | 4430 | 2.8562076749435668 |
| 5:1 | XGBoost | 1 | 5 | 9438 | 0.1912927155539341 | 4430 | 2.130474040632054 |
| 10:1 | Logistic | 1 | 10 | 34642 | 0.7021362844055292 | 8860 | 3.9099322799097065 |
| 10:1 | Random Forest | 1 | 10 | 30027 | 0.6085978353399003 | 8860 | 3.3890519187358916 |
| 10:1 | XGBoost | 1 | 10 | 23844 | 0.48327860878025053 | 8860 | 2.691196388261851 |

## B. Raw common threshold

Classifications

*Column panel 1 of 2 — Classifications*

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 0.5 | 49338 | 886 | 313 | 0.006343994487007985 5 | 33 | 280 | 48172 | 853 | 0.03724604966139955 |
| 1:1 | Random Forest | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 1:1 | XGBoost | 0.5 | 49338 | 886 | 66 | 0.001337711297579958 6 | 9 | 57 | 48395 | 877 | 0.010158013544018058 |
| 2:1 | Logistic | 0.3333333333333333 | 49338 | 886 | 1940 | 0.03932060480765333 | 154 | 1786 | 46666 | 732 | 0.17381489841986456 |
| 2:1 | Random Forest | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 2:1 | XGBoost | 0.3333333333333333 | 49338 | 886 | 576 | 0.011674571324334184 | 48 | 528 | 47924 | 838 | 0.05417607223476298 |
| 5:1 | Logistic | 0.16666666666666666 | 49338 | 886 | 10503 | 0.21287851149215614 | 500 | 10003 | 38449 | 386 | 0.5643340857787811 |
| 5:1 | Random Forest | 0.16666666666666666 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 | 0.0 |
| 5:1 | XGBoost | 0.16666666666666666 | 49338 | 886 | 4999 | 0.10132149661518505 | 304 | 4695 | 43757 | 582 | 0.3431151241534989 |
| 10:1 | Logistic | 0.09090909090909091 | 49338 | 886 | 22181 | 0.4495723377518343 | 720 | 21461 | 26991 | 166 | 0.8126410835214447 |

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10:1 | Random Forest | 0.09090909090909091 | 49338 | 886 | 822 | 0.016660586160768577 | 65 | 757 | 47695 | 821 | 0.07336343115124154 |
| 10:1 | XGBoost | 0.09090909090909091 | 49338 | 886 | 14337 | 0.29058737686975555 | 603 | 13734 | 34718 | 283 | 0.6805869074492099 |

*Column panel 2 of 2 — Classifications*

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 0.9942210847849418 | 0.10543130990415335 | 0.9826007139214686 | 0.005778915215058202 | 0.9627539503386005 |
| 1:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 1:1 | 0.9988235779740774 | 0.13636363636363635 | 0.9822008442929047 | 0.0011764220259225626 | 0.989841986455982 |
| 2:1 | 0.963138776521093 | 0.07938144329896907 | 0.9845563103928435 | 0.03686122347890696 | 0.8261851015801355 |
| 2:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 2:1 | 0.9891026170230331 | 0.08333333333333333 | 0.9828144866904557 | 0.010897382976966895 | 0.945823927765237 |
| 5:1 | 0.7935482539420458 | 0.04760544606302961 | 0.9900605124243594 | 0.20645174605795427 | 0.435665914221219 |
| 5:1 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 |
| 5:1 | 0.9030999752332205 | 0.060812162432486494 | 0.9868738582286475 | 0.09690002476677949 | 0.6568848758465011 |
| 10:1 | 0.5570667877486998 | 0.03246021369640684 | 0.9938873955149685 | 0.4429332122513003 | 0.1873589164785553 |
| 10:1 | 0.9843762899364319 | 0.07907542579075426 | 0.9830777475472009 | 0.015623710063568068 | 0.9266365688487584 |
| 10:1 | 0.7165442087013952 | 0.04205900816070308 | 0.9919145167280935 | 0.2834557912986048 | 0.3194130925507901 |

Illustrative loss

| FN:FP ratio | Model / benchmark | FP loss | FN loss | Total cost | Cost per loan | All non-risky cost | Relative cost versus all non-risky |
|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 1 | 1 | 1133 | 0.02296404394178929 | 886 | 1.2787810383747178 |
| 1:1 | Random Forest | 1 | 1 | 886 | 0.017957760752361263 | 886 | 1.0 |
| 1:1 | XGBoost | 1 | 1 | 934 | 0.018930641696055777 | 886 | 1.054176072234763 |
| 2:1 | Logistic | 1 | 2 | 3250 | 0.06587214722931614 | 1772 | 1.8340857787810383 |
| 2:1 | Random Forest | 1 | 2 | 1772 | 0.035915521504722527 | 1772 | 1.0 |
| 2:1 | XGBoost | 1 | 2 | 2204 | 0.044671449997973164 | 1772 | 1.2437923250564333 |
| 5:1 | Logistic | 1 | 5 | 11933 | 0.24186225627305524 | 4430 | 2.6936794582392776 |
| 5:1 | Random Forest | 1 | 5 | 4430 | 0.08978880376180631 | 4430 | 1.0 |
| 5:1 | XGBoost | 1 | 5 | 7605 | 0.15414082451659977 | 4430 | 1.716704288939052 |
| 10:1 | Logistic | 1 | 10 | 23121 | 0.4686245895658519 | 8860 | 2.6095936794582393 |
| 10:1 | Random Forest | 1 | 10 | 8967 | 0.18174632129393165 | 8860 | 1.012076749435666 |
| 10:1 | XGBoost | 1 | 10 | 16564 | 0.33572499898658237 | 8860 | 1.8695259593679459 |

## C. 2020 validation-optimal threshold transported unchanged

Classifications

*Column panel 1 of 2 — Classifications*

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 0.2528735632183909 | 49338 | 886 | 4260 | 0.08634318375288824 | 287 | 3973 | 44479 | 599 | 0.3239277652370203 |
| 1:1 | Random Forest | 0.336647567017351 | 49338 | 886 | 439 | 0.00889780696420609 | 36 | 403 | 48049 | 850 | 0.040632054176072234 |
| 1:1 | XGBoost | 0.26562500000000006 | 49338 | 886 | 471 | 0.009546394260002432 | 45 | 426 | 48026 | 841 | 0.050790067720090294 |

| FN:FP ratio | Model / benchmark | Threshold | N | Events | Risky N | Risky fraction | TP | FP | TN | FN | Recall |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2:1 | Logistic | 0.25287356321839083 | 49338 | 886 | 8531 | 0.17290931938870646 | 433 | 8098 | 40354 | 453 | 0.4887133182844244 |
| 2:1 | Random Forest | 0.336647567017351 | 49338 | 886 | 439 | 0.00889780696420609 | 36 | 403 | 48049 | 850 | 0.040632054176072234 |
| 2:1 | XGBoost | 0.26562500000000006 | 49338 | 886 | 471 | 0.009546394260002432 | 45 | 426 | 48026 | 841 | 0.050790067720090294 |
| 5:1 | Logistic | 0.1402439024390244 | 49338 | 886 | 15983 | 0.32394908589728 | 624 | 15359 | 33093 | 262 | 0.7042889390519187 |
| 5:1 | Random Forest | 0.2857142857142857 | 49338 | 886 | 792 | 0.016052535570959505 | 62 | 730 | 47722 | 824 | 0.06997742663656885 |
| 5:1 | XGBoost | 0.265625 | 49338 | 886 | 2226 | 0.04511735376383315 | 153 | 2073 | 46379 | 733 | 0.172686230248307 |
| 10:1 | Logistic | 0.1 | 49338 | 886 | 34229 | 0.6937654546191576 | 834 | 33395 | 15057 | 52 | 0.9413092550790068 |
| 10:1 | Random Forest | 0.17105263157894737 | 49338 | 886 | 11611 | 0.23533584660910453 | 567 | 11044 | 37408 | 319 | 0.6399548532731377 |
| 10:1 | XGBoost | 0.1065015479876161 | 49338 | 886 | 19361 | 0.3924155823097815 | 717 | 18644 | 29808 | 169 | 0.809255079006772 |

*Column panel 2 of 2 — Classifications*

| FN:FP ratio | Specificity | Precision | NPV | FP rate | FN rate |
|---|---|---|---|---|---|
| 1:1 | 0.9180013208949063 | 0.06737089201877934 | 0.986711921558188 | 0.0819986791050937 | 0.6760722347629797 |
| 1:1 | 0.9916824898868983 | 0.08200455580865604 | 0.9826172314362257 | 0.008317510113101627 | 0.9593679458239278 |
| 1:1 | 0.9912077932799471 | 0.09554140127388536 | 0.9827900218961672 | 0.008792206720052835 | 0.9492099322799097 |
| 2:1 | 0.8328655163873524 | 0.05075606611182745 | 0.9888989634131399 | 0.16713448361264757 | 0.5112866817155757 |
| 2:1 | 0.9916824898868983 | 0.08200455580865604 | 0.9826172314362257 | 0.008317510113101627 | 0.9593679458239278 |
| 2:1 | 0.9912077932799471 | 0.09554140127388536 | 0.9827900218961672 | 0.008792206720052835 | 0.9492099322799097 |
| 5:1 | 0.6830058614711467 | 0.03904148157417256 | 0.9921451056813072 | 0.3169941385288533 | 0.29571106094808125 |
| 5:1 | 0.9849335424750268 | 0.07828282828282829 | 0.9830264079429819 | 0.01506645752497317 | 0.9300225733634312 |
| 5:1 | 0.9572153884256583 | 0.06873315363881402 | 0.9844413312956359 | 0.042784611574341616 | 0.827313769751693 |
| 10:1 | 0.3107611656897548 | 0.024365304274153496 | 0.9965583427096433 | 0.6892388343102452 | 0.05869074492099323 |
| 10:1 | 0.7720630727317758 | 0.04883300318663337 | 0.9915445171892809 | 0.22793692726822423 | 0.3600451467268623 |
| 10:1 | 0.6152068026087675 | 0.03703321109446826 | 0.9943623444640891 | 0.3847931973912326 | 0.19074492099322798 |

Illustrative loss

| FN:FP ratio | Model / benchmark | FP loss | FN loss | Total cost | Cost per loan | All non-risky cost | Relative cost versus all non-risky |
|---|---|---|---|---|---|---|---|
| 1:1 | Logistic | 1 | 1 | 4572 | 0.09266690988690258 | 886 | 5.160270880361174 |
| 1:1 | Random Forest | 1 | 1 | 1253 | 0.02539624630102558 | 886 | 1.4142212189616252 |
| 1:1 | XGBoost | 1 | 1 | 1267 | 0.02568000324293648 | 886 | 1.430022573363431 |
| 2:1 | Logistic | 1 | 2 | 9004 | 0.1824962503546962 | 1772 | 5.081264108352144 |
| 2:1 | Random Forest | 1 | 2 | 2103 | 0.04262434634561595 | 1772 | 1.1867945823927766 |
| 2:1 | XGBoost | 1 | 2 | 2108 | 0.04272568811058414 | 1772 | 1.1896162528216705 |
| 5:1 | Logistic | 1 | 5 | 16669 | 0.3378531760509141 | 4430 | 3.7627539503386003 |
| 5:1 | Random Forest | 1 | 5 | 4850 | 0.09830151201913333 | 4430 | 1.0948081264108351 |
| 5:1 | XGBoost | 1 | 5 | 5738 | 0.11629980947748186 | 4430 | 1.2952595936794582 |
| 10:1 | Logistic | 1 | 10 | 33915 | 0.687401191779156 | 8860 | 3.8278781038374716 |
| 10:1 | Random Forest | 1 | 10 | 14234 | 0.2884997365114111 | 8860 | 1.606546275395034 |
| 10:1 | XGBoost | 1 | 10 | 20334 | 0.4121366897725891 | 8860 | 2.2950338600451468 |

All three frozen regimes are shown for illustrative FN:FP ratios 1:1, 2:1, 5:1 and 10:1. Calibrated common 5:1 remains the primary rule. The 2020-optimal thresholds were selected on validation and transported unchanged to 2023; they are robustness evidence, not primary, test-optimal or economically optimal thresholds. No threshold was

selected on 2023. Comparisons remain descriptive point estimates. Cost = FP + k × FN; relative cost is versus the ratio-specific all-non-risky benchmark. Classified as risky/non-risky does not denote approval decisions; illustrative loss is not measured lender economics. Undefined precision remains NA. Thresholds are rounded for display only: some saved Logistic and XGBoost 2020 thresholds differ only at floating-point precision but yield distinct classifications. Full-precision thresholds, confusion cells and rates remain in the accompanying tables. Artwork displays every ratio/model/regime in compact form.

## S.D Complete RQ3 profiles and adjusted associations

### Table S6. Complete numeric profiles of model-induced reclassification

All four joint classification cells are profiled for the seven numeric origination predictors. Direction 1 is Logistic classified as non-risky/XGBoost classified as risky; direction 0 is the reverse. The calibrated common illustrative 5:1 threshold is 1/6. The disagreement sample contains 5,950 mortgages (1,670 and 4,280 by direction). Observed and missing counts are separate; descriptive means use observed values, not zero imputation. Signed SMDs are direction 1 minus direction 0. Observed delinquency rates are descriptive only; the association outcome is disagreement direction. Adjusted ORs use separate full-cohort standard-deviation scaling.

Classification-cell context

| group | group_n | group_event_n | event_rate | wilson_95_lower | wilson_95_upper | wilson_95_width |
|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 343 | 0.009123310990530908 | 0.008211115403555114 | 0.010135809265694032 | 0.0019246938621389183 |
| Both classified as risky | 5792 | 349 | 0.060255524861878455 | 0.05441382808056738 | 0.06668014314125777 | 0.012266315060690385 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 60 | 0.03592814371257485 | 0.028013944072495044 | 0.04597242911003246 | 0.017958485037537417 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 134 | 0.03130841121495327 | 0.026496788969082256 | 0.036960615193472775 | 0.010463822622439052 |

Classic FICO

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37587 | 9 | 761.1190039109266 | 38.50395339521357 | 770.0 | 739.0 | 791.0 | 52.0 |
| Both classified as risky | 5792 | 0 | 705.2601864640884 | 38.334194002282494 | 706.0 | 680.0 | 733.0 | 53.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 724.5718562874251 | 39.21184935151508 | 730.0 | 702.25 | 751.0 | 48.75 |
| Direction 0: Logistic risky / XGBoost non-risky | 4276 | 4 | 719.5498129092609 | 43.77347427675106 | 718.0 | 688.0 | 755.0 | 67.0 |

DTI

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37594 | 2 | 36.590652763738895 | 9.45389594972535 | 38.0 | 30.0 | 44.0 | 14.0 |
| Both classified as risky | 5792 | 0 | 44.49602900552486 | 5.397734654491478 | 46.0 | 42.0 | 49.0 | 7.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 42.720359281437126 | 7.804701529098793 | 45.0 | 40.0 | 48.0 | 8.0 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 41.91144859813084 | 6.404409540371902 | 43.0 | 38.0 | 47.0 | 9.0 |

Original UPB

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 0 | 294117.432705607 | 161043.74521119395 | 262000.0 | 175000.0 | 380000.0 | 205000.0 |

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as risky | 5792 | 0 | 385762.0856353591 | 192459.02225347393 | 354000.0 | 225000.0 | 527000.0 | 302000.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 431199.4011976048 | 185890.9699953835 | 465000.0 | 271000.0 | 570000.0 | 299000.0 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 311127.33644859813 | 191313.34477048818 | 270000.0 | 182000.0 | 383000.0 | 201000.0 |

LTV

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 0 | 72.88256729439303 | 19.99574667807095 | 80.0 | 60.0 | 90.0 | 30.0 |
| Both classified as risky | 5792 | 0 | 86.65417817679558 | 12.68175284703878 | 92.0 | 80.0 | 95.0 | 15.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 84.10179640718563 | 16.402748140518444 | 90.0 | 80.0 | 95.0 | 15.0 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 80.95560747663552 | 14.1888538012966 | 80.0 | 75.0 | 95.0 | 20.0 |

CLTV

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 0 | 73.00526651771465 | 20.028303524584118 | 80.0 | 61.0 | 90.0 | 29.0 |
| Both classified as risky | 5792 | 0 | 87.43128453038673 | 13.020862152342735 | 94.0 | 80.0 | 95.0 | 15.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 84.31676646706588 | 16.39801224525459 | 90.0 | 80.0 | 95.0 | 15.0 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 81.2766355140187 | 14.373969483989645 | 80.0 | 75.0 | 95.0 | 20.0 |

Interest rate

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 0 | 6.626487950845835 | 0.6788306214531495 | 6.625 | 6.125 | 7.0 | 0.875 |
| Both classified as risky | 5792 | 0 | 7.022184392265193 | 0.5736004063723094 | 6.999 | 6.625 | 7.49 | 0.8650000000000002 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 6.2584095808383235 | 0.4335045822088959 | 6.25 | 5.99 | 6.5 | 0.5099999999999998 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 7.494653738317758 | 0.4871511042515407 | 7.5 | 7.125 | 7.875 | 0.75 |

Loan term

| Classification group | Observed N | Missing N | Mean | SD | Median | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|---|
| Both classified as non-risky | 37596 | 0 | 346.7933556761358 | 46.31002392290539 | 360.0 | 360.0 | 360.0 | 0.0 |
| Both classified as risky | 5792 | 0 | 357.542817679558 | 19.33948758028453 | 360.0 | 360.0 | 360.0 | 0.0 |
| Direction 1: Logistic non-risky / XGBoost risky | 1670 | 0 | 351.874251497006 | 36.66957369536253 | 360.0 | 360.0 | 360.0 | 0.0 |
| Direction 0: Logistic risky / XGBoost non-risky | 4280 | 0 | 357.2018691588785 | 19.69751144799563 | 360.0 | 360.0 | 360.0 | 0.0 |

Directional mean comparison and signed SMD

| Variable | Direction 1 mean | Direction 0 mean | Signed SMD (Direction 1 minus 0) |
|---|---|---|---|
| classic_fico | 724.5718562874251 | 719.5498129092609 | 0.12085204983433813 |
| original_dti | 42.720359281437126 | 41.91144859813084 | 0.1133091350528446 |
| original_upb | 431199.4011976048 | 311127.33644859813 | 0.6365762839991018 |
| original_ltv | 84.10179640718563 | 80.95560747663552 | 0.205153189061564 |
| original_cltv | 84.31676646706588 | 81.2766355140187 | 0.19716479779553037 |
| original_interest_rate | 6.2584095808383235 | 7.494653738317758 | -2.6810255873958826 |
| original_loan_term | 351.874251497006 | 357.2018691588785 | -0.1810058776120427 |

## Table S7. Complete categorical profiles of model-induced reclassification

Complete categorical profiles and directional percentage-point contrasts retain every recorded category, including unknown/unavailable categories and detailed geography. Direction 1 is Logistic classified as non-risky/XGBoost classified as risky; direction 0 is the reverse, under calibrated probabilities and the common illustrative 5:1 threshold of 1/6. Differences are direction 1 minus direction 0. Cells below 100 mortgages in either disagreement direction are suppressed from interpretation, not deleted or newly pooled. Geography is descriptive only and excluded from the adjusted model. All rows and feature/group section breaks are retained.

First time homebuyer

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 24498 | 37596 | 65.16118736035749 | False |
| yes | 13098 | 37596 | 34.83881263964251 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 2325 | 5792 | 40.14157458563536 | False |
| yes | 3467 | 5792 | 59.85842541436464 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 857 | 1670 | 51.31736526946108 | False |
| yes | 813 | 1670 | 48.68263473053892 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 2370 | 4280 | 55.373831775700936 | False |
| yes | 1910 | 4280 | 44.626168224299064 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| no | 857 | 2370 | 51.31736526946108 | 55.373831775700936 | -4.056466506239853 | False | False | Descriptive only |
| yes | 813 | 1910 | 48.68263473053892 | 44.626168224299064 | 4.056466506239853 | False | False | Descriptive only |

## Borrower structure

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| multiple | 18137 | 37596 | 48.24183423768486 | False |
| one | 19459 | 37596 | 51.75816576231514 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| multiple | 1834 | 5792 | 31.664364640883978 | False |
| one | 3958 | 5792 | 68.33563535911603 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| multiple | 655 | 1670 | 39.221556886227546 | False |
| one | 1015 | 1670 | 60.778443113772454 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| multiple | 1439 | 4280 | 33.62149532710281 | False |
| one | 2841 | 4280 | 66.3785046728972 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| multiple | 655 | 1439 | 39.221556886227546 | 33.62149532710281 | 5.600061559124741 | False | False | Descriptive only |
| one | 1015 | 2841 | 60.778443113772454 | 66.3785046728972 | -5.600061559124747 | False | False | Descriptive only |

## Loan purpose

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| cash_out_refinance | 4625 | 37596 | 12.301840621342697 | False |
| no_cash_out_refinance | 1537 | 37596 | 4.088200872433237 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| purchase | 31434 | 37596 | 83.60995850622406 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| cash_out_refinance | 390 | 5792 | 6.733425414364641 | False |
| no_cash_out_refinance | 163 | 5792 | 2.8142265193370166 | False |
| purchase | 5239 | 5792 | 90.45234806629834 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| cash_out_refinance | 164 | 1670 | 9.820359281437126 | False |
| no_cash_out_refinance | 37 | 1670 | 2.215568862275449 | True |
| purchase | 1469 | 1670 | 87.96407185628742 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| cash_out_refinance | 791 | 4280 | 18.481308411214954 | False |
| no_cash_out_refinance | 160 | 4280 | 3.7383177570093458 | False |
| purchase | 3329 | 4280 | 77.7803738317757 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| cash_out_refinance | 164 | 791 | 9.820359281437126 | 18.481308411214954 | -8.660949129777826 | False | False | Descriptive only |
| no_cash_out_refinance | 37 | 160 | 2.215568862275449 | 3.7383177570093458 | -1.5227488947338965 | True | False | suppressed from interpretation |
| purchase | 1469 | 3329 | 87.96407185628742 | 77.7803738317757 | 10.183698024511735 | False | False | Descriptive only |

## Channel

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| broker | 5714 | 37596 | 15.198425364400467 | False |
| correspondent | 13001 | 37596 | 34.580806468773275 | False |
| retail | 18881 | 37596 | 50.22076816682626 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| broker | 1088 | 5792 | 18.784530386740332 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| correspondent | 3059 | 5792 | 52.81422651933702 | False |
| retail | 1645 | 5792 | 28.401243093922652 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| broker | 379 | 1670 | 22.694610778443113 | False |
| correspondent | 674 | 1670 | 40.35928143712575 | False |
| retail | 617 | 1670 | 36.946107784431135 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| broker | 736 | 4280 | 17.19626168224299 | False |
| correspondent | 1770 | 4280 | 41.35514018691589 | False |
| retail | 1774 | 4280 | 41.44859813084112 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| broker | 379 | 736 | 22.694610778443113 | 17.19626168224299 | 5.498349096200123 | False | False | Descriptive only |
| correspondent | 674 | 1770 | 40.35928143712575 | 41.35514018691589 | -0.9958587497901406 | False | False | Descriptive only |
| retail | 617 | 1774 | 36.946107784431135 | 41.44859813084112 | -4.5024903464099815 | False | False | Descriptive only |

## Super conforming flag

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 37319 | 37596 | 99.26321949143527 | False |
| yes | 277 | 37596 | 0.736780508564741 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 5730 | 5792 | 98.92955801104972 | False |
| yes | 62 | 5792 | 1.0704419889502763 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 1663 | 1670 | 99.58083832335329 | False |
| yes | 7 | 1670 | 0.41916167664670656 | True |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 4135 | 4280 | 96.61214953271028 | False |
| yes | 145 | 4280 | 3.3878504672897196 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| no | 1663 | 4135 | 99.58083832335329 | 96.61214953271028 | 2.968688790643015 | False | False | Descriptive only |
| yes | 7 | 145 | 0.41916167664670656 | 3.3878504672897196 | -2.9686887906430126 | True | False | suppressed from interpretation |

## Harp indicator

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 37596 | 37596 | 100.0 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 5792 | 5792 | 100.0 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 1670 | 1670 | 100.0 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| no | 4280 | 4280 | 100.0 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| no | 1670 | 4280 | 100.0 | 100.0 | 0.0 | False | False | Descriptive only |

Special eligibility program

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| hfa_advantage | 263 | 37596 | 0.6995425045217576 | False |
| home_possible | 2631 | 37596 | 6.998084902649218 | False |
| not_available_or_not_applicable | 34702 | 37596 | 92.30237259282903 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| hfa_advantage | 631 | 5792 | 10.894337016574585 | False |
| home_possible | 1219 | 5792 | 21.046270718232044 | False |
| not_available_or_not_applicable | 3942 | 5792 | 68.05939226519337 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| hfa_advantage | 35 | 1670 | 2.095808383233533 | True |
| home_possible | 187 | 1670 | 11.197604790419161 | False |
| not_available_or_not_applicable | 1448 | 1670 | 86.7065868263473 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| hfa_advantage | 211 | 4280 | 4.929906542056075 | False |
| home_possible | 578 | 4280 | 13.504672897196262 | False |
| not_available_or_not_applicable | 3491 | 4280 | 81.56542056074767 | False |

Directional comparison

*Column panel 1 of 2 — Directional comparison*

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse |
|---|---|---|---|---|---|---|---|
| hfa_advantage | 35 | 211 | 2.095808383233533 | 4.929906542056075 | -2.8340981588225422 | True | False |
| home_possible | 187 | 578 | 11.197604790419161 | 13.504672897196262 | -2.3070681067771006 | False | False |
| not_available_or_not_applicable | 1448 | 3491 | 86.7065868263473 | 81.56542056074767 | 5.141166265599651 | False | False |

*Column panel 2 of 2 — Directional comparison*

| Category | Interpretation |
|---|---|
| hfa_advantage | suppressed from interpretation |
| home_possible | Descriptive only |
| not_available_or_not_applicable | Descriptive only |

Occupancy status

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| investment_property | 3061 | 37596 | 8.141823598255133 | False |
| primary_residence | 33672 | 37596 | 89.5627194382381 | False |
| second_home | 863 | 37596 | 2.295456963506756 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| investment_property | 138 | 5792 | 2.382596685082873 | False |
| primary_residence | 5617 | 5792 | 96.978591160221 | False |
| second_home | 37 | 5792 | 0.6388121546961326 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| investment_property | 28 | 1670 | 1.6766467065868262 | True |
| primary_residence | 1636 | 1670 | 97.96407185628742 | False |
| second_home | 6 | 1670 | 0.3592814371257485 | True |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| investment_property | 426 | 4280 | 9.953271028037383 | False |
| primary_residence | 3725 | 4280 | 87.03271028037383 | False |
| second_home | 129 | 4280 | 3.014018691588785 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| investment_property | 28 | 426 | 1.6766467065868262 | 9.953271028037383 | -8.276624321450557 | True | False | suppressed from interpretation |
| primary_residence | 1636 | 3725 | 97.96407185628742 | 87.03271028037383 | 10.93136157591359 | False | False | Descriptive only |
| second_home | 6 | 129 | 0.3592814371257485 | 3.014018691588785 | -2.6547372544630363 | True | False | suppressed from interpretation |

Property type

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| condominium | 3641 | 37596 | 9.684540908607298 | False |
| cooperative | 93 | 37596 | 0.24736674114267476 | False |
| manufactured_housing | 446 | 37596 | 1.1862964145121822 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| planned_unit_development | 11542 | 37596 | 30.700074476008087 | False |
| single_family | 21874 | 37596 | 58.18172145972976 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| condominium | 687 | 5792 | 11.861187845303867 | False |
| cooperative | 10 | 5792 | 0.17265193370165746 | False |
| manufactured_housing | 56 | 5792 | 0.9668508287292817 | False |
| planned_unit_development | 1648 | 5792 | 28.45303867403315 | False |
| single_family | 3391 | 5792 | 58.546270718232044 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| condominium | 138 | 1670 | 8.263473053892216 | False |
| cooperative | 4 | 1670 | 0.23952095808383234 | True |
| manufactured_housing | 5 | 1670 | 0.2994011976047904 | True |
| planned_unit_development | 554 | 1670 | 33.17365269461078 | False |
| single_family | 969 | 1670 | 58.02395209580838 | False |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| condominium | 479 | 4280 | 11.19158878504673 | False |
| cooperative | 8 | 4280 | 0.18691588785046728 | True |
| manufactured_housing | 118 | 4280 | 2.7570093457943927 | False |
| planned_unit_development | 1013 | 4280 | 23.66822429906542 | False |
| single_family | 2662 | 4280 | 62.19626168224299 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| condominium | 138 | 479 | 8.263473053892216 | 11.19158878504673 | -2.9281157311545125 | False | False | Descriptive only |
| cooperative | 4 | 8 | 0.23952095808383234 | 0.18691588785046728 | 0.05260507023336503 | True | True | suppressed from interpretation |
| manufactured_housing | 5 | 118 | 0.2994011976047904 | 2.7570093457943927 | -2.4576081481896024 | True | False | suppressed from interpretation |
| planned_unit_development | 554 | 1013 | 33.17365269461078 | 23.66822429906542 | 9.505428395545357 | False | False | Descriptive only |
| single_family | 969 | 2662 | 58.02395209580838 | 62.19626168224299 | -4.172309586434608 | False | False | Descriptive only |

Number of units

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| four | 57 | 37596 | 0.15161187360357484 | False |
| one | 36972 | 37596 | 98.34024896265561 | False |
| three | 91 | 37596 | 0.24204702627939143 | False |
| two | 476 | 37596 | 1.266092137461432 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| four | 17 | 5792 | 0.2935082872928177 | False |
| one | 5596 | 5792 | 96.61602209944752 | False |
| three | 34 | 5792 | 0.5870165745856354 | False |
| two | 145 | 5792 | 2.5034530386740332 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| four | 1 | 1670 | 0.059880239520958084 | True |
| one | 1655 | 1670 | 99.10179640718563 | False |
| three | 2 | 1670 | 0.11976047904191617 | True |
| two | 12 | 1670 | 0.718562874251497 | True |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| four | 24 | 4280 | 0.5607476635514018 | True |
| one | 4055 | 4280 | 94.74299065420561 | False |
| three | 47 | 4280 | 1.0981308411214954 | True |
| two | 154 | 4280 | 3.5981308411214954 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| four | 1 | 24 | 0.059880239520958084 | 0.5607476635514018 | -0.5008674240304437 | True | True | suppressed from interpretation |
| one | 1655 | 4055 | 99.10179640718563 | 94.74299065420561 | 4.358805752980022 | False | False | Descriptive only |
| three | 2 | 47 | 0.11976047904191617 | 1.0981308411214954 | -0.9783703620795791 | True | True | suppressed from interpretation |
| two | 12 | 154 | 0.718562874251497 | 3.5981308411214954 | -2.8795679668699985 | True | False | suppressed from interpretation |

Property state

Detailed geography: descriptive only, not an adjusted association.

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| AK | 54 | 37596 | 0.14363230130864985 | False |
| AL | 551 | 37596 | 1.4655814448345568 | False |
| AR | 347 | 37596 | 0.9229705287796575 | False |
| AZ | 1185 | 37596 | 3.1519310564953718 | False |
| CA | 2548 | 37596 | 6.77731673582296 | False |
| CO | 912 | 37596 | 2.4257899776571974 | False |
| CT | 281 | 37596 | 0.7474199382913076 | False |
| DC | 61 | 37596 | 0.1622513033301415 | False |
| DE | 181 | 37596 | 0.4814341951271412 | False |
| FL | 2470 | 37596 | 6.56984785615491 | False |
| GA | 1322 | 37596 | 3.5163315246302798 | False |
| GU | 6 | 37596 | 0.015959144589849983 | False |
| HI | 55 | 37596 | 0.14629215874029153 | False |
| IA | 393 | 37596 | 1.045323970635174 | False |
| ID | 304 | 37596 | 0.8085966592190659 | False |
| IL | 1667 | 37596 | 4.433982338546654 | False |
| IN | 1113 | 37596 | 2.960421321417172 | False |
| KS | 368 | 37596 | 0.9788275348441323 | False |
| KY | 565 | 37596 | 1.5028194488775402 | False |
| LA | 358 | 37596 | 0.9522289605277157 | False |
| MA | 541 | 37596 | 1.4389828705181402 | False |
| MD | 610 | 37596 | 1.622513033301415 | False |
| ME | 158 | 37596 | 0.4202574741993829 | False |
| MI | 1555 | 37596 | 4.136078306202788 | False |
| MN | 811 | 37596 | 2.1571443770613894 | False |
| MO | 936 | 37596 | 2.4896265560165975 | False |
| MS | 152 | 37596 | 0.40429832960953294 | False |
| MT | 129 | 37596 | 0.34312160868177466 | False |
| NC | 1388 | 37596 | 3.69188211511863 | False |
| ND | 42 | 37596 | 0.1117140121289499 | False |
| NE | 220 | 37596 | 0.5851686349611661 | False |
| NH | 172 | 37596 | 0.4574954782423662 | False |
| NJ | 803 | 37596 | 2.135865517608256 | False |
| NM | 177 | 37596 | 0.4707947654005745 | False |
| NV | 437 | 37596 | 1.1623576976274073 | False |
| NY | 1051 | 37596 | 2.7955101606553887 | False |
| OH | 1899 | 37596 | 5.05106926268752 | False |
| OK | 500 | 37596 | 1.329928715820832 | False |
| OR | 534 | 37596 | 1.4203638684966486 | False |
| PA | 1556 | 37596 | 4.13873816363443 | False |
| PR | 4 | 37596 | 0.010639429726566656 | False |
| RI | 78 | 37596 | 0.2074688796680498 | False |
| SC | 857 | 37596 | 2.279497818916906 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| SD | 83 | 37596 | 0.2207681668262581 | False |
| TN | 1032 | 37596 | 2.7449728694541973 | False |
| TX | 3668 | 37596 | 9.756357059261624 | False |
| UT | 544 | 37596 | 1.4469624428130652 | False |
| VA | 1137 | 37596 | 3.024257899776572 | False |
| VI | 0 | 37596 | 0.0 | False |
| VT | 78 | 37596 | 0.2074688796680498 | False |
| WA | 887 | 37596 | 2.359293541866156 | False |
| WI | 618 | 37596 | 1.6437918927545483 | False |
| WV | 126 | 37596 | 0.33514203638684964 | False |
| WY | 72 | 37596 | 0.19150973507819982 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| AK | 3 | 5792 | 0.05179558011049724 | False |
| AL | 29 | 5792 | 0.5006906077348067 | False |
| AR | 22 | 5792 | 0.3798342541436464 | False |
| AZ | 86 | 5792 | 1.4848066298342542 | False |
| CA | 396 | 5792 | 6.837016574585635 | False |
| CO | 113 | 5792 | 1.9509668508287292 | False |
| CT | 168 | 5792 | 2.9005524861878453 | False |
| DC | 19 | 5792 | 0.3280386740331492 | False |
| DE | 16 | 5792 | 0.27624309392265195 | False |
| FL | 1192 | 5792 | 20.58011049723757 | False |
| GA | 270 | 5792 | 4.661602209944752 | False |
| GU | 0 | 5792 | 0.0 | False |
| HI | 11 | 5792 | 0.1899171270718232 | False |
| IA | 43 | 5792 | 0.7424033149171271 | False |
| ID | 62 | 5792 | 1.0704419889502763 | False |
| IL | 321 | 5792 | 5.542127071823204 | False |
| IN | 107 | 5792 | 1.8473756906077348 | False |
| KS | 24 | 5792 | 0.4143646408839779 | False |
| KY | 32 | 5792 | 0.5524861878453039 | False |
| LA | 59 | 5792 | 1.018646408839779 | False |
| MA | 91 | 5792 | 1.5711325966850829 | False |
| MD | 161 | 5792 | 2.779696132596685 | False |
| ME | 18 | 5792 | 0.31077348066298344 | False |
| MI | 128 | 5792 | 2.2099447513812156 | False |
| MN | 119 | 5792 | 2.054558011049724 | False |
| MO | 66 | 5792 | 1.1395027624309393 | False |
| MS | 14 | 5792 | 0.24171270718232044 | False |
| MT | 10 | 5792 | 0.17265193370165746 | False |
| NC | 132 | 5792 | 2.2790055248618786 | False |
| ND | 3 | 5792 | 0.05179558011049724 | False |
| NE | 14 | 5792 | 0.24171270718232044 | False |
| NH | 16 | 5792 | 0.27624309392265195 | False |
| NJ | 292 | 5792 | 5.041436464088398 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| NM | 12 | 5792 | 0.20718232044198895 | False |
| NV | 51 | 5792 | 0.880524861878453 | False |
| NY | 401 | 5792 | 6.923342541436464 | False |
| OH | 137 | 5792 | 2.3653314917127073 | False |
| OK | 35 | 5792 | 0.6042817679558011 | False |
| OR | 43 | 5792 | 0.7424033149171271 | False |
| PA | 126 | 5792 | 2.175414364640884 | False |
| PR | 0 | 5792 | 0.0 | False |
| RI | 24 | 5792 | 0.4143646408839779 | False |
| SC | 52 | 5792 | 0.8977900552486188 | False |
| SD | 6 | 5792 | 0.10359116022099447 | False |
| TN | 65 | 5792 | 1.1222375690607735 | False |
| TX | 524 | 5792 | 9.046961325966851 | False |
| UT | 38 | 5792 | 0.6560773480662984 | False |
| VA | 78 | 5792 | 1.3466850828729282 | False |
| VI | 0 | 5792 | 0.0 | False |
| VT | 7 | 5792 | 0.12085635359116022 | False |
| WA | 89 | 5792 | 1.5366022099447514 | False |
| WI | 47 | 5792 | 0.81146408839779 | False |
| WV | 11 | 5792 | 0.1899171270718232 | False |
| WY | 9 | 5792 | 0.15538674033149172 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| AK | 1 | 1670 | 0.059880239520958084 | True |
| AL | 8 | 1670 | 0.47904191616766467 | True |
| AR | 4 | 1670 | 0.23952095808383234 | True |
| AZ | 45 | 1670 | 2.694610778443114 | True |
| CA | 221 | 1670 | 13.233532934131736 | False |
| CO | 51 | 1670 | 3.053892215568862 | True |
| CT | 44 | 1670 | 2.6347305389221556 | True |
| DC | 4 | 1670 | 0.23952095808383234 | True |
| DE | 3 | 1670 | 0.17964071856287425 | True |
| FL | 204 | 1670 | 12.215568862275449 | False |
| GA | 100 | 1670 | 5.9880239520958085 | False |
| GU | 0 | 1670 | 0.0 | True |
| HI | 4 | 1670 | 0.23952095808383234 | True |
| IA | 4 | 1670 | 0.23952095808383234 | True |
| ID | 12 | 1670 | 0.718562874251497 | True |
| IL | 60 | 1670 | 3.592814371257485 | True |
| IN | 16 | 1670 | 0.9580838323353293 | True |
| KS | 5 | 1670 | 0.2994011976047904 | True |
| KY | 15 | 1670 | 0.8982035928143712 | True |
| LA | 30 | 1670 | 1.7964071856287425 | True |
| MA | 37 | 1670 | 2.215568862275449 | True |
| MD | 29 | 1670 | 1.7365269461077844 | True |
| ME | 1 | 1670 | 0.059880239520958084 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| MI | 31 | 1670 | 1.8562874251497006 | True |
| MN | 12 | 1670 | 0.718562874251497 | True |
| MO | 19 | 1670 | 1.1377245508982037 | True |
| MS | 2 | 1670 | 0.11976047904191617 | True |
| MT | 3 | 1670 | 0.17964071856287425 | True |
| NC | 35 | 1670 | 2.095808383233533 | True |
| ND | 0 | 1670 | 0.0 | True |
| NE | 6 | 1670 | 0.3592814371257485 | True |
| NH | 6 | 1670 | 0.3592814371257485 | True |
| NJ | 123 | 1670 | 7.365269461077844 | False |
| NM | 3 | 1670 | 0.17964071856287425 | True |
| NV | 29 | 1670 | 1.7365269461077844 | True |
| NY | 177 | 1670 | 10.598802395209582 | False |
| OH | 23 | 1670 | 1.377245508982036 | True |
| OK | 5 | 1670 | 0.2994011976047904 | True |
| OR | 20 | 1670 | 1.1976047904191616 | True |
| PA | 26 | 1670 | 1.5568862275449102 | True |
| PR | 0 | 1670 | 0.0 | True |
| RI | 0 | 1670 | 0.0 | True |
| SC | 16 | 1670 | 0.9580838323353293 | True |
| SD | 2 | 1670 | 0.11976047904191617 | True |
| TN | 24 | 1670 | 1.437125748502994 | True |
| TX | 114 | 1670 | 6.826347305389222 | False |
| UT | 16 | 1670 | 0.9580838323353293 | True |
| VA | 34 | 1670 | 2.035928143712575 | True |
| VI | 0 | 1670 | 0.0 | True |
| VT | 1 | 1670 | 0.059880239520958084 | True |
| WA | 34 | 1670 | 2.035928143712575 | True |
| WI | 10 | 1670 | 0.5988023952095808 | True |
| WV | 1 | 1670 | 0.059880239520958084 | True |
| WY | 0 | 1670 | 0.0 | True |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| AK | 6 | 4280 | 0.14018691588785046 | True |
| AL | 49 | 4280 | 1.144859813084112 | True |
| AR | 24 | 4280 | 0.5607476635514018 | True |
| AZ | 78 | 4280 | 1.8224299065420562 | True |
| CA | 236 | 4280 | 5.514018691588785 | False |
| CO | 66 | 4280 | 1.5420560747663552 | True |
| CT | 24 | 4280 | 0.5607476635514018 | True |
| DC | 8 | 4280 | 0.18691588785046728 | True |
| DE | 17 | 4280 | 0.397196261682243 | True |
| FL | 498 | 4280 | 11.63551401869159 | False |
| GA | 136 | 4280 | 3.177570093457944 | False |
| GU | 0 | 4280 | 0.0 | True |
| HI | 11 | 4280 | 0.2570093457943925 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| IA | 38 | 4280 | 0.8878504672897196 | True |
| ID | 38 | 4280 | 0.8878504672897196 | True |
| IL | 239 | 4280 | 5.58411214953271 | False |
| IN | 145 | 4280 | 3.3878504672897196 | False |
| KS | 35 | 4280 | 0.8177570093457944 | True |
| KY | 55 | 4280 | 1.2850467289719627 | True |
| LA | 36 | 4280 | 0.8411214953271028 | True |
| MA | 76 | 4280 | 1.7757009345794392 | True |
| MD | 92 | 4280 | 2.149532710280374 | True |
| ME | 19 | 4280 | 0.4439252336448598 | True |
| MI | 193 | 4280 | 4.509345794392523 | False |
| MN | 91 | 4280 | 2.1261682242990654 | True |
| MO | 102 | 4280 | 2.383177570093458 | False |
| MS | 15 | 4280 | 0.35046728971962615 | True |
| MT | 12 | 4280 | 0.2803738317757009 | True |
| NC | 136 | 4280 | 3.177570093457944 | False |
| ND | 4 | 4280 | 0.09345794392523364 | True |
| NE | 20 | 4280 | 0.4672897196261682 | True |
| NH | 27 | 4280 | 0.6308411214953271 | True |
| NJ | 112 | 4280 | 2.616822429906542 | False |
| NM | 24 | 4280 | 0.5607476635514018 | True |
| NV | 36 | 4280 | 0.8411214953271028 | True |
| NY | 103 | 4280 | 2.4065420560747666 | False |
| OH | 180 | 4280 | 4.205607476635514 | False |
| OK | 46 | 4280 | 1.074766355140187 | True |
| OR | 34 | 4280 | 0.794392523364486 | True |
| PA | 157 | 4280 | 3.6682242990654204 | False |
| PR | 1 | 4280 | 0.02336448598130841 | True |
| RI | 31 | 4280 | 0.7242990654205608 | True |
| SC | 71 | 4280 | 1.658878504672897 | True |
| SD | 3 | 4280 | 0.07009345794392523 | True |
| TN | 60 | 4280 | 1.4018691588785046 | True |
| TX | 600 | 4280 | 14.018691588785046 | False |
| UT | 38 | 4280 | 0.8878504672897196 | True |
| VA | 80 | 4280 | 1.8691588785046729 | True |
| VI | 2 | 4280 | 0.04672897196261682 | True |
| VT | 12 | 4280 | 0.2803738317757009 | True |
| WA | 78 | 4280 | 1.8224299065420562 | True |
| WI | 65 | 4280 | 1.5186915887850467 | True |
| WV | 12 | 4280 | 0.2803738317757009 | True |
| WY | 9 | 4280 | 0.2102803738317757 | True |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| AK | 1 | 6 | 0.059880239520958084 | 0.14018691588785046 | -0.08030667636689237 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| AL | 8 | 49 | 0.47904191616766467 | 1.144859813084112 | -0.6658178969164475 | True | True | suppressed from interpretation |
| AR | 4 | 24 | 0.23952095808383234 | 0.5607476635514018 | -0.3212267054675695 | True | True | suppressed from interpretation |
| AZ | 45 | 78 | 2.694610778443114 | 1.8224299065420562 | 0.8721808719010578 | True | True | suppressed from interpretation |
| CA | 221 | 236 | 13.233532934131736 | 5.514018691588785 | 7.719514242542952 | False | False | Descriptive only |
| CO | 51 | 66 | 3.053892215568862 | 1.5420560747663552 | 1.5118361408025072 | True | True | suppressed from interpretation |
| CT | 44 | 24 | 2.6347305389221556 | 0.5607476635514018 | 2.0739828753707537 | True | True | suppressed from interpretation |
| DC | 4 | 8 | 0.23952095808383234 | 0.18691588785046728 | 0.05260507023336503 | True | True | suppressed from interpretation |
| DE | 3 | 17 | 0.17964071856287425 | 0.397196261682243 | -0.21755554311936876 | True | True | suppressed from interpretation |
| FL | 204 | 498 | 12.215568862275449 | 11.63551401869159 | 0.5800548435838604 | False | False | Descriptive only |
| GA | 100 | 136 | 5.9880239520958085 | 3.177570093457944 | 2.8104538586378647 | False | False | Descriptive only |
| GU | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| HI | 4 | 11 | 0.23952095808383234 | 0.2570093457943925 | -0.017488387710560195 | True | True | suppressed from interpretation |
| IA | 4 | 38 | 0.23952095808383234 | 0.8878504672897196 | -0.6483295092058874 | True | True | suppressed from interpretation |
| ID | 12 | 38 | 0.718562874251497 | 0.8878504672897196 | -0.1692875930382227 | True | True | suppressed from interpretation |
| IL | 60 | 239 | 3.592814371257485 | 5.58411214953271 | -1.9912977782752257 | True | False | suppressed from interpretation |
| IN | 16 | 145 | 0.9580838323353293 | 3.3878504672897196 | -2.42976663495439 | True | False | suppressed from interpretation |
| KS | 5 | 35 | 0.2994011976047904 | 0.8177570093457944 | -0.5183558117410039 | True | True | suppressed from interpretation |
| KY | 15 | 55 | 0.8982035928143712 | 1.2850467289719627 | -0.38684313615759136 | True | True | suppressed from interpretation |
| LA | 30 | 36 | 1.7964071856287425 | 0.8411214953271028 | 0.9552856903016396 | True | True | suppressed from interpretation |
| MA | 37 | 76 | 2.215568862275449 | 1.7757009345794392 | 0.43986792769600963 | True | True | suppressed from interpretation |
| MD | 29 | 92 | 1.7365269461077844 | 2.149532710280374 | -0.4130057641725894 | True | True | suppressed from interpretation |
| ME | 1 | 19 | 0.059880239520958084 | 0.4439252336448598 | -0.38404499412390175 | True | True | suppressed from interpretation |
| MI | 31 | 193 | 1.8562874251497006 | 4.509345794392523 | -2.6530583692428227 | True | False | suppressed from interpretation |
| MN | 12 | 91 | 0.718562874251497 | 2.1261682242990654 | -1.4076053500475687 | True | True | suppressed from interpretation |
| MO | 19 | 102 | 1.1377245508982037 | 2.383177570093458 | -1.2454530191952544 | True | False | suppressed from interpretation |
| MS | 2 | 15 | 0.11976047904191617 | 0.35046728971962615 | -0.23070681067770998 | True | True | suppressed from interpretation |
| MT | 3 | 12 | 0.17964071856287425 | 0.2803738317757009 | -0.10073311321282666 | True | True | suppressed from interpretation |
| NC | 35 | 136 | 2.095808383233533 | 3.177570093457944 | -1.081761710224411 | True | False | suppressed from interpretation |
| ND | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| NE | 6 | 20 | 0.3592814371257485 | 0.4672897196261682 | -0.10800828250041969 | True | True | suppressed from interpretation |
| NH | 6 | 27 | 0.3592814371257485 | 0.6308411214953271 | -0.2715596843695786 | True | True | suppressed from interpretation |
| NJ | 123 | 112 | 7.365269461077844 | 2.616822429906542 | 4.7484470311713025 | False | False | Descriptive only |
| NM | 3 | 24 | 0.17964071856287425 | 0.5607476635514018 | -0.3811069449885276 | True | True | suppressed from interpretation |
| NV | 29 | 36 | 1.7365269461077844 | 0.8411214953271028 | 0.8954054507806817 | True | True | suppressed from interpretation |
| NY | 177 | 103 | 10.598802395209582 | 2.4065420560747666 | 8.192260339134814 | False | False | Descriptive only |
| OH | 23 | 180 | 1.377245508982036 | 4.205607476635514 | -2.828361967653478 | True | False | suppressed from interpretation |
| OK | 5 | 46 | 0.2994011976047904 | 1.074766355140187 | -0.7753651575353965 | True | True | suppressed from interpretation |
| OR | 20 | 34 | 1.1976047904191616 | 0.794392523364486 | 0.4032122670546758 | True | True | suppressed from interpretation |
| PA | 26 | 157 | 1.5568862275449102 | 3.6682242990654204 | -2.1113380715205103 | True | False | suppressed from interpretation |
| PR | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| RI | 0 | 31 | 0.0 | 0.7242990654205608 | -0.7242990654205608 | True | True | suppressed from interpretation |
| SC | 16 | 71 | 0.9580838323353293 | 1.658878504672897 | -0.700794672337568 | True | True | suppressed from interpretation |
| SD | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| TN | 24 | 60 | 1.437125748502994 | 1.4018691588785046 | 0.03525658962448934 | True | True | suppressed from interpretation |
| TX | 114 | 600 | 6.826347305389222 | 14.018691588785046 | -7.192344283395824 | False | False | Descriptive only |
| UT | 16 | 38 | 0.9580838323353293 | 0.8878504672897196 | 0.07023336504560965 | True | True | suppressed from interpretation |
| VA | 34 | 80 | 2.035928143712575 | 1.8691588785046729 | 0.1667692652079021 | True | True | suppressed from interpretation |
| VI | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| VT | 1 | 12 | 0.059880239520958084 | 0.2803738317757009 | -0.2204935922547428 | True | True | suppressed from interpretation |
| WA | 34 | 78 | 2.035928143712575 | 1.8224299065420562 | 0.2134982371705188 | True | True | suppressed from interpretation |
| WI | 10 | 65 | 0.5988023952095808 | 1.5186915887850467 | -0.9198891935754658 | True | True | suppressed from interpretation |
| WV | 1 | 12 | 0.059880239520958084 | 0.2803738317757009 | -0.2204935922547428 | True | True | suppressed from interpretation |
| WY | 0 | 9 | 0.0 | 0.2102803738317757 | -0.2102803738317757 | True | True | suppressed from interpretation |

Msa or division

Detailed geography: descriptive only, not an adjusted association.

Both classified as non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 10180 | 15 | 37596 | 0.03989786147462496 | False |
| 10420 | 123 | 37596 | 0.3271624640919247 | False |
| 10500 | 6 | 37596 | 0.015959144589849983 | False |
| 10540 | 19 | 37596 | 0.05053729120119162 | False |
| 10580 | 55 | 37596 | 0.14629215874029153 | False |
| 10740 | 94 | 37596 | 0.2500265985743164 | False |
| 10780 | 8 | 37596 | 0.021278859453133313 | False |
| 10900 | 112 | 37596 | 0.29790403234386637 | False |
| 11020 | 18 | 37596 | 0.047877433769549956 | False |
| 11100 | 25 | 37596 | 0.0664964357910416 | False |
| 11180 | 8 | 37596 | 0.021278859453133313 | False |
| 11200 | 4 | 37596 | 0.010639429726566656 | False |
| 11244 | 180 | 37596 | 0.47877433769549954 | False |
| 11260 | 36 | 37596 | 0.09575486753909991 | False |
| 11460 | 47 | 37596 | 0.1250132992871582 | False |
| 11500 | 13 | 37596 | 0.034578146611341634 | False |
| 11540 | 26 | 37596 | 0.06915629322268327 | False |
| 11694 | 89 | 37596 | 0.2367273114161081 | False |
| 11700 | 69 | 37596 | 0.1835301627832748 | False |
| 12020 | 25 | 37596 | 0.0664964357910416 | False |
| 12054 | 140 | 37596 | 0.37238004042983297 | False |
| 12060 | 652 | 37596 | 1.734227045430365 | False |
| 12100 | 34 | 37596 | 0.09043515267581657 | False |
| 12220 | 25 | 37596 | 0.0664964357910416 | False |
| 12260 | 61 | 37596 | 0.1622513033301415 | False |
| 12420 | 461 | 37596 | 1.2261942759868072 | False |
| 12540 | 63 | 37596 | 0.16757101819342482 | False |
| 12580 | 315 | 37596 | 0.8378550909671242 | False |
| 12620 | 11 | 37596 | 0.029258431748058304 | False |
| 12700 | 21 | 37596 | 0.05585700606447495 | False |
| 12940 | 88 | 37596 | 0.23406745398446643 | False |
| 12980 | 23 | 37596 | 0.06117672092775827 | False |
| 13020 | 9 | 37596 | 0.023938716884774978 | False |
| 13140 | 24 | 37596 | 0.06383657835939993 | False |
| 13220 | 9 | 37596 | 0.023938716884774978 | False |
| 13380 | 20 | 37596 | 0.05319714863283328 | False |
| 13460 | 48 | 37596 | 0.12767315671879986 | False |
| 13740 | 25 | 37596 | 0.0664964357910416 | False |
| 13780 | 13 | 37596 | 0.034578146611341634 | False |
| 13820 | 162 | 37596 | 0.4308969039259496 | False |
| 13900 | 8 | 37596 | 0.021278859453133313 | False |
| 13980 | 25 | 37596 | 0.0664964357910416 | False |
| 14010 | 31 | 37596 | 0.08245558038089158 | False |
| 14020 | 19 | 37596 | 0.05053729120119162 | False |
| 14100 | 3 | 37596 | 0.007979572294924991 | False |
| 14260 | 165 | 37596 | 0.43887647622087456 | False |
| 14454 | 138 | 37596 | 0.3670603255665496 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 14500 | 41 | 37596 | 0.10905415469730823 | False |
| 14540 | 30 | 37596 | 0.07979572294924991 | False |
| 14580 | 7 | 37596 | 0.018619002021491648 | False |
| 14740 | 27 | 37596 | 0.07181615065432492 | False |
| 14860 | 50 | 37596 | 0.1329928715820832 | False |
| 15180 | 33 | 37596 | 0.0877752952441749 | False |
| 15260 | 19 | 37596 | 0.05053729120119162 | False |
| 15380 | 138 | 37596 | 0.3670603255665496 | False |
| 15500 | 17 | 37596 | 0.04521757633790829 | False |
| 15540 | 35 | 37596 | 0.09309501010745824 | False |
| 15680 | 6 | 37596 | 0.015959144589849983 | False |
| 15764 | 170 | 37596 | 0.4521757633790829 | False |
| 15804 | 149 | 37596 | 0.3963187573146079 | False |
| 15940 | 71 | 37596 | 0.18884987764655814 | False |
| 15980 | 123 | 37596 | 0.3271624640919247 | False |
| 16020 | 12 | 37596 | 0.031918289179699966 | False |
| 16060 | 15 | 37596 | 0.03989786147462496 | False |
| 16180 | 10 | 37596 | 0.02659857431641664 | False |
| 16220 | 14 | 37596 | 0.037238004042983296 | False |
| 16300 | 49 | 37596 | 0.13033301415044155 | False |
| 16540 | 14 | 37596 | 0.037238004042983296 | False |
| 16580 | 26 | 37596 | 0.06915629322268327 | False |
| 16620 | 8 | 37596 | 0.021278859453133313 | False |
| 16700 | 156 | 37596 | 0.4149377593360996 | False |
| 16740 | 444 | 37596 | 1.1809766996488988 | False |
| 16820 | 46 | 37596 | 0.12235344185551654 | False |
| 16860 | 108 | 37596 | 0.2872646026172997 | False |
| 16940 | 13 | 37596 | 0.034578146611341634 | False |
| 16984 | 988 | 37596 | 2.627939142461964 | False |
| 17020 | 17 | 37596 | 0.04521757633790829 | False |
| 17140 | 436 | 37596 | 1.1596978401957656 | False |
| 17300 | 34 | 37596 | 0.09043515267581657 | False |
| 17410 | 74 | 37596 | 0.19682944994148313 | False |
| 17420 | 20 | 37596 | 0.05319714863283328 | False |
| 17460 | 275 | 37596 | 0.7314607937014576 | False |
| 17660 | 27 | 37596 | 0.07181615065432492 | False |
| 17780 | 44 | 37596 | 0.11703372699223322 | False |
| 17820 | 103 | 37596 | 0.2739653154590914 | False |
| 17860 | 39 | 37596 | 0.1037344398340249 | False |
| 17900 | 104 | 37596 | 0.2766251728907331 | False |
| 17980 | 26 | 37596 | 0.06915629322268327 | False |
| 18020 | 12 | 37596 | 0.031918289179699966 | False |
| 18140 | 407 | 37596 | 1.0825619746781572 | False |
| 18580 | 35 | 37596 | 0.09309501010745824 | False |
| 18700 | 8 | 37596 | 0.021278859453133313 | False |
| 18880 | 27 | 37596 | 0.07181615065432492 | False |
| 19060 | 5 | 37596 | 0.01329928715820832 | False |
| 19124 | 932 | 37596 | 2.478987126290031 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 19140 | 10 | 37596 | 0.02659857431641664 | False |
| 19180 | 5 | 37596 | 0.01329928715820832 | False |
| 19300 | 75 | 37596 | 0.1994893073731248 | False |
| 19340 | 42 | 37596 | 0.1117140121289499 | False |
| 19430 | 127 | 37596 | 0.33780189381849135 | False |
| 19460 | 20 | 37596 | 0.05319714863283328 | False |
| 19500 | 6 | 37596 | 0.015959144589849983 | False |
| 19660 | 143 | 37596 | 0.38035961272475793 | False |
| 19740 | 500 | 37596 | 1.329928715820832 | False |
| 19780 | 128 | 37596 | 0.340461751250133 | False |
| 19804 | 219 | 37596 | 0.5825087775295245 | False |
| 20020 | 14 | 37596 | 0.037238004042983296 | False |
| 20100 | 23 | 37596 | 0.06117672092775827 | False |
| 20220 | 10 | 37596 | 0.02659857431641664 | False |
| 20260 | 43 | 37596 | 0.11437386956059155 | False |
| 20500 | 92 | 37596 | 0.24470688371103308 | False |
| 20580 | 0 | 37596 | 0.0 | False |
| 20700 | 25 | 37596 | 0.0664964357910416 | False |
| 20740 | 18 | 37596 | 0.047877433769549956 | False |
| 20940 | 10 | 37596 | 0.02659857431641664 | False |
| 20994 | 132 | 37596 | 0.3511011809766996 | False |
| 21060 | 11 | 37596 | 0.029258431748058304 | False |
| 21140 | 19 | 37596 | 0.05053729120119162 | False |
| 21300 | 7 | 37596 | 0.018619002021491648 | False |
| 21340 | 44 | 37596 | 0.11703372699223322 | False |
| 21420 | 5 | 37596 | 0.01329928715820832 | False |
| 21500 | 27 | 37596 | 0.07181615065432492 | False |
| 21660 | 41 | 37596 | 0.10905415469730823 | False |
| 21780 | 69 | 37596 | 0.1835301627832748 | False |
| 21794 | 13 | 37596 | 0.034578146611341634 | False |
| 21820 | 3 | 37596 | 0.007979572294924991 | False |
| 22020 | 28 | 37596 | 0.07447600808596659 | False |
| 22140 | 9 | 37596 | 0.023938716884774978 | False |
| 22180 | 36 | 37596 | 0.09575486753909991 | False |
| 22220 | 116 | 37596 | 0.30854346207043304 | False |
| 22380 | 18 | 37596 | 0.047877433769549956 | False |
| 22420 | 49 | 37596 | 0.13033301415044155 | False |
| 22500 | 12 | 37596 | 0.031918289179699966 | False |
| 22520 | 15 | 37596 | 0.03989786147462496 | False |
| 22540 | 8 | 37596 | 0.021278859453133313 | False |
| 22660 | 66 | 37596 | 0.1755505904883498 | False |
| 22744 | 162 | 37596 | 0.4308969039259496 | False |
| 22900 | 22 | 37596 | 0.05851686349611661 | False |
| 23060 | 63 | 37596 | 0.16757101819342482 | False |
| 23104 | 367 | 37596 | 0.9761676774124907 | False |
| 23224 | 143 | 37596 | 0.38035961272475793 | False |
| 23420 | 112 | 37596 | 0.29790403234386637 | False |
| 23460 | 8 | 37596 | 0.021278859453133313 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 23540 | 39 | 37596 | 0.1037344398340249 | False |
| 23580 | 34 | 37596 | 0.09043515267581657 | False |
| 23844 | 88 | 37596 | 0.23406745398446643 | False |
| 23900 | 14 | 37596 | 0.037238004042983296 | False |
| 24020 | 7 | 37596 | 0.018619002021491648 | False |
| 24140 | 9 | 37596 | 0.023938716884774978 | False |
| 24220 | 6 | 37596 | 0.015959144589849983 | False |
| 24260 | 4 | 37596 | 0.010639429726566656 | False |
| 24300 | 30 | 37596 | 0.07979572294924991 | False |
| 24340 | 205 | 37596 | 0.5452707734865411 | False |
| 24420 | 7 | 37596 | 0.018619002021491648 | False |
| 24500 | 6 | 37596 | 0.015959144589849983 | False |
| 24540 | 59 | 37596 | 0.15693158846685817 | False |
| 24580 | 43 | 37596 | 0.11437386956059155 | False |
| 24660 | 77 | 37596 | 0.20480902223640812 | False |
| 24780 | 18 | 37596 | 0.047877433769549956 | False |
| 24860 | 153 | 37596 | 0.4069581870411746 | False |
| 25060 | 36 | 37596 | 0.09575486753909991 | False |
| 25180 | 37 | 37596 | 0.09841472497074157 | False |
| 25220 | 8 | 37596 | 0.021278859453133313 | False |
| 25260 | 9 | 37596 | 0.023938716884774978 | False |
| 25420 | 94 | 37596 | 0.2500265985743164 | False |
| 25500 | 17 | 37596 | 0.04521757633790829 | False |
| 25540 | 95 | 37596 | 0.25268645600595807 | False |
| 25620 | 18 | 37596 | 0.047877433769549956 | False |
| 25740 | 4 | 37596 | 0.010639429726566656 | False |
| 25860 | 41 | 37596 | 0.10905415469730823 | False |
| 25940 | 57 | 37596 | 0.15161187360357484 | False |
| 25980 | 6 | 37596 | 0.015959144589849983 | False |
| 26140 | 27 | 37596 | 0.07181615065432492 | False |
| 26300 | 13 | 37596 | 0.034578146611341634 | False |
| 26380 | 18 | 37596 | 0.047877433769549956 | False |
| 26420 | 875 | 37596 | 2.327375252686456 | False |
| 26580 | 25 | 37596 | 0.0664964357910416 | False |
| 26620 | 66 | 37596 | 0.1755505904883498 | False |
| 26820 | 23 | 37596 | 0.06117672092775827 | False |
| 26900 | 439 | 37596 | 1.1676774124906906 | False |
| 26980 | 24 | 37596 | 0.06383657835939993 | False |
| 27060 | 6 | 37596 | 0.015959144589849983 | False |
| 27100 | 24 | 37596 | 0.06383657835939993 | False |
| 27140 | 27 | 37596 | 0.07181615065432492 | False |
| 27180 | 19 | 37596 | 0.05053729120119162 | False |
| 27260 | 199 | 37596 | 0.5293116288966911 | False |
| 27340 | 17 | 37596 | 0.04521757633790829 | False |
| 27500 | 15 | 37596 | 0.03989786147462496 | False |
| 27620 | 19 | 37596 | 0.05053729120119162 | False |
| 27740 | 23 | 37596 | 0.06117672092775827 | False |
| 27780 | 7 | 37596 | 0.018619002021491648 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 27860 | 6 | 37596 | 0.015959144589849983 | False |
| 27900 | 24 | 37596 | 0.06383657835939993 | False |
| 27980 | 7 | 37596 | 0.018619002021491648 | False |
| 28020 | 40 | 37596 | 0.10639429726566656 | False |
| 28100 | 19 | 37596 | 0.05053729120119162 | False |
| 28140 | 360 | 37596 | 0.9575486753909991 | False |
| 28420 | 37 | 37596 | 0.09841472497074157 | False |
| 28450 | 5 | 37596 | 0.01329928715820832 | False |
| 28660 | 46 | 37596 | 0.12235344185551654 | False |
| 28700 | 29 | 37596 | 0.07713586551760826 | False |
| 28740 | 18 | 37596 | 0.047877433769549956 | False |
| 28880 | 17 | 37596 | 0.04521757633790829 | False |
| 28940 | 130 | 37596 | 0.3457814661134163 | False |
| 29020 | 9 | 37596 | 0.023938716884774978 | False |
| 29100 | 13 | 37596 | 0.034578146611341634 | False |
| 29180 | 53 | 37596 | 0.1409724438770082 | False |
| 29200 | 25 | 37596 | 0.0664964357910416 | False |
| 29340 | 16 | 37596 | 0.042557718906266626 | False |
| 29404 | 118 | 37596 | 0.31386317693371635 | False |
| 29414 | 31 | 37596 | 0.08245558038089158 | False |
| 29420 | 51 | 37596 | 0.13565272901372485 | False |
| 29460 | 93 | 37596 | 0.24736674114267476 | False |
| 29484 | 51 | 37596 | 0.13565272901372485 | False |
| 29540 | 67 | 37596 | 0.1782104479199915 | False |
| 29620 | 67 | 37596 | 0.1782104479199915 | False |
| 29700 | 9 | 37596 | 0.023938716884774978 | False |
| 29740 | 15 | 37596 | 0.03989786147462496 | False |
| 29820 | 321 | 37596 | 0.8538142355569741 | False |
| 29940 | 14 | 37596 | 0.037238004042983296 | False |
| 30020 | 5 | 37596 | 0.01329928715820832 | False |
| 30140 | 18 | 37596 | 0.047877433769549956 | False |
| 30300 | 8 | 37596 | 0.021278859453133313 | False |
| 30340 | 13 | 37596 | 0.034578146611341634 | False |
| 30460 | 71 | 37596 | 0.18884987764655814 | False |
| 30500 | 6 | 37596 | 0.015959144589849983 | False |
| 30620 | 12 | 37596 | 0.031918289179699966 | False |
| 30700 | 48 | 37596 | 0.12767315671879986 | False |
| 30780 | 102 | 37596 | 0.2713054580274497 | False |
| 30860 | 35 | 37596 | 0.09309501010745824 | False |
| 30980 | 14 | 37596 | 0.037238004042983296 | False |
| 31020 | 16 | 37596 | 0.042557718906266626 | False |
| 31084 | 430 | 37596 | 1.1437386956059155 | False |
| 31140 | 208 | 37596 | 0.5532503457814661 | False |
| 31180 | 45 | 37596 | 0.11969358442387489 | False |
| 31340 | 36 | 37596 | 0.09575486753909991 | False |
| 31420 | 26 | 37596 | 0.06915629322268327 | False |
| 31460 | 15 | 37596 | 0.03989786147462496 | False |
| 31540 | 80 | 37596 | 0.21278859453133311 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 31700 | 45 | 37596 | 0.11969358442387489 | False |
| 31740 | 17 | 37596 | 0.04521757633790829 | False |
| 31860 | 8 | 37596 | 0.021278859453133313 | False |
| 31900 | 9 | 37596 | 0.023938716884774978 | False |
| 31924 | 63 | 37596 | 0.16757101819342482 | False |
| 32580 | 30 | 37596 | 0.07979572294924991 | False |
| 32780 | 25 | 37596 | 0.0664964357910416 | False |
| 32820 | 154 | 37596 | 0.40961804447281625 | False |
| 32900 | 19 | 37596 | 0.05053729120119162 | False |
| 33124 | 94 | 37596 | 0.2500265985743164 | False |
| 33140 | 17 | 37596 | 0.04521757633790829 | False |
| 33220 | 5 | 37596 | 0.01329928715820832 | False |
| 33260 | 24 | 37596 | 0.06383657835939993 | False |
| 33340 | 182 | 37596 | 0.48409405255878285 | False |
| 33460 | 569 | 37596 | 1.5134588786041068 | False |
| 33540 | 13 | 37596 | 0.034578146611341634 | False |
| 33660 | 38 | 37596 | 0.10107458240238323 | False |
| 33700 | 38 | 37596 | 0.10107458240238323 | False |
| 33740 | 11 | 37596 | 0.029258431748058304 | False |
| 33780 | 24 | 37596 | 0.06383657835939993 | False |
| 33860 | 30 | 37596 | 0.07979572294924991 | False |
| 33874 | 285 | 37596 | 0.7580593680178742 | False |
| 34060 | 16 | 37596 | 0.042557718906266626 | False |
| 34100 | 19 | 37596 | 0.05053729120119162 | False |
| 34580 | 13 | 37596 | 0.034578146611341634 | False |
| 34620 | 14 | 37596 | 0.037238004042983296 | False |
| 34740 | 18 | 37596 | 0.047877433769549956 | False |
| 34820 | 181 | 37596 | 0.4814341951271412 | False |
| 34900 | 10 | 37596 | 0.02659857431641664 | False |
| 34940 | 57 | 37596 | 0.15161187360357484 | False |
| 34980 | 385 | 37596 | 1.0240451111820406 | False |
| 35004 | 162 | 37596 | 0.4308969039259496 | False |
| 35084 | 160 | 37596 | 0.42557718906266623 | False |
| 35100 | 11 | 37596 | 0.029258431748058304 | False |
| 35154 | 197 | 37596 | 0.5239919140334078 | False |
| 35300 | 63 | 37596 | 0.16757101819342482 | False |
| 35380 | 95 | 37596 | 0.25268645600595807 | False |
| 35614 | 411 | 37596 | 1.0932014044047238 | False |
| 35660 | 24 | 37596 | 0.06383657835939993 | False |
| 35840 | 161 | 37596 | 0.4282370464943079 | False |
| 35980 | 15 | 37596 | 0.03989786147462496 | False |
| 36084 | 168 | 37596 | 0.4468560485157996 | False |
| 36100 | 78 | 37596 | 0.2074688796680498 | False |
| 36140 | 7 | 37596 | 0.018619002021491648 | False |
| 36220 | 9 | 37596 | 0.023938716884774978 | False |
| 36260 | 95 | 37596 | 0.25268645600595807 | False |
| 36420 | 217 | 37596 | 0.577189062666241 | False |
| 36500 | 33 | 37596 | 0.0877752952441749 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 36540 | 135 | 37596 | 0.35908075327162464 | False |
| 36740 | 278 | 37596 | 0.7394403659963826 | False |
| 36780 | 14 | 37596 | 0.037238004042983296 | False |
| 36980 | 20 | 37596 | 0.05319714863283328 | False |
| 37100 | 55 | 37596 | 0.14629215874029153 | False |
| 37140 | 3 | 37596 | 0.007979572294924991 | False |
| 37340 | 82 | 37596 | 0.21810830939461645 | False |
| 37460 | 20 | 37596 | 0.05319714863283328 | False |
| 37620 | 11 | 37596 | 0.029258431748058304 | False |
| 37860 | 57 | 37596 | 0.15161187360357484 | False |
| 37900 | 39 | 37596 | 0.1037344398340249 | False |
| 37964 | 220 | 37596 | 0.5851686349611661 | False |
| 38060 | 806 | 37596 | 2.143845089903181 | False |
| 38220 | 3 | 37596 | 0.007979572294924991 | False |
| 38240 | 4 | 37596 | 0.010639429726566656 | False |
| 38300 | 355 | 37596 | 0.9442493882327907 | False |
| 38340 | 6 | 37596 | 0.015959144589849983 | False |
| 38540 | 12 | 37596 | 0.031918289179699966 | False |
| 38860 | 73 | 37596 | 0.19416959250984148 | False |
| 38900 | 349 | 37596 | 0.9282902436429408 | False |
| 38940 | 78 | 37596 | 0.2074688796680498 | False |
| 39100 | 51 | 37596 | 0.13565272901372485 | False |
| 39150 | 53 | 37596 | 0.1409724438770082 | False |
| 39300 | 129 | 37596 | 0.34312160868177466 | False |
| 39340 | 132 | 37596 | 0.3511011809766996 | False |
| 39380 | 16 | 37596 | 0.042557718906266626 | False |
| 39460 | 56 | 37596 | 0.14895201617193318 | False |
| 39540 | 18 | 37596 | 0.047877433769549956 | False |
| 39580 | 266 | 37596 | 0.7075220768166827 | False |
| 39660 | 20 | 37596 | 0.05319714863283328 | False |
| 39740 | 48 | 37596 | 0.12767315671879986 | False |
| 39820 | 21 | 37596 | 0.05585700606447495 | False |
| 39900 | 68 | 37596 | 0.18087030535163315 | False |
| 40060 | 220 | 37596 | 0.5851686349611661 | False |
| 40140 | 373 | 37596 | 0.9921268220023407 | False |
| 40220 | 54 | 37596 | 0.14363230130864985 | False |
| 40340 | 34 | 37596 | 0.09043515267581657 | False |
| 40380 | 155 | 37596 | 0.4122779019044579 | False |
| 40420 | 37 | 37596 | 0.09841472497074157 | False |
| 40484 | 55 | 37596 | 0.14629215874029153 | False |
| 40580 | 16 | 37596 | 0.042557718906266626 | False |
| 40660 | 11 | 37596 | 0.029258431748058304 | False |
| 40900 | 312 | 37596 | 0.8298755186721992 | False |
| 40980 | 23 | 37596 | 0.06117672092775827 | False |
| 41060 | 23 | 37596 | 0.06117672092775827 | False |
| 41100 | 53 | 37596 | 0.1409724438770082 | False |
| 41140 | 20 | 37596 | 0.05319714863283328 | False |
| 41180 | 494 | 37596 | 1.313969571230982 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 41304 | 25 | 37596 | 0.0664964357910416 | False |
| 41420 | 46 | 37596 | 0.12235344185551654 | False |
| 41500 | 26 | 37596 | 0.06915629322268327 | False |
| 41540 | 78 | 37596 | 0.2074688796680498 | False |
| 41620 | 178 | 37596 | 0.4734546228322162 | False |
| 41660 | 14 | 37596 | 0.037238004042983296 | False |
| 41700 | 289 | 37596 | 0.7686987977444409 | False |
| 41740 | 210 | 37596 | 0.5585700606447495 | False |
| 41780 | 4 | 37596 | 0.010639429726566656 | False |
| 41884 | 44 | 37596 | 0.11703372699223322 | False |
| 41940 | 70 | 37596 | 0.18619002021491649 | False |
| 41980 | 1 | 37596 | 0.002659857431641664 | False |
| 42020 | 24 | 37596 | 0.06383657835939993 | False |
| 42034 | 15 | 37596 | 0.03989786147462496 | False |
| 42100 | 11 | 37596 | 0.029258431748058304 | False |
| 42140 | 13 | 37596 | 0.034578146611341634 | False |
| 42200 | 12 | 37596 | 0.031918289179699966 | False |
| 42220 | 43 | 37596 | 0.11437386956059155 | False |
| 42340 | 69 | 37596 | 0.1835301627832748 | False |
| 42540 | 48 | 37596 | 0.12767315671879986 | False |
| 42644 | 299 | 37596 | 0.7952973720608575 | False |
| 42680 | 26 | 37596 | 0.06915629322268327 | False |
| 42700 | 17 | 37596 | 0.04521757633790829 | False |
| 43100 | 11 | 37596 | 0.029258431748058304 | False |
| 43300 | 32 | 37596 | 0.08511543781253325 | False |
| 43340 | 27 | 37596 | 0.07181615065432492 | False |
| 43420 | 17 | 37596 | 0.04521757633790829 | False |
| 43580 | 12 | 37596 | 0.031918289179699966 | False |
| 43620 | 34 | 37596 | 0.09043515267581657 | False |
| 43640 | 9 | 37596 | 0.023938716884774978 | False |
| 43780 | 56 | 37596 | 0.14895201617193318 | False |
| 43900 | 46 | 37596 | 0.12235344185551654 | False |
| 44060 | 91 | 37596 | 0.24204702627939143 | False |
| 44100 | 24 | 37596 | 0.06383657835939993 | False |
| 44140 | 41 | 37596 | 0.10905415469730823 | False |
| 44180 | 70 | 37596 | 0.18619002021491649 | False |
| 44220 | 18 | 37596 | 0.047877433769549956 | False |
| 44300 | 15 | 37596 | 0.03989786147462496 | False |
| 44420 | 13 | 37596 | 0.034578146611341634 | False |
| 44700 | 70 | 37596 | 0.18619002021491649 | False |
| 44940 | 8 | 37596 | 0.021278859453133313 | False |
| 45060 | 51 | 37596 | 0.13565272901372485 | False |
| 45104 | 82 | 37596 | 0.21810830939461645 | False |
| 45220 | 30 | 37596 | 0.07979572294924991 | False |
| 45294 | 50 | 37596 | 0.1329928715820832 | False |
| 45300 | 359 | 37596 | 0.9548888179593574 | False |
| 45460 | 26 | 37596 | 0.06915629322268327 | False |
| 45500 | 5 | 37596 | 0.01329928715820832 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 45540 | 18 | 37596 | 0.047877433769549956 | False |
| 45780 | 124 | 37596 | 0.32982232152356633 | False |
| 45820 | 29 | 37596 | 0.07713586551760826 | False |
| 45900 | 2 | 37596 | 0.005319714863283328 | False |
| 45940 | 37 | 37596 | 0.09841472497074157 | False |
| 46060 | 181 | 37596 | 0.4814341951271412 | False |
| 46140 | 146 | 37596 | 0.38833918501968295 | False |
| 46220 | 23 | 37596 | 0.06117672092775827 | False |
| 46300 | 7 | 37596 | 0.018619002021491648 | False |
| 46340 | 27 | 37596 | 0.07181615065432492 | False |
| 46520 | 25 | 37596 | 0.0664964357910416 | False |
| 46540 | 12 | 37596 | 0.031918289179699966 | False |
| 46660 | 10 | 37596 | 0.02659857431641664 | False |
| 46700 | 48 | 37596 | 0.12767315671879986 | False |
| 47020 | 5 | 37596 | 0.01329928715820832 | False |
| 47220 | 9 | 37596 | 0.023938716884774978 | False |
| 47260 | 200 | 37596 | 0.5319714863283328 | False |
| 47300 | 35 | 37596 | 0.09309501010745824 | False |
| 47380 | 26 | 37596 | 0.06915629322268327 | False |
| 47460 | 4 | 37596 | 0.010639429726566656 | False |
| 47580 | 23 | 37596 | 0.06117672092775827 | False |
| 47664 | 511 | 37596 | 1.3591871475688904 | False |
| 47764 | 30 | 37596 | 0.07979572294924991 | False |
| 47894 | 430 | 37596 | 1.1437386956059155 | False |
| 47930 | 1 | 37596 | 0.002659857431641664 | False |
| 47940 | 28 | 37596 | 0.07447600808596659 | False |
| 48060 | 4 | 37596 | 0.010639429726566656 | False |
| 48140 | 12 | 37596 | 0.031918289179699966 | False |
| 48260 | 8 | 37596 | 0.021278859453133313 | False |
| 48300 | 19 | 37596 | 0.05053729120119162 | False |
| 48424 | 120 | 37596 | 0.31918289179699966 | False |
| 48540 | 10 | 37596 | 0.02659857431641664 | False |
| 48620 | 103 | 37596 | 0.2739653154590914 | False |
| 48660 | 14 | 37596 | 0.037238004042983296 | False |
| 48680 | 4 | 37596 | 0.010639429726566656 | False |
| 48700 | 8 | 37596 | 0.021278859453133313 | False |
| 48864 | 111 | 37596 | 0.2952441749122247 | False |
| 48900 | 55 | 37596 | 0.14629215874029153 | False |
| 49020 | 26 | 37596 | 0.06915629322268327 | False |
| 49180 | 84 | 37596 | 0.2234280242578998 | False |
| 49340 | 110 | 37596 | 0.29258431748058306 | False |
| 49420 | 21 | 37596 | 0.05585700606447495 | False |
| 49500 | 1 | 37596 | 0.002659857431641664 | False |
| 49620 | 66 | 37596 | 0.1755505904883498 | False |
| 49660 | 74 | 37596 | 0.19682944994148313 | False |
| 49700 | 11 | 37596 | 0.029258431748058304 | False |
| 49740 | 20 | 37596 | 0.05319714863283328 | False |
| nonmetropolitan_or_unknown | 4178 | 37596 | 11.112884349398872 | False |

Both classified as risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 10180 | 3 | 5792 | 0.05179558011049724 | False |
| 10420 | 15 | 5792 | 0.2589779005524862 | False |
| 10500 | 0 | 5792 | 0.0 | False |
| 10540 | 0 | 5792 | 0.0 | False |
| 10580 | 15 | 5792 | 0.2589779005524862 | False |
| 10740 | 9 | 5792 | 0.15538674033149172 | False |
| 10780 | 2 | 5792 | 0.034530386740331494 | False |
| 10900 | 14 | 5792 | 0.24171270718232044 | False |
| 11020 | 0 | 5792 | 0.0 | False |
| 11100 | 4 | 5792 | 0.06906077348066299 | False |
| 11180 | 2 | 5792 | 0.034530386740331494 | False |
| 11200 | 1 | 5792 | 0.017265193370165747 | False |
| 11244 | 22 | 5792 | 0.3798342541436464 | False |
| 11260 | 3 | 5792 | 0.05179558011049724 | False |
| 11460 | 3 | 5792 | 0.05179558011049724 | False |
| 11500 | 0 | 5792 | 0.0 | False |
| 11540 | 1 | 5792 | 0.017265193370165747 | False |
| 11694 | 11 | 5792 | 0.1899171270718232 | False |
| 11700 | 2 | 5792 | 0.034530386740331494 | False |
| 12020 | 1 | 5792 | 0.017265193370165747 | False |
| 12054 | 64 | 5792 | 1.1049723756906078 | False |
| 12060 | 148 | 5792 | 2.5552486187845305 | False |
| 12100 | 15 | 5792 | 0.2589779005524862 | False |
| 12220 | 0 | 5792 | 0.0 | False |
| 12260 | 6 | 5792 | 0.10359116022099447 | False |
| 12420 | 54 | 5792 | 0.9323204419889503 | False |
| 12540 | 12 | 5792 | 0.20718232044198895 | False |
| 12580 | 73 | 5792 | 1.2603591160220995 | False |
| 12620 | 3 | 5792 | 0.05179558011049724 | False |
| 12700 | 1 | 5792 | 0.017265193370165747 | False |
| 12940 | 17 | 5792 | 0.2935082872928177 | False |
| 12980 | 1 | 5792 | 0.017265193370165747 | False |
| 13020 | 0 | 5792 | 0.0 | False |
| 13140 | 2 | 5792 | 0.034530386740331494 | False |
| 13220 | 0 | 5792 | 0.0 | False |
| 13380 | 3 | 5792 | 0.05179558011049724 | False |
| 13460 | 3 | 5792 | 0.05179558011049724 | False |
| 13740 | 1 | 5792 | 0.017265193370165747 | False |
| 13780 | 4 | 5792 | 0.06906077348066299 | False |
| 13820 | 12 | 5792 | 0.20718232044198895 | False |
| 13900 | 0 | 5792 | 0.0 | False |
| 13980 | 0 | 5792 | 0.0 | False |
| 14010 | 7 | 5792 | 0.12085635359116022 | False |
| 14020 | 3 | 5792 | 0.05179558011049724 | False |
| 14100 | 0 | 5792 | 0.0 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 14260 | 34 | 5792 | 0.5870165745856354 | False |
| 14454 | 20 | 5792 | 0.3453038674033149 | False |
| 14500 | 2 | 5792 | 0.034530386740331494 | False |
| 14540 | 2 | 5792 | 0.034530386740331494 | False |
| 14580 | 0 | 5792 | 0.0 | False |
| 14740 | 2 | 5792 | 0.034530386740331494 | False |
| 14860 | 24 | 5792 | 0.4143646408839779 | False |
| 15180 | 2 | 5792 | 0.034530386740331494 | False |
| 15260 | 2 | 5792 | 0.034530386740331494 | False |
| 15380 | 36 | 5792 | 0.6215469613259669 | False |
| 15500 | 5 | 5792 | 0.08632596685082873 | False |
| 15540 | 3 | 5792 | 0.05179558011049724 | False |
| 15680 | 0 | 5792 | 0.0 | False |
| 15764 | 30 | 5792 | 0.5179558011049724 | False |
| 15804 | 39 | 5792 | 0.6733425414364641 | False |
| 15940 | 4 | 5792 | 0.06906077348066299 | False |
| 15980 | 51 | 5792 | 0.880524861878453 | False |
| 16020 | 0 | 5792 | 0.0 | False |
| 16060 | 0 | 5792 | 0.0 | False |
| 16180 | 1 | 5792 | 0.017265193370165747 | False |
| 16220 | 1 | 5792 | 0.017265193370165747 | False |
| 16300 | 5 | 5792 | 0.08632596685082873 | False |
| 16540 | 1 | 5792 | 0.017265193370165747 | False |
| 16580 | 1 | 5792 | 0.017265193370165747 | False |
| 16620 | 2 | 5792 | 0.034530386740331494 | False |
| 16700 | 7 | 5792 | 0.12085635359116022 | False |
| 16740 | 40 | 5792 | 0.6906077348066298 | False |
| 16820 | 2 | 5792 | 0.034530386740331494 | False |
| 16860 | 4 | 5792 | 0.06906077348066299 | False |
| 16940 | 2 | 5792 | 0.034530386740331494 | False |
| 16984 | 240 | 5792 | 4.143646408839779 | False |
| 17020 | 1 | 5792 | 0.017265193370165747 | False |
| 17140 | 26 | 5792 | 0.4488950276243094 | False |
| 17300 | 5 | 5792 | 0.08632596685082873 | False |
| 17410 | 12 | 5792 | 0.20718232044198895 | False |
| 17420 | 0 | 5792 | 0.0 | False |
| 17460 | 4 | 5792 | 0.06906077348066299 | False |
| 17660 | 2 | 5792 | 0.034530386740331494 | False |
| 17780 | 5 | 5792 | 0.08632596685082873 | False |
| 17820 | 12 | 5792 | 0.20718232044198895 | False |
| 17860 | 1 | 5792 | 0.017265193370165747 | False |
| 17900 | 10 | 5792 | 0.17265193370165746 | False |
| 17980 | 3 | 5792 | 0.05179558011049724 | False |
| 18020 | 2 | 5792 | 0.034530386740331494 | False |
| 18140 | 38 | 5792 | 0.6560773480662984 | False |
| 18580 | 2 | 5792 | 0.034530386740331494 | False |
| 18700 | 1 | 5792 | 0.017265193370165747 | False |
| 18880 | 5 | 5792 | 0.08632596685082873 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 19060 | 1 | 5792 | 0.017265193370165747 | False |
| 19124 | 97 | 5792 | 1.6747237569060773 | False |
| 19140 | 0 | 5792 | 0.0 | False |
| 19180 | 0 | 5792 | 0.0 | False |
| 19300 | 1 | 5792 | 0.017265193370165747 | False |
| 19340 | 5 | 5792 | 0.08632596685082873 | False |
| 19430 | 13 | 5792 | 0.2244475138121547 | False |
| 19460 | 0 | 5792 | 0.0 | False |
| 19500 | 1 | 5792 | 0.017265193370165747 | False |
| 19660 | 19 | 5792 | 0.3280386740331492 | False |
| 19740 | 68 | 5792 | 1.1740331491712708 | False |
| 19780 | 18 | 5792 | 0.31077348066298344 | False |
| 19804 | 27 | 5792 | 0.4661602209944751 | False |
| 20020 | 0 | 5792 | 0.0 | False |
| 20100 | 4 | 5792 | 0.06906077348066299 | False |
| 20220 | 1 | 5792 | 0.017265193370165747 | False |
| 20260 | 2 | 5792 | 0.034530386740331494 | False |
| 20500 | 9 | 5792 | 0.15538674033149172 | False |
| 20580 | 0 | 5792 | 0.0 | False |
| 20700 | 3 | 5792 | 0.05179558011049724 | False |
| 20740 | 2 | 5792 | 0.034530386740331494 | False |
| 20940 | 0 | 5792 | 0.0 | False |
| 20994 | 14 | 5792 | 0.24171270718232044 | False |
| 21060 | 3 | 5792 | 0.05179558011049724 | False |
| 21140 | 3 | 5792 | 0.05179558011049724 | False |
| 21300 | 0 | 5792 | 0.0 | False |
| 21340 | 5 | 5792 | 0.08632596685082873 | False |
| 21420 | 0 | 5792 | 0.0 | False |
| 21500 | 2 | 5792 | 0.034530386740331494 | False |
| 21660 | 2 | 5792 | 0.034530386740331494 | False |
| 21780 | 2 | 5792 | 0.034530386740331494 | False |
| 21794 | 4 | 5792 | 0.06906077348066299 | False |
| 21820 | 0 | 5792 | 0.0 | False |
| 22020 | 2 | 5792 | 0.034530386740331494 | False |
| 22140 | 0 | 5792 | 0.0 | False |
| 22180 | 5 | 5792 | 0.08632596685082873 | False |
| 22220 | 6 | 5792 | 0.10359116022099447 | False |
| 22380 | 0 | 5792 | 0.0 | False |
| 22420 | 3 | 5792 | 0.05179558011049724 | False |
| 22500 | 1 | 5792 | 0.017265193370165747 | False |
| 22520 | 1 | 5792 | 0.017265193370165747 | False |
| 22540 | 1 | 5792 | 0.017265193370165747 | False |
| 22660 | 3 | 5792 | 0.05179558011049724 | False |
| 22744 | 139 | 5792 | 2.3998618784530388 | False |
| 22900 | 1 | 5792 | 0.017265193370165747 | False |
| 23060 | 5 | 5792 | 0.08632596685082873 | False |
| 23104 | 48 | 5792 | 0.8287292817679558 | False |
| 23224 | 29 | 5792 | 0.5006906077348067 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 23420 | 8 | 5792 | 0.13812154696132597 | False |
| 23460 | 0 | 5792 | 0.0 | False |
| 23540 | 8 | 5792 | 0.13812154696132597 | False |
| 23580 | 6 | 5792 | 0.10359116022099447 | False |
| 23844 | 11 | 5792 | 0.1899171270718232 | False |
| 23900 | 0 | 5792 | 0.0 | False |
| 24020 | 1 | 5792 | 0.017265193370165747 | False |
| 24140 | 0 | 5792 | 0.0 | False |
| 24220 | 0 | 5792 | 0.0 | False |
| 24260 | 0 | 5792 | 0.0 | False |
| 24300 | 2 | 5792 | 0.034530386740331494 | False |
| 24340 | 18 | 5792 | 0.31077348066298344 | False |
| 24420 | 0 | 5792 | 0.0 | False |
| 24500 | 1 | 5792 | 0.017265193370165747 | False |
| 24540 | 11 | 5792 | 0.1899171270718232 | False |
| 24580 | 4 | 5792 | 0.06906077348066299 | False |
| 24660 | 8 | 5792 | 0.13812154696132597 | False |
| 24780 | 2 | 5792 | 0.034530386740331494 | False |
| 24860 | 8 | 5792 | 0.13812154696132597 | False |
| 25060 | 1 | 5792 | 0.017265193370165747 | False |
| 25180 | 10 | 5792 | 0.17265193370165746 | False |
| 25220 | 3 | 5792 | 0.05179558011049724 | False |
| 25260 | 1 | 5792 | 0.017265193370165747 | False |
| 25420 | 7 | 5792 | 0.12085635359116022 | False |
| 25500 | 1 | 5792 | 0.017265193370165747 | False |
| 25540 | 45 | 5792 | 0.7769337016574586 | False |
| 25620 | 1 | 5792 | 0.017265193370165747 | False |
| 25740 | 1 | 5792 | 0.017265193370165747 | False |
| 25860 | 0 | 5792 | 0.0 | False |
| 25940 | 2 | 5792 | 0.034530386740331494 | False |
| 25980 | 1 | 5792 | 0.017265193370165747 | False |
| 26140 | 7 | 5792 | 0.12085635359116022 | False |
| 26300 | 1 | 5792 | 0.017265193370165747 | False |
| 26380 | 1 | 5792 | 0.017265193370165747 | False |
| 26420 | 209 | 5792 | 3.608425414364641 | False |
| 26580 | 2 | 5792 | 0.034530386740331494 | False |
| 26620 | 4 | 5792 | 0.06906077348066299 | False |
| 26820 | 3 | 5792 | 0.05179558011049724 | False |
| 26900 | 43 | 5792 | 0.7424033149171271 | False |
| 26980 | 1 | 5792 | 0.017265193370165747 | False |
| 27060 | 0 | 5792 | 0.0 | False |
| 27100 | 5 | 5792 | 0.08632596685082873 | False |
| 27140 | 4 | 5792 | 0.06906077348066299 | False |
| 27180 | 0 | 5792 | 0.0 | False |
| 27260 | 63 | 5792 | 1.087707182320442 | False |
| 27340 | 4 | 5792 | 0.06906077348066299 | False |
| 27500 | 2 | 5792 | 0.034530386740331494 | False |
| 27620 | 0 | 5792 | 0.0 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 27740 | 1 | 5792 | 0.017265193370165747 | False |
| 27780 | 0 | 5792 | 0.0 | False |
| 27860 | 0 | 5792 | 0.0 | False |
| 27900 | 2 | 5792 | 0.034530386740331494 | False |
| 27980 | 1 | 5792 | 0.017265193370165747 | False |
| 28020 | 1 | 5792 | 0.017265193370165747 | False |
| 28100 | 3 | 5792 | 0.05179558011049724 | False |
| 28140 | 29 | 5792 | 0.5006906077348067 | False |
| 28420 | 3 | 5792 | 0.05179558011049724 | False |
| 28450 | 0 | 5792 | 0.0 | False |
| 28660 | 6 | 5792 | 0.10359116022099447 | False |
| 28700 | 0 | 5792 | 0.0 | False |
| 28740 | 2 | 5792 | 0.034530386740331494 | False |
| 28880 | 10 | 5792 | 0.17265193370165746 | False |
| 28940 | 8 | 5792 | 0.13812154696132597 | False |
| 29020 | 1 | 5792 | 0.017265193370165747 | False |
| 29100 | 0 | 5792 | 0.0 | False |
| 29180 | 5 | 5792 | 0.08632596685082873 | False |
| 29200 | 0 | 5792 | 0.0 | False |
| 29340 | 4 | 5792 | 0.06906077348066299 | False |
| 29404 | 14 | 5792 | 0.24171270718232044 | False |
| 29414 | 6 | 5792 | 0.10359116022099447 | False |
| 29420 | 3 | 5792 | 0.05179558011049724 | False |
| 29460 | 44 | 5792 | 0.7596685082872928 | False |
| 29484 | 29 | 5792 | 0.5006906077348067 | False |
| 29540 | 3 | 5792 | 0.05179558011049724 | False |
| 29620 | 4 | 5792 | 0.06906077348066299 | False |
| 29700 | 1 | 5792 | 0.017265193370165747 | False |
| 29740 | 1 | 5792 | 0.017265193370165747 | False |
| 29820 | 43 | 5792 | 0.7424033149171271 | False |
| 29940 | 0 | 5792 | 0.0 | False |
| 30020 | 0 | 5792 | 0.0 | False |
| 30140 | 0 | 5792 | 0.0 | False |
| 30300 | 1 | 5792 | 0.017265193370165747 | False |
| 30340 | 2 | 5792 | 0.034530386740331494 | False |
| 30460 | 3 | 5792 | 0.05179558011049724 | False |
| 30500 | 0 | 5792 | 0.0 | False |
| 30620 | 0 | 5792 | 0.0 | False |
| 30700 | 5 | 5792 | 0.08632596685082873 | False |
| 30780 | 11 | 5792 | 0.1899171270718232 | False |
| 30860 | 3 | 5792 | 0.05179558011049724 | False |
| 30980 | 2 | 5792 | 0.034530386740331494 | False |
| 31020 | 2 | 5792 | 0.034530386740331494 | False |
| 31084 | 81 | 5792 | 1.3984806629834254 | False |
| 31140 | 16 | 5792 | 0.27624309392265195 | False |
| 31180 | 2 | 5792 | 0.034530386740331494 | False |
| 31340 | 2 | 5792 | 0.034530386740331494 | False |
| 31420 | 3 | 5792 | 0.05179558011049724 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 31460 | 0 | 5792 | 0.0 | False |
| 31540 | 5 | 5792 | 0.08632596685082873 | False |
| 31700 | 3 | 5792 | 0.05179558011049724 | False |
| 31740 | 0 | 5792 | 0.0 | False |
| 31860 | 1 | 5792 | 0.017265193370165747 | False |
| 31900 | 1 | 5792 | 0.017265193370165747 | False |
| 31924 | 8 | 5792 | 0.13812154696132597 | False |
| 32580 | 7 | 5792 | 0.12085635359116022 | False |
| 32780 | 3 | 5792 | 0.05179558011049724 | False |
| 32820 | 9 | 5792 | 0.15538674033149172 | False |
| 32900 | 3 | 5792 | 0.05179558011049724 | False |
| 33124 | 186 | 5792 | 3.2113259668508287 | False |
| 33140 | 2 | 5792 | 0.034530386740331494 | False |
| 33220 | 3 | 5792 | 0.05179558011049724 | False |
| 33260 | 4 | 5792 | 0.06906077348066299 | False |
| 33340 | 17 | 5792 | 0.2935082872928177 | False |
| 33460 | 93 | 5792 | 1.6056629834254144 | False |
| 33540 | 1 | 5792 | 0.017265193370165747 | False |
| 33660 | 2 | 5792 | 0.034530386740331494 | False |
| 33700 | 4 | 5792 | 0.06906077348066299 | False |
| 33740 | 1 | 5792 | 0.017265193370165747 | False |
| 33780 | 4 | 5792 | 0.06906077348066299 | False |
| 33860 | 5 | 5792 | 0.08632596685082873 | False |
| 33874 | 26 | 5792 | 0.4488950276243094 | False |
| 34060 | 1 | 5792 | 0.017265193370165747 | False |
| 34100 | 0 | 5792 | 0.0 | False |
| 34580 | 2 | 5792 | 0.034530386740331494 | False |
| 34620 | 1 | 5792 | 0.017265193370165747 | False |
| 34740 | 3 | 5792 | 0.05179558011049724 | False |
| 34820 | 12 | 5792 | 0.20718232044198895 | False |
| 34900 | 0 | 5792 | 0.0 | False |
| 34940 | 21 | 5792 | 0.36256906077348067 | False |
| 34980 | 34 | 5792 | 0.5870165745856354 | False |
| 35004 | 125 | 5792 | 2.1581491712707184 | False |
| 35084 | 89 | 5792 | 1.5366022099447514 | False |
| 35100 | 1 | 5792 | 0.017265193370165747 | False |
| 35154 | 43 | 5792 | 0.7424033149171271 | False |
| 35300 | 30 | 5792 | 0.5179558011049724 | False |
| 35380 | 16 | 5792 | 0.27624309392265195 | False |
| 35614 | 193 | 5792 | 3.332182320441989 | False |
| 35660 | 1 | 5792 | 0.017265193370165747 | False |
| 35840 | 47 | 5792 | 0.81146408839779 | False |
| 35980 | 8 | 5792 | 0.13812154696132597 | False |
| 36084 | 25 | 5792 | 0.43162983425414364 | False |
| 36100 | 20 | 5792 | 0.3453038674033149 | False |
| 36140 | 2 | 5792 | 0.034530386740331494 | False |
| 36220 | 2 | 5792 | 0.034530386740331494 | False |
| 36260 | 7 | 5792 | 0.12085635359116022 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 36420 | 20 | 5792 | 0.3453038674033149 | False |
| 36500 | 1 | 5792 | 0.017265193370165747 | False |
| 36540 | 8 | 5792 | 0.13812154696132597 | False |
| 36740 | 171 | 5792 | 2.9523480662983426 | False |
| 36780 | 2 | 5792 | 0.034530386740331494 | False |
| 36980 | 0 | 5792 | 0.0 | False |
| 37100 | 16 | 5792 | 0.27624309392265195 | False |
| 37140 | 0 | 5792 | 0.0 | False |
| 37340 | 28 | 5792 | 0.48342541436464087 | False |
| 37460 | 7 | 5792 | 0.12085635359116022 | False |
| 37620 | 1 | 5792 | 0.017265193370165747 | False |
| 37860 | 12 | 5792 | 0.20718232044198895 | False |
| 37900 | 8 | 5792 | 0.13812154696132597 | False |
| 37964 | 20 | 5792 | 0.3453038674033149 | False |
| 38060 | 64 | 5792 | 1.1049723756906078 | False |
| 38220 | 0 | 5792 | 0.0 | False |
| 38240 | 1 | 5792 | 0.017265193370165747 | False |
| 38300 | 20 | 5792 | 0.3453038674033149 | False |
| 38340 | 0 | 5792 | 0.0 | False |
| 38540 | 4 | 5792 | 0.06906077348066299 | False |
| 38860 | 7 | 5792 | 0.12085635359116022 | False |
| 38900 | 30 | 5792 | 0.5179558011049724 | False |
| 38940 | 40 | 5792 | 0.6906077348066298 | False |
| 39100 | 21 | 5792 | 0.36256906077348067 | False |
| 39150 | 4 | 5792 | 0.06906077348066299 | False |
| 39300 | 37 | 5792 | 0.6388121546961326 | False |
| 39340 | 10 | 5792 | 0.17265193370165746 | False |
| 39380 | 2 | 5792 | 0.034530386740331494 | False |
| 39460 | 8 | 5792 | 0.13812154696132597 | False |
| 39540 | 0 | 5792 | 0.0 | False |
| 39580 | 34 | 5792 | 0.5870165745856354 | False |
| 39660 | 2 | 5792 | 0.034530386740331494 | False |
| 39740 | 7 | 5792 | 0.12085635359116022 | False |
| 39820 | 2 | 5792 | 0.034530386740331494 | False |
| 39900 | 5 | 5792 | 0.08632596685082873 | False |
| 40060 | 19 | 5792 | 0.3280386740331492 | False |
| 40140 | 79 | 5792 | 1.363950276243094 | False |
| 40220 | 1 | 5792 | 0.017265193370165747 | False |
| 40340 | 2 | 5792 | 0.034530386740331494 | False |
| 40380 | 29 | 5792 | 0.5006906077348067 | False |
| 40420 | 10 | 5792 | 0.17265193370165746 | False |
| 40484 | 6 | 5792 | 0.10359116022099447 | False |
| 40580 | 1 | 5792 | 0.017265193370165747 | False |
| 40660 | 0 | 5792 | 0.0 | False |
| 40900 | 31 | 5792 | 0.5352209944751382 | False |
| 40980 | 1 | 5792 | 0.017265193370165747 | False |
| 41060 | 2 | 5792 | 0.034530386740331494 | False |
| 41100 | 4 | 5792 | 0.06906077348066299 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 41140 | 3 | 5792 | 0.05179558011049724 | False |
| 41180 | 32 | 5792 | 0.5524861878453039 | False |
| 41304 | 15 | 5792 | 0.2589779005524862 | False |
| 41420 | 6 | 5792 | 0.10359116022099447 | False |
| 41500 | 5 | 5792 | 0.08632596685082873 | False |
| 41540 | 9 | 5792 | 0.15538674033149172 | False |
| 41620 | 12 | 5792 | 0.20718232044198895 | False |
| 41660 | 0 | 5792 | 0.0 | False |
| 41700 | 36 | 5792 | 0.6215469613259669 | False |
| 41740 | 31 | 5792 | 0.5352209944751382 | False |
| 41780 | 0 | 5792 | 0.0 | False |
| 41884 | 2 | 5792 | 0.034530386740331494 | False |
| 41940 | 8 | 5792 | 0.13812154696132597 | False |
| 41980 | 0 | 5792 | 0.0 | False |
| 42020 | 2 | 5792 | 0.034530386740331494 | False |
| 42034 | 2 | 5792 | 0.034530386740331494 | False |
| 42100 | 0 | 5792 | 0.0 | False |
| 42140 | 1 | 5792 | 0.017265193370165747 | False |
| 42200 | 4 | 5792 | 0.06906077348066299 | False |
| 42220 | 4 | 5792 | 0.06906077348066299 | False |
| 42340 | 5 | 5792 | 0.08632596685082873 | False |
| 42540 | 9 | 5792 | 0.15538674033149172 | False |
| 42644 | 28 | 5792 | 0.48342541436464087 | False |
| 42680 | 3 | 5792 | 0.05179558011049724 | False |
| 42700 | 2 | 5792 | 0.034530386740331494 | False |
| 43100 | 0 | 5792 | 0.0 | False |
| 43300 | 1 | 5792 | 0.017265193370165747 | False |
| 43340 | 5 | 5792 | 0.08632596685082873 | False |
| 43420 | 3 | 5792 | 0.05179558011049724 | False |
| 43580 | 4 | 5792 | 0.06906077348066299 | False |
| 43620 | 1 | 5792 | 0.017265193370165747 | False |
| 43640 | 1 | 5792 | 0.017265193370165747 | False |
| 43780 | 10 | 5792 | 0.17265193370165746 | False |
| 43900 | 4 | 5792 | 0.06906077348066299 | False |
| 44060 | 9 | 5792 | 0.15538674033149172 | False |
| 44100 | 1 | 5792 | 0.017265193370165747 | False |
| 44140 | 10 | 5792 | 0.17265193370165746 | False |
| 44180 | 8 | 5792 | 0.13812154696132597 | False |
| 44220 | 1 | 5792 | 0.017265193370165747 | False |
| 44300 | 1 | 5792 | 0.017265193370165747 | False |
| 44420 | 1 | 5792 | 0.017265193370165747 | False |
| 44700 | 27 | 5792 | 0.4661602209944751 | False |
| 44940 | 2 | 5792 | 0.034530386740331494 | False |
| 45060 | 5 | 5792 | 0.08632596685082873 | False |
| 45104 | 20 | 5792 | 0.3453038674033149 | False |
| 45220 | 11 | 5792 | 0.1899171270718232 | False |
| 45294 | 46 | 5792 | 0.7941988950276243 | False |
| 45300 | 89 | 5792 | 1.5366022099447514 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 45460 | 2 | 5792 | 0.034530386740331494 | False |
| 45500 | 0 | 5792 | 0.0 | False |
| 45540 | 3 | 5792 | 0.05179558011049724 | False |
| 45780 | 7 | 5792 | 0.12085635359116022 | False |
| 45820 | 3 | 5792 | 0.05179558011049724 | False |
| 45900 | 0 | 5792 | 0.0 | False |
| 45940 | 12 | 5792 | 0.20718232044198895 | False |
| 46060 | 10 | 5792 | 0.17265193370165746 | False |
| 46140 | 11 | 5792 | 0.1899171270718232 | False |
| 46220 | 1 | 5792 | 0.017265193370165747 | False |
| 46300 | 1 | 5792 | 0.017265193370165747 | False |
| 46340 | 3 | 5792 | 0.05179558011049724 | False |
| 46520 | 8 | 5792 | 0.13812154696132597 | False |
| 46540 | 2 | 5792 | 0.034530386740331494 | False |
| 46660 | 0 | 5792 | 0.0 | False |
| 46700 | 12 | 5792 | 0.20718232044198895 | False |
| 47020 | 1 | 5792 | 0.017265193370165747 | False |
| 47220 | 0 | 5792 | 0.0 | False |
| 47260 | 10 | 5792 | 0.17265193370165746 | False |
| 47300 | 5 | 5792 | 0.08632596685082873 | False |
| 47380 | 3 | 5792 | 0.05179558011049724 | False |
| 47460 | 1 | 5792 | 0.017265193370165747 | False |
| 47580 | 3 | 5792 | 0.05179558011049724 | False |
| 47664 | 33 | 5792 | 0.5697513812154696 | False |
| 47764 | 17 | 5792 | 0.2935082872928177 | False |
| 47894 | 76 | 5792 | 1.3121546961325967 | False |
| 47930 | 3 | 5792 | 0.05179558011049724 | False |
| 47940 | 0 | 5792 | 0.0 | False |
| 48060 | 1 | 5792 | 0.017265193370165747 | False |
| 48140 | 2 | 5792 | 0.034530386740331494 | False |
| 48260 | 0 | 5792 | 0.0 | False |
| 48300 | 1 | 5792 | 0.017265193370165747 | False |
| 48424 | 125 | 5792 | 2.1581491712707184 | False |
| 48540 | 1 | 5792 | 0.017265193370165747 | False |
| 48620 | 5 | 5792 | 0.08632596685082873 | False |
| 48660 | 0 | 5792 | 0.0 | False |
| 48680 | 0 | 5792 | 0.0 | False |
| 48700 | 0 | 5792 | 0.0 | False |
| 48864 | 9 | 5792 | 0.15538674033149172 | False |
| 48900 | 4 | 5792 | 0.06906077348066299 | False |
| 49020 | 1 | 5792 | 0.017265193370165747 | False |
| 49180 | 9 | 5792 | 0.15538674033149172 | False |
| 49340 | 21 | 5792 | 0.36256906077348067 | False |
| 49420 | 2 | 5792 | 0.034530386740331494 | False |
| 49500 | 0 | 5792 | 0.0 | False |
| 49620 | 6 | 5792 | 0.10359116022099447 | False |
| 49660 | 3 | 5792 | 0.05179558011049724 | False |
| 49700 | 1 | 5792 | 0.017265193370165747 | False |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 49740 | 1 | 5792 | 0.017265193370165747 | False |
| nonmetropolitan_or_unknown | 401 | 5792 | 6.923342541436464 | False |

Direction 1: Logistic non-risky / XGBoost risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 10180 | 0 | 1670 | 0.0 | True |
| 10420 | 3 | 1670 | 0.17964071856287425 | True |
| 10500 | 0 | 1670 | 0.0 | True |
| 10540 | 1 | 1670 | 0.059880239520958084 | True |
| 10580 | 9 | 1670 | 0.5389221556886228 | True |
| 10740 | 3 | 1670 | 0.17964071856287425 | True |
| 10780 | 0 | 1670 | 0.0 | True |
| 10900 | 4 | 1670 | 0.23952095808383234 | True |
| 11020 | 0 | 1670 | 0.0 | True |
| 11100 | 0 | 1670 | 0.0 | True |
| 11180 | 0 | 1670 | 0.0 | True |
| 11200 | 0 | 1670 | 0.0 | True |
| 11244 | 11 | 1670 | 0.6586826347305389 | True |
| 11260 | 0 | 1670 | 0.0 | True |
| 11460 | 1 | 1670 | 0.059880239520958084 | True |
| 11500 | 0 | 1670 | 0.0 | True |
| 11540 | 1 | 1670 | 0.059880239520958084 | True |
| 11694 | 4 | 1670 | 0.23952095808383234 | True |
| 11700 | 0 | 1670 | 0.0 | True |
| 12020 | 2 | 1670 | 0.11976047904191617 | True |
| 12054 | 10 | 1670 | 0.5988023952095808 | True |
| 12060 | 56 | 1670 | 3.3532934131736525 | True |
| 12100 | 3 | 1670 | 0.17964071856287425 | True |
| 12220 | 0 | 1670 | 0.0 | True |
| 12260 | 3 | 1670 | 0.17964071856287425 | True |
| 12420 | 14 | 1670 | 0.8383233532934131 | True |
| 12540 | 5 | 1670 | 0.2994011976047904 | True |
| 12580 | 13 | 1670 | 0.7784431137724551 | True |
| 12620 | 0 | 1670 | 0.0 | True |
| 12700 | 2 | 1670 | 0.11976047904191617 | True |
| 12940 | 10 | 1670 | 0.5988023952095808 | True |
| 12980 | 1 | 1670 | 0.059880239520958084 | True |
| 13020 | 0 | 1670 | 0.0 | True |
| 13140 | 0 | 1670 | 0.0 | True |
| 13220 | 0 | 1670 | 0.0 | True |
| 13380 | 0 | 1670 | 0.0 | True |
| 13460 | 2 | 1670 | 0.11976047904191617 | True |
| 13740 | 0 | 1670 | 0.0 | True |
| 13780 | 1 | 1670 | 0.059880239520958084 | True |
| 13820 | 5 | 1670 | 0.2994011976047904 | True |
| 13900 | 0 | 1670 | 0.0 | True |
| 13980 | 0 | 1670 | 0.0 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 14010 | 1 | 1670 | 0.059880239520958084 | True |
| 14020 | 0 | 1670 | 0.0 | True |
| 14100 | 0 | 1670 | 0.0 | True |
| 14260 | 5 | 1670 | 0.2994011976047904 | True |
| 14454 | 10 | 1670 | 0.5988023952095808 | True |
| 14500 | 0 | 1670 | 0.0 | True |
| 14540 | 1 | 1670 | 0.059880239520958084 | True |
| 14580 | 0 | 1670 | 0.0 | True |
| 14740 | 2 | 1670 | 0.11976047904191617 | True |
| 14860 | 9 | 1670 | 0.5389221556886228 | True |
| 15180 | 0 | 1670 | 0.0 | True |
| 15260 | 0 | 1670 | 0.0 | True |
| 15380 | 17 | 1670 | 1.0179640718562875 | True |
| 15500 | 0 | 1670 | 0.0 | True |
| 15540 | 0 | 1670 | 0.0 | True |
| 15680 | 0 | 1670 | 0.0 | True |
| 15764 | 12 | 1670 | 0.718562874251497 | True |
| 15804 | 16 | 1670 | 0.9580838323353293 | True |
| 15940 | 0 | 1670 | 0.0 | True |
| 15980 | 7 | 1670 | 0.41916167664670656 | True |
| 16020 | 0 | 1670 | 0.0 | True |
| 16060 | 0 | 1670 | 0.0 | True |
| 16180 | 0 | 1670 | 0.0 | True |
| 16220 | 0 | 1670 | 0.0 | True |
| 16300 | 0 | 1670 | 0.0 | True |
| 16540 | 0 | 1670 | 0.0 | True |
| 16580 | 0 | 1670 | 0.0 | True |
| 16620 | 0 | 1670 | 0.0 | True |
| 16700 | 3 | 1670 | 0.17964071856287425 | True |
| 16740 | 19 | 1670 | 1.1377245508982037 | True |
| 16820 | 1 | 1670 | 0.059880239520958084 | True |
| 16860 | 3 | 1670 | 0.17964071856287425 | True |
| 16940 | 0 | 1670 | 0.0 | True |
| 16984 | 42 | 1670 | 2.5149700598802394 | True |
| 17020 | 1 | 1670 | 0.059880239520958084 | True |
| 17140 | 2 | 1670 | 0.11976047904191617 | True |
| 17300 | 1 | 1670 | 0.059880239520958084 | True |
| 17410 | 0 | 1670 | 0.0 | True |
| 17420 | 0 | 1670 | 0.0 | True |
| 17460 | 4 | 1670 | 0.23952095808383234 | True |
| 17660 | 3 | 1670 | 0.17964071856287425 | True |
| 17780 | 0 | 1670 | 0.0 | True |
| 17820 | 5 | 1670 | 0.2994011976047904 | True |
| 17860 | 1 | 1670 | 0.059880239520958084 | True |
| 17900 | 5 | 1670 | 0.2994011976047904 | True |
| 17980 | 2 | 1670 | 0.11976047904191617 | True |
| 18020 | 0 | 1670 | 0.0 | True |
| 18140 | 9 | 1670 | 0.5389221556886228 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 18580 | 0 | 1670 | 0.0 | True |
| 18700 | 0 | 1670 | 0.0 | True |
| 18880 | 1 | 1670 | 0.059880239520958084 | True |
| 19060 | 0 | 1670 | 0.0 | True |
| 19124 | 38 | 1670 | 2.2754491017964074 | True |
| 19140 | 0 | 1670 | 0.0 | True |
| 19180 | 0 | 1670 | 0.0 | True |
| 19300 | 1 | 1670 | 0.059880239520958084 | True |
| 19340 | 0 | 1670 | 0.0 | True |
| 19430 | 1 | 1670 | 0.059880239520958084 | True |
| 19460 | 0 | 1670 | 0.0 | True |
| 19500 | 0 | 1670 | 0.0 | True |
| 19660 | 6 | 1670 | 0.3592814371257485 | True |
| 19740 | 38 | 1670 | 2.2754491017964074 | True |
| 19780 | 0 | 1670 | 0.0 | True |
| 19804 | 9 | 1670 | 0.5389221556886228 | True |
| 20020 | 0 | 1670 | 0.0 | True |
| 20100 | 1 | 1670 | 0.059880239520958084 | True |
| 20220 | 1 | 1670 | 0.059880239520958084 | True |
| 20260 | 0 | 1670 | 0.0 | True |
| 20500 | 1 | 1670 | 0.059880239520958084 | True |
| 20580 | 0 | 1670 | 0.0 | True |
| 20700 | 0 | 1670 | 0.0 | True |
| 20740 | 0 | 1670 | 0.0 | True |
| 20940 | 1 | 1670 | 0.059880239520958084 | True |
| 20994 | 4 | 1670 | 0.23952095808383234 | True |
| 21060 | 0 | 1670 | 0.0 | True |
| 21140 | 0 | 1670 | 0.0 | True |
| 21300 | 0 | 1670 | 0.0 | True |
| 21340 | 1 | 1670 | 0.059880239520958084 | True |
| 21420 | 0 | 1670 | 0.0 | True |
| 21500 | 0 | 1670 | 0.0 | True |
| 21660 | 1 | 1670 | 0.059880239520958084 | True |
| 21780 | 0 | 1670 | 0.0 | True |
| 21794 | 1 | 1670 | 0.059880239520958084 | True |
| 21820 | 0 | 1670 | 0.0 | True |
| 22020 | 0 | 1670 | 0.0 | True |
| 22140 | 0 | 1670 | 0.0 | True |
| 22180 | 0 | 1670 | 0.0 | True |
| 22220 | 2 | 1670 | 0.11976047904191617 | True |
| 22380 | 0 | 1670 | 0.0 | True |
| 22420 | 2 | 1670 | 0.11976047904191617 | True |
| 22500 | 1 | 1670 | 0.059880239520958084 | True |
| 22520 | 0 | 1670 | 0.0 | True |
| 22540 | 0 | 1670 | 0.0 | True |
| 22660 | 3 | 1670 | 0.17964071856287425 | True |
| 22744 | 20 | 1670 | 1.1976047904191616 | True |
| 22900 | 0 | 1670 | 0.0 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 23060 | 2 | 1670 | 0.11976047904191617 | True |
| 23104 | 6 | 1670 | 0.3592814371257485 | True |
| 23224 | 7 | 1670 | 0.41916167664670656 | True |
| 23420 | 6 | 1670 | 0.3592814371257485 | True |
| 23460 | 1 | 1670 | 0.059880239520958084 | True |
| 23540 | 1 | 1670 | 0.059880239520958084 | True |
| 23580 | 2 | 1670 | 0.11976047904191617 | True |
| 23844 | 2 | 1670 | 0.11976047904191617 | True |
| 23900 | 0 | 1670 | 0.0 | True |
| 24020 | 0 | 1670 | 0.0 | True |
| 24140 | 0 | 1670 | 0.0 | True |
| 24220 | 0 | 1670 | 0.0 | True |
| 24260 | 0 | 1670 | 0.0 | True |
| 24300 | 0 | 1670 | 0.0 | True |
| 24340 | 4 | 1670 | 0.23952095808383234 | True |
| 24420 | 2 | 1670 | 0.11976047904191617 | True |
| 24500 | 1 | 1670 | 0.059880239520958084 | True |
| 24540 | 4 | 1670 | 0.23952095808383234 | True |
| 24580 | 0 | 1670 | 0.0 | True |
| 24660 | 1 | 1670 | 0.059880239520958084 | True |
| 24780 | 0 | 1670 | 0.0 | True |
| 24860 | 0 | 1670 | 0.0 | True |
| 25060 | 0 | 1670 | 0.0 | True |
| 25180 | 1 | 1670 | 0.059880239520958084 | True |
| 25220 | 0 | 1670 | 0.0 | True |
| 25260 | 0 | 1670 | 0.0 | True |
| 25420 | 4 | 1670 | 0.23952095808383234 | True |
| 25500 | 1 | 1670 | 0.059880239520958084 | True |
| 25540 | 12 | 1670 | 0.718562874251497 | True |
| 25620 | 0 | 1670 | 0.0 | True |
| 25740 | 0 | 1670 | 0.0 | True |
| 25860 | 0 | 1670 | 0.0 | True |
| 25940 | 0 | 1670 | 0.0 | True |
| 25980 | 1 | 1670 | 0.059880239520958084 | True |
| 26140 | 1 | 1670 | 0.059880239520958084 | True |
| 26300 | 0 | 1670 | 0.0 | True |
| 26380 | 3 | 1670 | 0.17964071856287425 | True |
| 26420 | 29 | 1670 | 1.7365269461077844 | True |
| 26580 | 0 | 1670 | 0.0 | True |
| 26620 | 0 | 1670 | 0.0 | True |
| 26820 | 0 | 1670 | 0.0 | True |
| 26900 | 9 | 1670 | 0.5389221556886228 | True |
| 26980 | 1 | 1670 | 0.059880239520958084 | True |
| 27060 | 0 | 1670 | 0.0 | True |
| 27100 | 0 | 1670 | 0.0 | True |
| 27140 | 1 | 1670 | 0.059880239520958084 | True |
| 27180 | 0 | 1670 | 0.0 | True |
| 27260 | 16 | 1670 | 0.9580838323353293 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 27340 | 0 | 1670 | 0.0 | True |
| 27500 | 0 | 1670 | 0.0 | True |
| 27620 | 1 | 1670 | 0.059880239520958084 | True |
| 27740 | 0 | 1670 | 0.0 | True |
| 27780 | 0 | 1670 | 0.0 | True |
| 27860 | 0 | 1670 | 0.0 | True |
| 27900 | 1 | 1670 | 0.059880239520958084 | True |
| 27980 | 0 | 1670 | 0.0 | True |
| 28020 | 0 | 1670 | 0.0 | True |
| 28100 | 0 | 1670 | 0.0 | True |
| 28140 | 9 | 1670 | 0.5389221556886228 | True |
| 28420 | 3 | 1670 | 0.17964071856287425 | True |
| 28450 | 0 | 1670 | 0.0 | True |
| 28660 | 0 | 1670 | 0.0 | True |
| 28700 | 0 | 1670 | 0.0 | True |
| 28740 | 3 | 1670 | 0.17964071856287425 | True |
| 28880 | 1 | 1670 | 0.059880239520958084 | True |
| 28940 | 0 | 1670 | 0.0 | True |
| 29020 | 0 | 1670 | 0.0 | True |
| 29100 | 0 | 1670 | 0.0 | True |
| 29180 | 6 | 1670 | 0.3592814371257485 | True |
| 29200 | 0 | 1670 | 0.0 | True |
| 29340 | 3 | 1670 | 0.17964071856287425 | True |
| 29404 | 3 | 1670 | 0.17964071856287425 | True |
| 29414 | 0 | 1670 | 0.0 | True |
| 29420 | 1 | 1670 | 0.059880239520958084 | True |
| 29460 | 6 | 1670 | 0.3592814371257485 | True |
| 29484 | 9 | 1670 | 0.5389221556886228 | True |
| 29540 | 0 | 1670 | 0.0 | True |
| 29620 | 0 | 1670 | 0.0 | True |
| 29700 | 0 | 1670 | 0.0 | True |
| 29740 | 0 | 1670 | 0.0 | True |
| 29820 | 29 | 1670 | 1.7365269461077844 | True |
| 29940 | 0 | 1670 | 0.0 | True |
| 30020 | 0 | 1670 | 0.0 | True |
| 30140 | 1 | 1670 | 0.059880239520958084 | True |
| 30300 | 0 | 1670 | 0.0 | True |
| 30340 | 0 | 1670 | 0.0 | True |
| 30460 | 1 | 1670 | 0.059880239520958084 | True |
| 30500 | 0 | 1670 | 0.0 | True |
| 30620 | 0 | 1670 | 0.0 | True |
| 30700 | 3 | 1670 | 0.17964071856287425 | True |
| 30780 | 2 | 1670 | 0.11976047904191617 | True |
| 30860 | 0 | 1670 | 0.0 | True |
| 30980 | 0 | 1670 | 0.0 | True |
| 31020 | 3 | 1670 | 0.17964071856287425 | True |
| 31084 | 49 | 1670 | 2.934131736526946 | True |
| 31140 | 8 | 1670 | 0.47904191616766467 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 31180 | 0 | 1670 | 0.0 | True |
| 31340 | 2 | 1670 | 0.11976047904191617 | True |
| 31420 | 0 | 1670 | 0.0 | True |
| 31460 | 0 | 1670 | 0.0 | True |
| 31540 | 2 | 1670 | 0.11976047904191617 | True |
| 31700 | 3 | 1670 | 0.17964071856287425 | True |
| 31740 | 0 | 1670 | 0.0 | True |
| 31860 | 0 | 1670 | 0.0 | True |
| 31900 | 0 | 1670 | 0.0 | True |
| 31924 | 5 | 1670 | 0.2994011976047904 | True |
| 32580 | 1 | 1670 | 0.059880239520958084 | True |
| 32780 | 0 | 1670 | 0.0 | True |
| 32820 | 2 | 1670 | 0.11976047904191617 | True |
| 32900 | 0 | 1670 | 0.0 | True |
| 33124 | 30 | 1670 | 1.7964071856287425 | True |
| 33140 | 0 | 1670 | 0.0 | True |
| 33220 | 0 | 1670 | 0.0 | True |
| 33260 | 1 | 1670 | 0.059880239520958084 | True |
| 33340 | 1 | 1670 | 0.059880239520958084 | True |
| 33460 | 13 | 1670 | 0.7784431137724551 | True |
| 33540 | 1 | 1670 | 0.059880239520958084 | True |
| 33660 | 0 | 1670 | 0.0 | True |
| 33700 | 2 | 1670 | 0.11976047904191617 | True |
| 33740 | 0 | 1670 | 0.0 | True |
| 33780 | 2 | 1670 | 0.11976047904191617 | True |
| 33860 | 0 | 1670 | 0.0 | True |
| 33874 | 4 | 1670 | 0.23952095808383234 | True |
| 34060 | 0 | 1670 | 0.0 | True |
| 34100 | 0 | 1670 | 0.0 | True |
| 34580 | 1 | 1670 | 0.059880239520958084 | True |
| 34620 | 0 | 1670 | 0.0 | True |
| 34740 | 0 | 1670 | 0.0 | True |
| 34820 | 2 | 1670 | 0.11976047904191617 | True |
| 34900 | 2 | 1670 | 0.11976047904191617 | True |
| 34940 | 3 | 1670 | 0.17964071856287425 | True |
| 34980 | 15 | 1670 | 0.8982035928143712 | True |
| 35004 | 46 | 1670 | 2.754491017964072 | True |
| 35084 | 33 | 1670 | 1.9760479041916168 | True |
| 35100 | 0 | 1670 | 0.0 | True |
| 35154 | 32 | 1670 | 1.9161676646706587 | True |
| 35300 | 15 | 1670 | 0.8982035928143712 | True |
| 35380 | 6 | 1670 | 0.3592814371257485 | True |
| 35614 | 81 | 1670 | 4.850299401197605 | True |
| 35660 | 1 | 1670 | 0.059880239520958084 | True |
| 35840 | 3 | 1670 | 0.17964071856287425 | True |
| 35980 | 5 | 1670 | 0.2994011976047904 | True |
| 36084 | 18 | 1670 | 1.0778443113772456 | True |
| 36100 | 3 | 1670 | 0.17964071856287425 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 36140 | 2 | 1670 | 0.11976047904191617 | True |
| 36220 | 2 | 1670 | 0.11976047904191617 | True |
| 36260 | 3 | 1670 | 0.17964071856287425 | True |
| 36420 | 3 | 1670 | 0.17964071856287425 | True |
| 36500 | 2 | 1670 | 0.11976047904191617 | True |
| 36540 | 3 | 1670 | 0.17964071856287425 | True |
| 36740 | 31 | 1670 | 1.8562874251497006 | True |
| 36780 | 0 | 1670 | 0.0 | True |
| 36980 | 1 | 1670 | 0.059880239520958084 | True |
| 37100 | 7 | 1670 | 0.41916167664670656 | True |
| 37140 | 0 | 1670 | 0.0 | True |
| 37340 | 8 | 1670 | 0.47904191616766467 | True |
| 37460 | 0 | 1670 | 0.0 | True |
| 37620 | 0 | 1670 | 0.0 | True |
| 37860 | 2 | 1670 | 0.11976047904191617 | True |
| 37900 | 3 | 1670 | 0.17964071856287425 | True |
| 37964 | 5 | 1670 | 0.2994011976047904 | True |
| 38060 | 41 | 1670 | 2.4550898203592815 | True |
| 38220 | 0 | 1670 | 0.0 | True |
| 38240 | 0 | 1670 | 0.0 | True |
| 38300 | 6 | 1670 | 0.3592814371257485 | True |
| 38340 | 0 | 1670 | 0.0 | True |
| 38540 | 1 | 1670 | 0.059880239520958084 | True |
| 38860 | 1 | 1670 | 0.059880239520958084 | True |
| 38900 | 13 | 1670 | 0.7784431137724551 | True |
| 38940 | 8 | 1670 | 0.47904191616766467 | True |
| 39100 | 9 | 1670 | 0.5389221556886228 | True |
| 39150 | 0 | 1670 | 0.0 | True |
| 39300 | 3 | 1670 | 0.17964071856287425 | True |
| 39340 | 5 | 1670 | 0.2994011976047904 | True |
| 39380 | 0 | 1670 | 0.0 | True |
| 39460 | 3 | 1670 | 0.17964071856287425 | True |
| 39540 | 0 | 1670 | 0.0 | True |
| 39580 | 7 | 1670 | 0.41916167664670656 | True |
| 39660 | 1 | 1670 | 0.059880239520958084 | True |
| 39740 | 0 | 1670 | 0.0 | True |
| 39820 | 1 | 1670 | 0.059880239520958084 | True |
| 39900 | 0 | 1670 | 0.0 | True |
| 40060 | 6 | 1670 | 0.3592814371257485 | True |
| 40140 | 44 | 1670 | 2.6347305389221556 | True |
| 40220 | 0 | 1670 | 0.0 | True |
| 40340 | 0 | 1670 | 0.0 | True |
| 40380 | 14 | 1670 | 0.8383233532934131 | True |
| 40420 | 3 | 1670 | 0.17964071856287425 | True |
| 40484 | 1 | 1670 | 0.059880239520958084 | True |
| 40580 | 0 | 1670 | 0.0 | True |
| 40660 | 1 | 1670 | 0.059880239520958084 | True |
| 40900 | 18 | 1670 | 1.0778443113772456 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 40980 | 0 | 1670 | 0.0 | True |
| 41060 | 1 | 1670 | 0.059880239520958084 | True |
| 41100 | 4 | 1670 | 0.23952095808383234 | True |
| 41140 | 0 | 1670 | 0.0 | True |
| 41180 | 8 | 1670 | 0.47904191616766467 | True |
| 41304 | 1 | 1670 | 0.059880239520958084 | True |
| 41420 | 1 | 1670 | 0.059880239520958084 | True |
| 41500 | 3 | 1670 | 0.17964071856287425 | True |
| 41540 | 2 | 1670 | 0.11976047904191617 | True |
| 41620 | 4 | 1670 | 0.23952095808383234 | True |
| 41660 | 0 | 1670 | 0.0 | True |
| 41700 | 12 | 1670 | 0.718562874251497 | True |
| 41740 | 16 | 1670 | 0.9580838323353293 | True |
| 41780 | 0 | 1670 | 0.0 | True |
| 41884 | 1 | 1670 | 0.059880239520958084 | True |
| 41940 | 4 | 1670 | 0.23952095808383234 | True |
| 41980 | 0 | 1670 | 0.0 | True |
| 42020 | 2 | 1670 | 0.11976047904191617 | True |
| 42034 | 0 | 1670 | 0.0 | True |
| 42100 | 1 | 1670 | 0.059880239520958084 | True |
| 42140 | 0 | 1670 | 0.0 | True |
| 42200 | 3 | 1670 | 0.17964071856287425 | True |
| 42220 | 2 | 1670 | 0.11976047904191617 | True |
| 42340 | 5 | 1670 | 0.2994011976047904 | True |
| 42540 | 2 | 1670 | 0.11976047904191617 | True |
| 42644 | 15 | 1670 | 0.8982035928143712 | True |
| 42680 | 0 | 1670 | 0.0 | True |
| 42700 | 1 | 1670 | 0.059880239520958084 | True |
| 43100 | 0 | 1670 | 0.0 | True |
| 43300 | 2 | 1670 | 0.11976047904191617 | True |
| 43340 | 1 | 1670 | 0.059880239520958084 | True |
| 43420 | 0 | 1670 | 0.0 | True |
| 43580 | 0 | 1670 | 0.0 | True |
| 43620 | 1 | 1670 | 0.059880239520958084 | True |
| 43640 | 0 | 1670 | 0.0 | True |
| 43780 | 0 | 1670 | 0.0 | True |
| 43900 | 4 | 1670 | 0.23952095808383234 | True |
| 44060 | 0 | 1670 | 0.0 | True |
| 44100 | 0 | 1670 | 0.0 | True |
| 44140 | 5 | 1670 | 0.2994011976047904 | True |
| 44180 | 0 | 1670 | 0.0 | True |
| 44220 | 0 | 1670 | 0.0 | True |
| 44300 | 0 | 1670 | 0.0 | True |
| 44420 | 0 | 1670 | 0.0 | True |
| 44700 | 7 | 1670 | 0.41916167664670656 | True |
| 44940 | 0 | 1670 | 0.0 | True |
| 45060 | 4 | 1670 | 0.23952095808383234 | True |
| 45104 | 3 | 1670 | 0.17964071856287425 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 45220 | 2 | 1670 | 0.11976047904191617 | True |
| 45294 | 6 | 1670 | 0.3592814371257485 | True |
| 45300 | 24 | 1670 | 1.437125748502994 | True |
| 45460 | 0 | 1670 | 0.0 | True |
| 45500 | 0 | 1670 | 0.0 | True |
| 45540 | 0 | 1670 | 0.0 | True |
| 45780 | 1 | 1670 | 0.059880239520958084 | True |
| 45820 | 0 | 1670 | 0.0 | True |
| 45900 | 0 | 1670 | 0.0 | True |
| 45940 | 2 | 1670 | 0.11976047904191617 | True |
| 46060 | 2 | 1670 | 0.11976047904191617 | True |
| 46140 | 2 | 1670 | 0.11976047904191617 | True |
| 46220 | 1 | 1670 | 0.059880239520958084 | True |
| 46300 | 1 | 1670 | 0.059880239520958084 | True |
| 46340 | 1 | 1670 | 0.059880239520958084 | True |
| 46520 | 4 | 1670 | 0.23952095808383234 | True |
| 46540 | 3 | 1670 | 0.17964071856287425 | True |
| 46660 | 2 | 1670 | 0.11976047904191617 | True |
| 46700 | 10 | 1670 | 0.5988023952095808 | True |
| 47020 | 0 | 1670 | 0.0 | True |
| 47220 | 2 | 1670 | 0.11976047904191617 | True |
| 47260 | 3 | 1670 | 0.17964071856287425 | True |
| 47300 | 1 | 1670 | 0.059880239520958084 | True |
| 47380 | 1 | 1670 | 0.059880239520958084 | True |
| 47460 | 0 | 1670 | 0.0 | True |
| 47580 | 0 | 1670 | 0.0 | True |
| 47664 | 8 | 1670 | 0.47904191616766467 | True |
| 47764 | 2 | 1670 | 0.11976047904191617 | True |
| 47894 | 23 | 1670 | 1.377245508982036 | True |
| 47930 | 0 | 1670 | 0.0 | True |
| 47940 | 2 | 1670 | 0.11976047904191617 | True |
| 48060 | 0 | 1670 | 0.0 | True |
| 48140 | 0 | 1670 | 0.0 | True |
| 48260 | 0 | 1670 | 0.0 | True |
| 48300 | 0 | 1670 | 0.0 | True |
| 48424 | 19 | 1670 | 1.1377245508982037 | True |
| 48540 | 0 | 1670 | 0.0 | True |
| 48620 | 0 | 1670 | 0.0 | True |
| 48660 | 0 | 1670 | 0.0 | True |
| 48680 | 1 | 1670 | 0.059880239520958084 | True |
| 48700 | 1 | 1670 | 0.059880239520958084 | True |
| 48864 | 2 | 1670 | 0.11976047904191617 | True |
| 48900 | 2 | 1670 | 0.11976047904191617 | True |
| 49020 | 2 | 1670 | 0.11976047904191617 | True |
| 49180 | 2 | 1670 | 0.11976047904191617 | True |
| 49340 | 6 | 1670 | 0.3592814371257485 | True |
| 49420 | 1 | 1670 | 0.059880239520958084 | True |
| 49500 | 0 | 1670 | 0.0 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 49620 | 1 | 1670 | 0.059880239520958084 | True |
| 49660 | 1 | 1670 | 0.059880239520958084 | True |
| 49700 | 2 | 1670 | 0.11976047904191617 | True |
| 49740 | 1 | 1670 | 0.059880239520958084 | True |
| nonmetropolitan_or_unknown | 80 | 1670 | 4.790419161676646 | True |

Direction 0: Logistic risky / XGBoost non-risky

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 10180 | 3 | 4280 | 0.07009345794392523 | True |
| 10420 | 12 | 4280 | 0.2803738317757009 | True |
| 10500 | 1 | 4280 | 0.02336448598130841 | True |
| 10540 | 3 | 4280 | 0.07009345794392523 | True |
| 10580 | 4 | 4280 | 0.09345794392523364 | True |
| 10740 | 12 | 4280 | 0.2803738317757009 | True |
| 10780 | 0 | 4280 | 0.0 | True |
| 10900 | 13 | 4280 | 0.3037383177570093 | True |
| 11020 | 0 | 4280 | 0.0 | True |
| 11100 | 4 | 4280 | 0.09345794392523364 | True |
| 11180 | 2 | 4280 | 0.04672897196261682 | True |
| 11200 | 0 | 4280 | 0.0 | True |
| 11244 | 21 | 4280 | 0.49065420560747663 | True |
| 11260 | 5 | 4280 | 0.11682242990654206 | True |
| 11460 | 4 | 4280 | 0.09345794392523364 | True |
| 11500 | 2 | 4280 | 0.04672897196261682 | True |
| 11540 | 2 | 4280 | 0.04672897196261682 | True |
| 11694 | 20 | 4280 | 0.4672897196261682 | True |
| 11700 | 7 | 4280 | 0.16355140186915887 | True |
| 12020 | 3 | 4280 | 0.07009345794392523 | True |
| 12054 | 32 | 4280 | 0.7476635514018691 | True |
| 12060 | 51 | 4280 | 1.191588785046729 | True |
| 12100 | 7 | 4280 | 0.16355140186915887 | True |
| 12220 | 1 | 4280 | 0.02336448598130841 | True |
| 12260 | 4 | 4280 | 0.09345794392523364 | True |
| 12420 | 59 | 4280 | 1.3785046728971964 | True |
| 12540 | 13 | 4280 | 0.3037383177570093 | True |
| 12580 | 41 | 4280 | 0.9579439252336449 | True |
| 12620 | 1 | 4280 | 0.02336448598130841 | True |
| 12700 | 0 | 4280 | 0.0 | True |
| 12940 | 9 | 4280 | 0.2102803738317757 | True |
| 12980 | 2 | 4280 | 0.04672897196261682 | True |
| 13020 | 0 | 4280 | 0.0 | True |
| 13140 | 7 | 4280 | 0.16355140186915887 | True |
| 13220 | 3 | 4280 | 0.07009345794392523 | True |
| 13380 | 1 | 4280 | 0.02336448598130841 | True |
| 13460 | 2 | 4280 | 0.04672897196261682 | True |
| 13740 | 1 | 4280 | 0.02336448598130841 | True |
| 13780 | 1 | 4280 | 0.02336448598130841 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 13820 | 13 | 4280 | 0.3037383177570093 | True |
| 13900 | 0 | 4280 | 0.0 | True |
| 13980 | 0 | 4280 | 0.0 | True |
| 14010 | 0 | 4280 | 0.0 | True |
| 14020 | 2 | 4280 | 0.04672897196261682 | True |
| 14100 | 0 | 4280 | 0.0 | True |
| 14260 | 19 | 4280 | 0.4439252336448598 | True |
| 14454 | 22 | 4280 | 0.514018691588785 | True |
| 14500 | 1 | 4280 | 0.02336448598130841 | True |
| 14540 | 3 | 4280 | 0.07009345794392523 | True |
| 14580 | 1 | 4280 | 0.02336448598130841 | True |
| 14740 | 2 | 4280 | 0.04672897196261682 | True |
| 14860 | 0 | 4280 | 0.0 | True |
| 15180 | 7 | 4280 | 0.16355140186915887 | True |
| 15260 | 1 | 4280 | 0.02336448598130841 | True |
| 15380 | 12 | 4280 | 0.2803738317757009 | True |
| 15500 | 4 | 4280 | 0.09345794392523364 | True |
| 15540 | 3 | 4280 | 0.07009345794392523 | True |
| 15680 | 0 | 4280 | 0.0 | True |
| 15764 | 20 | 4280 | 0.4672897196261682 | True |
| 15804 | 21 | 4280 | 0.49065420560747663 | True |
| 15940 | 4 | 4280 | 0.09345794392523364 | True |
| 15980 | 26 | 4280 | 0.6074766355140186 | True |
| 16020 | 2 | 4280 | 0.04672897196261682 | True |
| 16060 | 1 | 4280 | 0.02336448598130841 | True |
| 16180 | 0 | 4280 | 0.0 | True |
| 16220 | 1 | 4280 | 0.02336448598130841 | True |
| 16300 | 6 | 4280 | 0.14018691588785046 | True |
| 16540 | 1 | 4280 | 0.02336448598130841 | True |
| 16580 | 10 | 4280 | 0.2336448598130841 | True |
| 16620 | 2 | 4280 | 0.04672897196261682 | True |
| 16700 | 14 | 4280 | 0.32710280373831774 | True |
| 16740 | 45 | 4280 | 1.0514018691588785 | True |
| 16820 | 0 | 4280 | 0.0 | True |
| 16860 | 3 | 4280 | 0.07009345794392523 | True |
| 16940 | 3 | 4280 | 0.07009345794392523 | True |
| 16984 | 134 | 4280 | 3.130841121495327 | False |
| 17020 | 1 | 4280 | 0.02336448598130841 | True |
| 17140 | 46 | 4280 | 1.074766355140187 | True |
| 17300 | 3 | 4280 | 0.07009345794392523 | True |
| 17410 | 24 | 4280 | 0.5607476635514018 | True |
| 17420 | 1 | 4280 | 0.02336448598130841 | True |
| 17460 | 16 | 4280 | 0.37383177570093457 | True |
| 17660 | 5 | 4280 | 0.11682242990654206 | True |
| 17780 | 8 | 4280 | 0.18691588785046728 | True |
| 17820 | 4 | 4280 | 0.09345794392523364 | True |
| 17860 | 1 | 4280 | 0.02336448598130841 | True |
| 17900 | 9 | 4280 | 0.2102803738317757 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 17980 | 1 | 4280 | 0.02336448598130841 | True |
| 18020 | 1 | 4280 | 0.02336448598130841 | True |
| 18140 | 31 | 4280 | 0.7242990654205608 | True |
| 18580 | 7 | 4280 | 0.16355140186915887 | True |
| 18700 | 0 | 4280 | 0.0 | True |
| 18880 | 4 | 4280 | 0.09345794392523364 | True |
| 19060 | 0 | 4280 | 0.0 | True |
| 19124 | 107 | 4280 | 2.5 | False |
| 19140 | 1 | 4280 | 0.02336448598130841 | True |
| 19180 | 0 | 4280 | 0.0 | True |
| 19300 | 3 | 4280 | 0.07009345794392523 | True |
| 19340 | 8 | 4280 | 0.18691588785046728 | True |
| 19430 | 11 | 4280 | 0.2570093457943925 | True |
| 19460 | 4 | 4280 | 0.09345794392523364 | True |
| 19500 | 0 | 4280 | 0.0 | True |
| 19660 | 26 | 4280 | 0.6074766355140186 | True |
| 19740 | 28 | 4280 | 0.6542056074766355 | True |
| 19780 | 11 | 4280 | 0.2570093457943925 | True |
| 19804 | 39 | 4280 | 0.9112149532710281 | True |
| 20020 | 0 | 4280 | 0.0 | True |
| 20100 | 2 | 4280 | 0.04672897196261682 | True |
| 20220 | 0 | 4280 | 0.0 | True |
| 20260 | 9 | 4280 | 0.2102803738317757 | True |
| 20500 | 9 | 4280 | 0.2102803738317757 | True |
| 20580 | 1 | 4280 | 0.02336448598130841 | True |
| 20700 | 1 | 4280 | 0.02336448598130841 | True |
| 20740 | 2 | 4280 | 0.04672897196261682 | True |
| 20940 | 1 | 4280 | 0.02336448598130841 | True |
| 20994 | 30 | 4280 | 0.7009345794392523 | True |
| 21060 | 1 | 4280 | 0.02336448598130841 | True |
| 21140 | 2 | 4280 | 0.04672897196261682 | True |
| 21300 | 1 | 4280 | 0.02336448598130841 | True |
| 21340 | 15 | 4280 | 0.35046728971962615 | True |
| 21420 | 0 | 4280 | 0.0 | True |
| 21500 | 1 | 4280 | 0.02336448598130841 | True |
| 21660 | 1 | 4280 | 0.02336448598130841 | True |
| 21780 | 8 | 4280 | 0.18691588785046728 | True |
| 21794 | 3 | 4280 | 0.07009345794392523 | True |
| 21820 | 1 | 4280 | 0.02336448598130841 | True |
| 22020 | 1 | 4280 | 0.02336448598130841 | True |
| 22140 | 0 | 4280 | 0.0 | True |
| 22180 | 5 | 4280 | 0.11682242990654206 | True |
| 22220 | 7 | 4280 | 0.16355140186915887 | True |
| 22380 | 2 | 4280 | 0.04672897196261682 | True |
| 22420 | 5 | 4280 | 0.11682242990654206 | True |
| 22500 | 0 | 4280 | 0.0 | True |
| 22520 | 1 | 4280 | 0.02336448598130841 | True |
| 22540 | 2 | 4280 | 0.04672897196261682 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 22660 | 5 | 4280 | 0.11682242990654206 | True |
| 22744 | 40 | 4280 | 0.9345794392523364 | True |
| 22900 | 0 | 4280 | 0.0 | True |
| 23060 | 9 | 4280 | 0.2102803738317757 | True |
| 23104 | 62 | 4280 | 1.4485981308411215 | True |
| 23224 | 19 | 4280 | 0.4439252336448598 | True |
| 23420 | 10 | 4280 | 0.2336448598130841 | True |
| 23460 | 1 | 4280 | 0.02336448598130841 | True |
| 23540 | 4 | 4280 | 0.09345794392523364 | True |
| 23580 | 2 | 4280 | 0.04672897196261682 | True |
| 23844 | 5 | 4280 | 0.11682242990654206 | True |
| 23900 | 0 | 4280 | 0.0 | True |
| 24020 | 4 | 4280 | 0.09345794392523364 | True |
| 24140 | 0 | 4280 | 0.0 | True |
| 24220 | 1 | 4280 | 0.02336448598130841 | True |
| 24260 | 1 | 4280 | 0.02336448598130841 | True |
| 24300 | 1 | 4280 | 0.02336448598130841 | True |
| 24340 | 24 | 4280 | 0.5607476635514018 | True |
| 24420 | 1 | 4280 | 0.02336448598130841 | True |
| 24500 | 1 | 4280 | 0.02336448598130841 | True |
| 24540 | 8 | 4280 | 0.18691588785046728 | True |
| 24580 | 1 | 4280 | 0.02336448598130841 | True |
| 24660 | 6 | 4280 | 0.14018691588785046 | True |
| 24780 | 3 | 4280 | 0.07009345794392523 | True |
| 24860 | 8 | 4280 | 0.18691588785046728 | True |
| 25060 | 4 | 4280 | 0.09345794392523364 | True |
| 25180 | 2 | 4280 | 0.04672897196261682 | True |
| 25220 | 0 | 4280 | 0.0 | True |
| 25260 | 0 | 4280 | 0.0 | True |
| 25420 | 7 | 4280 | 0.16355140186915887 | True |
| 25500 | 1 | 4280 | 0.02336448598130841 | True |
| 25540 | 5 | 4280 | 0.11682242990654206 | True |
| 25620 | 1 | 4280 | 0.02336448598130841 | True |
| 25740 | 0 | 4280 | 0.0 | True |
| 25860 | 6 | 4280 | 0.14018691588785046 | True |
| 25940 | 3 | 4280 | 0.07009345794392523 | True |
| 25980 | 0 | 4280 | 0.0 | True |
| 26140 | 10 | 4280 | 0.2336448598130841 | True |
| 26300 | 1 | 4280 | 0.02336448598130841 | True |
| 26380 | 1 | 4280 | 0.02336448598130841 | True |
| 26420 | 170 | 4280 | 3.97196261682243 | False |
| 26580 | 2 | 4280 | 0.04672897196261682 | True |
| 26620 | 7 | 4280 | 0.16355140186915887 | True |
| 26820 | 3 | 4280 | 0.07009345794392523 | True |
| 26900 | 41 | 4280 | 0.9579439252336449 | True |
| 26980 | 0 | 4280 | 0.0 | True |
| 27060 | 0 | 4280 | 0.0 | True |
| 27100 | 4 | 4280 | 0.09345794392523364 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 27140 | 5 | 4280 | 0.11682242990654206 | True |
| 27180 | 0 | 4280 | 0.0 | True |
| 27260 | 33 | 4280 | 0.7710280373831776 | True |
| 27340 | 1 | 4280 | 0.02336448598130841 | True |
| 27500 | 1 | 4280 | 0.02336448598130841 | True |
| 27620 | 2 | 4280 | 0.04672897196261682 | True |
| 27740 | 5 | 4280 | 0.11682242990654206 | True |
| 27780 | 4 | 4280 | 0.09345794392523364 | True |
| 27860 | 1 | 4280 | 0.02336448598130841 | True |
| 27900 | 3 | 4280 | 0.07009345794392523 | True |
| 27980 | 1 | 4280 | 0.02336448598130841 | True |
| 28020 | 1 | 4280 | 0.02336448598130841 | True |
| 28100 | 0 | 4280 | 0.0 | True |
| 28140 | 35 | 4280 | 0.8177570093457944 | True |
| 28420 | 2 | 4280 | 0.04672897196261682 | True |
| 28450 | 3 | 4280 | 0.07009345794392523 | True |
| 28660 | 10 | 4280 | 0.2336448598130841 | True |
| 28700 | 2 | 4280 | 0.04672897196261682 | True |
| 28740 | 4 | 4280 | 0.09345794392523364 | True |
| 28880 | 2 | 4280 | 0.04672897196261682 | True |
| 28940 | 4 | 4280 | 0.09345794392523364 | True |
| 29020 | 1 | 4280 | 0.02336448598130841 | True |
| 29100 | 3 | 4280 | 0.07009345794392523 | True |
| 29180 | 4 | 4280 | 0.09345794392523364 | True |
| 29200 | 3 | 4280 | 0.07009345794392523 | True |
| 29340 | 0 | 4280 | 0.0 | True |
| 29404 | 14 | 4280 | 0.32710280373831774 | True |
| 29414 | 12 | 4280 | 0.2803738317757009 | True |
| 29420 | 7 | 4280 | 0.16355140186915887 | True |
| 29460 | 15 | 4280 | 0.35046728971962615 | True |
| 29484 | 17 | 4280 | 0.397196261682243 | True |
| 29540 | 5 | 4280 | 0.11682242990654206 | True |
| 29620 | 8 | 4280 | 0.18691588785046728 | True |
| 29700 | 4 | 4280 | 0.09345794392523364 | True |
| 29740 | 3 | 4280 | 0.07009345794392523 | True |
| 29820 | 26 | 4280 | 0.6074766355140186 | True |
| 29940 | 1 | 4280 | 0.02336448598130841 | True |
| 30020 | 1 | 4280 | 0.02336448598130841 | True |
| 30140 | 1 | 4280 | 0.02336448598130841 | True |
| 30300 | 1 | 4280 | 0.02336448598130841 | True |
| 30340 | 4 | 4280 | 0.09345794392523364 | True |
| 30460 | 8 | 4280 | 0.18691588785046728 | True |
| 30500 | 0 | 4280 | 0.0 | True |
| 30620 | 0 | 4280 | 0.0 | True |
| 30700 | 1 | 4280 | 0.02336448598130841 | True |
| 30780 | 6 | 4280 | 0.14018691588785046 | True |
| 30860 | 2 | 4280 | 0.04672897196261682 | True |
| 30980 | 2 | 4280 | 0.04672897196261682 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 31020 | 0 | 4280 | 0.0 | True |
| 31084 | 47 | 4280 | 1.0981308411214954 | True |
| 31140 | 29 | 4280 | 0.677570093457944 | True |
| 31180 | 9 | 4280 | 0.2102803738317757 | True |
| 31340 | 4 | 4280 | 0.09345794392523364 | True |
| 31420 | 2 | 4280 | 0.04672897196261682 | True |
| 31460 | 0 | 4280 | 0.0 | True |
| 31540 | 5 | 4280 | 0.11682242990654206 | True |
| 31700 | 6 | 4280 | 0.14018691588785046 | True |
| 31740 | 2 | 4280 | 0.04672897196261682 | True |
| 31860 | 2 | 4280 | 0.04672897196261682 | True |
| 31900 | 2 | 4280 | 0.04672897196261682 | True |
| 31924 | 17 | 4280 | 0.397196261682243 | True |
| 32580 | 8 | 4280 | 0.18691588785046728 | True |
| 32780 | 1 | 4280 | 0.02336448598130841 | True |
| 32820 | 9 | 4280 | 0.2102803738317757 | True |
| 32900 | 0 | 4280 | 0.0 | True |
| 33124 | 23 | 4280 | 0.5373831775700935 | True |
| 33140 | 5 | 4280 | 0.11682242990654206 | True |
| 33220 | 5 | 4280 | 0.11682242990654206 | True |
| 33260 | 6 | 4280 | 0.14018691588785046 | True |
| 33340 | 22 | 4280 | 0.514018691588785 | True |
| 33460 | 64 | 4280 | 1.4953271028037383 | True |
| 33540 | 1 | 4280 | 0.02336448598130841 | True |
| 33660 | 2 | 4280 | 0.04672897196261682 | True |
| 33700 | 2 | 4280 | 0.04672897196261682 | True |
| 33740 | 1 | 4280 | 0.02336448598130841 | True |
| 33780 | 3 | 4280 | 0.07009345794392523 | True |
| 33860 | 0 | 4280 | 0.0 | True |
| 33874 | 24 | 4280 | 0.5607476635514018 | True |
| 34060 | 0 | 4280 | 0.0 | True |
| 34100 | 1 | 4280 | 0.02336448598130841 | True |
| 34580 | 0 | 4280 | 0.0 | True |
| 34620 | 1 | 4280 | 0.02336448598130841 | True |
| 34740 | 2 | 4280 | 0.04672897196261682 | True |
| 34820 | 15 | 4280 | 0.35046728971962615 | True |
| 34900 | 1 | 4280 | 0.02336448598130841 | True |
| 34940 | 10 | 4280 | 0.2336448598130841 | True |
| 34980 | 23 | 4280 | 0.5373831775700935 | True |
| 35004 | 16 | 4280 | 0.37383177570093457 | True |
| 35084 | 23 | 4280 | 0.5373831775700935 | True |
| 35100 | 3 | 4280 | 0.07009345794392523 | True |
| 35154 | 21 | 4280 | 0.49065420560747663 | True |
| 35300 | 3 | 4280 | 0.07009345794392523 | True |
| 35380 | 13 | 4280 | 0.3037383177570093 | True |
| 35614 | 47 | 4280 | 1.0981308411214954 | True |
| 35660 | 2 | 4280 | 0.04672897196261682 | True |
| 35840 | 25 | 4280 | 0.5841121495327103 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 35980 | 0 | 4280 | 0.0 | True |
| 36084 | 28 | 4280 | 0.6542056074766355 | True |
| 36100 | 13 | 4280 | 0.3037383177570093 | True |
| 36140 | 0 | 4280 | 0.0 | True |
| 36220 | 2 | 4280 | 0.04672897196261682 | True |
| 36260 | 9 | 4280 | 0.2102803738317757 | True |
| 36420 | 18 | 4280 | 0.4205607476635514 | True |
| 36500 | 1 | 4280 | 0.02336448598130841 | True |
| 36540 | 11 | 4280 | 0.2570093457943925 | True |
| 36740 | 64 | 4280 | 1.4953271028037383 | True |
| 36780 | 4 | 4280 | 0.09345794392523364 | True |
| 36980 | 0 | 4280 | 0.0 | True |
| 37100 | 3 | 4280 | 0.07009345794392523 | True |
| 37140 | 0 | 4280 | 0.0 | True |
| 37340 | 12 | 4280 | 0.2803738317757009 | True |
| 37460 | 4 | 4280 | 0.09345794392523364 | True |
| 37620 | 0 | 4280 | 0.0 | True |
| 37860 | 11 | 4280 | 0.2570093457943925 | True |
| 37900 | 4 | 4280 | 0.09345794392523364 | True |
| 37964 | 29 | 4280 | 0.677570093457944 | True |
| 38060 | 52 | 4280 | 1.2149532710280373 | True |
| 38220 | 0 | 4280 | 0.0 | True |
| 38240 | 1 | 4280 | 0.02336448598130841 | True |
| 38300 | 39 | 4280 | 0.9112149532710281 | True |
| 38340 | 0 | 4280 | 0.0 | True |
| 38540 | 1 | 4280 | 0.02336448598130841 | True |
| 38860 | 6 | 4280 | 0.14018691588785046 | True |
| 38900 | 23 | 4280 | 0.5373831775700935 | True |
| 38940 | 29 | 4280 | 0.677570093457944 | True |
| 39100 | 9 | 4280 | 0.2102803738317757 | True |
| 39150 | 2 | 4280 | 0.04672897196261682 | True |
| 39300 | 43 | 4280 | 1.0046728971962617 | True |
| 39340 | 9 | 4280 | 0.2102803738317757 | True |
| 39380 | 2 | 4280 | 0.04672897196261682 | True |
| 39460 | 6 | 4280 | 0.14018691588785046 | True |
| 39540 | 1 | 4280 | 0.02336448598130841 | True |
| 39580 | 16 | 4280 | 0.37383177570093457 | True |
| 39660 | 2 | 4280 | 0.04672897196261682 | True |
| 39740 | 5 | 4280 | 0.11682242990654206 | True |
| 39820 | 1 | 4280 | 0.02336448598130841 | True |
| 39900 | 7 | 4280 | 0.16355140186915887 | True |
| 40060 | 11 | 4280 | 0.2570093457943925 | True |
| 40140 | 32 | 4280 | 0.7476635514018691 | True |
| 40220 | 3 | 4280 | 0.07009345794392523 | True |
| 40340 | 4 | 4280 | 0.09345794392523364 | True |
| 40380 | 9 | 4280 | 0.2102803738317757 | True |
| 40420 | 10 | 4280 | 0.2336448598130841 | True |
| 40484 | 9 | 4280 | 0.2102803738317757 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 40580 | 2 | 4280 | 0.04672897196261682 | True |
| 40660 | 1 | 4280 | 0.02336448598130841 | True |
| 40900 | 16 | 4280 | 0.37383177570093457 | True |
| 40980 | 3 | 4280 | 0.07009345794392523 | True |
| 41060 | 3 | 4280 | 0.07009345794392523 | True |
| 41100 | 0 | 4280 | 0.0 | True |
| 41140 | 4 | 4280 | 0.09345794392523364 | True |
| 41180 | 54 | 4280 | 1.2616822429906542 | True |
| 41304 | 8 | 4280 | 0.18691588785046728 | True |
| 41420 | 3 | 4280 | 0.07009345794392523 | True |
| 41500 | 3 | 4280 | 0.07009345794392523 | True |
| 41540 | 7 | 4280 | 0.16355140186915887 | True |
| 41620 | 14 | 4280 | 0.32710280373831774 | True |
| 41660 | 2 | 4280 | 0.04672897196261682 | True |
| 41700 | 35 | 4280 | 0.8177570093457944 | True |
| 41740 | 12 | 4280 | 0.2803738317757009 | True |
| 41780 | 2 | 4280 | 0.04672897196261682 | True |
| 41884 | 13 | 4280 | 0.3037383177570093 | True |
| 41940 | 8 | 4280 | 0.18691588785046728 | True |
| 41980 | 0 | 4280 | 0.0 | True |
| 42020 | 3 | 4280 | 0.07009345794392523 | True |
| 42034 | 1 | 4280 | 0.02336448598130841 | True |
| 42100 | 2 | 4280 | 0.04672897196261682 | True |
| 42140 | 0 | 4280 | 0.0 | True |
| 42200 | 2 | 4280 | 0.04672897196261682 | True |
| 42220 | 2 | 4280 | 0.04672897196261682 | True |
| 42340 | 4 | 4280 | 0.09345794392523364 | True |
| 42540 | 5 | 4280 | 0.11682242990654206 | True |
| 42644 | 25 | 4280 | 0.5841121495327103 | True |
| 42680 | 3 | 4280 | 0.07009345794392523 | True |
| 42700 | 2 | 4280 | 0.04672897196261682 | True |
| 43100 | 0 | 4280 | 0.0 | True |
| 43300 | 4 | 4280 | 0.09345794392523364 | True |
| 43340 | 4 | 4280 | 0.09345794392523364 | True |
| 43420 | 0 | 4280 | 0.0 | True |
| 43580 | 2 | 4280 | 0.04672897196261682 | True |
| 43620 | 0 | 4280 | 0.0 | True |
| 43640 | 3 | 4280 | 0.07009345794392523 | True |
| 43780 | 6 | 4280 | 0.14018691588785046 | True |
| 43900 | 7 | 4280 | 0.16355140186915887 | True |
| 44060 | 9 | 4280 | 0.2102803738317757 | True |
| 44100 | 5 | 4280 | 0.11682242990654206 | True |
| 44140 | 6 | 4280 | 0.14018691588785046 | True |
| 44180 | 8 | 4280 | 0.18691588785046728 | True |
| 44220 | 0 | 4280 | 0.0 | True |
| 44300 | 0 | 4280 | 0.0 | True |
| 44420 | 2 | 4280 | 0.04672897196261682 | True |
| 44700 | 3 | 4280 | 0.07009345794392523 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 44940 | 2 | 4280 | 0.04672897196261682 | True |
| 45060 | 2 | 4280 | 0.04672897196261682 | True |
| 45104 | 14 | 4280 | 0.32710280373831774 | True |
| 45220 | 3 | 4280 | 0.07009345794392523 | True |
| 45294 | 27 | 4280 | 0.6308411214953271 | True |
| 45300 | 66 | 4280 | 1.5420560747663552 | True |
| 45460 | 7 | 4280 | 0.16355140186915887 | True |
| 45500 | 2 | 4280 | 0.04672897196261682 | True |
| 45540 | 3 | 4280 | 0.07009345794392523 | True |
| 45780 | 11 | 4280 | 0.2570093457943925 | True |
| 45820 | 3 | 4280 | 0.07009345794392523 | True |
| 45900 | 1 | 4280 | 0.02336448598130841 | True |
| 45940 | 3 | 4280 | 0.07009345794392523 | True |
| 46060 | 8 | 4280 | 0.18691588785046728 | True |
| 46140 | 9 | 4280 | 0.2102803738317757 | True |
| 46220 | 4 | 4280 | 0.09345794392523364 | True |
| 46300 | 2 | 4280 | 0.04672897196261682 | True |
| 46340 | 7 | 4280 | 0.16355140186915887 | True |
| 46520 | 10 | 4280 | 0.2336448598130841 | True |
| 46540 | 2 | 4280 | 0.04672897196261682 | True |
| 46660 | 0 | 4280 | 0.0 | True |
| 46700 | 2 | 4280 | 0.04672897196261682 | True |
| 47020 | 1 | 4280 | 0.02336448598130841 | True |
| 47220 | 1 | 4280 | 0.02336448598130841 | True |
| 47260 | 13 | 4280 | 0.3037383177570093 | True |
| 47300 | 3 | 4280 | 0.07009345794392523 | True |
| 47380 | 5 | 4280 | 0.11682242990654206 | True |
| 47460 | 0 | 4280 | 0.0 | True |
| 47580 | 3 | 4280 | 0.07009345794392523 | True |
| 47664 | 56 | 4280 | 1.308411214953271 | True |
| 47764 | 9 | 4280 | 0.2102803738317757 | True |
| 47894 | 34 | 4280 | 0.794392523364486 | True |
| 47930 | 0 | 4280 | 0.0 | True |
| 47940 | 1 | 4280 | 0.02336448598130841 | True |
| 48060 | 0 | 4280 | 0.0 | True |
| 48140 | 3 | 4280 | 0.07009345794392523 | True |
| 48260 | 0 | 4280 | 0.0 | True |
| 48300 | 0 | 4280 | 0.0 | True |
| 48424 | 19 | 4280 | 0.4439252336448598 | True |
| 48540 | 4 | 4280 | 0.09345794392523364 | True |
| 48620 | 16 | 4280 | 0.37383177570093457 | True |
| 48660 | 3 | 4280 | 0.07009345794392523 | True |
| 48680 | 0 | 4280 | 0.0 | True |
| 48700 | 0 | 4280 | 0.0 | True |
| 48864 | 13 | 4280 | 0.3037383177570093 | True |
| 48900 | 9 | 4280 | 0.2102803738317757 | True |
| 49020 | 1 | 4280 | 0.02336448598130841 | True |
| 49180 | 7 | 4280 | 0.16355140186915887 | True |

| Category | N | Group denominator | Percent | Sparse direction cell (<100) |
|---|---|---|---|---|
| 49340 | 18 | 4280 | 0.4205607476635514 | True |
| 49420 | 2 | 4280 | 0.04672897196261682 | True |
| 49500 | 0 | 4280 | 0.0 | True |
| 49620 | 7 | 4280 | 0.16355140186915887 | True |
| 49660 | 6 | 4280 | 0.14018691588785046 | True |
| 49700 | 3 | 4280 | 0.07009345794392523 | True |
| 49740 | 5 | 4280 | 0.11682242990654206 | True |
| nonmetropolitan_or_unknown | 501 | 4280 | 11.705607476635514 | False |

Directional comparison

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 10180 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 10420 | 3 | 12 | 0.17964071856287425 | 0.2803738317757009 | -0.10073311321282666 | True | True | suppressed from interpretation |
| 10500 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.02336448598130841 | True | True | suppressed from interpretation |
| 10540 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.01021321842296714 5 | True | True | suppressed from interpretation |
| 10580 | 9 | 4 | 0.5389221556886228 | 0.09345794392523364 | 0.44546421176338913 | True | True | suppressed from interpretation |
| 10740 | 3 | 12 | 0.17964071856287425 | 0.2803738317757009 | -0.10073311321282666 | True | True | suppressed from interpretation |
| 10780 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 10900 | 4 | 13 | 0.23952095808383234 | 0.3037383177570093 | -0.06421735967317702 | True | True | suppressed from interpretation |
| 11020 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 11100 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 11180 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 11200 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 11244 | 11 | 21 | 0.6586826347305389 | 0.49065420560747663 | 0.16802842912306226 | True | True | suppressed from interpretation |
| 11260 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.11682242990654204 | True | True | suppressed from interpretation |
| 11460 | 1 | 4 | 0.05988023952095808 4 | 0.09345794392523364 | -0.03357770440427556 4 | True | True | suppressed from interpretation |
| 11500 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 11540 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 11694 | 4 | 20 | 0.23952095808383234 | 0.4672897196261682 | -0.22776876154233586 | True | True | suppressed from interpretation |
| 11700 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 12020 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 12054 | 10 | 32 | 0.5988023952095808 | 0.7476635514018691 | -0.14886115619228832 | True | True | suppressed from interpretation |
| 12060 | 56 | 51 | 3.3532934131736525 | 1.191588785046729 | 2.1617046281269237 | True | True | suppressed from interpretation |
| 12100 | 3 | 7 | 0.17964071856287425 | 0.16355140186915887 | 0.016089316693715362 | True | True | suppressed from interpretation |
| 12220 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 12260 | 3 | 4 | 0.17964071856287425 | 0.09345794392523364 | 0.0861827746376406 | True | True | suppressed from interpretation |
| 12420 | 14 | 59 | 0.8383233532934131 | 1.3785046728971964 | -0.5401813196037832 | True | True | suppressed from interpretation |
| 12540 | 5 | 13 | 0.2994011976047904 | 0.3037383177570093 | -0.004337120152218907 | True | True | suppressed from interpretation |
| 12580 | 13 | 41 | 0.7784431137724551 | 0.9579439252336449 | -0.1795008114611898 | True | True | suppressed from interpretation |
| 12620 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 12700 | 2 | 0 | 0.11976047904191617 | 0.0 | 0.11976047904191617 | True | True | suppressed from interpretation |
| 12940 | 10 | 9 | 0.5988023952095808 | 0.2102803738317757 | 0.38852202137780517 | True | True | suppressed from interpretation |
| 12980 | 1 | 2 | 0.059880239520958084 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 13020 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 13140 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 13220 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 13380 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 13460 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 13740 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 13780 | 1 | 1 | 0.059880239520958084 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 13820 | 5 | 13 | 0.2994011976047904 | 0.3037383177570093 | -0.004337120152218907 | True | True | suppressed from interpretation |
| 13900 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 13980 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 14010 | 1 | 0 | 0.059880239520958084 | 0.0 | 0.059880239520958084 | True | True | suppressed from interpretation |
| 14020 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 14100 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 14260 | 5 | 19 | 0.2994011976047904 | 0.4439252336448598 | -0.14452403604006941 | True | True | suppressed from interpretation |
| 14454 | 10 | 22 | 0.5988023952095808 | 0.514018691588785 | 0.08478370362079582 | True | True | suppressed from interpretation |
| 14500 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 14540 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 14580 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 14740 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 14860 | 9 | 0 | 0.5389221556886228 | 0.0 | 0.5389221556886228 | True | True | suppressed from interpretation |
| 15180 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 15260 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 15380 | 17 | 12 | 1.0179640718562875 | 0.2803738317757009 | 0.7375902400805865 | True | True | suppressed from interpretation |
| 15500 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 15540 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 15680 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 15764 | 12 | 20 | 0.718562874251497 | 0.4672897196261682 | 0.2512731546253288 | True | True | suppressed from interpretation |
| 15804 | 16 | 21 | 0.9580838323353293 | 0.49065420560747663 | 0.4674296267278527 | True | True | suppressed from interpretation |
| 15940 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 15980 | 7 | 26 | 0.41916167664670656 | 0.6074766355140186 | -0.1883149588673121 4 | True | True | suppressed from interpretation |
| 16020 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 16060 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 16180 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 16220 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 16300 | 0 | 6 | 0.0 | 0.14018691588785046 | -0.1401869158878504 6 | True | True | suppressed from interpretation |
| 16540 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 16580 | 0 | 10 | 0.0 | 0.2336448598130841 | -0.2336448598130840 8 | True | True | suppressed from interpretation |
| 16620 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 16700 | 3 | 14 | 0.17964071856287425 | 0.32710280373831774 | -0.1474620851754435 | True | True | suppressed from interpretation |
| 16740 | 19 | 45 | 1.1377245508982037 | 1.0514018691588785 | 0.0863226817393252 | True | True | suppressed from interpretation |
| 16820 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 16860 | 3 | 3 | 0.17964071856287425 | 0.07009345794392523 | 0.10954726061894902 | True | True | suppressed from interpretation |
| 16940 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 16984 | 42 | 134 | 2.5149700598802394 | 3.130841121495327 | -0.6158710616150878 | True | False | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 17020 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 17140 | 2 | 46 | 0.11976047904191617 | 1.074766355140187 | -0.9550058760982707 | True | True | suppressed from interpretation |
| 17300 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 17410 | 0 | 24 | 0.0 | 0.5607476635514018 | -0.5607476635514018 | True | True | suppressed from interpretation |
| 17420 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 17460 | 4 | 16 | 0.23952095808383234 | 0.37383177570093457 | -0.1343108176171022 6 | True | True | suppressed from interpretation |
| 17660 | 3 | 5 | 0.17964071856287425 | 0.11682242990654206 | 0.0628182886563322 | True | True | suppressed from interpretation |
| 17780 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |
| 17820 | 5 | 4 | 0.2994011976047904 | 0.09345794392523364 | 0.2059432536795568 | True | True | suppressed from interpretation |
| 17860 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 17900 | 5 | 9 | 0.2994011976047904 | 0.2102803738317757 | 0.08912082377301472 | True | True | suppressed from interpretation |
| 17980 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 18020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 18140 | 9 | 31 | 0.5389221556886228 | 0.7242990654205608 | -0.185376909731938 | True | True | suppressed from interpretation |
| 18580 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 18700 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 18880 | 1 | 4 | 0.05988023952095808 4 | 0.09345794392523364 | -0.0335777044042755 64 | True | True | suppressed from interpretation |
| 19060 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 19124 | 38 | 107 | 2.2754491017964074 | 2.5 | -0.2245508982035928 6 | True | False | suppressed from interpretation |
| 19140 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 19180 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 19300 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 19340 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |
| 19430 | 1 | 11 | 0.05988023952095808 4 | 0.2570093457943925 | -0.1971291062734344 7 | True | True | suppressed from interpretation |
| 19460 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 19500 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 19660 | 6 | 26 | 0.3592814371257485 | 0.6074766355140186 | -0.2481951983882702 | True | True | suppressed from interpretation |
| 19740 | 38 | 28 | 2.2754491017964074 | 0.6542056074766355 | 1.621243494319772 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 19780 | 0 | 11 | 0.0 | 0.2570093457943925 | -0.25700934579439255 | True | True | suppressed from interpretation |
| 19804 | 9 | 39 | 0.5389221556886228 | 0.9112149532710281 | -0.3722927975824052 4 | True | True | suppressed from interpretation |
| 20020 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 20100 | 1 | 2 | 0.059880239520958084 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 20220 | 1 | 0 | 0.059880239520958084 | 0.0 | 0.059880239520958084 | True | True | suppressed from interpretation |
| 20260 | 0 | 9 | 0.0 | 0.2102803738317757 | -0.2102803738317757 | True | True | suppressed from interpretation |
| 20500 | 1 | 9 | 0.059880239520958084 | 0.2102803738317757 | -0.1504001343108176 | True | True | suppressed from interpretation |
| 20580 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 20700 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 20740 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 20940 | 1 | 1 | 0.059880239520958084 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 20994 | 4 | 30 | 0.23952095808383234 | 0.7009345794392523 | -0.4614136213554199 7 | True | True | suppressed from interpretation |
| 21060 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 21140 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 21300 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 21340 | 1 | 15 | 0.059880239520958084 | 0.35046728971962615 | -0.2905870501986681 | True | True | suppressed from interpretation |
| 21420 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 21500 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 21660 | 1 | 1 | 0.059880239520958084 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 21780 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |
| 21794 | 1 | 3 | 0.059880239520958084 | 0.07009345794392523 | -0.010213218422967145 | True | True | suppressed from interpretation |
| 21820 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 22020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 22140 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 22180 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.11682242990654204 | True | True | suppressed from interpretation |
| 22220 | 2 | 7 | 0.11976047904191617 | 0.16355140186915887 | -0.043790922827242 72 | True | True | suppressed from interpretation |
| 22380 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 22420 | 2 | 5 | 0.11976047904191617 | 0.11682242990654206 | 0.002938049135374119 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 22500 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 22520 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 22540 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 22660 | 3 | 5 | 0.17964071856287425 | 0.11682242990654206 | 0.0628182886563322 | True | True | suppressed from interpretation |
| 22744 | 20 | 40 | 1.1976047904191616 | 0.9345794392523364 | 0.26302535116682535 | True | True | suppressed from interpretation |
| 22900 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 23060 | 2 | 9 | 0.11976047904191617 | 0.2102803738317757 | -0.0905198947898595 4 | True | True | suppressed from interpretation |
| 23104 | 6 | 62 | 0.3592814371257485 | 1.4485981308411215 | -1.089316693715373 | True | True | suppressed from interpretation |
| 23224 | 7 | 19 | 0.41916167664670656 | 0.4439252336448598 | -0.0247635569981532 85 | True | True | suppressed from interpretation |
| 23420 | 6 | 10 | 0.3592814371257485 | 0.2336448598130841 | 0.1256365773126644 | True | True | suppressed from interpretation |
| 23460 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 23540 | 1 | 4 | 0.05988023952095808 4 | 0.09345794392523364 | -0.0335777044042755 64 | True | True | suppressed from interpretation |
| 23580 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 23844 | 2 | 5 | 0.11976047904191617 | 0.11682242990654206 | 0.00293804913537411 9 | True | True | suppressed from interpretation |
| 23900 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 24020 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 24140 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 24220 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 24260 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 24300 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 24340 | 4 | 24 | 0.23952095808383234 | 0.5607476635514018 | -0.3212267054675695 | True | True | suppressed from interpretation |
| 24420 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 24500 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 24540 | 4 | 8 | 0.23952095808383234 | 0.18691588785046728 | 0.05260507023336503 | True | True | suppressed from interpretation |
| 24580 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 24660 | 1 | 6 | 0.05988023952095808 4 | 0.14018691588785046 | -0.0803066763668923 7 | True | True | suppressed from interpretation |
| 24780 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 24860 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 25060 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 25180 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 25220 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 25260 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 25420 | 4 | 7 | 0.23952095808383234 | 0.16355140186915887 | 0.07596955621467344 | True | True | suppressed from interpretation |
| 25500 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 25540 | 12 | 5 | 0.718562874251497 | 0.11682242990654206 | 0.6017404443449549 | True | True | suppressed from interpretation |
| 25620 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 25740 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 25860 | 0 | 6 | 0.0 | 0.14018691588785046 | -0.14018691588785046 | True | True | suppressed from interpretation |
| 25940 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 25980 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 26140 | 1 | 10 | 0.05988023952095808 4 | 0.2336448598130841 | -0.173764620292126 | True | True | suppressed from interpretation |
| 26300 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 26380 | 3 | 1 | 0.17964071856287425 | 0.02336448598130841 | 0.15627623258156584 | True | True | suppressed from interpretation |
| 26420 | 29 | 170 | 1.7365269461077844 | 3.97196261682243 | -2.235435670714645 | True | False | suppressed from interpretation |
| 26580 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 26620 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 26820 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 26900 | 9 | 41 | 0.5389221556886228 | 0.9579439252336449 | -0.4190217695450221 | True | True | suppressed from interpretation |
| 26980 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 27060 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 27100 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 27140 | 1 | 5 | 0.05988023952095808 4 | 0.11682242990654206 | -0.05694219038558396 4 | True | True | suppressed from interpretation |
| 27180 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 27260 | 16 | 33 | 0.9580838323353293 | 0.7710280373831776 | 0.1870557949521518 | True | True | suppressed from interpretation |
| 27340 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 27500 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 27620 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 27740 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.1168224299065420 4 | True | True | suppressed from interpretation |
| 27780 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 27860 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 27900 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 27980 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 28020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 28100 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 28140 | 9 | 35 | 0.5389221556886228 | 0.8177570093457944 | -0.2788348536571715 3 | True | True | suppressed from interpretation |
| 28420 | 3 | 2 | 0.17964071856287425 | 0.04672897196261682 | 0.13291174660025742 | True | True | suppressed from interpretation |
| 28450 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 28660 | 0 | 10 | 0.0 | 0.2336448598130841 | -0.2336448598130840 8 | True | True | suppressed from interpretation |
| 28700 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 28740 | 3 | 4 | 0.17964071856287425 | 0.09345794392523364 | 0.0861827746376406 | True | True | suppressed from interpretation |
| 28880 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 28940 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 29020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 29100 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 29180 | 6 | 4 | 0.3592814371257485 | 0.09345794392523364 | 0.26582349320051485 | True | True | suppressed from interpretation |
| 29200 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 29340 | 3 | 0 | 0.17964071856287425 | 0.0 | 0.17964071856287425 | True | True | suppressed from interpretation |
| 29404 | 3 | 14 | 0.17964071856287425 | 0.32710280373831774 | -0.1474620851754435 | True | True | suppressed from interpretation |
| 29414 | 0 | 12 | 0.0 | 0.2803738317757009 | -0.2803738317757009 | True | True | suppressed from interpretation |
| 29420 | 1 | 7 | 0.05988023952095808 4 | 0.16355140186915887 | -0.1036711623482008 | True | True | suppressed from interpretation |
| 29460 | 6 | 15 | 0.3592814371257485 | 0.35046728971962615 | 0.00881414740612233 5 | True | True | suppressed from interpretation |
| 29484 | 9 | 17 | 0.5389221556886228 | 0.397196261682243 | 0.14172589400637978 | True | True | suppressed from interpretation |
| 29540 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.1168224299065420 4 | True | True | suppressed from interpretation |
| 29620 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 29700 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 29740 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 29820 | 29 | 26 | 1.7365269461077844 | 0.6074766355140186 | 1.129050310593766 | True | True | suppressed from interpretation |
| 29940 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.02336448598130841 | True | True | suppressed from interpretation |
| 30020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.02336448598130841 | True | True | suppressed from interpretation |
| 30140 | 1 | 1 | 0.05988023952095808 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 30300 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.02336448598130841 | True | True | suppressed from interpretation |
| 30340 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 30460 | 1 | 8 | 0.05988023952095808 | 0.18691588785046728 | -0.12703564832950923 | True | True | suppressed from interpretation |
| 30500 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 30620 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 30700 | 3 | 1 | 0.17964071856287425 | 0.02336448598130841 | 0.15627623258156584 | True | True | suppressed from interpretation |
| 30780 | 2 | 6 | 0.11976047904191617 | 0.14018691588785046 | -0.02042643684593429 | True | True | suppressed from interpretation |
| 30860 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 30980 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 31020 | 3 | 0 | 0.17964071856287425 | 0.0 | 0.17964071856287425 | True | True | suppressed from interpretation |
| 31084 | 49 | 47 | 2.934131736526946 | 1.0981308411214954 | 1.836000895405451 | True | True | suppressed from interpretation |
| 31140 | 8 | 29 | 0.47904191616766467 | 0.677570093457944 | -0.19852817729027922 | True | True | suppressed from interpretation |
| 31180 | 0 | 9 | 0.0 | 0.2102803738317757 | -0.2102803738317757 | True | True | suppressed from interpretation |
| 31340 | 2 | 4 | 0.11976047904191617 | 0.09345794392523364 | 0.026302535116682516 | True | True | suppressed from interpretation |
| 31420 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 31460 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 31540 | 2 | 5 | 0.11976047904191617 | 0.11682242990654206 | 0.002938049135374119 | True | True | suppressed from interpretation |
| 31700 | 3 | 6 | 0.17964071856287425 | 0.14018691588785046 | 0.03945380267502379 | True | True | suppressed from interpretation |
| 31740 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 31860 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 31900 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 31924 | 5 | 17 | 0.2994011976047904 | 0.397196261682243 | -0.09779506407745255 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 32580 | 1 | 8 | 0.05988023952095808 4 | 0.18691588785046728 | -0.1270356483295092 3 | True | True | suppressed from interpretation |
| 32780 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 32820 | 2 | 9 | 0.11976047904191617 | 0.2102803738317757 | -0.0905198947898595 4 | True | True | suppressed from interpretation |
| 32900 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 33124 | 30 | 23 | 1.7964071856287425 | 0.5373831775700935 | 1.259024008058649 | True | True | suppressed from interpretation |
| 33140 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.1168224299065420 4 | True | True | suppressed from interpretation |
| 33220 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.1168224299065420 4 | True | True | suppressed from interpretation |
| 33260 | 1 | 6 | 0.05988023952095808 4 | 0.14018691588785046 | -0.0803066763668923 7 | True | True | suppressed from interpretation |
| 33340 | 1 | 22 | 0.05988023952095808 4 | 0.514018691588785 | -0.4541384520678269 4 | True | True | suppressed from interpretation |
| 33460 | 13 | 64 | 0.7784431137724551 | 1.4953271028037383 | -0.7168839890312833 | True | True | suppressed from interpretation |
| 33540 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 33660 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 33700 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 33740 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 33780 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| 33860 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 33874 | 4 | 24 | 0.23952095808383234 | 0.5607476635514018 | -0.3212267054675695 | True | True | suppressed from interpretation |
| 34060 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 34100 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 34580 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 34620 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 34740 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 34820 | 2 | 15 | 0.11976047904191617 | 0.35046728971962615 | -0.2307068106777099 8 | True | True | suppressed from interpretation |
| 34900 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 34940 | 3 | 10 | 0.17964071856287425 | 0.2336448598130841 | -0.0540041412502098 44 | True | True | suppressed from interpretation |
| 34980 | 15 | 23 | 0.8982035928143712 | 0.5373831775700935 | 0.36082041524427777 | True | True | suppressed from interpretation |
| 35004 | 46 | 16 | 2.754491017964072 | 0.37383177570093457 | 2.380659242263137 | True | True | suppressed from interpretation |
| 35084 | 33 | 23 | 1.9760479041916168 | 0.5373831775700935 | 1.4386647266215236 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 35100 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 35154 | 32 | 21 | 1.9161676646706587 | 0.49065420560747663 | 1.425513459063182 | True | True | suppressed from interpretation |
| 35300 | 15 | 3 | 0.8982035928143712 | 0.07009345794392523 | 0.828110134870446 | True | True | suppressed from interpretation |
| 35380 | 6 | 13 | 0.3592814371257485 | 0.3037383177570093 | 0.05554311936873915 | True | True | suppressed from interpretation |
| 35614 | 81 | 47 | 4.850299401197605 | 1.0981308411214954 | 3.752168560076109 | True | True | suppressed from interpretation |
| 35660 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 35840 | 3 | 25 | 0.17964071856287425 | 0.5841121495327103 | -0.404471430969836 | True | True | suppressed from interpretation |
| 35980 | 5 | 0 | 0.2994011976047904 | 0.0 | 0.29940119760479045 | True | True | suppressed from interpretation |
| 36084 | 18 | 28 | 1.0778443113772456 | 0.6542056074766355 | 0.42363870390061 | True | True | suppressed from interpretation |
| 36100 | 3 | 13 | 0.17964071856287425 | 0.3037383177570093 | -0.1240975991941351 | True | True | suppressed from interpretation |
| 36140 | 2 | 0 | 0.11976047904191617 | 0.0 | 0.11976047904191617 | True | True | suppressed from interpretation |
| 36220 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 36260 | 3 | 9 | 0.17964071856287425 | 0.2102803738317757 | -0.03063965526890146 | True | True | suppressed from interpretation |
| 36420 | 3 | 18 | 0.17964071856287425 | 0.4205607476635514 | -0.2409200291006772 | True | True | suppressed from interpretation |
| 36500 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 36540 | 3 | 11 | 0.17964071856287425 | 0.2570093457943925 | -0.07736862723151827 | True | True | suppressed from interpretation |
| 36740 | 31 | 64 | 1.8562874251497006 | 1.4953271028037383 | 0.36096032234596237 | True | True | suppressed from interpretation |
| 36780 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 36980 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 37100 | 7 | 3 | 0.41916167664670656 | 0.07009345794392523 | 0.3490682187027814 | True | True | suppressed from interpretation |
| 37140 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 37340 | 8 | 12 | 0.47904191616766467 | 0.2803738317757009 | 0.19866808439196376 | True | True | suppressed from interpretation |
| 37460 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 37620 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 37860 | 2 | 11 | 0.11976047904191617 | 0.2570093457943925 | -0.13724886675247636 | True | True | suppressed from interpretation |
| 37900 | 3 | 4 | 0.17964071856287425 | 0.09345794392523364 | 0.0861827746376406 | True | True | suppressed from interpretation |
| 37964 | 5 | 29 | 0.2994011976047904 | 0.677570093457944 | -0.37816889585315344 | True | True | suppressed from interpretation |
| 38060 | 41 | 52 | 2.4550898203592815 | 1.2149532710280373 | 1.240136549331244 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 38220 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 38240 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 38300 | 6 | 39 | 0.3592814371257485 | 0.9112149532710281 | -0.5519335161452796 | True | True | suppressed from interpretation |
| 38340 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 38540 | 1 | 1 | 0.059880239520958084 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 38860 | 1 | 6 | 0.059880239520958084 | 0.14018691588785046 | -0.0803066763668923 7 | True | True | suppressed from interpretation |
| 38900 | 13 | 23 | 0.7784431137724551 | 0.5373831775700935 | 0.24105993620236163 | True | True | suppressed from interpretation |
| 38940 | 8 | 29 | 0.47904191616766467 | 0.677570093457944 | -0.1985281772902792 | True | True | suppressed from interpretation |
| 39100 | 9 | 9 | 0.5389221556886228 | 0.2102803738317757 | 0.32864178185684706 | True | True | suppressed from interpretation |
| 39150 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 39300 | 3 | 43 | 0.17964071856287425 | 1.0046728971962617 | -0.8250321786333874 | True | True | suppressed from interpretation |
| 39340 | 5 | 9 | 0.2994011976047904 | 0.2102803738317757 | 0.08912082377301472 | True | True | suppressed from interpretation |
| 39380 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 39460 | 3 | 6 | 0.17964071856287425 | 0.14018691588785046 | 0.03945380267502379 | True | True | suppressed from interpretation |
| 39540 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.023364485981308414 | True | True | suppressed from interpretation |
| 39580 | 7 | 16 | 0.41916167664670656 | 0.37383177570093457 | 0.04532990094577197 | True | True | suppressed from interpretation |
| 39660 | 1 | 2 | 0.059880239520958084 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 39740 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.1168224299065420 4 | True | True | suppressed from interpretation |
| 39820 | 1 | 1 | 0.059880239520958084 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 39900 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 40060 | 6 | 11 | 0.3592814371257485 | 0.2570093457943925 | 0.10227209133135598 | True | True | suppressed from interpretation |
| 40140 | 44 | 32 | 2.6347305389221556 | 0.7476635514018691 | 1.8870669875202866 | True | True | suppressed from interpretation |
| 40220 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 40340 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 40380 | 14 | 9 | 0.8383233532934131 | 0.2102803738317757 | 0.6280429794616375 | True | True | suppressed from interpretation |
| 40420 | 3 | 10 | 0.17964071856287425 | 0.2336448598130841 | -0.0540041412502098 44 | True | True | suppressed from interpretation |
| 40484 | 1 | 9 | 0.059880239520958084 | 0.2102803738317757 | -0.1504001343108176 | True | True | suppressed from interpretation |
| 40580 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 40660 | 1 | 1 | 0.05988023952095808 4 | 0.02336448598130841 | 0.03651575353964967 | True | True | suppressed from interpretation |
| 40900 | 18 | 16 | 1.0778443113772456 | 0.37383177570093457 | 0.704012535676311 | True | True | suppressed from interpretation |
| 40980 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 41060 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 41100 | 4 | 0 | 0.23952095808383234 | 0.0 | 0.23952095808383234 | True | True | suppressed from interpretation |
| 41140 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.0934579439252336 6 | True | True | suppressed from interpretation |
| 41180 | 8 | 54 | 0.47904191616766467 | 1.2616822429906542 | -0.7826403268229896 | True | True | suppressed from interpretation |
| 41304 | 1 | 8 | 0.05988023952095808 4 | 0.18691588785046728 | -0.1270356483295092 3 | True | True | suppressed from interpretation |
| 41420 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 41500 | 3 | 3 | 0.17964071856287425 | 0.07009345794392523 | 0.10954726061894902 | True | True | suppressed from interpretation |
| 41540 | 2 | 7 | 0.11976047904191617 | 0.16355140186915887 | -0.0437909228272427 2 | True | True | suppressed from interpretation |
| 41620 | 4 | 14 | 0.23952095808383234 | 0.32710280373831774 | -0.0875818456544854 4 | True | True | suppressed from interpretation |
| 41660 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 41700 | 12 | 35 | 0.718562874251497 | 0.8177570093457944 | -0.0991941350942973 1 | True | True | suppressed from interpretation |
| 41740 | 16 | 12 | 0.9580838323353293 | 0.2803738317757009 | 0.6777100005596284 | True | True | suppressed from interpretation |
| 41780 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.0467289719626168 3 | True | True | suppressed from interpretation |
| 41884 | 1 | 13 | 0.05988023952095808 4 | 0.3037383177570093 | -0.2438580782360512 4 | True | True | suppressed from interpretation |
| 41940 | 4 | 8 | 0.23952095808383234 | 0.18691588785046728 | 0.05260507023336503 | True | True | suppressed from interpretation |
| 41980 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 42020 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| 42034 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 42100 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 42140 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 42200 | 3 | 2 | 0.17964071856287425 | 0.04672897196261682 | 0.13291174660025742 | True | True | suppressed from interpretation |
| 42220 | 2 | 2 | 0.11976047904191617 | 0.04672897196261682 | 0.07303150707929934 | True | True | suppressed from interpretation |
| 42340 | 5 | 4 | 0.2994011976047904 | 0.09345794392523364 | 0.2059432536795568 | True | True | suppressed from interpretation |
| 42540 | 2 | 5 | 0.11976047904191617 | 0.11682242990654206 | 0.00293804913537411 9 | True | True | suppressed from interpretation |
| 42644 | 15 | 25 | 0.8982035928143712 | 0.5841121495327103 | 0.31409144328166094 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 42680 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 42700 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 43100 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 43300 | 2 | 4 | 0.11976047904191617 | 0.09345794392523364 | 0.02630253511668251 6 | True | True | suppressed from interpretation |
| 43340 | 1 | 4 | 0.05988023952095808 4 | 0.09345794392523364 | -0.03357770440427556 4 | True | True | suppressed from interpretation |
| 43420 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 43580 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 43620 | 1 | 0 | 0.05988023952095808 4 | 0.0 | 0.05988023952095808 4 | True | True | suppressed from interpretation |
| 43640 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 43780 | 0 | 6 | 0.0 | 0.14018691588785046 | -0.14018691588785046 | True | True | suppressed from interpretation |
| 43900 | 4 | 7 | 0.23952095808383234 | 0.16355140186915887 | 0.07596955621467344 | True | True | suppressed from interpretation |
| 44060 | 0 | 9 | 0.0 | 0.2102803738317757 | -0.2102803738317757 | True | True | suppressed from interpretation |
| 44100 | 0 | 5 | 0.0 | 0.11682242990654206 | -0.11682242990654204 | True | True | suppressed from interpretation |
| 44140 | 5 | 6 | 0.2994011976047904 | 0.14018691588785046 | 0.15921428171694 | True | True | suppressed from interpretation |
| 44180 | 0 | 8 | 0.0 | 0.18691588785046728 | -0.1869158878504673 | True | True | suppressed from interpretation |
| 44220 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 44300 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 44420 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 44700 | 7 | 3 | 0.41916167664670656 | 0.07009345794392523 | 0.3490682187027814 | True | True | suppressed from interpretation |
| 44940 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 45060 | 4 | 2 | 0.23952095808383234 | 0.04672897196261682 | 0.1927919861212155 | True | True | suppressed from interpretation |
| 45104 | 3 | 14 | 0.17964071856287425 | 0.32710280373831774 | -0.1474620851754435 | True | True | suppressed from interpretation |
| 45220 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| 45294 | 6 | 27 | 0.3592814371257485 | 0.6308411214953271 | -0.2715596843695786 | True | True | suppressed from interpretation |
| 45300 | 24 | 66 | 1.437125748502994 | 1.5420560747663552 | -0.10493032626336107 | True | True | suppressed from interpretation |
| 45460 | 0 | 7 | 0.0 | 0.16355140186915887 | -0.1635514018691589 | True | True | suppressed from interpretation |
| 45500 | 0 | 2 | 0.0 | 0.04672897196261682 | -0.04672897196261683 | True | True | suppressed from interpretation |
| 45540 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 45780 | 1 | 11 | 0.05988023952095808 4 | 0.2570093457943925 | -0.1971291062734344 7 | True | True | suppressed from interpretation |
| 45820 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 45900 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 45940 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| 46060 | 2 | 8 | 0.11976047904191617 | 0.18691588785046728 | -0.0671554088085511 3 | True | True | suppressed from interpretation |
| 46140 | 2 | 9 | 0.11976047904191617 | 0.2102803738317757 | -0.0905198947898595 4 | True | True | suppressed from interpretation |
| 46220 | 1 | 4 | 0.05988023952095808 4 | 0.09345794392523364 | -0.0335777044042755 64 | True | True | suppressed from interpretation |
| 46300 | 1 | 2 | 0.05988023952095808 4 | 0.04672897196261682 | 0.01315126755834125 8 | True | True | suppressed from interpretation |
| 46340 | 1 | 7 | 0.05988023952095808 4 | 0.16355140186915887 | -0.1036711623482008 | True | True | suppressed from interpretation |
| 46520 | 4 | 10 | 0.23952095808383234 | 0.2336448598130841 | 0.00587609827074823 8 | True | True | suppressed from interpretation |
| 46540 | 3 | 2 | 0.17964071856287425 | 0.04672897196261682 | 0.13291174660025742 | True | True | suppressed from interpretation |
| 46660 | 2 | 0 | 0.11976047904191617 | 0.0 | 0.11976047904191617 | True | True | suppressed from interpretation |
| 46700 | 10 | 2 | 0.5988023952095808 | 0.04672897196261682 | 0.552073423246964 | True | True | suppressed from interpretation |
| 47020 | 0 | 1 | 0.0 | 0.02336448598130841 | -0.0233644859813084 14 | True | True | suppressed from interpretation |
| 47220 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 47260 | 3 | 13 | 0.17964071856287425 | 0.3037383177570093 | -0.1240975991941351 | True | True | suppressed from interpretation |
| 47300 | 1 | 3 | 0.05988023952095808 4 | 0.07009345794392523 | -0.0102132184229671 45 | True | True | suppressed from interpretation |
| 47380 | 1 | 5 | 0.05988023952095808 4 | 0.11682242990654206 | -0.0569421903855839 64 | True | True | suppressed from interpretation |
| 47460 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 47580 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 47664 | 8 | 56 | 0.47904191616766467 | 1.308411214953271 | -0.8293692987856064 | True | True | suppressed from interpretation |
| 47764 | 2 | 9 | 0.11976047904191617 | 0.2102803738317757 | -0.0905198947898595 4 | True | True | suppressed from interpretation |
| 47894 | 23 | 34 | 1.377245508982036 | 0.794392523364486 | 0.58285298561755 | True | True | suppressed from interpretation |
| 47930 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 47940 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 48060 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 48140 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.0700934579439252 3 | True | True | suppressed from interpretation |
| 48260 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |

| Category | Direction 1 N | Direction 0 N | Direction 1 percent | Direction 0 percent | Difference (percentage points) | Direction 1 sparse | Direction 0 sparse | Interpretation |
|---|---|---|---|---|---|---|---|---|
| 48300 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 48424 | 19 | 19 | 1.1377245508982037 | 0.4439252336448598 | 0.6937993172533438 | True | True | suppressed from interpretation |
| 48540 | 0 | 4 | 0.0 | 0.09345794392523364 | -0.09345794392523366 | True | True | suppressed from interpretation |
| 48620 | 0 | 16 | 0.0 | 0.37383177570093457 | -0.3738317757009346 | True | True | suppressed from interpretation |
| 48660 | 0 | 3 | 0.0 | 0.07009345794392523 | -0.07009345794392523 | True | True | suppressed from interpretation |
| 48680 | 1 | 0 | 0.059880239520958084 | 0.0 | 0.059880239520958084 | True | True | suppressed from interpretation |
| 48700 | 1 | 0 | 0.059880239520958084 | 0.0 | 0.059880239520958084 | True | True | suppressed from interpretation |
| 48864 | 2 | 13 | 0.11976047904191617 | 0.3037383177570093 | -0.18397783871509318 | True | True | suppressed from interpretation |
| 48900 | 2 | 9 | 0.11976047904191617 | 0.2102803738317757 | -0.09051989478985954 | True | True | suppressed from interpretation |
| 49020 | 2 | 1 | 0.11976047904191617 | 0.02336448598130841 | 0.09639599306060775 | True | True | suppressed from interpretation |
| 49180 | 2 | 7 | 0.11976047904191617 | 0.16355140186915887 | -0.04379092282724272 | True | True | suppressed from interpretation |
| 49340 | 6 | 18 | 0.3592814371257485 | 0.4205607476635514 | -0.06127931053780292 | True | True | suppressed from interpretation |
| 49420 | 1 | 2 | 0.059880239520958084 | 0.04672897196261682 | 0.013151267558341258 | True | True | suppressed from interpretation |
| 49500 | 0 | 0 | 0.0 | 0.0 | 0.0 | True | True | suppressed from interpretation |
| 49620 | 1 | 7 | 0.059880239520958084 | 0.16355140186915887 | -0.1036711623482008 | True | True | suppressed from interpretation |
| 49660 | 1 | 6 | 0.059880239520958084 | 0.14018691588785046 | -0.08030667636689237 | True | True | suppressed from interpretation |
| 49700 | 2 | 3 | 0.11976047904191617 | 0.07009345794392523 | 0.04966702109799094 | True | True | suppressed from interpretation |
| 49740 | 1 | 5 | 0.059880239520958084 | 0.11682242990654206 | -0.05694219038558396 | True | True | suppressed from interpretation |
| nonmetropolitan_or_unknown | 80 | 501 | 4.790419161676646 | 11.705607476635514 | -6.9151883149588675 | True | False | suppressed from interpretation |

## Table S8. Complete adjusted associations with disagreement direction

Complete adjusted associations retain all 27 terms, including the intercept and missingness indicators, and all reference categories and full-cohort numeric scales. The outcome is disagreement direction among 5,950 mortgages, not delinquency: OR > 1 is associated with Logistic non-risky/XGBoost risky and OR < 1 with the reverse. Ordinary maximum-likelihood estimation failed the Classic FICO missingness-indicator estimability guard; the identical-design Firth fallback was used. Eleven sparse terms remain in the fitted design but are suppressed from substantive interpretation. Intervals are approximate expected-information Wald 95% intervals, not profile-likelihood intervals. Extreme ORs and wide intervals are retained untruncated. Section S.F reports the approved stopping rule and numerical diagnostics.

*Column panel 1 of 2 — Table S8. Complete adjusted associations with disagreement direction*

| Term | Reference / unit | Direction 1 N | Direction 0 N | Coefficient | Sign | OR | Approximate 95% CI lower | Approximate 95% CI upper |
|---|---|---|---|---|---|---|---|---|
| intercept | Intercept; not a characteristic contrast | 5950 | 5950 | 2.3851191249658643 | positive | 10.860356327442016 | 6.740705515772732 | 17.497773680072818 |

| Term | Reference / unit | Direction 1 N | Direction 0 N | Coefficient | Sign | OR | Approximate 95% CI lower | Approximate 95% CI upper |
|---|---|---|---|---|---|---|---|---|
| classic_fico__per_full_cohort_sd | Per full-cohort SD (44.173447841467649); frozen median 758 | 1670 | 4280 | 1.8134209989230619 | positive | 6.131387063121282 | 4.636822070528204 | 8.10768814200552 |
| classic_fico__missing | Missing versus not missing | 0 | 4 | -5.757622479671519 | negative | 0.00315861234338060 94 | 3.942756147040187e-0 5 | 0.25304207421617814 |
| original_dti__per_full_cohort_sd | Per full-cohort SD (9.2547926693979239); frozen median 40 | 1670 | 4280 | -1.5115392391975768 | negative | 0.22057020622423765 | 0.17340885299240483 | 0.2805578552320718 |
| original_upb__per_full_cohort_sd | Per full-cohort SD (172671.59917042078); frozen median 275000 | 1670 | 4280 | -0.4090689524625496 | negative | 0.6642684277991769 | 0.5364624503855299 | 0.8225227019219764 |
| original_ltv__per_full_cohort_sd | Per full-cohort SD (19.365198196235486); frozen median 80 | 1670 | 4280 | -0.8592963639306651 | negative | 0.42345993920117664 | 0.11124316299918584 | 1.6119491326363722 |
| original_cltv__per_full_cohort_sd | Per full-cohort SD (19.479849700081267); frozen median 80 | 1670 | 4280 | 0.5406789034073942 | positive | 1.7171722606424376 | 0.44500308938421035 | 6.6262024760327085 |
| original_interest_rate__per_full_cohort_sd | Per full-cohort SD (0.70220553076074377); frozen median 6.75 | 1670 | 4280 | -5.7911320127599 | negative | 0.00305452246017097 65 | 0.00204015667906366 64 | 0.0045732308481185 01 |
| original_loan_term__per_full_cohort_sd | Per full-cohort SD (42.137722737000161); frozen median 360 | 1670 | 4280 | -0.2710552529473691 5 | negative | 0.7625743607599966 | 0.6197637625094745 | 0.9382924444209138 |
| first_time_homebuyer==yes | Versus no | 813 | 1910 | -1.4184289191783024 | negative | 0.2420940676203471 | 0.17270201368760282 | 0.33936800345005125 |
| borrower_structure==multiple | Versus one | 655 | 1439 | 2.7948930008884094 | positive | 16.36087806008031 | 10.878835694277283 | 24.605420875841258 |
| loan_purpose==cash_out_refinance | Versus purchase | 164 | 791 | -1.0993147675507295 | negative | 0.3330992559326348 | 0.19168219863768482 | 0.5788493406870862 |
| loan_purpose==no_cash_out_refinance | Versus purchase | 37 | 160 | -1.57751851584564 | negative | 0.2064868568451426 | 0.08816622448726023 | 0.4835958701616776 |
| channel==broker | Versus correspondent | 379 | 736 | 0.7877632129344385 | positive | 2.198473405556473 | 1.5018227271125142 | 3.218279513076628 |
| channel==retail | Versus correspondent | 617 | 1774 | 1.3450123334219564 | positive | 3.838233878866018 | 2.740338399103068 | 5.375992729108485 |
| super_conforming_flag==yes | Versus no | 7 | 145 | -4.59422734625086 | negative | 0.01011002931599835 2 | 0.00293030243945693 | 0.03488127757532396 |
| special_eligibility_program==hfa_advantage | Versus not_available_or_not_applicable | 35 | 211 | -4.001103127717393 | negative | 0.018295445539783486 | 0.007874215550898536 | 0.04250878393352948 |
| special_eligibility_program==home_possible | Versus not_available_or_not_applicable | 187 | 578 | -2.707624355139308 | negative | 0.06669506244321108 | 0.04119222406315898 | 0.1079871615449427 |
| occupancy_status==investment_property | Versus primary_residence | 28 | 426 | 0.7538766490786909 | positive | 2.1252228109227014 | 1.0254951386254427 | 4.4042841608397385 |
| occupancy_status==second_home | Versus primary_residence | 6 | 129 | -1.7866380799255053 | negative | 0.1675224210055456 | 0.036337321085462136 | 0.772312341725902 |
| property_type==condominium | Versus single_family | 138 | 479 | -0.7496824873370793 | negative | 0.47251655891622 | 0.2989738541701436 | 0.7467940602022122 |
| property_type==cooperative | Versus single_family | 4 | 8 | -1.1833769607016067 | negative | 0.3062428204882701 | 0.04689888626606478 | 1.9997205172113388 |
| property_type==manufactured_housing | Versus single_family | 5 | 118 | -2.9171012888363763 | negative | 0.05409025229054295 | 0.01356794469195712 | 0.21563733190841589 |
| property_type==planned_unit_development | Versus single_family | 554 | 1013 | -0.8713075567371794 | negative | 0.41840410431252245 | 0.30626180051051544 | 0.5716089770704312 |
| number_of_units==four | Versus one | 1 | 24 | -8.679240173816488 | negative | 0.00017008024994195 312 | 5.859708641054365e-0 6 | 0.00493664330298719 9 |
| number_of_units==three | Versus one | 2 | 47 | -4.513580632137283 | negative | 0.010959149156989718 | 0.001369619290544012 3 | 0.08769075543426644 |
| number_of_units==two | Versus one | 12 | 154 | -3.9272849006386026 | negative | 0.019697079545706785 | 0.006780146236340608 5 | 0.05722220865243448 |

*Column panel 2 of 2 — Table S8. Complete adjusted associations with disagreement direction*

| Term | Association direction | Interpretation |
|---|---|---|
| intercept | Intercept; no characteristic direction | Descriptive association only |
| classic_fico__per_full_cohort_sd | Associated with Direction 1 | Descriptive association only |
| classic_fico__missing | Associated with Direction 0 | suppressed from interpretation |
| original_dti__per_full_cohort_sd | Associated with Direction 0 | Descriptive association only |
| original_upb__per_full_cohort_sd | Associated with Direction 0 | Descriptive association only |
| original_ltv__per_full_cohort_sd | Associated with Direction 0 | Descriptive association only |
| original_cltv__per_full_cohort_sd | Associated with Direction 1 | Descriptive association only |
| original_interest_rate__per_full_cohort_sd | Associated with Direction 0 | Descriptive association only |
| original_loan_term__per_full_cohort_sd | Associated with Direction 0 | Descriptive association only |
| first_time_homebuyer==yes | Associated with Direction 0 | Descriptive association only |
| borrower_structure==multiple | Associated with Direction 1 | Descriptive association only |
| loan_purpose==cash_out_refinance | Associated with Direction 0 | Descriptive association only |
| loan_purpose==no_cash_out_refinance | Associated with Direction 0 | suppressed from interpretation |
| channel==broker | Associated with Direction 1 | Descriptive association only |
| channel==retail | Associated with Direction 1 | Descriptive association only |
| super_conforming_flag==yes | Associated with Direction 0 | suppressed from interpretation |
| special_eligibility_program==hfa_advantage | Associated with Direction 0 | suppressed from interpretation |
| special_eligibility_program==home_possible | Associated with Direction 0 | Descriptive association only |
| occupancy_status==investment_property | Associated with Direction 1 | suppressed from interpretation |
| occupancy_status==second_home | Associated with Direction 0 | suppressed from interpretation |
| property_type==condominium | Associated with Direction 0 | Descriptive association only |
| property_type==cooperative | Associated with Direction 0 | suppressed from interpretation |
| property_type==manufactured_housing | Associated with Direction 0 | suppressed from interpretation |
| property_type==planned_unit_development | Associated with Direction 0 | Descriptive association only |
| number_of_units==four | Associated with Direction 0 | suppressed from interpretation |
| number_of_units==three | Associated with Direction 0 | suppressed from interpretation |
| number_of_units==two | Associated with Direction 0 | suppressed from interpretation |

## Frozen reference categories and numeric scales

| Variable | Reference type | Reference category | Frozen imputation median | Full-cohort mean | Full-cohort SD |
|---|---|---|---|---|---|
| classic_fico | numeric_scaling | NA | 758.0 | 749.7209250476307 | 44.17344784146765 |
| original_dti | numeric_scaling | NA | 40.0 | 38.187887632251005 | 9.254792669397924 |
| original_upb | numeric_scaling | NA | 275000.0 | 310991.5480968017 | 172671.59917042078 |
| original_ltv | numeric_scaling | NA | 80.0 | 75.57935060197008 | 19.365198196235486 |
| original_cltv | numeric_scaling | NA | 80.0 | 75.79920142689205 | 19.479849700081267 |
| original_interest_rate | numeric_scaling | NA | 6.75 | 6.7357938100449966 | 0.7022055307607438 |
| original_loan_term | numeric_scaling | NA | 360.0 | 349.1301836312781 | 42.13772273700016 |
| first_time_homebuyer | categorical_reference | no | NA | NA | NA |

| Variable | Reference type | Reference category | Frozen imputation median | Full-cohort mean | Full-cohort SD |
|---|---|---|---|---|---|
| borrower_structure | categorical_reference | one | NA | NA | NA |
| loan_purpose | categorical_reference | purchase | NA | NA | NA |
| channel | categorical_reference | correspondent | NA | NA | NA |
| super_conforming_flag | categorical_reference | no | NA | NA | NA |
| harp_indicator | categorical_reference | no | NA | NA | NA |
| special_eligibility_program | categorical_reference | not_available_or_not_applicable | NA | NA | NA |
| occupancy_status | categorical_reference | primary_residence | NA | NA | NA |
| property_type | categorical_reference | single_family | NA | NA | NA |
| number_of_units | categorical_reference | one | NA | NA | NA |

### Optional Figure S2

Figure S2, a directional-odds-ratio forest plot, is optional, not generated and not required. No Figure S2 asset is inserted or claimed as present; complete adjusted estimates remain in Table S8.

## S.E Robustness definitions and full evidence

### Table S9. Robustness cohort flows

Annual cohort flows preserve all seven sensitivity definitions and role-specific maturity restrictions. Event rate is events divided by eligible mortgages; exclusion rate uses the calendar-and-quarter eligible pool, not gross annual mortgages. Annual source reproduction ends in March 2026, while role-specific maturity cutoffs are applied separately; Section S.G explains the implementation correction without changing a scientific rule. R6 restricts test to 2023Q2–Q4. R2 changes the outcome to distress composite; R3 changes the remaining-on-book population and estimand; R7 adds a First Payment Date timing proxy. These are purchased/eligible mortgage cohorts, not the whole mortgage market. All annual rows and specification sections are retained.

R1

Candidate C: observed 03–99 event requires complete valid history through that event; Candidate A complete-path negatives remain.

*Column panel 1 of 2 — R1*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2784 | 47216 | 247 | 0.05568 | 0.0052312775330396475 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 1975 | 48025 | 324 | 0.0395 | 0.00674648620510151 | 202603 | 202212 | Q1–Q4 |
| 2017 | train | 50000 | 50000 | 1795 | 48205 | 391 | 0.0359 | 0.008111191785084534 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 2663 | 47336 | 1222 | 0.05326106522130443 | 0.025815447017069462 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 1482 | 48514 | 2227 | 0.029642371389711177 | 0.045904275054623404 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 385 | 45360 | 767 | 0.008416220351951033 | 0.016909171075837743 | 202603 | 202212 | Q1–Q4 |
| 2023 | test | 50000 | 49964 | 636 | 49328 | 876 | 0.012729164998799135 | 0.017758676613687966 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R1*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Candidate C complete-history 90+ outcome

R2

Distress composite: 03–99 or RA or first effective-date Zero Balance Code 02/03/09 in scheduled-payment months 1–24; Codes 15/16/96 are not recoded as events.

*Column panel 1 of 2 — R2*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2778 | 47222 | 253 | 0.05556 | 0.005357672271398924 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 1969 | 48031 | 330 | 0.03938 | 0.006870562761549832 | 202603 | 202212 | Q1–Q4 |
| 2017 | train | 50000 | 50000 | 1789 | 48211 | 397 | 0.03578 | 0.008234635249216984 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 2606 | 47393 | 1279 | 0.05212104242084842 | 0.026987107800730065 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 1417 | 48579 | 2292 | 0.02834226738139051 | 0.047180880627431 61 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 381 | 45364 | 771 | 0.008328779101541152 | 0.016995855744643 33 | 202603 | 202212 | Q1–Q4 |
| 2023 | test | 50000 | 49964 | 626 | 49338 | 886 | 0.012529020895044433 | 0.017957760752361263 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R2*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Candidate B distress-composite outcome

R3

Remove otherwise-valid within-window Code 01 non-events; earlier observed 03–99 events remain. Estimand conditional on remaining on book, not post-payoff counterfactual risk.

*Column panel 1 of 2 — R3*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 10457 | 39543 | 252 | 0.20914 | 0.006372809346787042 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 7845 | 42155 | 330 | 0.1569 | 0.007828252876289882 | 202603 | 202212 | Q1–Q4 |
| 2017 | train | 50000 | 50000 | 8438 | 41562 | 397 | 0.16876 | 0.009551994610461 48 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 20959 | 29040 | 1279 | 0.41918838376767537 | 0.044042699724517 91 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 25983 | 24013 | 2292 | 0.5197015761260901 | 0.095448298838129 34 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 12949 | 32796 | 771 | 0.28306918788938684 | 0.023508964507866 81 | 202603 | 202212 | Q1–Q4 |
| 2023 | test | 50000 | 49964 | 9450 | 40514 | 886 | 0.1891361780481947 | 0.021868983561238 09 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R3*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Conditional-on-remaining-on-book cohort

R4

Exclude every effective-date Code 96 exit in scheduled-payment months 1–24, regardless of prior 03–99. Code 96 is not recoded; no performance-Period substitution for missing effective dates.

*Column panel 1 of 2 — R4*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2781 | 47219 | 250 | 0.05562 | 0.005294478917384951 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 1970 | 48030 | 329 | 0.0394 | 0.006849885488236519 | 202603 | 202212 | Q1–Q4 |
| 2017 | train | 50000 | 50000 | 1791 | 48209 | 395 | 0.03582 | 0.008193490841958 97 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 2611 | 47388 | 1274 | 0.052221044420888416 | 0.026884443318983 71 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 1425 | 48571 | 2284 | 0.028502280182414592 | 0.047023944328920 55 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 396 | 45349 | 756 | 0.008656683790578205 | 0.016670709387197072 | 202603 | 202212 | Q1–Q4 |

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2023 | test | 50000 | 49964 | 656 | 49308 | 856 | 0.013129453206308542 | 0.017360266082582947 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R4*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Exclude all within-window Code 96 loans

R5

Exclude First Payment Dates at least four months after the end of the Loan Identifier quarter, using the frozen unusual-FPD flag.

*Column panel 1 of 2 — R5*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2868 | 47132 | 252 | 0.05736 | 0.005346685903420182 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 2054 | 47946 | 328 | 0.04108 | 0.006841029491511283 | 202603 | 202212 | Q1–Q4 |
| 2017 | train | 50000 | 50000 | 1882 | 48118 | 396 | 0.03764 | 0.008229768485805727 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 2706 | 47293 | 1279 | 0.05412108242164843 | 0.027044171441862 43 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 1499 | 48497 | 2291 | 0.029982398591887352 | 0.04724003546611131 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 394 | 45351 | 771 | 0.008612963165373265 | 0.017000727657604024 | 202603 | 202212 | Q1–Q4 |
| 2023 | test | 50000 | 49964 | 643 | 49321 | 886 | 0.012869265871427428 | 0.017963950447071227 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R5*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Exclude unusual First Payment Dates

R6

T1 temporal sensitivity: unchanged Candidate A/19 predictors; 2020 validation maturity through 2023-03; test 2023Q2–Q4.

*Column panel 1 of 2 — R6*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2779 | 47221 | 252 | 0.05558 | 0.005336608712225493 | 202603 | 202212 | Q2–Q4 |
| 2016 | train | 50000 | 50000 | 1969 | 48031 | 330 | 0.03938 | 0.006870562761549832 | 202603 | 202212 | Q2–Q4 |
| 2017 | train | 50000 | 50000 | 1789 | 48211 | 397 | 0.03578 | 0.008234635249216984 | 202603 | 202212 | Q2–Q4 |
| 2018 | train | 50000 | 49999 | 2606 | 47393 | 1279 | 0.05212104242084842 | 0.026987107800730065 | 202603 | 202212 | Q2–Q4 |
| 2019 | train | 50000 | 49996 | 1417 | 48579 | 2292 | 0.02834226738139051 | 0.04718088062743161 | 202603 | 202212 | Q2–Q4 |
| 2020 | validation | 50000 | 49965 | 432 | 49533 | 807 | 0.008646052236565597 | 0.016292168857125554 | 202603 | 202303 | Q2–Q4 |
| 2023 | test | 50000 | 37466 | 444 | 37022 | 638 | 0.011850744675172156 | 0.017232996596618227 | 202603 | 202603 | Q2–Q4 |

*Column panel 2 of 2 — R6*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | True |

Saved change description: T1 temporal design

R7

Add First Payment Date continuous month index relative to 2015-01 and calendar-quarter sine/cosine. Scheduled-payment-date proxy for origination timing, not exact origination month.

*Column panel 1 of 2 — R7*

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | train | 50000 | 50000 | 2779 | 47221 | 252 | 0.05558 | 0.005336608712225493 | 202603 | 202212 | Q1–Q4 |
| 2016 | train | 50000 | 50000 | 1969 | 48031 | 330 | 0.03938 | 0.006870562761549832 | 202603 | 202212 | Q1–Q4 |

| vintage | role | gross_vintage_n | calendar_and_quarter_pool_n | excluded_after_role_pool_n | eligible_n | positive_n | exclusion_rate_after_role_pool | event_rate | annual_source_endpoint | role_month24_cutoff | test_identifier_quarters |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2017 | train | 50000 | 50000 | 1789 | 48211 | 397 | 0.03578 | 0.008234635249216984 | 202603 | 202212 | Q1–Q4 |
| 2018 | train | 50000 | 49999 | 2606 | 47393 | 1279 | 0.05212104242084842 | 0.026987107800730065 | 202603 | 202212 | Q1–Q4 |
| 2019 | train | 50000 | 49996 | 1417 | 48579 | 2292 | 0.02834226738139051 | 0.04718088062743161 | 202603 | 202212 | Q1–Q4 |
| 2020 | validation | 50000 | 45745 | 381 | 45364 | 771 | 0.008328779101541152 | 0.01699585574464333 | 202603 | 202212 | Q1–Q4 |
| 2023 | test | 50000 | 49964 | 626 | 49338 | 886 | 0.012529020895044433 | 0.017957760752361263 | 202603 | 202603 | Q1–Q4 |

*Column panel 2 of 2 — R7*

| vintage | quarter_restriction_applies |
|---|---|
| 2015 | False |
| 2016 | False |
| 2017 | False |
| 2018 | False |
| 2019 | False |
| 2020 | False |
| 2023 | False |

Saved change description: Three time-feature sensitivity

## Table S10. Full robustness matrix

Complete R1–R7 evidence includes role counts, selected settings, all configuration-by-fold results, raw/calibrated validation and test metrics, calibration diagnostics, four illustrative loss ratios, three decision regimes, matching benchmarks and calibrated common-5:1 Logistic–XGBoost classification cells. No model/configuration or regime is averaged away. Each sensitivity's 2020-optimal thresholds are applied unchanged, not replaced with the primary analysis's thresholds or optimized on test. Empty/inapplicable fields and undefined quantities are distinguished from zero. R2 changes the outcome, R3 changes the estimand, and R7 adds the First Payment Date timing proxy with out-of-support extrapolation. The archival sections retain all rows; no FT-Transformer or directional-association sensitivity fit is included.

R1

Candidate C: observed 03–99 event requires complete valid history through that event; Candidate A complete-path negatives remain.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.01843323749665686 | 239296 | 4411 |
| validation | 0.016909171075837743 | 45360 | 767 |
| test | 0.017758676613687966 | 49328 | 876 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_200","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47216 | 247 | 48025 | 324 | 48025 | 324 | 0.00674648620510151 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.00811119178508453 4 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95241 | 571 | 48205 | 391 | 48205 | 391 | 0.008111191785084534 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143446 | 962 | 47336 | 1222 | 47336 | 1222 | 0.025815447017069462 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190782 | 2184 | 48514 | 2227 | 48514 | 2227 | 0.045904275054623404 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.006240781091955972 | 0.8175261486093544 | 0.028751146522144846 | 0.006636463087195855 | 0.036486226737839716 | 199 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | 1 | 0.006233428331700323 | 0.8174503161540471 | 0.028869074094682402 | 0.006636324207785411 | 0.036481398034576965 | 137 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.006244708849693645 | 0.8165375444415716 | 0.02842660620311325 | 0.006637512917790754 | 0.03651227312209151 | 341 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.0062438448633604015 | 0.816730360752848 | 0.028469190253517764 | 0.006637415833805948 | 0.03650868239533805 | 266 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.004835164811880833 | 0.8459002981794258 | 0.04189487901164629 | 0.006604887502442228 | 0.03504564717118194 | 199 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.004832820460978587 | 0.8454949957049843 | 0.04118660488081814 | 0.0066074230193705966 | 0.03507768959136974 | 137 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.004834948928659971 | 0.8455097480939007 | 0.041673916651770365 | 0.006606357229964519 | 0.035060984084876796 | 341 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.0048354800242644 | 0.8455309061253731 | 0.041727828018832964 | 0.006605979314456473 | 0.035058073721770576 | 266 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.004581153059767063 | 0.7975318088680493 | 0.03839146541636639 | 0.0066614105809015868 | 0.036589458609744656 | 199 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.004536088312086325 | 0.8109745997508658 | 0.039288937936838284 | 0.006608379887857252 | 0.03611542490935979 | 137 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.0045557039494889154 | 0.7908918103795219 | 0.036756566913336675 | 0.006629967868387866 | 0.037024765096765604 | 341 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.0045729748912721364 | 0.7962027997963653 | 0.03868065159329606 | 0.006622853969825981 | 0.03673279836196101 | 266 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.004596876153319074 | 0.8356963684018324 | 0.04249563629355892 | 0.006598988583321588 | 0.03525243728149121 | 199 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.0045744074362059975 | 0.837571215863425 | 0.042775019681436 | 0.0065995632693309015 | 0.035205589267994025 | 137 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.004589251780303731 | 0.8348767049685271 | 0.04255861665533609 | 0.0066013747599087775 | 0.03528686785449432 | 341 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.004593403781867233 | 0.8354294019252125 | 0.042775573922369095 | 0.006600361333178937 | 0.03525508900407054 | 266 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004799151599500918 | 0.8397633043901815 | 0.033210266367302475 | 0.00660823138982825 | 0.035197970556740274 | 137 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.0047912455705240884 | 0.8394055589589575 | 0.03391105319465716 | 0.006606103329717414 | 0.03518057983293476 | 266 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004716469944706302 | 0.8386144944550428 | 0.036626558352825765 | 0.006625600664676004 | 0.035809421004873936 | 137 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004746205236176394 | 0.8358457686913415 | 0.035466135957356495 | 0.006635738416375253 | 0.036231717774890745 | 266 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.004973812117362779 | 0.8226264959116472 | 0.030187978272369954 | 0.006628120814735972 | 0.036033475228626266 | 137 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.00485748894090513 | 0.8186929137546874 | 0.028461745934033306 | 0.006638758012406744 | 0.036268903924905393 | 266 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.004744957726290032 | 0.82551314373149 | 0.035036383180221814 | 0.006620504098982792 | 0.03588140119933338 | 137 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.0047629582457857705 | 0.8231268801207936 | 0.03495558607530488 | 0.006629886733974524 | 0.03626102284790434 | 266 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.004811682162659713 | 0.8381662612347853 | 0.03346715874498584 | 0.006606637252658074 | 0.03532111063993519 | 137 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004826664415069895 | 0.8359571233462766 | 0.03367662607323125 | 0.006605394561870798 | 0.03543448914386921 | 266 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.004695441233131515 | 0.8402119581829302 | 0.03711319995363005 | 0.0066193332553185266 | 0.03553711576034082 | 137 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004737978810156685 | 0.8387150436321312 | 0.03648716606332663 | 0.0066267957216996545 | 0.0358070015066651 | 266 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.005312430372251856 | 0.7897557146743047 | 0.027250554577889656 | 0.006649646994445032 | 0.05161172786537244 | 137 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.0051587147346652765 | 0.7763654630011381 | 0.02492767299727896 | 0.00666864567717023 | 0.05944065644630457 | 266 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.004842095404086337 | 0.8130463074899948 | 0.031067176337823216 | 0.0066195781987365395 | 0.03618812755305434 | 137 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.0047969547598880795 | 0.8113210220765618 | 0.03181605285392281 | 0.006622425740556942 5 | 0.0362223963347129 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.004539936766755197 | 0.8332496394076164 | 0.040551542728755434 | 0.006591397845502176 | 0.03531315656779106 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.004570753106706113 | 0.8347332897490826 | 0.04040101514731124 | 0.006592055499961959 | 0.03526963755650449 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.004494658000257952 | 0.8346165323552239 | 0.04144238595300805 | 0.006591171430476479 | 0.035372230068358906 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004561894549311671 | 0.835077059232912 | 0.04227716370390798 | 0.006589488999108645 | 0.03536459328182106 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.004496169499336393 | 0.8171501568023654 | 0.034369668913587094 | 0.006622727895811547 5 | 0.03631221216854876 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.004557945256177876 5 | 0.8209063544232967 | 0.035291834786280694 | 0.006616903187067485 | 0.036065330445753226 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.003785950875150975 | 0.8084296185523971 | 0.035145429453159134 | 0.00662989396619672 | 0.03763063371967297 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.003830781757613442 6 | 0.8114457703477501 | 0.04209120812840904 | 0.006605309297301625 | 0.037319033134362814 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004513977743765721 | 0.8049153148172736 | 0.03296034633030003 | 0.006637808129081913 | 0.036991233812499254 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.004565978100435264 | 0.8077922894698224 | 0.03260928951746522 | 0.006634182512386104 | 0.03692397515427195 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004180768019524306 | 0.8005618719073363 | 0.03412166805429886 | 0.006651950394412276 5 | 0.03804423950294074 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.004313989319415941 5 | 0.8017298664184125 | 0.035690708013019934 | 0.006651898556544447 | 0.037931040083804296 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004086898263765662 | 0.7717405567241001 | 0.02602112285660011 | 0.006722336399563798 | 0.039956724211514236 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.004258095483862776 | 0.7755189799835965 | 0.02566012504220946 | 0.006748690689703928 | 0.03976616271470865 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.001745447693697077 2 | 0.7604589584658137 | 0.025617429132310746 | 0.006752892290243813 | 0.048784280979995376 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.001718576338127917 7 | 0.7638315939749173 | 0.031123669812476096 | 0.006694637075229179 | 0.04805170681721483 | 266 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.008364975469975158 | 0.8253471973719133 | 0.04401539120615185 | 0.00790077191086575 | 0.041377279484959624 | 263 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.008351851643487463 | 0.8255291684946686 | 0.04401172293519624 | 0.007900656199547548 | 0.04137481432282368 | 200 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.008373896116034809 | 0.825527135895414 | 0.04438752073498319 | 0.007899721878741385 | 0.04136020347056207 | 443 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.00837334142252325 | 0.8255290615157604 | 0.04436809701422223 | 0.007899796628245163 | 0.0413610762521359 | 347 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.007567848104696633 | 0.8434548217907906 | 0.053159418074958886 | 0.00786520869367505 | 0.04019813344065755 | 263 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.00754739653196776 | 0.8438564206119684 | 0.05337566662097771 | 0.007863851590217645 | 0.04018463757068116 | 200 |
| logistic | lr_c_0p1_freq_25 | 2 | 0.007575220656942351 | 0.8433438311735896 | 0.0534560823131608 | 0.007863265289421975 | 0.0401859612146441 | 443 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.007573944303344010 5 | 0.843189567588044 | 0.05365251835864484 | 0.007862319627284602 | 0.04018103690193525 | 347 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.007373964018909516 5 | 0.8117677226875627 | 0.05083252004660567 | 0.007894714911111555 | 0.04151038461039811 | 263 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.007311035020890617 | 0.8228546957910325 | 0.05156278629976548 | 0.007874995360997054 | 0.0409047607929421 | 200 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.007471553770339434 | 0.8013313418139792 | 0.04491751027186815 | 0.007932385840053742 | 0.04219982980983035 | 443 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.007377897280798021 | 0.804852124659954 | 0.049863386941759257 | 0.007900391311853579 | 0.041845104335798015 | 347 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.007399479292865247 | 0.8349221840771097 | 0.050994938292361694 | 0.007877914869993579 | 0.04054574296453426 | 263 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.007353994355598845 | 0.8374369907603387 | 0.053715683545963275 | 0.007868613140958377 | 0.040383474517094085 | 200 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.00742454791375984 | 0.8332682901571808 | 0.05111291975709313 | 0.00787745623451226 | 0.04062047355669598 | 443 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.007401560952362434 | 0.8335261628152655 | 0.05147491009281579 | 0.007874262640783026 | 0.04057345754089794 | 347 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007291034647454672 | 0.8360108014463977 | 0.048614468884535855 | 0.00787108122984111 | 0.04039297514828553 | 200 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007290430030986792 | 0.8335795987798843 | 0.04819076318017479 | 0.007872628461752965 | 0.0405061942133801 | 347 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.0071104914294574575 | 0.8352981614497867 | 0.0527348458542723 | 0.007916409022064235 | 0.041369965290815866 | 200 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.006925637208595731 | 0.8334102511682899 | 0.050191046974216454 | 0.007934133829208172 | 0.04183400790837819 | 347 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.0076379917335919 | 0.8265617824055428 | 0.0425530227314529 | 0.007912663460670285 | 0.041119378877111805 | 200 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.007505492678302542 | 0.8258930305059985 | 0.041204622791438994 | 0.007917969777957935 | 0.04122181419716777 | 347 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007246266313437234 | 0.8302575827452435 | 0.047337575115855436 | 0.00790465655454113 | 0.04125904676510022 | 200 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007048734365025433 | 0.8304136649722277 | 0.045532599293272205 | 0.007922089939222977 | 0.04168597118439046 | 347 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007359068898132457 | 0.8332537142809462 | 0.048508497258484565 | 0.00787100246056049 | 0.04049761112464826 | 200 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.0073644806017774 | 0.8294696028525712 | 0.04853844557375548 | 0.00787379772562025 | 0.04069057838517892 | 347 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007314194115505672 | 0.8372851876896803 | 0.051932348576106256 | 0.007904980397100435 | 0.04090402955686037 | 200 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.0072009457896427456 | 0.8365608335026273 | 0.05126566510588522 | 0.007918567125373872 | 0.04124299716418392 | 347 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.00836569495667015 | 0.7941063607840142 | 0.03620088799336854 | 0.007996245367609317 | 0.05429027703904759 | 200 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.008345633388046772 | 0.7901343676482088 | 0.03519643874263244 | 0.008002696059906573 | 0.059101153517603734 | 347 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.007677298709790401 | 0.8181920735689673 | 0.043665037355918646 | 0.007895835424013441 | 0.041132269386756876 | 200 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.007601762872982137 | 0.8219119976524547 | 0.046024595608898225 | 0.007895576975993916 | 0.04101380137372878 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.007102465301399723 | 0.834652597228583 | 0.04417534649538307 | 0.007896571701192442 | 0.04068370724955315 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.007109413419263941 | 0.8354366991358352 | 0.04481944130997471 | 0.007892291588557318 | 0.040636489041327925 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.007100753330933217 | 0.8364087629846988 | 0.048405971444070814 | 0.00787771959476133 | 0.04046270335383503 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.007098185787642449 | 0.8351985105968491 | 0.0462573166964045 | 0.007888737538740788 | 0.04057401434830791 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.006934062211673512 | 0.8235445492802086 | 0.03740222989041549 | 0.00795383220839214 | 0.04164043209109428 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.006949253940590746 | 0.8219681615792314 | 0.03898087428446642 | 0.007947474663814876 | 0.04169167129285952 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006542124404534089 | 0.8163621458556852 | 0.03524879564174357 | 0.007985209414935876 | 0.042286013825769735 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.006649653816644001 | 0.8171649690718628 | 0.036357336432127005 | 0.007980363527935614 | 0.04215960220897431 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.006893160116326939 | 0.8154360027031431 | 0.03555978060339029 | 0.007980946470225988 | 0.04223990815626573 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.0069164312813993455 | 0.8177031264692884 | 0.035528071509944636 | 0.007985193868657222 | 0.042126062667060656 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.006793726316918238 | 0.8129883252847753 | 0.03481316418949301 | 0.008002426910444713 | 0.04252972895799776 | 200 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.00687271244112334 | 0.8133798948333146 | 0.03524795732491441 | 0.008017361480136228 | 0.04255875939540704 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.006363332617316041 | 0.7886155345998139 | 0.02758222586943032 | 0.008132050097642696 | 0.04513412675945346 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006564873676820584 | 0.7912191872662578 | 0.03190851732945189 | 0.008113003525530129 | 0.044782767046373455 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.00475083842099379 6 | 0.7675020970540469 | 0.02382974090084497 | 0.008241279445233035 | 0.0492587850462954 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.00474219211510686 | 0.7724585368473337 | 0.024906878767969064 | 0.008191445639467474 | 0.048589083548298447 | 347 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.01095111251592648 | 0.7447007618705141 | 0.07578771264998961 | 0.024907722863826448 | 0.11753325881642417 | 311 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.010942600969085059 | 0.7447814875920893 | 0.07576445185577553 | 0.02490823333076243 | 0.11754137670130792 | 242 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.010952752665056444 | 0.744723423278835 | 0.07582110709502192 | 0.024907341630239867 | 0.1175276961333347 | 489 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.010952344278059597 | 0.7447060146323489 | 0.07578989914151169 | 0.024907657216482658 | 0.11753173721499449 | 411 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.010658074638281181 | 0.7358133727792086 | 0.0720483301549641 | 0.024928972384688258 | 0.12055858316059885 | 311 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.010668419581065321 | 0.7357473938315683 | 0.07225408847122539 | 0.02492631981721001 | 0.12054342303363738 | 242 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.01065933109018761 | 0.7361155308054251 | 0.0724749553411154 | 0.024924248501887004 | 0.12050371256903639 | 489 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.010655782937072004 | 0.736033012756332 | 0.07228303196463126 | 0.02492705178514826 | 0.12053141219722825 | 411 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.010507998192729033 | 0.7111888689433793 | 0.06651451406362509 | 0.025005620897243976 | 0.12550053489730448 | 311 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.010542744710110463 | 0.7161705101858507 | 0.06789024598138688 | 0.024987895543394423 | 0.12459600618014369 | 242 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.01060924592293913 | 0.7093750867326807 | 0.06603863634734534 | 0.025014012102156917 | 0.12616270254241047 | 489 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.010571304845738308 | 0.7097878899279498 | 0.06631047153966921 | 0.025007770372895814 | 0.12591442234438574 | 411 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.01058151505538594 | 0.7269269064703875 | 0.06915529861024046 | 0.024970970819617282 | 0.12273195463904357 | 311 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.010602817405490616 | 0.7272485139120461 | 0.06974423612062175 | 0.024964437886903425 | 0.12260763301484635 | 242 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.010615229502117874 | 0.7272415220601445 | 0.06970205588220767 | 0.024963424043138692 | 0.12271287125818613 | 489 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.010596398956096603 | 0.7271623579704664 | 0.06939930635747882 | 0.024968234675730207 | 0.12273150245609502 | 411 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010211107206451261 | 0.7193719797950386 | 0.0669874447210973 | 0.024986555401021383 | 0.12345052145190297 | 242 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.010134319932831475 | 0.7197198687916879 | 0.06701886777737429 | 0.02498971439679778 | 0.12367832313703525 | 411 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.009762696746168567 | 0.7080661020326271 | 0.06452704505357641 | 0.025153171792380247 | 0.121921175201067 | 242 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009355177978367591 | 0.7050093495611495 | 0.06387990312877541 | 0.02520330595068734 | 0.12298597034610723 | 411 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011170648582996111 | 0.7110345246999413 | 0.06387102467327228 | 0.02502401162897863 | 0.1253369678065323 | 242 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.010582320205189556 | 0.7109124423518262 | 0.06402001959137373 | 0.025044604315872975 | 0.12649100773044267 | 411 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.010197726674553691 | 0.7082702499115017 | 0.06378674096258447 | 0.02510366537876863 | 0.12129826469821738 | 242 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.009531188617593027 | 0.7052279957725205 | 0.06381264366760321 | 0.025165696688681474 | 0.12261338549963395 | 411 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.010589765558758197 | 0.7171213665528402 | 0.06722049267575038 | 0.024962025323795944 | 0.12445231887166211 | 242 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.01061997444754999 | 0.7188618407934737 | 0.06769416552427564 | 0.024957183531254857 | 0.1244468566404692 | 411 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010724554261750221 | 0.7158538360813205 | 0.0673852497231543 | 0.025075470366386316 | 0.11977501282381993 | 242 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010421834787186364 | 0.7117727063229836 | 0.066078142249464 | 0.025116753434178123 | 0.12052419969391939 | 411 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.013690861396778983 | 0.7017949077597275 | 0.0638230477621505 | 0.024950783370960784 | 0.16497762622040393 | 242 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.013088474519831021 | 0.6985587450782864 | 0.06428206702997115 | 0.024960650034745697 | 0.1863722240988241 | 411 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.012218675591861 | 0.7093958493385815 | 0.06553182189298948 | 0.02494378304274956 | 0.11917171460855655 | 242 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.011855894656156926 | 0.7121924658785205 | 0.06583124537979348 | 0.024963917709898923 | 0.11900156161809024 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.009878815794126057 | 0.7227142180976527 | 0.0687017338469938 | 0.02498245169227035 | 0.12427365073492272 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.009844078392355654 | 0.7252458718438266 | 0.06880864065592596 | 0.024985612064290758 | 0.12396172131995817 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.009789092627062628 | 0.7238460374335943 | 0.06949449151328957 | 0.024974151056494906 | 0.12428162191655742 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.009715130888155439 | 0.7245396149455839 | 0.06824699851143269 | 0.024995421191637206 | 0.124461192631331 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.010347031097466134 | 0.7173483799169311 | 0.06649081927127452 | 0.025006465289855187 | 0.1255594097880728 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.010363755820797916 | 0.7177387967640432 | 0.06703081114625446 | 0.02500149620279824 | 0.12537601900512785 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.009948457238507093 | 0.7147501438653386 | 0.06488091291893425 | 0.02504811287292146 | 0.12747814908916136 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.00986937892202707 | 0.717086318564247 | 0.0658286919048239 | 0.025045742099184174 | 0.12732469883069153 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.010626494983522904 | 0.7122531565726922 | 0.06630161916559198 | 0.02502722866156185 | 0.12615405768785767 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.010269355607008618 | 0.7123732691351192 | 0.06632604059933404 | 0.0250408131390401 | 0.12665760891773198 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.010233136588473742 | 0.7135506952207746 | 0.0666644973812982 | 0.025038932278646495 | 0.1277540976802927 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.010116930413852055 | 0.713012358115982 | 0.06637017326145266 | 0.025046559200753892 | 0.12791045262286468 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.01005350011009918 | 0.6941388831648769 | 0.06066378616260837 | 0.025234964746486805 | 0.1351371812372737 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.010242998439597579 | 0.6976130172524122 | 0.06027909136062208 | 0.02526576388270134 | 0.13402287154655657 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008451088361828602 | 0.691420987069191 | 0.056849253052679904 | 0.025415013238082224 | 0.1437514537609428 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.008635570103795632 | 0.6937371533594215 | 0.057864662217822625 | 0.025413870106536448 | 0.14235694258547865 | 411 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.013123672559151558 | 0.6956295859682355 | 0.09647888408884335 | 0.04415604409887635 | 0.20663501719321045 | 349 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.013120122036945979 | 0.6956563997943019 | 0.09647383733475083 | 0.044156461387280174 | 0.20663921603315868 | 268 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.013127795121680408 | 0.6957132094055336 | 0.09653845195283445 | 0.044154881679279064 | 0.2066104922547777 | 512 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.013128446489125358 | 0.6956937490093363 | 0.09652456880570731 | 0.04415509855967371 | 0.20661196327716383 | 449 |
| logistic | lr_c_0p1_freq_100 | 4 | 0.012745177452667443 | 0.6951237126780571 | 0.09611901829033266 | 0.044211453324131136 | 0.21042399916145527 | 349 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.01274586458270537 | 0.695178950711929 | 0.0961278277902819 | 0.044210917759221126 | 0.21040170910057832 | 268 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.012754216919494847 | 0.6954828375069819 | 0.09637337580081408 | 0.04420599317381409 | 0.210330023370693 | 512 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.01275386899365956 | 0.6953008837726479 | 0.09626331214067929 | 0.04420790599584015 | 0.21035914714297912 | 449 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.012713295649369101 | 0.6872813476303025 | 0.09385716005363122 | 0.04427535510097667 | 0.2141037380923401 | 349 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.012689150766045524 | 0.6898580651249823 | 0.09433435736107386 | 0.044262010623030645 | 0.2131559065954926 | 268 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.012769750558808439 | 0.6866940821546333 | 0.09440767675148865 | 0.04427217762960364 | 0.21470074561392244 | 512 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.012744204040569612 | 0.6863326969706168 | 0.09421275676753152 | 0.04426801156946314 | 0.21454512013655336 | 449 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.01270335624770449 | 0.6910291424865667 | 0.09470027328298813 | 0.044255351077548626 | 0.21260591819870084 | 349 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.012691237186831952 | 0.6916867506007329 | 0.09488969817908166 | 0.04424951200413761 | 0.21234260085620216 | 268 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_25 | 4 | 0.0127430784166448 | 0.6920039375967763 | 0.09546422180199853 | 0.04424137242184743 | 0.21243261774942793 | 512 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.012728048826380524 | 0.6913488226639771 | 0.09515611990754765 | 0.0442438558438341 | 0.21251387798593857 | 449 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.012446300917673707 | 0.6721755497700166 | 0.09041345719112723 | 0.04425822780744642 | 0.21836191280409806 | 268 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012444441177597961 | 0.6737153075389177 | 0.09012936078470013 | 0.04425774305179946 | 0.21806026798556752 | 449 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.012340988975493421 | 0.6808275778920548 | 0.09131799496953276 | 0.04444599391500364 | 0.20519016184939654 | 268 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.012276979143910495 | 0.6770899982886298 | 0.08951816877472173 | 0.0445028527059411 | 0.20556234079305322 | 449 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.012464724344294317 | 0.6671613109395976 | 0.08886677179864028 | 0.04429426687389417 | 0.2183155210466235 | 268 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012415767470778034 | 0.66513228815484 | 0.08876010440541143 | 0.04430614773150306 | 0.2179777192592928 | 449 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012379045916241033 | 0.6800041877686094 | 0.09090581825986176 | 0.04442711862166361 | 0.20542919949313485 | 268 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012269445511921164 | 0.677515449502799 | 0.09006640694547033 | 0.04448566594117593 | 0.20566243139456075 | 449 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.012540168633218838 | 0.6707923676714158 | 0.0902111434936605 | 0.04424173586690827 | 0.23439390917607564 | 268 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.012499372088133821 | 0.6718575478820089 | 0.09007188050891524 | 0.044246306538223076 | 0.2353808712160519 | 449 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.012526124112938108 | 0.6831357496800894 | 0.0911901107124708 | 0.044345999556399215 | 0.20600840080312133 | 268 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012447160366641468 | 0.683113999825516 | 0.09103657636181443 | 0.04438696629782129 | 0.20529494147140048 | 449 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.01334315201124756 | 0.6538151461621756 | 0.0871121679315025 | 0.04425021978480034 | 0.3877218211662241 | 268 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.013318302394871608 | 0.6499621380821046 | 0.08641540568018799 | 0.04425407166546939 | 0.4120778290565292 | 449 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.012797142365002373 | 0.6756404315982159 | 0.09144949404517029 | 0.04423836708337633 | 0.2114339859557281 | 268 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.01273426245791373 | 0.6761751559443715 | 0.0907127786817743 | 0.04426881387774826 | 0.21003884792769892 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012470425273863661 | 0.6879491855489505 | 0.0973401150142671 | 0.04415927283615018 | 0.21435013637421357 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.012452274624702832 | 0.6875638629134799 | 0.09652313808536411 | 0.04417018367725457 | 0.21447089993396537 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.012432446562229534 | 0.6870115456318788 | 0.09676724835798041 | 0.04417264942810916 | 0.21438044227665393 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.01241003828929532 | 0.6862941351187305 | 0.09715500477694805 | 0.04416696280039467 | 0.21440964483358826 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.0126427166875971 | 0.6857568884879232 | 0.09656197687449002 | 0.044157325781569015 | 0.21774402432710663 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.01261417823732638 | 0.684284393259916 | 0.0958239284435355 | 0.04416538514148994 | 0.21811316837905817 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.012447472884576288 | 0.6843935984842389 | 0.09540571650895649 | 0.04419421950299674 | 0.21856010119043146 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.012432924859387578 | 0.6837832298512699 | 0.09595983368376476 | 0.04418468697437315 | 0.21860596016749417 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.012646579816550799 | 0.6860823796211274 | 0.09615699880281091 | 0.04416308812825139 | 0.2178230089002983 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.012683179432802201 | 0.6844335427421361 | 0.09560488563202836 | 0.04416874094070664 | 0.2177813502319237 | 449 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.012502717331456012 | 0.6832945711538392 | 0.09614793269418373 | 0.0441761398015394 | 0.21833583558105207 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.012517508745492501 | 0.6826989045300611 | 0.09531606486067454 | 0.04419338416796654 | 0.21871053324711323 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.012634996764752807 | 0.6736896044882076 | 0.09192299559270257 | 0.04429424562520777 | 0.22798852335373193 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.012731670502255925 | 0.6724578758818454 | 0.09128209306948969 | 0.0443023203644283 | 0.228019257333059 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.011874041438957608 | 0.6745113260233451 | 0.09157405751693465 | 0.04438378366153686 | 0.23180819375940706 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.012072501897225432 | 0.672730670668019 | 0.0914139107500589 | 0.04438413025730721 | 0.23146331370526374 | 449 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.28813991700008046 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.2879390419984702 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.28236816699791234 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.2931685420044232 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.30629800001042895 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 9 | 0.2991626669972902 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.30604945799859706 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.3061620420048712 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 18 | 0.4036173329950543 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 27 | 0.4444061659887666 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 19 | 0.4201182919932762 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 33 | 0.4707199580007 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.33791270799702033 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 12 | 0.3319076670013601 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 14 | 0.3483588749950286 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 13 | 0.3310421670030337 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.227123374992516 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.3010699589940486 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9031977920094505 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.854954791997443 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.584304207994137 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.518288167004357 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 0.9958342079917202 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.9031252920103725 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.4783693329954986 | NA | 50 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.5437483330024406 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9666869170032442 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.8860079589940142 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 2.8647944999975152 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.7684862909954973 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.1737032079981873 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.1348745419963961 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1563008329976583 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.1087922909937333 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.0924322920036502 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.0927217080025002 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.7578012500016484 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.717336832996807 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.7062752500060014 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7042016250052257 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.0994543749984587 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.1057951249968028 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.1618815419933526 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.0931353329942795 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.7007683330011787 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.7035267920000479 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7214934999938123 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.7165250420075608 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.45613229199079797 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.46352662499703 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.4430281660024775 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.461104666988831 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.4884021670004586 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.4882442920061294 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.48917433300812263 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.4859740829997463 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.7463148750102846 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7158947499992792 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 20 | 0.8264586250006687 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 19 | 0.7725725830096053 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 14 | 0.612911415999406 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.5762239159957971 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.5863570829969831 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.5908264169993345 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.327880625001853 | 12.0 | 50 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.232167791007669 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.5265582499996526 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.3699740000010934 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 5.984826542000519 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 5.902787665996584 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.6892529580072733 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.4408867080055643 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 7.020905792000121 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 7.125204790994758 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.0002498749963706 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.8309725839935709 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 9.005961000002571 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 10.666282833000878 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.761577415993088 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.6285003749944735 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.3880202919972362 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.3562667499936651 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.3098557909979718 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.3203817499888828 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 1.9793210829957388 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 1.992613207999966 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 2.0125903750013094 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.9604395000060322 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3293954170076177 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.3318077079893555 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.3203284999908647 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.307515250009601 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 2.107640582995373 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 1.9868032920057885 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.9955787920043804 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.9836252499953844 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.6215392500016605 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.6300887499965029 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6256156249874039 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6323815830110107 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_100 | 3 | 10 | 0.686012167003355 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 10 | 0.7195272089884384 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 10 | 0.6801686250109924 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.7205956660036463 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 18 | 1.129440375007107 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.0907522909983527 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 20 | 1.201684458006639 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 19 | 1.1953665409964742 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 14 | 0.8635145830048714 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 13 | 0.8189129170059459 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.8880003329977626 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 15 | 0.9027210409985855 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 10.121005832988885 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 9.605634958003066 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.2579660409974167 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 1.9921579579968238 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 10.157910042005824 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 10.257637083996087 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.6402041249966715 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.146190250001382 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 14.763284000000567 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 15.106514541999786 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.4993540839932393 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 3.1286035000084667 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 19.416451665994828 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 22.763904583000112 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.124099749999004 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 5.093152792003821 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.5494041249912698 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.5768032090127235 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.5366757090087049 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.606039124992094 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.25062683300348 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.2558217920013703 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.2028133749990957 | 6.0 | NA | NA | 0.03 | 1 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.217307041006279 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.548526624988881 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.557020249994821 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.5313599170040106 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.5340660419897176 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.2439637500065146 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.2731056659977185 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.3092567499988945 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.227735708002001 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 8 | 0.7956240000057733 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 9 | 0.8277651250100462 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 8 | 0.7947835000086343 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 8 | 0.8064713329949882 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 10 | 0.9133765000005951 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 0.9265878330043051 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 0.970590416007326 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 11 | 0.968806416989537 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 19 | 1.7261770839977544 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 15 | 1.3422440419963095 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 21 | 1.8023415419884259 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.6989874159917235 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 15 | 1.2437270829977933 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 14 | 1.2282387500017649 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.2720619589963462 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 16 | 1.2741037079977104 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 17.613753749988973 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 17.98145537500386 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.1967067499936093 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.717115208011819 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 16.82539570800145 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 16.63821900000039 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 3.7500972080015345 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 2.99035270899185 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 27.50703816600435 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 27.64830366600654 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.581874334005988 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 4.546880958005204 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 44.979121167009 | NA | 5 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 45.30104950000532 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 10.397097916997154 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.214889499999117 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.7724002080067294 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.8167698750039563 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.758888249998563 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.7703920000058133 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.5028624999977183 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.5343266670097364 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.4866277080000145 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.4734693329955917 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.7919001250120346 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7813412499963306 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.7478138750011567 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.7533426250010962 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.506977208002354 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.5230331659986405 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.4748939169949153 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.476070874996367 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45360 | 767 | logistic | raw | 45360 | 767 | 0.016909171075837743 | 0.009157974673287434 | 0.7750280671211104 |
| logistic | validation | calibrated | 45360 | 767 | logistic | calibrated | 45360 | 767 | 0.016909171075837743 | 0.02115247240457646 | 0.7742206193399606 |
| rf | validation | raw | 45360 | 767 | rf | raw | 45360 | 767 | 0.016909171075837743 | 0.014359487986722687 | 0.7351604023655234 |
| rf | validation | calibrated | 45360 | 767 | rf | calibrated | 45360 | 767 | 0.016909171075837743 | 0.03573784393588063 | 0.7357126227358198 |
| xgb | validation | raw | 45360 | 767 | xgb | raw | 45360 | 767 | 0.016909171075837743 | 0.011313179145510515 | 0.7646921829365528 |
| xgb | validation | calibrated | 45360 | 767 | xgb | calibrated | 45360 | 767 | 0.016909171075837743 | 0.02884413669072913 | 0.7647484648273706 |
| logistic | test | raw | 49328 | 876 | logistic | raw | 49328 | 876 | 0.017758676613687966 | 0.10499495184809703 | 0.7613768152409559 |
| logistic | test | calibrated | 49328 | 876 | logistic | calibrated | 49328 | 876 | 0.017758676613687966 | 0.13289528639549975 | 0.7573674689859229 |
| rf | test | raw | 49328 | 876 | rf | raw | 49328 | 876 | 0.017758676613687966 | 0.045954201100329266 | 0.7812215624972905 |
| rf | test | calibrated | 49328 | 876 | rf | calibrated | 49328 | 876 | 0.017758676613687966 | 0.11985339796195968 | 0.7688729480233132 |
| xgb | test | raw | 49328 | 876 | xgb | raw | 49328 | 876 | 0.017758676613687966 | 0.07471619877033242 | 0.7800545528842366 |

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 49328 | 876 | xgb | calibrated | 49328 | 876 | 0.017758676613687966 | 0.10168951971828737 | 0.7776104590826037 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.062679567310709 | 0.016373449884055563 | 0.07981257477827791 |
| logistic | 0.05909657271923585 | 0.016255271580265605 | 0.07761688275161717 |
| rf | 0.05011018038129096 | 0.01640343135933741 | 0.0799899924485693 |
| rf | 0.04689860718054216 | 0.016869324404119354 | 0.08611128809746466 |
| xgb | 0.05187796808368992 | 0.01638192731239651 | 0.07962089008136297 |
| xgb | 0.049093151746911 | 0.016509826977045082 | 0.08157630621152577 |
| logistic | 0.05583613742033927 | 0.030295309541858006 | 0.14541117610904195 |
| logistic | 0.047882684676502016 | 0.03380781747600887 | 0.17223225073571444 |
| rf | 0.05678760022460123 | 0.0179028033317105 | 0.09555657646426809 |
| rf | 0.04526106755914953 | 0.029209908063707848 | 0.15929796322849718 |
| xgb | 0.055935868806985034 | 0.02288119704316084 | 0.11535981352245538 |
| xgb | 0.05249373001109393 | 0.030175373729718104 | 0.34287277437558106 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45360 | 767 | logistic | raw | 0.8421894495866397 | 1.049438131887896 | 0.0077511964025503075 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 45360 | 767 | logistic | calibrated | 0.11459186758442509 | 1.1033363221124115 | 0.004265666689268328 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 45360 | 767 | rf | raw | 0.9768292655039192 | 1.2059054288723865 | 0.002549683089115056 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 45360 | 767 | rf | calibrated | -0.3276480088275166 | 1.1524777007192724 | 0.018872764570801264 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 45360 | 767 | xgb | raw | -0.1603030061140078 | 0.8535945356393047 | 0.005903501398660133 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 45360 | 767 | xgb | calibrated | 0.35463824558649293 | 1.2878289761361983 | 0.011934965614891387 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 49328 | 876 | logistic | raw | -2.018714077506166 | 0.9743701326761888 | 0.08723627523440906 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 49328 | 876 | logistic | calibrated | -1.4704244062787208 | 1.470734174416546 | 0.11513660978181178 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 49328 | 876 | rf | raw | 4.493038652574499 | 2.931419886814198 | 0.028195524486641296 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| rf | test | calibrated | 49328 | 876 | rf | calibrated | -1.231029339965776 | 1.4692591879244847 | 0.10209472134827169 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 49328 | 876 | xgb | raw | -1.412654848205772 | 1.0758595167214302 | 0.05695752215664446 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 49328 | 876 | xgb | calibrated | -3.697958805868478 | 0.15984782572977332 | 0.08393084310459942 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R1_candidate_c/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 251 | 0.005088387933830684 | 26 | 225 | 48227 | 850 |
| logistic | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 0 | 0.0 | 0 | 0 | 48452 | 876 |
| logistic | 1:1 | 1 | 1 | 0.24637681159420294 | 49328 | 876 | 4949 | 0.1003284138825819 | 315 | 4634 | 43818 | 561 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 1621 | 0.032861660720077845 | 135 | 1486 | 46966 | 741 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 0 | 0.0 | 0 | 0 | 48452 | 876 |
| logistic | 2:1 | 1 | 2 | 0.2463768115942029 | 49328 | 876 | 8152 | 0.1652611093091145 | 423 | 7729 | 40723 | 453 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 9526 | 0.19311547194291276 | 462 | 9064 | 39388 | 414 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 11455 | 0.23222105092442427 | 527 | 10928 | 37524 | 349 |
| logistic | 5:1 | 1 | 5 | 0.2463768115942029 | 49328 | 876 | 8152 | 0.1652611093091145 | 423 | 7729 | 40723 | 453 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 20980 | 0.42531625040544924 | 691 | 20289 | 28163 | 185 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 34772 | 0.7049140447615958 | 828 | 33944 | 14508 | 48 |
| logistic | 10:1 | 1 | 10 | 0.09919261822376009 | 49328 | 876 | 33913 | 0.6875 | 822 | 33091 | 15361 | 54 |
| rf | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 0 | 0.0 | 0 | 0 | 48452 | 876 |
| rf | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 350 | 0.007095361660720078 | 29 | 321 | 48131 | 847 |
| rf | 1:1 | 1 | 1 | 0.3333333333333333 | 49328 | 876 | 725 | 0.014697534868634447 | 52 | 673 | 47779 | 824 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 0 | 0.0 | 0 | 0 | 48452 | 876 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 725 | 0.014697534868634447 | 52 | 673 | 47779 | 824 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 725 | 0.014697534868634447 | 52 | 673 | 47779 | 824 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 0 | 0.0 | 0 | 0 | 48452 | 876 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 10403 | 0.2108944210184885 | 526 | 9877 | 38575 | 350 |
| rf | 5:1 | 1 | 5 | 0.3333333333333333 | 49328 | 876 | 725 | 0.014697534868634447 | 52 | 673 | 47779 | 824 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 636 | 0.01289328576062277 | 50 | 586 | 47866 | 826 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 29266 | 0.5932938696075252 | 817 | 28449 | 20003 | 59 |
| rf | 10:1 | 1 | 10 | 0.1687116564417178 | 49328 | 876 | 10403 | 0.2108944210184885 | 526 | 9877 | 38575 | 350 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 50 | 0.00101362309438858 26 | 4 | 46 | 48406 | 872 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49328 | 876 | 442 | 0.00896042815439506 9 | 40 | 402 | 48050 | 836 |
| xgb | 1:1 | 1 | 1 | 0.2272727272727273 | 49328 | 876 | 474 | 0.00960914693480376 3 | 41 | 433 | 48019 | 835 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 487 | 0.00987268893934479 4 | 41 | 446 | 48006 | 835 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49328 | 876 | 462 | 0.00936587739215050 3 | 41 | 421 | 48031 | 835 |
| xgb | 2:1 | 1 | 2 | 0.2272727272727273 | 49328 | 876 | 474 | 0.00960914693480376 3 | 41 | 433 | 48019 | 835 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 4747 | 0.09623337658125203 | 300 | 4447 | 44005 | 576 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49328 | 876 | 6139 | 0.12445264352903017 | 353 | 5786 | 42666 | 523 |
| xgb | 5:1 | 1 | 5 | 0.22727272727272727 | 49328 | 876 | 1965 | 0.03983538760947129 | 157 | 1808 | 46644 | 719 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 14008 | 0.2839766461239053 | 597 | 13411 | 35041 | 279 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49328 | 876 | 29189 | 0.5917328900421667 | 812 | 28377 | 20075 | 64 |
| xgb | 10:1 | 1 | 10 | 0.12745098039215685 | 49328 | 876 | 8145 | 0.16511920207590008 | 437 | 7708 | 40744 | 439 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.02968036529680365 | 0.9953562288450425 | 0.10358565737051793 | 0.9826802779305989 | 0.00464377115495748 4 | 0.9703196347031964 | 1075 | 0.02179289652935452 5 | 876 |
| logistic | 0.0 | 1.0 | NA | 0.982241323386312 | 0.0 | 1.0 | 876 | 0.01775867661368796 6 | 876 |
| logistic | 0.3595890410958904 | 0.9043589531907867 | 0.06364922206506365 | 0.9873588859595754 | 0.09564104680921325 | 0.6404109589041096 | 5195 | 0.10531543950697372 | 876 |
| logistic | 0.1541095890410959 | 0.9693304713943697 | 0.08328192473781616 | 0.9844676881799317 | 0.03066952860563031 5 | 0.8458904109589042 | 2968 | 0.06016866688290626 | 1752 |
| logistic | 0.0 | 1.0 | NA | 0.982241323386312 | 0.0 | 1.0 | 1752 | 0.03551735322737593 | 1752 |
| logistic | 0.4828767123287671 | 0.8404813010814827 | 0.05188910696761531 | 0.9889984456965223 | 0.1595186989185173 | 0.5171232876712328 | 8635 | 0.1750527084009082 | 1752 |
| logistic | 0.5273972602739726 | 0.8129282588954017 | 0.04849884526558892 | 0.9895985126375559 | 0.18707174110459837 | 0.4726027397260274 | 11134 | 0.22571359065844956 | 4380 |
| logistic | 0.6015981735159818 | 0.7744571947494427 | 0.04600611086861633 | 0.9907849919467695 | 0.22554280525055725 | 0.3984018264840183 | 12673 | 0.25691290950373014 | 4380 |
| logistic | 0.4828767123287671 | 0.8404813010814827 | 0.05188910696761531 | 0.9889984456965223 | 0.1595186989185173 | 0.5171232876712328 | 9994 | 0.20260298410638988 | 4380 |
| logistic | 0.7888127853881278 | 0.5812556757203005 | 0.03293612964728312 4 | 0.993473966417384 | 0.4187443242796995 | 0.21118721461187215 | 22139 | 0.4488120337333766 | 8760 |
| logistic | 0.9452054794520548 | 0.2994303640716586 | 0.02381226274013574 | 0.9967023907666942 | 0.7005696359283414 | 0.0547945205479452 | 34424 | 0.6978592280246513 | 8760 |
| logistic | 0.9383561643835616 | 0.31703541649467515 | 0.02423849261345206 7 | 0.996496918585793 | 0.6829645835053249 | 0.06164383561643835 | 33631 | 0.6817831657476484 | 8760 |
| rf | 0.0 | 1.0 | NA | 0.982241323386312 | 0.0 | 1.0 | 876 | 0.01775867661368796 6 | 876 |
| rf | 0.033105022831050226 | 0.993374886485594 | 0.08285714285714285 | 0.9827065212952755 | 0.00662511351440601 | 0.9668949771689498 | 1168 | 0.02367823548491729 | 876 |
| rf | 0.0593607305936073 | 0.9861099645009493 | 0.07172413793103448 | 0.9830463140135383 | 0.01389003549905060 7 | 0.9406392694063926 | 1497 | 0.03034787544599416 | 876 |
| rf | 0.0 | 1.0 | NA | 0.982241323386312 | 0.0 | 1.0 | 1752 | 0.03551735322737593 | 1752 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| rf | 0.0593607305936073 | 0.9861099645009493 | 0.07172413793103448 | 0.9830463140135383 | 0.013890035499050607 | 0.9406392694063926 | 2321 | 0.047052384041518 | 1752 |
| rf | 0.0593607305936073 | 0.9861099645009493 | 0.07172413793103448 | 0.9830463140135383 | 0.013890035499050607 | 0.9406392694063926 | 2321 | 0.047052384041518 | 1752 |
| rf | 0.0 | 1.0 | NA | 0.982241323386312 | 0.0 | 1.0 | 4380 | 0.08879338306843983 | 4380 |
| rf | 0.6004566210045662 | 0.7961487657888219 | 0.050562337787176774 | 0.9910083493898523 | 0.20385123421117807 | 0.3995433789954338 | 11627 | 0.23570791436912097 | 4380 |
| rf | 0.0593607305936073 | 0.9861099645009493 | 0.07172413793103448 | 0.9830463140135383 | 0.013890035499050607 | 0.9406392694063926 | 4793 | 0.09716590982808952 | 4380 |
| rf | 0.05707762557077625 | 0.9879055560141996 | 0.07861635220125786 | 0.9830362277170788 | 0.01209444398580038 | 0.9429223744292238 | 8846 | 0.17933019785922802 | 8760 |
| rf | 0.932648401826484 | 0.41284157516717573 | 0.027916353447686736 | 0.9970591167381119 | 0.5871584248328242 | 0.06735159817351598 | 29039 | 0.588692020759001 | 8760 |
| rf | 0.6004566210045662 | 0.7961487657888219 | 0.050562337787176774 | 0.9910083493898523 | 0.20385123421117807 | 0.3995433789954338 | 13377 | 0.27118472267272137 | 8760 |
| xgb | 0.0045662100456621 | 0.9990506067860976 | 0.08 | 0.9823044766427209 | 0.000949393213902418 9 | 0.9954337899543378 | 918 | 0.018610120012974376 | 876 |
| xgb | 0.045662100456621 | 0.9917031288698093 | 0.09049773755656108 | 0.9828989894857424 | 0.008296871130190705 | 0.954337899543379 | 1238 | 0.025097307817061303 | 876 |
| xgb | 0.04680365296803653 | 0.9910633203995707 | 0.08649789029535865 | 0.9829082572563147 | 0.00893667960042929 | 0.9531963470319634 | 1268 | 0.025705481673694453 | 876 |
| xgb | 0.04680365296803653 | 0.9907950136217287 | 0.08418891170431211 | 0.9829037079502877 | 0.009204986378272279 | 0.9531963470319634 | 2116 | 0.042896529354524815 | 1752 |
| xgb | 0.04680365296803653 | 0.9913109881945018 | 0.08874458874458875 | 0.9829124544673188 | 0.008689011805498225 | 0.9531963470319634 | 2091 | 0.04238971780733052 | 1752 |
| xgb | 0.04680365296803653 | 0.9910633203995707 | 0.08649789029535865 | 0.9829082572563147 | 0.00893667960042929 | 0.9531963470319634 | 2103 | 0.04263298734998378 | 1752 |
| xgb | 0.3424657534246575 | 0.9082184429951292 | 0.06319780914261638 | 0.9870796976290348 | 0.0917815570048708 | 0.6575342465753424 | 7327 | 0.14853632825170288 | 4380 |
| xgb | 0.4029680365296804 | 0.8805828448774045 | 0.05750122169734485 | 0.987890435064484 | 0.11941715512259556 | 0.5970319634703196 | 8401 | 0.17030895231916965 | 4380 |
| xgb | 0.17922374429223745 | 0.9626847188970528 | 0.07989821882951655 | 0.9848193737727762 | 0.03731528110294725 | 0.8207762557077626 | 5403 | 0.10953211157963023 | 4380 |
| xgb | 0.6815068493150684 | 0.7232106001816231 | 0.042618503712164475 | 0.9921007927519819 | 0.27678939981837697 | 0.3184931506849315 | 16201 | 0.32843415504378853 | 8760 |
| xgb | 0.9269406392694064 | 0.41432758193676217 | 0.02781869882489979 | 0.9968220864988331 | 0.5856724180632379 | 0.0730593607305936 | 29017 | 0.58824602659747 | 8760 |
| xgb | 0.4988584474885845 | 0.8409147197226121 | 0.05365254757519951 | 0.9893402617584925 | 0.15908528027738794 | 0.5011415525114156 | 12098 | 0.24525624391826142 | 8760 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 1.2271689497716896 |
| logistic | 1.0 |
| logistic | 5.930365296803653 |
| logistic | 1.6940639269406392 |
| logistic | 1.0 |
| logistic | 4.92865296803653 |
| logistic | 2.5420091324200915 |
| logistic | 2.89337899543379 |
| logistic | 2.2817351598173516 |
| logistic | 2.527283105022831 |
| logistic | 3.9296803652968038 |
| logistic | 3.8391552511415523 |
| rf | 1.0 |
| rf | 1.3333333333333333 |

| model | decision_relative_cost_vs_all_negative |
|---|---|
| rf | 1.7089041095890412 |
| rf | 1.0 |
| rf | 1.3247716894977168 |
| rf | 1.3247716894977168 |
| rf | 1.0 |
| rf | 2.654566210045662 |
| rf | 1.0942922374429225 |
| rf | 1.0098173515981734 |
| rf | 3.314954337899543 |
| rf | 1.527054794520548 |
| xgb | 1.047945205479452 |
| xgb | 1.41324200913242 |
| xgb | 1.447488584474886 |
| xgb | 1.2077625570776256 |
| xgb | 1.1934931506849316 |
| xgb | 1.2003424657534247 |
| xgb | 1.6728310502283106 |
| xgb | 1.9180365296803652 |
| xgb | 1.2335616438356165 |
| xgb | 1.8494292237442922 |
| xgb | 3.312442922374429 |
| xgb | 1.3810502283105024 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 49328 | 876 | 0 | 0 | 48452 | 876 | 876 | 0.017758676613687966 |
| test | all_non_risky | 2:1 | 2:1 | 49328 | 876 | 0 | 0 | 48452 | 876 | 1752 | 0.03551735322737593 |
| test | all_non_risky | 5:1 | 5:1 | 49328 | 876 | 0 | 0 | 48452 | 876 | 4380 | 0.08879338306843983 |
| test | all_non_risky | 10:1 | 10:1 | 49328 | 876 | 0 | 0 | 48452 | 876 | 8760 | 0.17758676613687965 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 37048 | 328 | 0.00885337939969769 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 5314 | 332 | 0.0624764772299586 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 825 | 21 | 0.025454545454545455 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 6141 | 195 | 0.03175378602833415 |

R2

Distress composite: 03–99 or RA or first effective-date Zero Balance Code 02/03/09 in scheduled-payment months 1–24; Codes 15/16/96 are not recoded as events.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.019007166842078885 | 239436 | 4551 |
| validation | 0.01699585574464333 | 45364 | 771 |
| test | 0.017957760752361263 | 49338 | 886 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_200","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47222 | 253 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95253 | 583 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143464 | 980 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190857 | 2259 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.006318342558882794 | 0.8173971322626487 | 0.029046801011317444 | 0.006756842689366335 | 0.037005284473667764 | 199 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.006309966573420216 | 0.8173423084326421 | 0.029165744112405297 | 0.006756657513018129 | 0.03699926369627041 | 137 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.006322462289318933 | 0.8162655252129268 | 0.02875842968260646 | 0.006758052419344768 | 0.03703560607778498 | 341 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.006321564867707471 | 0.8165080714272555 | 0.02880057499917637 | 0.006757936864076082 | 0.03703115637183833 | 266 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.004897860415936282 | 0.8454934875261495 | 0.04201680383838004 | 0.006724656443636665 5 | 0.03556947984980361 | 199 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.004894846547004032 | 0.8451574295183444 | 0.04147742814854968 | 0.006727078263517339 | 0.03559830591304373 | 137 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.004897498903562639 | 0.8449894640414756 | 0.041786981537839774 | 0.006726422381713363 | 0.0355901378983719 | 341 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.004898210355627273 | 0.8450305660322222 | 0.04189941348302613 | 0.006726010130591443 | 0.03558652244413021 | 266 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.004706272932841975 | 0.7979613539643727 | 0.038491989216477805 | 0.006742375711987206 | 0.03719455088345293 | 199 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.004634178079426322 | 0.8112048346613658 | 0.03995412572719714 | 0.006728630135958122 | 0.03665745673262529 | 137 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.004669817368252381 | 0.7917329730079988 | 0.037641625454845035 | 0.006756950311611644 | 0.03762838503328475 | 341 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.0046911087876500696 | 0.7968125310885422 | 0.038846383145183114 | 0.006750684512558426 | 0.03733936497859071 | 266 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.004674488347287632 | 0.8350377001180967 | 0.042917018133162434 | 0.006718982947864474 5 | 0.0358032598302772 | 199 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.004649534720290821 | 0.8370106592009697 | 0.04334990876767108 | 0.006718422305813591 | 0.03573850649130979 | 137 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.004665661436013696 | 0.8341724619203079 | 0.0428951812233841 | 0.00672163177423222 | 0.03584184228551954 | 341 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.004670447689939853 | 0.8346905884064434 | 0.04309144132776109 | 0.006720581870758887 | 0.03580908191143715 | 266 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004928390280555393 5 | 0.8388427153232922 | 0.03388325774860083 | 0.00673011645087964 | 0.03579495749552005 | 137 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.00490480276705361 | 0.8397435286599035 | 0.034041570392897506 | 0.006728853810451082 | 0.03574396031398468 | 266 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004829776075705375 | 0.8360053438940674 | 0.036450033112239405 | 0.006745231499360201 | 0.03637104309715455 | 137 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004873261830852322 | 0.8356506089383807 | 0.03663473300320271 | 0.006753307493939384 | 0.036690626516462335 | 266 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.005073324263374916 | 0.8204285787795568 | 0.03058547893020907 | 0.006747446756350591 | 0.0366301150872763 | 137 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.004992996042717035 | 0.8213475290842642 | 0.029336787445856962 | 0.006756391566362875 | 0.03669807964578549 | 266 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.004906325888436626 | 0.8275044738913423 | 0.03436144480213313 | 0.006740564696043241 | 0.03638530319988458 | 137 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.004930524838586199 | 0.82489001882306 | 0.035231527805835765 | 0.006750006551445144 | 0.03677224751303924 | 266 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.004927668963763375 | 0.8355905758916178 | 0.035111285879218045 | 0.006730387453215886 | 0.036098192937940636 | 137 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004932749109170226 | 0.8347974091134612 | 0.03559824929676638 | 0.006728847138636165 5 | 0.03664206127031461 | 266 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.004804977720299901 | 0.8392628195965651 | 0.035972666448112034 | 0.006740627617100972 | 0.03612696943955571 | 137 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004854499620304401 | 0.8386872646720449 | 0.03632969779558512 | 0.006745927256163462 | 0.03630168588478597 | 266 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.005393994558691035 | 0.7937711425908738 | 0.027640192240269635 | 0.006770871185270719 | 0.05147717453358178 | 137 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.005294184359515995 | 0.781332803517873 | 0.024671591552059 | 0.006796645763602439 6 | 0.056588981651448884 | 266 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.004958977472104601 | 0.8139634008053958 | 0.030900622665030367 | 0.006740105527009919 | 0.03667541746384808 | 137 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.004910019859659371 | 0.8115160535990288 | 0.03173745838068897 | 0.00674142295900632 | 0.03671201784012487 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.004703017622230171 | 0.832728492446318 | 0.03869934764956622 | 0.006718200421322506 | 0.03591666160111897 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.004721345596011579 | 0.8327054003696003 | 0.039275769849512164 | 0.006716342799260058 | 0.03589540771821688 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.004625470839012178 | 0.8364672489554568 | 0.04179194058002409 | 0.006715075405165929 | 0.035885842223230206 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004674466887329185 | 0.8364888799104014 | 0.04320517905901893 | 0.006713028989967742 | 0.03589051084617846 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.004626038305364283 | 0.816895713386353 | 0.03411348527204489 | 0.006749001790565008 | 0.03689615125953689 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.0046909166802573165 | 0.8183923467712066 | 0.034472629750802355 | 0.006746243420012278 | 0.036821971486127945 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.0038538337981459583 | 0.8120353235717694 | 0.03443337488819489 | 0.006759532987619157 | 0.038108555902231514 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.0040397103587177605 | 0.8131423456594836 | 0.035276181361004084 | 0.006756954407085635 | 0.03795309268135113 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004643506047684224 | 0.8027833416871382 | 0.03148504483344643 | 0.006773609416178627 | 0.037755882208974856 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.004704860707556827 | 0.8086493962073089 | 0.032520452351138415 | 0.006767316656991881 | 0.03747325336187948 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004338055383431145 | 0.8043536664309814 | 0.032565740810675606 | 0.006792042148760643 | 0.03839477666662266 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.00441559000697833 | 0.8062307314566176 | 0.03489914031283354 | 0.00678065202971136 | 0.038284589915838045 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004218193244238968 | 0.764532920661723 | 0.02488411728063133 | 0.006867447472093497 | 0.04104627457769328 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.004309467788617718 | 0.7738287997265797 | 0.024519826047451006 | 0.006882387273838144 | 0.04063004790669243 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.0017347934869435585 | 0.7602501186367353 | 0.02687081511154962 | 0.006872873968198459 | 0.04956640680769957 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.0019499866479313183 | 0.7659822264065362 | 0.026814909983224368 | 0.006855205128995387 | 0.04831821717139848 | 266 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.008586217023551694 | 0.8257641728616946 | 0.04476380621572951 | 0.008016832119450047 | 0.04186054979625317 | 263 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.008571585879659088 | 0.8260014483074053 | 0.04470155765583941 | 0.008016649082051413 | 0.04185606341878155 | 200 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.008596555007114554 | 0.8259453956710296 | 0.045172223181837744 | 0.008015668086119877 | 0.041842593287905946 | 443 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.008595650247985457 | 0.8259549309409394 | 0.04513987595438914 | 0.008015793513528532 | 0.041843739581877995 | 347 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_100 | 2 | 0.007804367230494871 | 0.8429729117205745 | 0.0540844734474983 | 0.007980041327429604 | 0.0406684729306493 | 263 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.007781495941497294 | 0.8434043168326804 | 0.05429782092604411 | 0.007978556467150842 | 0.04065336328859704 | 200 |
| logistic | lr_c_0p1_freq_25 | 2 | 0.007814324988668546 | 0.8429035307787448 | 0.05449726135961752 | 0.007977832474093208 | 0.04065225984929011 | 443 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.007812478618410595 | 0.8427527576158622 | 0.0546316745041734 | 0.007977049238025915 | 0.04064821824651498 | 347 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.007648038405121716 | 0.8120748441773585 | 0.04951189689842957 | 0.00802153286696868 | 0.042011318054786105 | 263 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.007584202790806142 | 0.8227176804660461 | 0.0520836112068966 | 0.008000424122716308 | 0.041403415030688374 | 200 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.007742917258603424 | 0.8020846207264737 | 0.04489061661829675 | 0.00806134155089192 | 0.04269669993192109 | 443 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.007647059147000239 | 0.8055417619008336 | 0.04885719257348827 | 0.008028081045613842 | 0.04234265646302438 | 347 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.007659806969550023 | 0.8344109241952363 | 0.0514324038578007 | 0.007998923126713664 | 0.0410335799923188 | 263 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.007613428800741573 | 0.8368973116755218 | 0.05446856611272506 | 0.00798930094157591 | 0.0408705917923368 | 200 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.007686189159139084 | 0.832845928265901 | 0.05193800124815127 | 0.007998764623008386 | 0.04110645706114105 | 443 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.007661582279649353 | 0.8331494237904881 | 0.05234992803217991 | 0.007995310336931072 | 0.041058772284606984 | 347 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007521140943754727 | 0.8349195597255065 | 0.05033913194172224 | 0.007983573604384564 | 0.04089382552633625 | 200 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007501627095293543 | 0.8347636765008489 | 0.05023115916015713 | 0.007985774283739804 | 0.040934744653538566 | 347 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.007293296211927023 | 0.8370666285677317 | 0.05107482030985584 | 0.008033452961114686 | 0.04182142598228577 | 200 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.007155100515401507 | 0.8336121741268827 | 0.05204970504774525 | 0.00805023754497037 | 0.04234406203200904 | 347 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.00786980364219958 | 0.8266514797737959 | 0.04315317049740202 | 0.00802907460986931 | 0.04158241761910365 | 200 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.007764047278979788 | 0.826893549194986 | 0.04310768133742621 | 0.008027510195545744 | 0.041586140831119126 | 347 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007436203432824745 | 0.8318297108263455 | 0.04829317410478615 | 0.008018917353814178 | 0.041690525373740184 | 200 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007258044341844161 4 | 0.8284775629830917 | 0.0460016450728624 | 0.008039929107181008 | 0.04224385626289182 | 347 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007585016597815727 5 | 0.8334186766330782 | 0.049098560980154915 | 0.00798649452222243 | 0.04101753275293259 | 200 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.00761222346518 7026 | 0.828676249560245 | 0.049511128576787924 | 0.007988304815387414 | 0.04176281484846368 | 347 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007523734377466839 | 0.8382543755035649 | 0.05246292276043308 | 0.008017961028469205 | 0.04132754424554966 | 200 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.007415769756886535 | 0.8370412889830544 | 0.05284208761504318 | 0.00803382000906513 | 0.04170480994206177 | 347 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.00865341618693673 | 0.7949978079415417 | 0.03760401852565671 | 0.008109674744404277 | 0.05805477543054612 | 200 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.008588107489248348 | 0.7935188665061159 | 0.035905623274016606 | 0.00812416255664851 | 0.058331678521100896 | 347 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.007951803120830051 | 0.8192189739438478 | 0.04559453163631914 | 0.008008649614964648 | 0.041624969716953536 | 200 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.007859103974471218 | 0.8221410863822753 | 0.04680572818131783 | 0.008009490491119818 | 0.04147906412131706 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.007341337248081294 | 0.8375017213532834 | 0.04716793123740348 | 0.008003548683890202 | 0.04098817267080135 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.007376381762338008 | 0.8380429137719747 | 0.04718296073872326 | 0.007999422589073095 | 0.04093104739053545 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.007332119108587533 | 0.837731252684758 | 0.04893905686791443 | 0.007991607870976303 | 0.04087097194662816 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.007381107691405915 | 0.8374974541882961 | 0.050461261674209196 | 0.007985299984197879 | 0.0408225275784946 | 347 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.007202213916058627 | 0.8272110842191915 | 0.039299555941678346 | 0.008064577529116554 | 0.04190118914067396 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.007224822354234798 | 0.827474831892138 | 0.039804209630322655 | 0.008060618091757078 | 0.04190238303133722 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006849883142817557 | 0.821914610551656 | 0.03794287223166187 | 0.008095225879436316 | 0.042410943240731 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.006843608584375001 | 0.8215357021051031 | 0.03683742623167975 | 0.00809570727680185 | 0.04244160409045476 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.00718163393652039 | 0.8200445123257324 | 0.03844169345205684 | 0.008088196840109887 | 0.04243348359304898 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.00722016197776072 | 0.8238779278941836 | 0.0368705023941454 | 0.008102945288157682 | 0.04234011975318695 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.007142442017056046 | 0.8177689807449712 | 0.037055117870308234 | 0.008124595645467532 | 0.04275575447058839 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.007149574502919164 | 0.8187537739386639 | 0.037818230542656164 | 0.00812501792423627 | 0.042655663996518586 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.006624960600716583 | 0.7944199495125898 | 0.02989176242105524 | 0.008234282744228012 | 0.045310389962344766 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006753691786391992 | 0.7953855668043643 | 0.030815510581769585 | 0.008259180730389296 | 0.045317210483733154 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.004805747392022073 | 0.7751095265353919 | 0.025129670515313108 | 0.008344809532145243 | 0.049454305037322156 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.004860276727667581 | 0.7828445269499916 | 0.0256233210250981 | 0.008330323796597984 | 0.04884500256892173 | 347 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.011225935070216294 | 0.7426378106431887 | 0.07816171370923716 | 0.02601523626481025 | 0.12221087703771483 | 311 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.011215008898224216 | 0.7427329618547744 | 0.07812431346640164 | 0.026016227887132287 | 0.12222402233088545 | 242 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.011228860940767726 | 0.7426307912915142 | 0.07816519318230329 | 0.026014936473018235 | 0.12220664487901957 | 489 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.011228153961763599 | 0.7426186345882522 | 0.07814381904777354 | 0.026015243398655012 | 0.12221048561245236 | 411 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.010907529597960947 | 0.732706377501479 | 0.07392807998494884 | 0.02604875423342674 | 0.12563077064366013 | 311 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.010913075492823695 | 0.7326955941496315 | 0.07433205044706018 | 0.026046695953719588 | 0.12561878312880584 | 242 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.010909889969036598 | 0.7329583450986596 | 0.07422151944183042 | 0.026044228628157754 | 0.1255823673653872 | 489 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.010905546864916363 | 0.7328970698886327 | 0.07404009318123361 | 0.026046908065749018 | 0.12560728222777912 | 411 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.01073766448396805 | 0.7081961578510448 | 0.06826328608498217 | 0.02612856080505582 | 0.1308645236710714 | 311 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.010771455629834292 | 0.71339515765786 | 0.06960630912104354 | 0.026113285675139992 | 0.12990417443044827 | 242 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.010838261571133358 | 0.7063453040181245 | 0.06772789767480604 | 0.02613772070219934 | 0.13157538505157265 | 489 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.010801672124558197 | 0.7068637696095508 | 0.06803334191935557 | 0.026130552295103547 | 0.13129517669015392 | 411 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.010814219439738634 | 0.7237833233971641 | 0.07092925575806415 | 0.026094636042280246 | 0.12797262592059658 | 311 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.010834430765821194 | 0.7242219481020336 | 0.0713462113991885 | 0.02608954411855059 | 0.12783587338417762 | 242 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.010848345752380643 | 0.7239972440736749 | 0.0713025863197219 | 0.02608751342858121 | 0.12796897778106434 | 489 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.010828902851561786 | 0.7239891226498778 | 0.07108902018786212 | 0.026091803973849782 | 0.12797659802502734 | 411 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010436119267568723 | 0.7174923396662239 | 0.06790611073017247 | 0.026111322915608057 | 0.12873783338320632 | 242 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.010421589779413272 | 0.7198169743047307 | 0.06876013811519276 | 0.026101592979544584 | 0.1285186660259266 | 411 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.010036131938768105 | 0.7066728398530168 | 0.06514761995780254 | 0.02628331655387841 | 0.1267496547764538 | 242 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.00968040435507195 | 0.7027963944133693 | 0.06396361160124192 | 0.02632544447732484 | 0.12771390207342395 | 411 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011357352678046585 | 0.7084508636735767 | 0.06555832232016008 | 0.026134713212609274 | 0.13049746795281025 | 242 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.01079541189410875 | 0.7065208386748509 | 0.06537937556191714 | 0.02616805943135036 | 0.13189199586191722 | 411 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.01036874346974837 | 0.7049204773579621 | 0.06424735674182776 | 0.02623449115730116 | 0.1264371183949427 | 242 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.010027060355271365 | 0.7016257394946331 | 0.06384713124468126 | 0.0262878322098883 | 0.12720745390835198 | 411 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.010810717260099858 | 0.7156566266087753 | 0.06890429831935654 | 0.026081033084122147 | 0.13030132174323786 | 242 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.010765968954288628 | 0.7171328606947266 | 0.07007282535960066 | 0.02607181061142716 | 0.12994081473808788 | 411 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010898030634430182 | 0.7123294708700805 | 0.06670988977833454 | 0.026204183522345867 | 0.12484539443046345 | 242 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010755180232593827 | 0.7098193914032204 | 0.06709074165055288 | 0.026236615891030452 | 0.12521788446144153 | 411 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.013913817599341417 | 0.69694470510805 | 0.06580759291742236 | 0.026053606959448885 | 0.17945463232362405 | 242 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.013331396307908151 | 0.6964177366063224 | 0.06580240055452771 | 0.02607516798301942 | 0.19031875005859591 | 411 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.01253158983052011 | 0.7078910703775457 | 0.06751020781139883 | 0.026052085196550667 | 0.12395762823658127 | 242 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.012087863141406904 | 0.7084269487085123 | 0.06768341188942752 | 0.02608168129576933 | 0.12407940720025878 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.010343314097617516 | 0.7233629676570994 | 0.07054249757522692 | 0.0260890865394367 | 0.12872829983706277 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.010267833175327196 | 0.7249338019185753 | 0.07002539889636568 | 0.026102439454652206 | 0.12860628494725257 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.010090746659836464 | 0.7242546626891244 | 0.07156807731154723 | 0.026089392903033953 | 0.12898233663810948 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.009951059034308608 | 0.724123694133548 | 0.07111923402582437 | 0.026105746885383994 | 0.12931472738072355 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.010974664464398425 | 0.7159271683599637 | 0.06881785926691474 | 0.026099145392038982 | 0.1300505228062551 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.010681128746944218 | 0.7152117370477618 | 0.06867541200807664 | 0.026115219706824882 | 0.13057754980541003 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.01040477253596848 | 0.7157454960092612 | 0.06790052485869005 | 0.026153358832002645 | 0.1323258529826662 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.010385014755347755 | 0.7163770681103969 | 0.06833396264835007 | 0.026139795819975244 | 0.13206067347931305 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.010818172755857123 | 0.7131478781059402 | 0.06784720547591479 | 0.026148452539726086 | 0.13160383322903255 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.010790865034835765 | 0.7142089022808925 | 0.06871081784169653 | 0.026120637179709322 | 0.13094273219852015 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.010721610867427697 | 0.7136861979505325 | 0.06732866561647499 | 0.026177544798259355 | 0.13219874833527362 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.010697172324160372 | 0.7129176043746228 | 0.06841070935044602 | 0.026157869692486718 | 0.13291843776586412 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.010717528377088724 | 0.6929894869440567 | 0.06153048416133393 | 0.02638582267251573 | 0.1401808941079742 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.010687241283063085 | 0.697382397290354 | 0.06342424186824763 | 0.026326693720780985 | 0.13909049726221 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008766947909560308 | 0.6913108785064501 | 0.05827435995109351 | 0.026572191246754 | 0.1489712338080922 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.008690387074613554 | 0.6927324582925891 | 0.05816142607084475 | 0.02657475558437225 | 0.14802447518544656 | 411 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.013618717788701162 | 0.6968023430413728 | 0.10034168677127644 | 0.04529831475930565 | 0.21059971785255674 | 349 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.01361429644326178 | 0.6968435534106558 | 0.10034825332346742 | 0.045298700114983215 | 0.2106033122180637 | 268 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_25 | 4 | 0.013624334297617884 | 0.6968995908409822 | 0.10039379631112233 | 0.04529701618982174 | 0.2105696536856206 | 512 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.013624385970866789 | 0.6968789573784112 | 0.10038013949164484 | 0.04529729705545222 | 0.21057324812957345 | 449 |
| logistic | lr_c_0p1_freq_100 | 4 | 0.013245376267425158 | 0.696156305463624 | 0.10001015893367438 | 0.045349760655795206 | 0.21435536932697186 | 349 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.013246286083727615 | 0.6962005038674594 | 0.10005297680708747 | 0.04534946943508767 | 0.21433687969668733 | 268 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.01325622248370586 | 0.6965621597340306 | 0.10029309033109238 | 0.045344024176478974 | 0.21425151450412735 | 512 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.013255045151798862 | 0.6963718115644741 | 0.10015936484781848 | 0.04534595434027772 | 0.21428298086091366 | 449 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.013220324997734179 | 0.6888057498909133 | 0.09771642316496477 | 0.04541427741175767 | 0.2180001282312158 | 349 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.01319612606690539 | 0.6909408372551994 | 0.0981669893017531 | 0.04540085266329532 | 0.21710464321324413 | 268 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.013275080712595422 | 0.6884894612492638 | 0.09819119469005644 | 0.04541145769542903 | 0.21856569452720503 | 512 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.013249749369241162 | 0.6880766129042899 | 0.09806817999419083 | 0.04540690596229639 | 0.2184175035773508 | 449 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.013207886462338553 | 0.6922348258839276 | 0.0986230083923219 | 0.04539331450031054 | 0.2165161533618808 | 349 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.013196111268209578 | 0.692760880206735 | 0.09875903922105791 | 0.04538793973494884 | 0.21626492847667284 | 268 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.013247564678842222 | 0.6933425477909263 | 0.09933193627205433 | 0.04537936620224822 | 0.2163265113643658 | 512 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.013232425594900817 | 0.69268582115582 | 0.0991036150104293 | 0.045381531612341694 | 0.21640717296015366 | 449 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.012973271978762585 | 0.6749479620115049 | 0.0940633777985381 | 0.04540211031035335 | 0.2218485601318556 | 268 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012980604592712098 | 0.6744544791505129 | 0.09387389287921205 | 0.04540507108271528 | 0.22204082259234384 | 449 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.012860784417065261 | 0.6807942071417156 | 0.09443682768814744 | 0.045618976646277 | 0.20914345374798754 | 268 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.012735242789694557 | 0.6808192708132442 | 0.09412370386756591 | 0.04567861127623623 | 0.2091689270765943 | 449 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.01299730017835145 | 0.6695286287832147 | 0.09224585277681512 | 0.04544350415396515 | 0.22223780025933354 | 268 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012918535837187495 | 0.6672272153504969 | 0.09228503832329508 | 0.04545383542631232 | 0.22196858901573457 | 449 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012892622206293898 | 0.6802272063769672 | 0.09498565516739957 | 0.0455908055514103 | 0.20923711457410443 | 268 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012781659052575408 | 0.6788966826633029 | 0.0944454895594285 | 0.045652082614849425 | 0.20935113584021742 | 449 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.013047820256278304 | 0.6730751995733727 | 0.0937756849940039 | 0.04538293170871189 | 0.2381920570511255 | 268 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.013040487931916592 | 0.6732083462045043 | 0.09375745074929359 | 0.045384293472462825 | 0.23590208333202123 | 449 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.013034946684087582 | 0.6841714214119955 | 0.09489913943433954 | 0.04551031509382872 | 0.20982946151232626 | 268 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012951594410353519 | 0.6846013967562802 | 0.09528210217316474 | 0.045556393106275996 | 0.20899705944752917 | 449 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.013942553050305305 | 0.6574780928052238 | 0.08996157326880058 | 0.04539544823667501 | 0.39198136744303447 | 268 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.013828015001803364 | 0.6559021072373741 | 0.09004789875799801 | 0.04539660833976514 | 0.4011435615000487 | 449 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.013310050047913559 | 0.6800203533225494 | 0.09564284315097807 | 0.04538404934316596 | 0.2144524481948459 | 268 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.013277182602385796 | 0.6781912897114976 | 0.09591709497581291 | 0.04540237421503567 | 0.2134540662889697 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012938650272028193 | 0.6882872033583924 | 0.09998838791125492 | 0.045321181045138687 | 0.21862466199568223 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.012945223862991918 | 0.6884223530095315 | 0.10053291157987233 | 0.04530869039962659 | 0.21857569315859035 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.012898940162136453 | 0.6873855804276913 | 0.0994476325356208 | 0.04533580718639613 | 0.21865300363583368 | 268 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.012925659217013906 | 0.6875389881953218 | 0.10015903307486285 | 0.0453189555673731 | 0.21852085799581264 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.013081191694365665 | 0.6858615791202706 | 0.09937174588878847 | 0.045316023528870084 | 0.22208231653605964 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.013074591414771013 | 0.6860134316017776 | 0.09947001171214886 | 0.045311068733754754 | 0.22189296270619926 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.01294080170314168 | 0.685577456623447 | 0.09872353515086973 | 0.04534017228494419 | 0.2226346602342448 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.01296734291972626 | 0.6851470712491843 | 0.09971183434714068 | 0.04532381096903148 | 0.22236912146301907 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.013151184264564714 | 0.6877106964963382 | 0.0997962156458414 | 0.04530840471103287 | 0.2216076125331384 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.013199416905363878 | 0.6875987960162506 | 0.09962757313840775 | 0.045304331552581796 | 0.22124608768870754 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.013086949826411553 | 0.6868066369507101 | 0.10064589255175453 | 0.045300431048125395 | 0.22167975202558332 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.013161722251451994 | 0.687085452622761 | 0.10120226615641967 | 0.04528434814582225 | 0.2211749311381331 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.013276741911928076 | 0.6784606228511837 | 0.09431275608258671 | 0.04547744081251024 | 0.23064766627791408 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.013290682692742074 | 0.6750306985202839 | 0.09332184056213544 | 0.04551121399447005 | 0.23184789983217852 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.012497037465991925 | 0.6781569131751813 | 0.09391503502996926 | 0.045559460768734045 | 0.23502136702931575 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.012563561003718735 | 0.677040066922925 | 0.09456116572388903 | 0.045549020284873704 | 0.23469213516441392 | 449 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.29058391600847244 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.29431745799956843 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.2890604589920258 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.2916197500017006 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.3097935420082649 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.3147292909998214 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.30932566699630115 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.30797475000144914 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 17 | 0.4032335410011001 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 22 | 0.4172974169923691 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 18 | 0.416876624993165 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 18 | 0.44681525000487454 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.3374631250044331 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 13 | 0.352904250001302 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 14 | 0.35398004099261016 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 21 | 0.37626283300051 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.240578084005392 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.2028771249897545 | 12.0 | 50 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.8989286669966532 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.826820583009976 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.541369208993274 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.472933249999187 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 0.9482335830107331 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.8672537079983158 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.514338666005642 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.473862208003993 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 1.0370941250002943 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.9250559159991099 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 2.972274291998474 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.8805615000019316 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.1577718750049826 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.131351750009344 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1057900830055587 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.2441891659982502 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.0973413750034524 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.109465999994427 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.694559625000693 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.7204984999989392 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.7287344169890275 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7219979579967912 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.0969802910112776 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.0925296250061365 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.0941453339910368 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.1554912090068683 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.7214624579937663 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.7044107919937233 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7173560420051217 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.712146166988532 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.46517812499951106 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.463431166994269 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.46597370800736826 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.45891033300722484 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.4955363339977339 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.5008401660015807 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.49451004099682905 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.49738329199317377 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.75029170801281 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7290520000096876 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 21 | 0.8133727499953238 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 19 | 0.7992053749912884 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 14 | 0.5934936249977909 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.5796025419986108 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.5821258749929257 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.5950275830109604 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.301095207993058 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.313267375007854 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.5263944170001196 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.346360833995277 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 5.782918375000008 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 5.753868083003908 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.6912311670021154 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.430925832988578 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 7.242160082998453 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 7.048513583999011 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.169722416991135 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.784964416990988 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 9.707416499993997 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 10.618942916000378 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.7934959169942886 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.566498749991297 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.3312672909960384 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.3748234169906937 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.3197061249957187 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.3389817090064753 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 1.9768571669992525 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 2.097129458008567 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 1.984611332998611 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.9857907499972498 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3290730830049142 | 3.0 | NA | NA | 0.1 | 10 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.310592916997848 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.318206124997232 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.3109123339963844 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 1.9844356250105193 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 1.9848658749979222 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.9768359580048127 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.9746746250020806 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.6538703749974957 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.6283006659941748 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6330147080007009 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6298681250045775 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 11 | 0.7286234159982996 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 10 | 0.6971493749879301 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.7287842499936232 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.72804879200703 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 19 | 1.1529248339938931 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.1099284590018215 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 21 | 1.248015499993926 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 21 | 1.17146691599919 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 14 | 0.8448475420009345 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 14 | 0.8873780409921892 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.9287795839918545 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 16 | 0.9075699169916334 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 10.084558709000703 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 9.61284145800164 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.251058291993104 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.0319194159965264 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 10.09668241700274 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 10.041570166999009 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.5182024579989957 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.145649957994465 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 15.088270332998945 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 17.54178983300517 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.4715608339902246 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.936698499994236 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 20.78111420800269 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 22.99868633299775 | NA | 5 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.31146979201003 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 4.959855000008247 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.5463169579888927 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.5740047080034856 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.540234166997834 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.5549160000082338 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.245576291999896 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.249237666008412 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.2350167499971576 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.263967833001516 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.5564064170030179 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.6497642500035 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.5250079580000602 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.5315677920007147 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.23929933400359 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.3395870420063147 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.221879999997327 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.2280496250023134 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 8 | 0.8165555839950684 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 8 | 0.802078458989854 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 9 | 0.8495249170082388 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 8 | 0.8101037909946172 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 11 | 0.958483416994568 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 0.9203465409955243 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 0.9780673750065034 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 11 | 0.9714120839926181 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 18 | 1.5398971249960596 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 16 | 1.559106583998073 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 22 | 1.796089833995211 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.7391355419968022 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 15 | 1.2336717919970397 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 14 | 1.234415083003114 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.29999245799263 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 16 | 1.3241955840057926 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 17.584440666993032 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 17.84021979199315 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.1744635829963954 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.7240306250023423 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 17.555769207989215 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 16.680283415989834 | 12.0 | 5 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 3.801206208008807 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 2.9428964589897078 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 28.13815766699554 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 28.859075832995586 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.395334790999186 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 4.575401333000627 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 45.50897158400039 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 49.277226125006564 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 10.42356683401158 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.21610525000142 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.7690607920085313 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.7859892500127899 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.7512894160026917 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.793694374995539 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.5115134170046076 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.508368374998099 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.4995225829916308 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.508139500001562 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.774955625005532 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7773962909996044 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.7546117499878164 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.7663178750080988 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.494540166007937 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.52150558399444 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.460953290996258 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.5477315839962102 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45364 | 771 | logistic | raw | 45364 | 771 | 0.01699585574464333 | 0.009289063940361261 | 0.7754246121056323 |
| logistic | validation | calibrated | 45364 | 771 | logistic | calibrated | 45364 | 771 | 0.01699585574464333 | 0.021569458650033516 | 0.7754137776970748 |
| rf | validation | raw | 45364 | 771 | rf | raw | 45364 | 771 | 0.01699585574464333 | 0.014754282979064292 | 0.7366188437327222 |
| rf | validation | calibrated | 45364 | 771 | rf | calibrated | 45364 | 771 | 0.01699585574464333 | 0.036686992001013906 | 0.7354802419217268 |

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | validation | raw | 45364 | 771 | xgb | raw | 45364 | 771 | 0.01699585574464333 | 0.011992013269523807 | 0.7620381840623786 |
| xgb | validation | calibrated | 45364 | 771 | xgb | calibrated | 45364 | 771 | 0.01699585574464333 | 0.030323960589356755 | 0.761766058622207 |
| logistic | test | raw | 49338 | 886 | logistic | raw | 49338 | 886 | 0.017957760752361263 | 0.1107797683765124 | 0.759395698966411 |
| logistic | test | calibrated | 49338 | 886 | logistic | calibrated | 49338 | 886 | 0.017957760752361263 | 0.1385884869626291 | 0.7565372464223744 |
| rf | test | raw | 49338 | 886 | rf | raw | 49338 | 886 | 0.017957760752361263 | 0.048022864530425535 | 0.7816090216302132 |
| rf | test | calibrated | 49338 | 886 | rf | calibrated | 49338 | 886 | 0.017957760752361263 | 0.1291129041403006 | 0.7781996293741831 |
| xgb | test | raw | 49338 | 886 | xgb | raw | 49338 | 886 | 0.017957760752361263 | 0.077531355877136 | 0.7778877734106167 |
| xgb | test | calibrated | 49338 | 886 | xgb | calibrated | 49338 | 886 | 0.017957760752361263 | 0.10505365090194416 | 0.7753218656373327 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.06310014352284227 | 0.016449383899881254 | 0.0800451298148371 |
| logistic | 0.06076901374197774 | 0.01633408333315959 | 0.07788884828724431 |
| rf | 0.05113916032424499 | 0.016477877495879152 | 0.0801748052395751 |
| rf | 0.04652185563195544 | 0.017006626045062655 | 0.08697479908430901 |
| xgb | 0.05126433431483225 | 0.016468980921789108 | 0.07985482438139051 |
| xgb | 0.04919679907657565 | 0.016657841854325246 | 0.08273607940830344 |
| logistic | 0.05600720172726668 | 0.03218489580373498 | 0.15190729771314704 |
| logistic | 0.048019853584799634 | 0.03559613890360951 | 0.17868159128543104 |
| rf | 0.05870511945920039 | 0.018192029271692296 | 0.09747312092648885 |
| rf | 0.050791558749169616 | 0.0370547321997401 | 0.47787439594169734 |
| xgb | 0.055958289296431374 | 0.023767460825287967 | 0.11857976670986968 |
| xgb | 0.05132418346372028 | 0.03196071271794026 | 0.40276125420816217 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45364 | 771 | logistic | raw | 0.8055974222702987 | 1.043075013249362 | 0.007706791804282067 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 45364 | 771 | logistic | calibrated | 0.11482355189065141 | 1.1085727247319879 | 0.004595393438569063 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 45364 | 771 | rf | raw | 0.9596585793431843 | 1.208913316995386 | 0.002241572765579038 4 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 45364 | 771 | rf | calibrated | -0.3330344603661019 | 1.1593057643708458 | 0.019691136256370586 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| xgb | validation | raw | 45364 | 771 | xgb | raw | -0.2769491408123634 | 0.8368092404813591 | 0.00555355911209293 7 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 45364 | 771 | xgb | calibrated | 0.260880554565169 | 1.2768290276331387 | 0.01332810484471342 5 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 49338 | 886 | logistic | raw | -2.081673626741808 | 0.9663955700041601 | 0.09282200762415117 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 49338 | 886 | logistic | calibrated | -1.5617200298940301 | 1.4501032881249731 | 0.12063072621026781 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 49338 | 886 | rf | raw | 4.13643025895314 | 2.845716533355477 | 0.03006510377806427 5 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 49338 | 886 | rf | calibrated | -3.7892017252306607 | 0.13379446188250338 | 0.11115514338793935 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 49338 | 886 | xgb | raw | -1.5212312580513645 | 1.0404308440293293 | 0.05957359512477474 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 49338 | 886 | xgb | calibrated | -3.721074507234551 | 0.1497068186956985 | 0.08709589014958287 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R2_candidate_b/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 312 | 0.00632372613401435 | 33 | 279 | 48173 | 853 |
| logistic | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 1:1 | 1 | 1 | 0.2528735632183909 | 49338 | 886 | 4254 | 0.08622157363492643 | 288 | 3966 | 44486 | 598 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 1935 | 0.03921926304268515 | 154 | 1781 | 46671 | 732 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 2:1 | 1 | 2 | 0.25287356321839083 | 49338 | 886 | 8545 | 0.17319307633061737 | 433 | 8112 | 40340 | 453 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 10500 | 0.21281770643317524 | 500 | 10000 | 38452 | 386 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 10070 | 0.2041023146459119 | 484 | 9586 | 38866 | 402 |
| logistic | 5:1 | 1 | 5 | 0.1402439024390244 | 49338 | 886 | 15964 | 0.3235639871904009 | 623 | 15341 | 33111 | 263 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 22176 | 0.44947099598686613 | 721 | 21455 | 26997 | 165 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 35371 | 0.7169119137378897 | 841 | 34530 | 13922 | 45 |
| logistic | 10:1 | 1 | 10 | 0.09651474530831099 | 49338 | 886 | 35017 | 0.7097369167781427 | 840 | 34177 | 14275 | 46 |
| rf | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 515 | 0.010438201791722404 | 41 | 474 | 47978 | 845 |
| rf | 1:1 | 1 | 1 | 0.2272727272727273 | 49338 | 886 | 533 | 0.010803032145607848 | 43 | 490 | 47962 | 843 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 525 | 0.010640885321658761 | 41 | 484 | 47968 | 845 |
| rf | 2:1 | 1 | 2 | 0.2272727272727273 | 49338 | 886 | 533 | 0.010803032145607848 | 43 | 490 | 47962 | 843 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 6033 | 0.1222789736106044 | 374 | 5659 | 42793 | 512 |
| rf | 5:1 | 1 | 5 | 0.22727272727272727 | 49338 | 886 | 1371 | 0.027787911954274595 | 102 | 1269 | 47183 | 784 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 994 | 0.020146742875673924 | 76 | 918 | 47534 | 810 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 31255 | 0.6334873728160849 | 833 | 30422 | 18030 | 53 |
| rf | 10:1 | 1 | 10 | 0.16417910447761194 | 49338 | 886 | 9360 | 0.1897117840204305 | 496 | 8864 | 39588 | 390 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 72 | 0.001459321415541773 | 10 | 62 | 48390 | 876 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 442 | 0.008958612023186995 | 41 | 401 | 48051 | 845 |
| xgb | 1:1 | 1 | 1 | 0.7262429304291373 | 49338 | 886 | 422 | 0.00855324496331428 | 39 | 383 | 48069 | 847 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 633 | 0.01282986744497142 | 51 | 582 | 47870 | 835 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 455 | 0.00922210061210426 | 43 | 412 | 48040 | 843 |
| xgb | 2:1 | 1 | 2 | 0.7262429304291373 | 49338 | 886 | 422 | 0.00855324496331428 | 39 | 383 | 48069 | 847 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 5125 | 0.10387530909238316 | 311 | 4814 | 43638 | 575 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 4780 | 0.09688272730957882 | 295 | 4485 | 43967 | 591 |
| xgb | 5:1 | 1 | 5 | 0.7262429304291373 | 49338 | 886 | 422 | 0.00855324496331428 | 39 | 383 | 48069 | 847 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 14536 | 0.29462077911548906 | 619 | 13917 | 34535 | 267 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 29187 | 0.5915724188252462 | 825 | 28362 | 20090 | 61 |
| xgb | 10:1 | 1 | 10 | 0.10617283950617286 | 49338 | 886 | 16548 | 0.33540070533868416 | 664 | 15884 | 32568 | 222 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.03724604966139955 | 0.9942417237678527 | 0.10576923076923077 | 0.9826010688206258 | 0.005758276232147279 | 0.9627539503386005 | 1132 | 0.022943775588795654 | 886 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| logistic | 0.32505643340857787 | 0.9181457937752827 | 0.06770098730606489 | 0.9867358708189158 | 0.08185420622471724 | 0.6749435665914221 | 4564 | 0.09250476306295351 | 886 |
| logistic | 0.17381489841986456 | 0.9632419714356476 | 0.07958656330749354 | 0.9845579393709258 | 0.03675802856435235 | 0.8261851015801355 | 3245 | 0.06577080546434796 | 1772 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 1772 | 0.035915521504722527 | 1772 |
| logistic | 0.4887133182844244 | 0.8325765706265995 | 0.050672908133411354 | 0.9888951535802711 | 0.1674234293734005 | 0.5112866817155757 | 9018 | 0.18278000729660707 | 1772 |
| logistic | 0.5643340857787811 | 0.7936101708907785 | 0.047619047619047616 | 0.9900612801895051 | 0.2063898291092215 | 0.435665914221219 | 11930 | 0.24180145121407434 | 4430 |
| logistic | 0.54627539503386 | 0.8021547098159003 | 0.048063555114200594 | 0.9897626566160742 | 0.19784529018409971 | 0.45372460496614 | 11596 | 0.2350318213142 | 4430 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.7031602708803611 | 0.6833773631635432 | 0.039025306940616386 | 0.9921196140708336 | 0.3166226368364567 | 0.29683972911963885 | 16656 | 0.33758968746199686 | 4430 |
| logistic | 0.8137697516930023 | 0.5571906216461653 | 0.032512626262626264 | 0.9939253368676828 | 0.4428093783538347 | 0.18623024830699775 | 23105 | 0.4683002959179537 | 8860 |
| logistic | 0.9492099322799097 | 0.2873359200858582 | 0.0237765401034746 | 0.9967781198539414 | 0.7126640799141418 | 0.050790067720090294 | 34980 | 0.7089869877173781 | 8860 |
| logistic | 0.9480812641083521 | 0.2946214810534137 | 0.023983484851643487 4 | 0.9967879338035054 | 0.7053785189465863 | 0.05191873589164785 | 34637 | 0.702034942640561 | 8860 |
| rf | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| rf | 0.046275395033860044 | 0.990217122100223 | 0.07961165048543689 | 0.982692583413555 | 0.009782877899777099 | 0.95372460496614 | 1319 | 0.026733957598605536 | 886 |
| rf | 0.04853273137697517 | 0.9898868983736482 | 0.08067542213883677 | 0.9827271795922549 | 0.010113101626351853 | 0.9514672686230248 | 1333 | 0.02701771454051644 | 886 |
| rf | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 1772 | 0.035915521504722527 | 1772 |
| rf | 0.046275395033860044 | 0.9900107322711137 | 0.07809523809523809 | 0.9826890377563354 | 0.009989267728886 32 | 0.95372460496614 | 2174 | 0.044063399408164096 | 1772 |
| rf | 0.04853273137697517 | 0.9898868983736482 | 0.08067542213883677 | 0.9827271795922549 | 0.010113101626351853 | 0.9514672686230248 | 2176 | 0.044103936114151365 | 1772 |
| rf | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 4430 | 0.08978880376180631 | 4430 |
| rf | 0.4221218961625282 | 0.8832039957070915 | 0.0619923752693519 | 0.9881768848862718 | 0.11679600429290844 | 0.5778781038374717 | 8219 | 0.16658559325469213 | 4430 |
| rf | 0.11512415349887133 | 0.9738091306860398 | 0.07439824945295405 | 0.9836554297746367 | 0.026190869313960207 | 0.8848758465011287 | 5189 | 0.10517248368397585 | 4430 |
| rf | 0.08577878103837472 | 0.9810534136877734 | 0.07645875251509054 | 0.9832450769485355 | 0.018946586312226534 | 0.9142212189616253 | 9018 | 0.18278000729660707 | 8860 |
| rf | 0.9401805869074492 | 0.37212086188392635 | 0.026651735722284433 | 0.9970690703976111 | 0.6278791381160737 | 0.05981941309255079 | 30952 | 0.6273460618590133 | 8860 |
| rf | 0.5598194130925508 | 0.8170560554775861 | 0.05299145299145299 | 0.9902446345490019 | 0.18294394452241394 | 0.4401805869074492 | 12764 | 0.25870525761076657 | 8860 |
| xgb | 0.011286681715575621 | 0.9987203830595228 | 0.1388888888888889 | 0.9822189745463403 | 0.001279616940477173 3 | 0.9887133182844243 | 938 | 0.019011715108030323 | 886 |
| xgb | 0.046275395033860044 | 0.9917237678527202 | 0.09276018099547512 | 0.9827184227748691 | 0.008276232147279782 | 0.95372460496614 | 1246 | 0.025254367830070127 | 886 |
| xgb | 0.04401805869074492 | 0.9920952695451168 | 0.0924170616113744 | 0.9826846021751574 | 0.007904730454883184 | 0.9559819413092551 | 1230 | 0.024930074182171958 | 886 |
| xgb | 0.05756207674943566 | 0.9879881119458434 | 0.08056872037914692 | 0.982855969612976 | 0.012011888054156691 | 0.9424379232505643 | 2252 | 0.04564433094166768 | 1772 |
| xgb | 0.04853273137697517 | 0.9914967390407001 | 0.0945054945054945 | 0.9827547409119735 | 0.008503260959299926 | 0.9514672686230248 | 2098 | 0.042523004580647776 | 1772 |
| xgb | 0.04401805869074492 | 0.9920952695451168 | 0.0924170616113744 | 0.9826846021751574 | 0.007904730454883184 | 0.9559819413092551 | 2077 | 0.04209736916778142 | 1772 |
| xgb | 0.3510158013544018 | 0.9006439362668208 | 0.06068292682926829 | 0.9869947752923348 | 0.09935606373317923 | 0.6489841986455982 | 7689 | 0.15584336616806518 | 4430 |
| xgb | 0.3329571106094808 | 0.9074341616445142 | 0.06171548117154812 | 0.9867363885273127 | 0.09256583835548585 | 0.6670428893905191 | 7440 | 0.15079654627264988 | 4430 |
| xgb | 0.04401805869074492 | 0.9920952695451168 | 0.0924170616113744 | 0.9826846021751574 | 0.007904730454883184 | 0.9559819413092551 | 4618 | 0.09359925412460983 | 4430 |
| xgb | 0.698645598194131 | 0.7127672748286964 | 0.04258392955421024 | 0.9923280271248779 | 0.28723272517130355 | 0.3013544018058691 | 16587 | 0.33619117110543595 | 8860 |
| xgb | 0.9311512415349887 | 0.414637166680426 | 0.028266008839551854 | 0.996972854945164 | 0.5853628333195741 | 0.06884875846501129 | 28972 | 0.5872147229316146 | 8860 |
| xgb | 0.7494356659142212 | 0.6721703954429126 | 0.04012569494802997 | 0.9932296431838975 | 0.32782960455708743 | 0.2505643340857788 | 18104 | 0.36693826259678136 | 8860 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 1.2776523702031604 |
| logistic | 1.0 |
| logistic | 5.1512415349887135 |
| logistic | 1.8312641083521444 |
| logistic | 1.0 |
| logistic | 5.089164785553048 |
| logistic | 2.6930022573363432 |
| logistic | 2.617607223476298 |
| logistic | 3.7598194130925506 |
| logistic | 2.6077878103837473 |
| logistic | 3.9480812641083523 |
| logistic | 3.9093679458239277 |
| rf | 1.0 |
| rf | 1.4887133182844243 |
| rf | 1.5045146726862302 |
| rf | 1.0 |
| rf | 1.22686230248307 |
| rf | 1.2279909706546275 |
| rf | 1.0 |
| rf | 1.8553047404063205 |
| rf | 1.1713318284424379 |
| rf | 1.0178329571106095 |
| rf | 3.493453724604966 |
| rf | 1.4406320541760722 |
| xgb | 1.0586907449209932 |
| xgb | 1.4063205417607223 |
| xgb | 1.3882618510158014 |
| xgb | 1.2708803611738149 |
| xgb | 1.1839729119638827 |
| xgb | 1.1721218961625282 |
| xgb | 1.735665914221219 |
| xgb | 1.6794582392776525 |
| xgb | 1.0424379232505643 |
| xgb | 1.872121896162528 |
| xgb | 3.2699774266365687 |
| xgb | 2.0433408577878103 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 886 | 0.017957760752361263 |
| test | all_non_risky | 2:1 | 2:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 1772 | 0.035915521504722527 |
| test | all_non_risky | 5:1 | 5:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 4430 | 0.08978880376180631 |
| test | all_non_risky | 10:1 | 10:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 8860 | 0.17957760752361263 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 38568 | 375 | 0.009723086496577474 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 4080 | 268 | 0.06568627450980392 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 700 | 27 | 0.03857142857142857 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 5990 | 216 | 0.03606010016694491 |

R3

Remove otherwise-valid within-window Code 01 non-events; earlier observed 03–99 events remain. Estimand conditional on remaining on book, not post-payoff counterfactual risk.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.025806378429270672 | 176313 | 4550 |
| validation | 0.02350896450786681 | 32796 | 771 |
| test | 0.02186898356123809 | 40514 | 886 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 16 | 50 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_50","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 39543 | 252 | 42155 | 330 | 42155 | 330 | 0.007828252876289882 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.009551994610461480 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 81698 | 582 | 41562 | 397 | 41562 | 397 | 0.00955199461046148 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 123260 | 979 | 29040 | 1279 | 29040 | 1279 | 0.04404269972451791 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 152300 | 2258 | 24013 | 2292 | 24013 | 2292 | 0.09544829883812934 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.007386785986890098 | 0.8196780959626148 | 0.03299597075862059 | 0.007678954158306219 | 0.041083315017082844 | 183 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.007376562484198799 | 0.8187428861236393 | 0.03305740189702942 | 0.007678899612961211 | 0.041091744062567555 | 130 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.007390019674050171 | 0.8187510007426325 | 0.03275173250289221 | 0.007680105194459478 5 | 0.04111076949642141 | 322 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.007388824204864318 | 0.8190277672118675 | 0.03285729647789153 | 0.007679781087846363 | 0.041101696070242374 | 253 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.005704579398838518 | 0.8465387165136119 | 0.04709919720175177 | 0.007639574717064827 | 0.0394586798786985 | 183 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.005704354901864698 | 0.8461935553985763 | 0.0466867248070488 | 0.007640861150751054 | 0.03948340859438978 | 130 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.005703813972486508 | 0.8461879765980184 | 0.04717162246910358 | 0.007639037986378392 | 0.03946484588972281 | 322 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.005705593131967191 | 0.8462954952996795 | 0.047336224958705674 | 0.007638455217288113 | 0.03945709192064289 | 253 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.005421596013488262 | 0.80351841185314 | 0.04354919019489146 | 0.007658153175873945 | 0.04106194382144187 | 183 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.005386177007672709 | 0.8131073556847616 | 0.044311286566058664 | 0.007647054420772462 | 0.04066024651650596 | 130 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.005393560348121869 | 0.7915883279900017 | 0.04237355710996579 | 0.007677923830843429 | 0.041791656736346085 | 322 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.005441386996470842 | 0.7971690122987194 | 0.043453391365691915 | 0.007672937439696719 5 | 0.04146983707318374 | 253 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.005419077008604322 6 | 0.8363131373507943 | 0.04789598823416116 | 0.007634943498707448 5 | 0.03970338269521998 | 183 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.005409658546885024 | 0.8378162980673438 | 0.048031599831336556 | 0.007633433152082826 | 0.03966380913104257 | 130 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.005413042908137214 | 0.8347210780850948 | 0.04811499795336514 | 0.007635142540912603 | 0.03977573619604155 | 322 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.005422685857070356 5 | 0.835392417902878 | 0.04827736930255871 | 0.007634492062409631 | 0.03973898124323224 | 253 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.005790615140815795 5 | 0.8419067905595101 | 0.03833512672556325 | 0.007642319723119644 | 0.039576491719254925 | 130 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.00575891894764330 1 | 0.839662808600047 | 0.03722744116815214 | 0.007649543979277305 | 0.039752742287770045 | 253 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.005711343564635904 | 0.8400250683765328 | 0.042033287791344597 | 0.007660218193855932 | 0.04022303144789757 | 130 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.00577326750900906 | 0.8372322628556939 | 0.041006046406676884 | 0.00767703309524236 | 0.04081392041466887 | 253 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.005962252489695458 | 0.8234600155771704 | 0.0323505604173036 | 0.007683298156272274 | 0.04069600115110035 | 130 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.005832508517601064 | 0.8229355720987519 | 0.031741085158380936 | 0.007688977150973371 4 | 0.04083518816218813 | 253 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.005760063708937821 | 0.825912586715934 | 0.03838190109739013 | 0.007659350596429779 | 0.04043725280216657 | 130 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.005809634942815953 | 0.822778677389556 | 0.03908428511272761 | 0.007671136378227266 | 0.04086927951338607 | 253 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.005809581753545616 | 0.8379395388433045 | 0.03783808595638938 | 0.007643504878014264 | 0.03983655454481583 | 130 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.0057916079673357 | 0.8367869731384375 | 0.03698859467419922 | 0.007648847736708701 | 0.03997435284410137 | 253 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.005701383569799 | 0.8424079407343007 | 0.04213730608103242 | 0.007654094933116634 | 0.03995262752558225 | 130 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.005737152025620265 | 0.8379305548008478 | 0.04098000451564158 | 0.007668815265641927 6 | 0.04045071135147684 | 253 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.006356189219046263 | 0.7939366769910704 | 0.030604139759744887 | 0.007702244833473562 | 0.05628256469810621 | 130 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.00618174510968949 | 0.7860344509047439 | 0.028109275269601032 | 0.007723790163701246 | 0.05979434094432758 | 253 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.005883162232191296 | 0.8163609918672678 | 0.03598099468459385 | 0.007654115429669778 | 0.040576579884229864 | 130 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.005820849325028827 | 0.814605517216396 | 0.03815323696305978 | 0.007659291561601917 | 0.040627318352910384 | 253 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.005451355428005714 | 0.8317260953829991 | 0.04440658963688528 | 0.007628560163918862 | 0.039898457179081546 | 130 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.005444541351281467 | 0.8310382365194081 | 0.043704912502500255 | 0.00762931898908453 | 0.03992500640313837 | 253 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.005410379525601081 | 0.8331111594124146 | 0.0453036619790939 | 0.007631449660415368 4 | 0.039987764155611887 | 130 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.005442008694934894 | 0.8369510768171856 | 0.048188118159444945 | 0.007624452905190496 | 0.039854735164254015 | 253 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.005398564981854692 | 0.8168260971942981 | 0.03920897002998359 | 0.007668512982353379 | 0.040964661907338495 | 130 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.00543623963576876 | 0.815227915738376 | 0.03821285387802011 | 0.007671367766708226 5 | 0.04102481693434685 | 253 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.00452681650822946 | 0.810148236700538 | 0.03809696718922217 | 0.00769948869044633 | 0.04249440838363303 | 130 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.004589472822106417 5 | 0.8074471191291275 | 0.04111403152621903 | 0.007686841988688302 | 0.04252484149212152 | 253 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.00542703611347279 | 0.804132152366462 | 0.03559171119886008 | 0.007704298462140847 | 0.04191358242141776 | 130 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.005370210073794943 | 0.8040663297650746 | 0.037704158540280114 | 0.007685513348833283 | 0.04184033239278223 | 253 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.005099244169772418 | 0.79666076183231 | 0.03607207564055482 | 0.007737227842846713 | 0.04323667619689577 | 130 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.005126310690807059 | 0.799412124834719 | 0.0409534506302427 | 0.007720608126788367 5 | 0.043076970197037864 | 253 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004944277174097419 | 0.7620427466536253 | 0.027919168274005108 | 0.007817482006813774 | 0.04569860228064479 | 130 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.00495860788834579 | 0.7741172634896485 | 0.028995828846373185 | 0.007811120820511375 | 0.04507536307481642 | 253 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.0019792668974232255 | 0.7587009726674998 | 0.028762798025592274 | 0.007839693366396212 | 0.0559189119094700 3 | 130 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.002143104282380998 6 | 0.7665345505261825 | 0.02960770058937172 | 0.007813150403187743 | 0.0544621776154801 | 253 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.010190189570863895 | 0.8286476277657555 | 0.05118518859105037 | 0.00925937732137304 | 0.04706494941442165 | 255 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.010177487864215807 | 0.829036185089128 | 0.0513960330480562 | 0.00925842068893018 | 0.0470469930001309 | 183 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.010207479398144214 | 0.8291019032883881 | 0.05167110291713002 | 0.00925709917284532 | 0.047031227179190525 | 420 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.010203265944715528 | 0.8291647455515542 | 0.05156327080860337 | 0.009257570351241574 | 0.04703373924595563 | 323 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.009356348631678295 | 0.8458634707469878 | 0.06099308805576905 | 0.009215975506663835 | 0.045692702489755516 | 255 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.009348397366903852 | 0.8464589730888868 | 0.06162982504694344 | 0.009212280453557563 | 0.045646497883286584 | 183 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 2 | 0.009383940239824 | 0.8455910522897193 | 0.06178919866329619 | 0.009211824197907456 | 0.045667037758110866 | 420 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.009377639840573387 | 0.8458708135625476 | 0.06168653478144228 | 0.009212697860342758 | 0.04565783992890515 | 323 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.009185936538440565 | 0.818103038671244 | 0.05567605188234915 | 0.009276182208358399 | 0.04711022248501424 | 255 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.009129220731911839 | 0.8258775505958237 | 0.05924725912095199 | 0.009242992431677439 | 0.046490135209010415 | 183 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.00924722906441818 | 0.8031088257277571 | 0.05191704369226916 | 0.009315865474037314 | 0.048035277195260045 | 420 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.009200775691742952 | 0.8085689433780197 | 0.05538576493350065 | 0.0092847876903234 | 0.04758760906102807 | 323 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.00919617811045567 | 0.8382419341465704 | 0.0590553242038169 | 0.009242729093686338 | 0.04607300517276155 | 255 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.009163037600301663 | 0.8400214043073568 | 0.06095103311537765 | 0.009227485057757714 | 0.04588246988421568 | 183 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.009232085799414555 | 0.8350254749807328 | 0.05904855109043258 | 0.009243281029769243 | 0.04621850926791226 | 420 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.009214591645030249 | 0.8359949713951442 | 0.05936234614722263 | 0.009240771357499941 | 0.04613707906588643 | 323 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.008921551750894636 | 0.8380111861676041 | 0.05628843480184138 | 0.00922092397980176 | 0.04587853504505969 | 183 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.008892124153200223 | 0.8383544627950243 | 0.056564171380860576 | 0.009219601724654409 | 0.04587247472314108 | 323 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.008554337428479464 | 0.8408300930610086 | 0.06202653802726333 | 0.009279499934625124 | 0.04700024323560924 | 183 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.00837910410893319 | 0.8363028801276182 | 0.05894114073516671 | 0.00930489867364704 | 0.04763189125144165 | 323 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.009266962482387568 | 0.8278199088817778 | 0.05053407193676005 | 0.009271798911427813 | 0.04671618995230531 | 183 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.00910813059371549 | 0.8279323763401021 | 0.05070343003281957 | 0.009267237061814655 | 0.046733915529484756 | 323 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.00875576639828647 | 0.8347022075257129 | 0.05821827324186958 | 0.009260164022358857 | 0.04685229947386349 | 183 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.008531017250233398 | 0.8305800732507043 | 0.05665092991909734 | 0.009286776078738684 | 0.04748637700788975 | 323 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.008990871405978174 | 0.8369934719310167 | 0.05753709794539969 | 0.009220971559824839 | 0.045950766961700466 | 183 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.008949328786335933 | 0.8336740603720176 | 0.05547934625258192 | 0.009224739446781264 | 0.046129713682036895 | 323 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.008786891345329357 | 0.8408659504769922 | 0.05887996776906124 | 0.009265908929498523 | 0.04654466679277087 | 183 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.008735564216009113 | 0.839723285995629 | 0.059610261409702686 | 0.009286171615909505 | 0.04699675555049994 | 323 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.010073325474455514 | 0.8012852068884176 | 0.04403317620296095 | 0.009357061781389533 | 0.0592842461200763 | 183 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.010063644797838268 | 0.8011984392845528 | 0.043301729229825135 | 0.009373813094868122 | 0.06024961189978927 | 323 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.00930686246642603 | 0.8261150294890532 | 0.05457061205747684 | 0.009241890903738816 | 0.04651207817037929 | 183 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.009162931841125788 | 0.8235769852908107 | 0.052436910774115206 | 0.00925475916976951 | 0.046700029580958105 | 323 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.00888292479282589 | 0.8399312406512954 | 0.0519371259055958 | 0.009248727563183555 | 0.046021993085379874 | 183 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.008887370721200445 | 0.8397309653569023 | 0.053298004840070794 | 0.009241258638085094 | 0.04598625397093397 | 323 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.008785986047323852 | 0.8383403890652014 | 0.055153214250702305 | 0.009231150876263087 | 0.04598183811397162 | 183 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.00881543230214419 | 0.8388307361692715 | 0.05653766046654226 | 0.009230343419203821 | 0.04594572702467636 | 323 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.008602539175400933 | 0.8282423349419198 | 0.044375513284118207 | 0.009326031572919833 | 0.047105353146881815 | 183 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.008650168815465443 | 0.8279158244100278 | 0.04442103052528944 | 0.00932868832997812 | 0.04716102180069774 | 323 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.008116120213630368 | 0.8205393236838538 | 0.04210160564329344 | 0.009372107607008488 | 0.04784437361498735 | 183 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.008137436157796588 | 0.8208718308484532 | 0.04026522909567131 | 0.00938865899331917 | 0.04792315821932764 | 323 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.008643911204307108 | 0.8247488680590889 | 0.04399414317747102 | 0.009341214502123284 | 0.04735981652676752 | 183 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.008695332599077435 | 0.82299362918965 | 0.04341677614347878 | 0.009352506953935386 | 0.047525338436539964 | 323 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.008525331023607906 | 0.8214774907518768 | 0.04479197127344946 | 0.009376752957736776 | 0.047724816857238034 | 183 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.008621571897860723 | 0.8186658348888375 | 0.0407128714319206 | 0.009420735752061263 | 0.04809943804627051 | 323 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.008013535528898405 | 0.7971609921490005 | 0.0336519129925703 | 0.009578205025180812 | 0.05090343156518663 | 183 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.008093645449977194 | 0.7972603343245115 | 0.03456961844131117 | 0.00959448689264739 | 0.05093647509199784 | 323 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.00575901198883911 | 0.7800424720690003 | 0.02981868931185756 | 0.009683566873791129 | 0.055221044090560646 | 183 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.005662238023199974 | 0.7748742619323047 | 0.027871061598278558 | 0.009686920847326879 | 0.05564447559047452 | 323 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.014862407366012872 | 0.7318918922572627 | 0.11144640934880404 | 0.04200009180855709 | 0.1863416525721331 | 294 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.014812927985168164 | 0.7324879833361493 | 0.1116580333277637 | 0.04200247736318611 | 0.18636520393701972 | 227 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.014857244572373668 | 0.7320188837372864 | 0.1115477741382219 | 0.04199944136024546 | 0.186331071953593 | 474 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.014857900222854149 | 0.7319856783802342 | 0.11151364217954772 | 0.041999797445789085 | 0.18633607991930354 | 387 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.014789574270503113 | 0.7219524220463406 | 0.10589211707197532 | 0.042031300027320646 | 0.19192106070565476 | 294 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.014754720425915652 | 0.7227152721745107 | 0.1063373594948099 | 0.04202630667446587 | 0.1918199446285502 | 227 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.014759118856034127 | 0.722723045438757 | 0.10648668878380477 | 0.04202234212037521 | 0.1917889758043325 | 474 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.014765863341314427 | 0.7224367020416844 | 0.10624329532134656 | 0.04202638026732137 | 0.19184950236323398 | 387 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.014576985098944884 | 0.6989299285008959 | 0.09784005423538336 | 0.04220608337253586 | 0.20053002518088686 | 294 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.014619106449488335 | 0.7060750510352819 | 0.10119055824220664 | 0.04213359426663834 | 0.19839090916706253 | 227 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.014708034690802663 | 0.6980998509025957 | 0.0986024050943607 | 0.042182110290211385 | 0.20152724455579013 | 474 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.01460858088842863 | 0.6973654182513259 | 0.09878421273482016 | 0.042173075962190276 | 0.20130274395198194 | 387 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.014698581708207861 | 0.712933689352591 | 0.10160321119807342 | 0.04212636525071565 | 0.19592434852281684 | 294 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.014702082230893037 | 0.7146548759391251 | 0.10317874502065191 | 0.04209420333055459 | 0.19540384924478435 | 227 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.014709707797797051 | 0.7140634600843867 | 0.10287923921265181 | 0.04209276391193523 | 0.19570891527778528 | 474 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.014684487118897636 | 0.7134998139345281 | 0.10268522042827466 | 0.0421003177902843 | 0.19582969419479854 | 387 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.013806527619892235 | 0.7099581626583145 | 0.10071786747160615 | 0.042167837988909156 | 0.19588606635444863 | 227 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.013776207561242503 | 0.712542829348207 | 0.10138492205873093 | 0.04216329140702583 | 0.19537577859038036 | 387 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.012580630155658924 | 0.7038483206327302 | 0.09799509430177386 | 0.042585703063122204 | 0.19376124133965514 | 227 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.011642948148282902 | 0.6967020152103067 | 0.09567706271797395 | 0.04274199996170532 | 0.19716330288044456 | 387 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.015183140176124455 | 0.7014342714602435 | 0.09803066328413672 | 0.0421186937506658 | 0.19838316021876448 | 227 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.014606373471438745 | 0.6996940882551076 | 0.09688509487538095 | 0.04219009420964474 | 0.1997752232786002 | 387 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.012920690861957847 | 0.7053396889719826 | 0.0977205049547812 | 0.04247993258346811 | 0.19322324114322767 | 227 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.012252273769140572 | 0.7001512603995924 | 0.09785554068872948 | 0.042600092087584085 | 0.1951910226167483 | 387 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.014262735181172702 | 0.7110928480082658 | 0.10178470562974068 | 0.04210531825264487 | 0.19544821042695543 | 227 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.014168109729811322 | 0.7102480406374989 | 0.10196471848842399 | 0.04211291518558724 | 0.19665524750632657 | 387 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.013566304051021293 | 0.7108612413469275 | 0.10013658345879414 | 0.04242824290980556 | 0.1903747240267595 | 227 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.012920656665248994 | 0.7044480448677319 | 0.09791460905674015 | 0.04254383473800552 | 0.19261952491547626 | 387 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.018131802005719653 | 0.696962602628563 | 0.1006818191147876 | 0.041816327095089166 | 0.25241580681825493 | 227 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.017599385169090046 | 0.6993722018889088 | 0.10098305118548222 | 0.041853115189659264 | 0.2579362374869158 | 387 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.015725518776676408 | 0.7126907466808936 | 0.10371300271699398 | 0.042033538494353276 | 0.18719452908061188 | 227 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.015124700052216226 | 0.7092744252086509 | 0.10300534830899685 | 0.04213594514529624 | 0.1882669757894376 | 387 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.013733134369662862 | 0.7120801511415475 | 0.10246150085483194 | 0.042123920176495924 | 0.1977157129384953 | 227 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.013772953533539207 | 0.7130366710218539 | 0.10257353404991379 | 0.04212664644020615 | 0.19725568786650036 | 387 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.013579435393600233 | 0.7122043403034822 | 0.10204520856207695 | 0.0421490490958794 | 0.1981515825725644 | 227 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.013647997680894428 | 0.7145283491651163 | 0.1046781286098397 | 0.04210126464526369 | 0.19743396917061132 | 387 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.014448631854212197 | 0.7089103210051146 | 0.1020452996876571 | 0.042084502682695105 | 0.1991002665612703 | 227 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.014277796404465927 | 0.7069554013751749 | 0.10059366286982516 | 0.04213339912296164 | 0.19979800961833236 | 387 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.01387720762854093 | 0.7069435443307992 | 0.10129724054176405 | 0.04216919133342763 | 0.2021014579935475 | 227 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.013953911186051498 | 0.7088829315142469 | 0.10139834705184082 | 0.0421694583943325 | 0.20162845154182663 | 387 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.0147573489071357 | 0.7055104613913935 | 0.10205415236552297 | 0.042093596032588124 | 0.1996865549583293 | 227 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.01483800518148283 | 0.7073372066532722 | 0.10146938049351806 | 0.04210311529060147 | 0.19954091425541048 | 387 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.014329532227616718 | 0.6993863965453586 | 0.10003724025643354 | 0.04217809987527529 | 0.2035194636924032 | 227 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.014493214586064753 | 0.7045750926757572 | 0.10329096746979999 | 0.042090083831598284 | 0.20151205485712206 | 387 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.014160708015021515 | 0.6927313557905003 | 0.09383660155827155 | 0.0424255506149724 | 0.21254687609482806 | 227 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.014428444662167826 | 0.6943765277386259 | 0.09342699984190173 | 0.04248646082536413 | 0.21065354669381448 | 387 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.011659354001058168 | 0.6868231257653039 | 0.08839871698263012 | 0.042932416180718476 | 0.22946495688937932 | 227 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.011614430999814296 | 0.6849195631909915 | 0.08829439077131444 | 0.0429552904849425 | 0.2305670936856144 | 387 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.02092598569190502 | 0.6850249189848766 | 0.17489461336845935 | 0.08993222123258973 | 0.3912731321115665 | 324 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.020920109910007364 | 0.6851691204006899 | 0.1749013170619449 | 0.08993119483133467 | 0.3912619454731339 | 251 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.020948007389116254 | 0.6850020203062268 | 0.174915443345426 | 0.08992893722029052 | 0.391205244558029 | 502 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.02094473079098192 | 0.684977735654922 | 0.1749252512239274 | 0.08992918185873082 | 0.3912201996677795 | 420 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_100 | 4 | 0.02062016277345658 | 0.6845693959722269 | 0.17440344882479847 | 0.09005965891750772 | 0.39958090560889586 | 324 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.02062289944981225 | 0.6846298364319263 | 0.1746624278664859 | 0.09005873257483785 | 0.3995462004786345 | 251 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.020657575052421016 | 0.6846387146915431 | 0.17473160953560352 | 0.0900503357412691 | 0.39943407978688067 | 502 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.020645372237229613 | 0.6844954171709398 | 0.1747204660717333 | 0.09005158281856043 | 0.39950431736105546 | 420 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.020595255619658963 | 0.6787667301964393 | 0.17197183117806658 | 0.09020553238087789 | 0.40685205940181685 | 324 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.020557236666991174 | 0.680293007474691 | 0.17248006247752645 | 0.09017819503131952 | 0.4053580399718076 | 251 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.02074805272700708 | 0.6781112253902476 | 0.17162931171646645 | 0.09021237828214151 | 0.40855652743965415 | 502 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.02068824421801774 | 0.6775188124697045 | 0.17188929176809897 | 0.09020313771284853 | 0.4080239875179882 | 420 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.020580704691322747 | 0.6812181944383849 | 0.1728938513491245 | 0.0901626159474794 | 0.40409475385423416 | 324 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.02056481269688569 | 0.6816355329000583 | 0.17305577455445875 | 0.09015167621747655 | 0.4036731883043572 | 251 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.020689812436213834 | 0.6816557600661988 | 0.173201573047416 | 0.090147067692947 | 0.4040006130681324 | 502 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.020645591686585238 | 0.6810718638471885 | 0.17314565879737048 | 0.09014416213044049 | 0.40409932806297577 | 420 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.020272732832859332 | 0.6574166751231085 | 0.1687487059970162 | 0.09004149539599765 | 0.41645917672070637 | 251 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.020280774466667158 | 0.6575854323587896 | 0.16852508992867887 | 0.09004901491685179 | 0.41616400723806746 | 420 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.01872427606407177 | 0.6570070398572794 | 0.16067404539730762 | 0.09103156547167687 | 0.38820721278402476 | 251 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.01832587246787331 | 0.6562418021726104 | 0.1590897783120277 | 0.0912156137569732 | 0.3885010987304337 | 420 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.020109615328723556 | 0.6523596927656165 | 0.16682118690345143 | 0.09014663057529955 | 0.41753659288096107 | 251 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.02003962584523731 | 0.6502784137852295 | 0.1661711787493365 | 0.09018216890734741 | 0.41731865200336554 | 420 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.0188318171388381 | 0.6575997942493464 | 0.1624642094327935 | 0.09092141361164228 | 0.3883145937126263 | 251 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.018405333219342183 | 0.6554185344154686 | 0.16013626651802887 | 0.09114048727579924 | 0.38883427905425594 | 420 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.020378701506614718 | 0.6573231922718485 | 0.16920081586990307 | 0.08998117947986402 | 0.44175714260732024 | 251 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.020398313877340087 | 0.6563416323768997 | 0.16872376196585126 | 0.08998959225821454 | 0.4427520053362611 | 420 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.019471107335881022 | 0.6601739672876707 | 0.1618626751676721 | 0.09070028540621325 | 0.38838829596704944 | 251 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.019038762986363256 | 0.6622887807803437 | 0.16208279741745807 | 0.0908636357405251 | 0.38684707038212224 | 420 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.02139290825038551 | 0.6477219570930184 | 0.16588502410261258 | 0.08985078249127487 | 0.7030561336437299 | 251 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.021305480949453188 | 0.6444658553785341 | 0.16523617408529603 | 0.08987527745515428 | 0.7421745229214376 | 420 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.02028189826926895 | 0.6581171637809109 | 0.16898101058576903 | 0.09018313721395535 | 0.3977046762452934 | 251 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.0201539064128565 | 0.6615815530815877 | 0.16726305432683025 | 0.09028066817419857 | 0.39322644296867415 | 420 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.020297745479196685 | 0.670261447873006 | 0.17516907333431064 | 0.08986274835197719 | 0.4095020939465069 | 251 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.020282273368786897 | 0.6707163582455692 | 0.17586366223524785 | 0.0898449188201904 | 0.4094525850490859 | 420 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.020231835321885546 | 0.6694442763868906 | 0.17555914230715913 | 0.08986345203265722 | 0.4093635699164019 | 251 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.020200375295319255 | 0.669185661924069 | 0.17575699668132339 | 0.0898673701638504 | 0.4096368491591266 | 420 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.020648954164107367 | 0.670009652797379 | 0.1742701621251565 | 0.08982846954303418 | 0.4159979558684039 | 251 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.020718046879752733 | 0.6680408384676589 | 0.17260687747987524 | 0.0898637488800507 | 0.4174465697158981 | 420 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.0202249399254118 | 0.669148612263745 | 0.17400260044483518 | 0.08990830735516187 | 0.41785781566818214 | 251 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.02027844332727932 | 0.6690101656474344 | 0.17482295390268124 | 0.08987481493949859 | 0.4174839753631257 | 420 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.020739286147930122 | 0.671080507495782 | 0.17209484607094386 | 0.08991537878492133 | 0.415417072176877 | 251 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.020752897737139075 | 0.6715242798707036 | 0.1721197907194166 | 0.08992372882816986 | 0.41469995665687054 | 420 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.02062788294386877 | 0.6702667607681839 | 0.1719465866224948 | 0.0899595846909313 | 0.41652890771523565 | 251 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.020635261461482656 | 0.6701496962952267 | 0.17397683046295087 | 0.08989086778929953 | 0.4159206358933327 | 420 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.020897852521498932 | 0.6638391318010181 | 0.16760480070019165 | 0.0902137807851456 | 0.43491152421568996 | 251 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.020828964842174755 | 0.6626601511489552 | 0.1664713832072792 | 0.09026020119665928 | 0.4368487817349441 | 420 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.019539214346500477 | 0.6628139438972732 | 0.16999248103280476 | 0.09032617577172763 | 0.44525236449606764 | 251 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.019563156322152876 | 0.665925090949936 | 0.16809011784001068 | 0.09046642621443299 | 0.443420255340665 | 420 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.24669883299793582 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.25327216599544045 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.25615816700155847 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.24862799998663832 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.2680117089912528 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.2698008750012377 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 9 | 0.2626323330041487 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 9 | 0.26158216600015294 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 28 | 0.3987235829990823 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 15 | 0.32336787499662023 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 19 | 0.3578629580006236 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 27 | 0.3756020000000717 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.2931054170039715 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 12 | 0.290909041010309 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 13 | 0.3104401250020601 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 13 | 0.2891119580017403 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.006638999999268 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 1.9296087080001598 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.8201762910030084 | 12.0 | 50 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.7822651250025956 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.0779644590074895 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 1.928783041992574 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 0.8649247080029454 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.8266767090099165 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.077175124999485 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.032318750003469 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9286862920125714 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.7870803749974584 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 2.4503302500088466 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.0037983329966664 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.0374768329929793 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.9588463330001105 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1526222500106087 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.0947663330007344 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.0938418750010896 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.0874612090119626 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.8098725839954568 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.6614907500043046 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.6460875420016237 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7690432920062449 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.0852820839936612 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.0477382079989184 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.0539434590027668 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.0489500830008183 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.6676184999960242 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.636457166998298 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7360449590050848 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.7079532500065397 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.40405020900652744 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.40308337500027847 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.4042867079988355 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.402714666997781 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.43642516699037515 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.4379478329938138 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.429136708000442 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.4269862500077579 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.6299018749996321 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 15 | 0.5724807920050807 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 20 | 0.6587617920013145 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 34 | 0.7827682089991868 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 13 | 0.48929937499633525 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.4943158329988364 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.5346708750003017 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.49752545899536926 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 4.6748840829968685 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 4.617468042008113 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.3811335420032265 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.2337667499959935 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 4.961916416999884 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 4.526954083004966 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.3793670420127455 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.3179306670062942 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 5.833615374998772 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 5.889470709007583 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.7168443330010632 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.4761326670122799 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 7.7493074590020115 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 8.269927040993934 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.359176291996846 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.146365708002122 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.4388259159895824 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.277208667001105 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.294606625000597 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.2917280410038074 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 1.9470105840009637 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 1.9584555000037653 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 1.9570638750010403 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.9924929999979213 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.2663068330002716 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.2393542079953477 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.2333438749919878 | 3.0 | NA | NA | 0.1 | 1 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.2539608749939362 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 1.952044958001352 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 1.9706845420005266 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.8917303750058636 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.8923890840087552 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.5228153750067577 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.524598749994766 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.5311112500057789 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.5295190839970019 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 10 | 0.5795237909915159 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 10 | 0.5871603330015205 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.6004487089958275 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 10 | 0.5708318749966566 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 17 | 0.8974307500029681 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 15 | 0.8169874589948449 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 20 | 1.0319257089868188 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 19 | 1.0344204580032965 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 13 | 0.6770715000020573 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 13 | 0.6717530000023544 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.7679085840063635 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 14 | 0.6977337090065703 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 8.519675167000969 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 10.278211500000907 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 1.958414584005368 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 1.7407275839941576 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 8.592371334001655 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 7.8851327919983305 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 1.9768677499960177 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 1.681546541993157 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 11.854512334000901 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 11.825713541998994 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.9233606249908917 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.4497604589996627 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 16.18833329199697 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 18.202131416997872 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 4.27207045799878 | NA | 5 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 3.829336083988892 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.444707541988464 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.4300129579933127 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.4575702080037445 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.4964575420017354 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.118735625001136 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.087113125002361 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.06651287501154 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.0644417500006966 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.4260302920010872 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.3843648330075666 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.3811487080092775 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.4196417909988668 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.10030012500647 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.2392730000137817 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.032196333995671 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.1142694169975584 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 8 | 0.6050329160061665 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 8 | 0.6088127909897594 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 8 | 0.6184945830027573 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 8 | 0.6416613750043325 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 11 | 0.7590445420064498 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 9 | 0.7061335839971434 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 10 | 0.7392244580114493 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 9 | 0.6957526670012157 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 17 | 1.2677566669881344 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 14 | 1.1718816670036176 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 20 | 1.3601647079922259 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.4047416250105016 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 13 | 0.922788625000976 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 12 | 0.8733323749911506 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 14 | 0.9478103330038721 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 14 | 0.9592947500059381 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 12.473825958993984 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 13.222140208003111 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 2.4865932499960763 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.19769775000168 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 14.122174167001504 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 12.257453333993908 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 2.558869209009572 | 12.0 | 5 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 2.217345999990357 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 18.82239433300856 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 19.764956583996536 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.8820051250077086 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 3.284538332998636 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 31.53095349999785 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 34.0693471669947 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 6.755686124990461 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 6.604865209010313 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.5697808749973774 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.5668437499989523 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.5513914589973865 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.5623273749952205 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.1914999590080697 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.296681790990988 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.2107925830059685 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.196185583001352 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.5248233750025975 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.50184879200242 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.5039851659967098 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.5909421250107698 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.243993666008464 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.2860373750008876 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.154259542003274 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.2580092499993043 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 32796 | 771 | logistic | raw | 32796 | 771 | 0.02350896450786681 | 0.010094364946299379 | 0.8034677026601502 |
| logistic | validation | calibrated | 32796 | 771 | logistic | calibrated | 32796 | 771 | 0.02350896450786681 | 0.02696794247177296 | 0.803152955851814 |
| rf | validation | raw | 32796 | 771 | rf | raw | 32796 | 771 | 0.02350896450786681 | 0.020909705462400865 | 0.7341858206998221 |
| rf | validation | calibrated | 32796 | 771 | rf | calibrated | 32796 | 771 | 0.02350896450786681 | 0.06339831458808098 | 0.7343843321173168 |
| xgb | validation | raw | 32796 | 771 | xgb | raw | 32796 | 771 | 0.02350896450786681 | 0.015548518311444454 | 0.7760182088612273 |

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | validation | calibrated | 32796 | 771 | xgb | calibrated | 32796 | 771 | 0.02350896450786681 | 0.04355107168118394 6 | 0.7751482456859762 |
| logistic | test | raw | 40514 | 886 | logistic | raw | 40514 | 886 | 0.02186898356123809 | 0.25889728012665575 | 0.7320761410690528 |
| logistic | test | calibrated | 40514 | 886 | logistic | calibrated | 40514 | 886 | 0.02186898356123809 | 0.28088551841383247 | 0.724437537154225 |
| rf | test | raw | 40514 | 886 | rf | raw | 40514 | 886 | 0.02186898356123809 | 0.07677560336238072 | 0.735387637762569 |
| rf | test | calibrated | 40514 | 886 | rf | calibrated | 40514 | 886 | 0.02186898356123809 | 0.24310925265923605 | 0.7303252215126638 |
| xgb | test | raw | 40514 | 886 | xgb | raw | 40514 | 886 | 0.02186898356123809 | 0.1923440726637232 | 0.7023504113082366 |
| xgb | test | calibrated | 40514 | 886 | xgb | calibrated | 40514 | 886 | 0.02186898356123809 | 0.2590549501322911 | 0.7002872481572985 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.10872940508294741 | 0.02245067885644681 | 0.1031181886179106 |
| logistic | 0.1027829310926452 | 0.021929824064305263 | 0.09660349221637897 |
| rf | 0.06610097862063304 | 0.022540543749975443 | 0.10330301553162242 |
| rf | 0.062418916294538745 | 0.025085783879919124 | 0.12118582460180154 |
| xgb | 0.07384075420313325 | 0.022501716638679857 | 0.10249003415025658 |
| xgb | 0.06992091026364367 | 0.02292451509162242 | 0.10643277866462562 |
| logistic | 0.059948745395506944 | 0.1043408787001384 | 0.3502495327709984 |
| logistic | 0.05017526945700283 | 0.0988587097169352 | 0.35659931596146843 |
| rf | 0.05774421642370984 | 0.024116378370736137 | 0.1295661638667795 |
| rf | 0.04841058143937946 | 0.07551764566256154 | 0.30433527791880066 |
| xgb | 0.04606791028878641 | 0.06806441503127349 | 0.25719011723915325 |
| xgb | 0.04201166924070083 | 0.11313061685509004 | 1.9707851225390218 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 32796 | 771 | logistic | raw | 1.428150047043462 | 1.1329020919692465 | 0.01343317421694144 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 32796 | 771 | logistic | calibrated | 0.3297882199172016 | 1.1563439473452228 | 0.003776873224908229 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 32796 | 771 | rf | raw | 0.5621400286656376 | 1.1245971307551146 | 0.002599259045465947 4 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 32796 | 771 | rf | calibrated | -1.0361260486787216 | 1.0134640734635822 | 0.039919841603570684 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 32796 | 771 | xgb | raw | -0.2505289887496417 5 | 0.8089609493138094 | 0.008921193980366681 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 32796 | 771 | xgb | calibrated | -0.0493601095503287 8 | 1.2248641258129536 | 0.020042107173317135 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 40514 | 886 | logistic | raw | -3.1389770425346155 | 0.7734752398465414 | 0.23702829656541766 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 40514 | 886 | logistic | calibrated | -2.699817882783303 | 1.4260865938679124 | 0.2590165348525944 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 40514 | 886 | rf | raw | 2.593577154028027 | 2.6744983387398786 | 0.05490661980114261 4 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 40514 | 886 | rf | calibrated | -2.30161901309863 | 1.4819987446789615 | 0.221240269097998 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 40514 | 886 | xgb | raw | -2.8423660282341894 | 0.6924925731718646 | 0.1704750891024851 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 40514 | 886 | xgb | calibrated | -3.805999173862651 | 0.08919771536647365 | 0.23718596657105295 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R3_early_payoff/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 4955 | 0.12230340129338006 | 311 | 4644 | 34984 | 575 |
| logistic | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 3905 | 0.09638643431900083 | 285 | 3620 | 36008 | 601 |
| logistic | 1:1 | 1 | 1 | 0.35 | 40514 | 886 | 16985 | 0.41923779434269637 | 659 | 16326 | 23302 | 227 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 11934 | 0.29456484178308734 | 558 | 11376 | 28252 | 328 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 17293 | 0.4268401046551809 | 664 | 16629 | 22999 | 222 |
| logistic | 2:1 | 1 | 2 | 0.3298154369258028 | 40514 | 886 | 17319 | 0.42748185812311795 | 664 | 16655 | 22973 | 222 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 24764 | 0.6112454953843116 | 790 | 23974 | 15654 | 96 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 35070 | 0.8656266969442662 | 870 | 34200 | 5428 | 16 |
| logistic | 5:1 | 1 | 5 | 0.19375 | 40514 | 886 | 34262 | 0.8456829737868391 | 865 | 33397 | 6231 | 21 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 32807 | 0.8097694624080565 | 854 | 31953 | 7675 | 32 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 38698 | 0.9551759885471689 | 879 | 37819 | 1809 | 7 |
| logistic | 10:1 | 1 | 10 | 0.08878807517822422 | 40514 | 886 | 38981 | 0.9621612282174064 | 879 | 38102 | 1526 | 7 |
| rf | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 0 | 0.0 | 0 | 0 | 39628 | 886 |
| rf | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 2175 | 0.05368514587549983 | 151 | 2024 | 37604 | 735 |
| rf | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 2175 | 0.05368514587549983 | 151 | 2024 | 37604 | 735 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 0 | 0.0 | 0 | 0 | 39628 | 886 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 5664 | 0.1398035247075085 | 340 | 5324 | 34304 | 546 |
| rf | 2:1 | 1 | 2 | 0.5 | 40514 | 886 | 2175 | 0.05368514587549983 | 151 | 2024 | 37604 | 735 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 2 | 4.936565137976995e-05 | 1 | 1 | 39627 | 885 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 33895 | 0.8366243767586513 | 864 | 33031 | 6597 | 22 |
| rf | 5:1 | 1 | 5 | 0.34615384615384615 | 40514 | 886 | 5602 | 0.13827318951473563 | 336 | 5266 | 34362 | 550 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 11313 | 0.27923680702966874 | 528 | 10785 | 28843 | 358 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 39268 | 0.9692451991904033 | 883 | 38385 | 1243 | 3 |
| rf | 10:1 | 1 | 10 | 0.22772277227722773 | 40514 | 886 | 23973 | 0.5917213802636125 | 787 | 23186 | 16442 | 99 |
| xgb | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 2041 | 0.05037764723305524 | 124 | 1917 | 37711 | 762 |
| xgb | 1:1 | 1 | 1 | 0.5 | 40514 | 886 | 2070 | 0.051093449178061905 | 126 | 1944 | 37684 | 760 |
| xgb | 1:1 | 1 | 1 | 0.4285714285714286 | 40514 | 886 | 2073 | 0.05116749765513156 | 126 | 1947 | 37681 | 760 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 6746 | 0.16651034210396407 | 301 | 6445 | 33183 | 585 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 40514 | 886 | 11482 | 0.2834082045712593 | 476 | 11006 | 28622 | 410 |
| xgb | 2:1 | 1 | 2 | 0.4285714285714286 | 40514 | 886 | 2073 | 0.05116749765513156 | 126 | 1947 | 37681 | 760 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 18733 | 0.4623833736486153 | 677 | 18056 | 21572 | 209 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 40514 | 886 | 23329 | 0.5758256405193266 | 769 | 22560 | 17068 | 117 |
| xgb | 5:1 | 1 | 5 | 0.36363636363636365 | 40514 | 886 | 9344 | 0.23063632324628525 | 411 | 8933 | 30695 | 475 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 28393 | 0.7008194698129042 | 817 | 27576 | 12052 | 69 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 40514 | 886 | 36456 | 0.8998370933504467 | 873 | 35583 | 4045 | 13 |
| xgb | 10:1 | 1 | 10 | 0.12399999999999999 | 40514 | 886 | 30939 | 0.7636619440193514 | 839 | 30100 | 9528 | 47 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.3510158013544018 | 0.8828101342485112 | 0.0627648839556004 | 0.9838296914986361 | 0.11718986575148885 | 0.6489841986455982 | 5219 | 0.1288196672755097 | 886 |
| logistic | 0.3216704288939052 | 0.9086504491773494 | 0.07298335467349552 | 0.9835832718730367 | 0.09134955082265066 | 0.6783295711060948 | 4221 | 0.10418620723700449 | 886 |
| logistic | 0.7437923250564334 | 0.588018572726355 | 0.03879894024138946 | 0.9903523311657954 | 0.4119814272736449 | 0.2562076749435666 | 16553 | 0.408574813644666 | 886 |
| logistic | 0.6297968397291196 | 0.7129302513374381 | 0.04675716440422323 | 0.9885234429671099 | 0.28706974866256185 | 0.37020316027088035 | 12032 | 0.29698375870069604 | 1772 |
| logistic | 0.7494356659142212 | 0.580372463914404 | 0.03839703926444226 | 0.9904396882132552 | 0.41962753608559605 | 0.2505643340857788 | 17073 | 0.4214098830034062 | 1772 |
| logistic | 0.7494356659142212 | 0.5797163621681639 | 0.038339396039032275 | 0.9904289717611554 | 0.4202836378318361 | 0.2505643340857788 | 17099 | 0.42205163647134325 | 1772 |
| logistic | 0.891647855530474 | 0.39502372060159485 | 0.0319011468260378 | 0.993904761904762 | 0.6049762793984051 | 0.10835214446952596 | 24454 | 0.6035938194204472 | 4430 |
| logistic | 0.981941309255079 | 0.13697385686888058 | 0.02480752780153978 | 0.997060984570169 | 0.8630261431311194 | 0.01805869074492099 | 34280 | 0.846127264649257 | 4430 |
| logistic | 0.9762979683972912 | 0.1572373069546785 | 0.025246628918335182 | 0.996641074856046 | 0.8427626930453215 | 0.023702031602708805 | 33502 | 0.8269240262625265 | 4430 |
| logistic | 0.963882618510158 | 0.19367618855354798 | 0.026031029963117628 | 0.995847930452835 | 0.806323811446452 | 0.03611738148984198 | 32273 | 0.7965888334896579 | 8860 |
| logistic | 0.9920993227990971 | 0.04564954072877763 | 0.022714352162902476 | 0.9961453744493393 | 0.9543504592712224 | 0.007900677200902935 | 37889 | 0.9352075825640519 | 8860 |
| logistic | 0.9920993227990971 | 0.038508125567780355 | 0.02254944716656833 | 0.9954337899543378 | 0.9614918744322196 | 0.007900677200902935 | 38172 | 0.9421928222342894 | 8860 |
| rf | 0.0 | 1.0 | NA | 0.9781310164387619 | 0.0 | 1.0 | 886 | 0.02186898356123809 | 886 |
| rf | 0.17042889390519186 | 0.9489250025234682 | 0.06942528735632184 | 0.9808289209421216 | 0.05107499747653175 | 0.8295711060948081 | 2759 | 0.06809991607839265 | 886 |
| rf | 0.17042889390519186 | 0.9489250025234682 | 0.06942528735632184 | 0.9808289209421216 | 0.05107499747653175 | 0.8295711060948081 | 2759 | 0.06809991607839265 | 886 |
| rf | 0.0 | 1.0 | NA | 0.9781310164387619 | 0.0 | 1.0 | 1772 | 0.04373796712247618 | 1772 |
| rf | 0.3837471783295711 | 0.8656505501160795 | 0.06002824858757062 | 0.9843328550932569 | 0.13434944988392045 | 0.6162528216704289 | 6416 | 0.158365009626302 | 1772 |
| rf | 0.17042889390519186 | 0.9489250025234682 | 0.06942528735632184 | 0.9808289209421216 | 0.05107499747653175 | 0.8295711060948081 | 3494 | 0.08624179296045811 | 1772 |
| rf | 0.0011286681715575562 | 0.9999747653174523 | 0.5 | 0.9781546208530806 | 2.523468254769355e-05 | 0.9988713318284425 | 4426 | 0.10924618650343092 | 4430 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| rf | 0.9751693002257337 | 0.16647320076713434 | 0.025490485322318925 | 0.9966762350808279 | 0.8335267992328657 | 0.024830699774266364 | 33141 | 0.818013526188478 | 4430 |
| rf | 0.3792325056433409 | 0.8671141617038458 | 0.05997857907890039 | 0.9842461044912924 | 0.13288583829615425 | 0.6207674943566591 | 8016 | 0.197857530730118 | 4430 |
| rf | 0.5959367945823928 | 0.7278439487231251 | 0.046671970299655266 | 0.9877401458854149 | 0.2721560512768749 | 0.4040632054176072 | 14365 | 0.35456879103519773 | 8860 |
| rf | 0.9966139954853274 | 0.031366710406783085 | 0.022486503004991342 | 0.9975922953451043 | 0.9686332895932169 | 0.003386004514672686 | 38415 | 0.9481907488769314 | 8860 |
| rf | 0.8882618510158014 | 0.41490865044917735 | 0.03282859884036207 | 0.9940148721359047 | 0.5850913495508226 | 0.11173814898419865 | 24176 | 0.5967319938786593 | 8860 |
| xgb | 0.1399548532731377 | 0.9516251135560715 | 0.06075453209211171 | 0.9801939022171393 | 0.04837488644392854 | 0.8600451467268623 | 2679 | 0.06612529002320186 | 886 |
| xgb | 0.14221218961625282 | 0.9509437771272837 | 0.06086956521739131 | 0.9802309853293102 | 0.049056222872716264 | 0.8577878103837472 | 2704 | 0.06674236066544897 | 886 |
| xgb | 0.14221218961625282 | 0.9508680730796406 | 0.060781476121562955 | 0.9802294425223069 | 0.049131926920359344 | 0.8577878103837472 | 2707 | 0.06681640914251863 | 886 |
| xgb | 0.3397291196388262 | 0.8373624709801151 | 0.04461903350133412 | 0.9826759061833689 | 0.16263752901988493 | 0.6602708803611738 | 7615 | 0.1879597176284741 | 1772 |
| xgb | 0.5372460496613995 | 0.7222670838800848 | 0.04145619230099286 | 0.9858776522457977 | 0.2777329161199152 | 0.46275395033860045 | 11826 | 0.29189909660857977 | 1772 |
| xgb | 0.14221218961625282 | 0.9508680730796406 | 0.060781476121562955 | 0.9802294425223069 | 0.049131926920359344 | 0.8577878103837472 | 3467 | 0.08557535666683122 | 1772 |
| xgb | 0.7641083521444695 | 0.5443625719188453 | 0.03613943308599797 | 0.9904044809696525 | 0.45563742808115476 | 0.23589164785553046 | 19101 | 0.47146665350249295 | 4430 |
| xgb | 0.8679458239277652 | 0.4307055617240335 | 0.03296326460628402 | 0.9931917369799244 | 0.5692944382759665 | 0.13205417607223477 | 23145 | 0.5712840005923878 | 4430 |
| xgb | 0.463882618510158 | 0.7745785808014535 | 0.04398544520547945 | 0.9847609881296118 | 0.22542141919854647 | 0.536117381489842 | 11308 | 0.27911339290121934 | 4430 |
| xgb | 0.9221218961625283 | 0.3041283940648027 | 0.028774697989489934 | 0.9943074003795066 | 0.6958716059351974 | 0.07787810383747178 | 28266 | 0.6976847509502888 | 8860 |
| xgb | 0.9853273137697517 | 0.1020742909054204 | 0.023946675444371297 | 0.9967964514539182 | 0.8979257090945796 | 0.014672686230248307 | 35713 | 0.8814977538628622 | 8860 |
| xgb | 0.9469525959367946 | 0.24043605531442414 | 0.027117877113028863 | 0.9950913838120105 | 0.7595639446855759 | 0.05304740406320542 | 30570 | 0.7545539813397838 | 8860 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 5.890519187358916 |
| logistic | 4.764108352144469 |
| logistic | 18.682844243792324 |
| logistic | 6.790067720090294 |
| logistic | 9.634875846501128 |
| logistic | 9.649548532731377 |
| logistic | 5.520090293453725 |
| logistic | 7.738148984198646 |
| logistic | 7.562528216704289 |
| logistic | 3.64255079006772 |
| logistic | 4.276410835214447 |
| logistic | 4.308352144469526 |
| rf | 1.0 |
| rf | 3.113995485327314 |
| rf | 3.113995485327314 |
| rf | 1.0 |
| rf | 3.620767494356659 |
| rf | 1.971783295711061 |
| rf | 0.9990970654627539 |

| model | decision_relative_cost_vs_all_negative |
|---|---|
| rf | 7.481038374717833 |
| rf | 1.8094808126410835 |
| rf | 1.6213318284424378 |
| rf | 4.335778781038375 |
| rf | 2.7286681715575622 |
| xgb | 3.023702031602709 |
| xgb | 3.0519187358916477 |
| xgb | 3.0553047404063207 |
| xgb | 4.297404063205418 |
| xgb | 6.673814898419865 |
| xgb | 1.9565462753950338 |
| xgb | 4.311738148984198 |
| xgb | 5.224604966139955 |
| xgb | 2.5525959367945825 |
| xgb | 3.190293453724605 |
| xgb | 4.030812641083521 |
| xgb | 3.450338600451467 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 40514 | 886 | 0 | 0 | 39628 | 886 | 886 | 0.02186898356123809 |
| test | all_non_risky | 2:1 | 2:1 | 40514 | 886 | 0 | 0 | 39628 | 886 | 1772 | 0.04373796712247618 |
| test | all_non_risky | 5:1 | 5:1 | 40514 | 886 | 0 | 0 | 39628 | 886 | 4430 | 0.10934491780619045 |
| test | all_non_risky | 10:1 | 10:1 | 40514 | 886 | 0 | 0 | 39628 | 886 | 8860 | 0.2186898356123809 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 5323 | 15 | 0.0028179597971068945 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 23208 | 768 | 0.033092037228541885 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 121 | 1 | 0.008264462809917356 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 11862 | 102 | 0.008598887202832575 |

R4

Exclude every effective-date Code 96 exit in scheduled-payment months 1–24, regardless of prior 03–99. Code 96 is not recoded; no performance-Period substitution for missing effective dates.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.018929315796288485 | 239417 | 4532 |
| validation | 0.016670709387197072 | 45349 | 756 |
| test | 0.017360266082582947 | 49308 | 856 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 16 | 50 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_50","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47219 | 250 | 48030 | 329 | 48030 | 329 | 0.006849885488236519 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95249 | 579 | 48209 | 395 | 48209 | 395 | 0.00819349084195897 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143458 | 974 | 47388 | 1274 | 47388 | 1274 | 0.02688444331898371 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190846 | 2248 | 48571 | 2284 | 48571 | 2284 | 0.04702394432892055 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.006270666206326599 | 0.8169669360732308 | 0.028681556233063683 | 0.006737576139461493 | 0.03694690509449609 | 199 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.006262491238615629 | 0.8168907905239763 | 0.028799733150442722 | 0.006737430704034359 | 0.03694177358389302 | 137 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.006274879532945507 | 0.8158410014662639 | 0.02834084339819601 | 0.006738882623835998 | 0.03697929877967305 | 341 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.006274033919595624 5 | 0.8160549099255501 | 0.02841083298568542 | 0.006738718111619777 | 0.03697421420603719 | 266 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.004855344689347582 | 0.845886569639183 | 0.04202231899958729 | 0.00670448090567895 | 0.0354900853746717 | 199 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.004852392351538265 | 0.8455765712315487 | 0.041381375560377895 | 0.00670710087593963 | 0.035522117188495565 | 137 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.004855142228702039 | 0.8453871950203488 | 0.041700470013489206 | 0.006706476766801085 | 0.03551487842924705 | 341 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.004855656817267136 | 0.8454275298594098 | 0.04179828713448435 | 0.00670599165433008 | 0.035509575317076755 | 266 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_100 | 1 | 0.0046749471491019014 | 0.7983669678950612 | 0.03860466716294746 | 0.006721290693895305 | 0.037077630040405145 | 199 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.004600734525763426 | 0.8118116593682697 | 0.04001191336877642 | 0.006707842384212142 | 0.036545238534931815 | 137 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.004645376683576284 | 0.7919010956611757 | 0.03764706645141745 | 0.006735866844350964 | 0.037515162947187014 | 341 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.0046665488449327476 | 0.7970790567306006 | 0.03891052979057563 | 0.006729276298149414 | 0.03722228868715665 | 266 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.004639829341440588 | 0.8357681324058316 | 0.04294366801391222 | 0.006698165412447981 | 0.03569896235160137 | 199 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.004613839776042555 | 0.837693181099158 | 0.043335384489247016 | 0.006697833482482595 | 0.035638217880418335 | 137 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.004632387932887498 | 0.834821952271205 | 0.04293691670101459 | 0.006700879482698036 | 0.03574352397889598 | 341 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.004636787697321694 | 0.8353911004268039 | 0.043236864588140404 | 0.006699695121437246 | 0.03570720337504476 | 266 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004879553947741693 | 0.8402971040031595 | 0.03390729299534487 | 0.006707475637038598 | 0.035625638078972136 | 137 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.004845870926033872 | 0.8414456592544657 | 0.036168253954560366 | 0.006709036445819942 | 0.03561503340401883 | 266 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004804338010333742 | 0.8400338124470765 | 0.03653381576853916 | 0.006724752880325047 | 0.03621624791194144 | 137 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004835926083962838 | 0.8359847808304885 | 0.03637215429648166 | 0.0067357126417622185 | 0.036675320085528984 | 266 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.005016475713674366 | 0.8214328884670333 | 0.030555295116197687 | 0.006727730658654154 | 0.03654777820279125 | 137 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.004949212485942907 | 0.8194132791083566 | 0.028879642503797133 | 0.006738871059713717 | 0.0367165254088303 | 266 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.00485138629014208 | 0.8258497763646636 | 0.035267426817954334 | 0.006719567123575132 | 0.03631067130106988 | 137 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.00484854713905518 | 0.8186358935845877 | 0.03653266346441341 | 0.006729742633004086 | 0.03677420390515579 | 266 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.00490193942407169 | 0.835530169599396 | 0.034036103905065965 | 0.006709016148596164 | 0.03594676194642525 | 137 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004858748000212001 | 0.8362853805197 | 0.033141395562624465 | 0.006709749705133074 | 0.03592593170823606 | 266 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.004782785915380142 | 0.8399998496205052 | 0.03680688183952222 | 0.0067192387256788735 | 0.03598808987300818 | 137 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004796332731745088 | 0.8408343920963086 | 0.03716371871802233 | 0.006726231833418084 | 0.036209185274022136 | 266 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.005344838291365183 | 0.7955512711559576 | 0.028282978311428123 | 0.006746714663262143 | 0.050195159452741264 | 137 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.00528749803126843 | 0.7834576693510469 | 0.025176901251954452 | 0.006771390159994266 | 0.056899219746287866 | 266 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.0049248499719738185 | 0.8146286623699336 | 0.03248419074588972 | 0.006714788940889391 | 0.03707970162094752 | 137 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.004855622350582308 | 0.8121823830549326 | 0.03235094616118292 | 0.006719161197672586 | 0.036573966731328574 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.004584130008892763 | 0.8326353643252303 | 0.039345646196087006 | 0.0066698010651803136 | 0.03588006603306037 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.0046185979689383864 | 0.8327427964558103 | 0.038816696083389134 | 0.006698993448691047 | 0.03586706842567379 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.00458117555590127 | 0.8347693831681634 | 0.041523403003929815 | 0.006693579402403884 | 0.035827018475501106 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004629073550121695 | 0.8347629474355486 | 0.04238828947402075 | 0.006692936598508355 | 0.03583018326953834 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.00455605224763813 | 0.8164178278969128 | 0.03472859406602648 | 0.0067207148852028544 | 0.03675679236294298 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.004561242972436822 | 0.819491686722045 | 0.03424092862358733 | 0.006724187667948367 | 0.03668182222727818 | 266 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.0038121971821136968 | 0.8088892951400852 | 0.0340341112134161 | 0.00674324450184688 | 0.03814296195871596 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.003963597097969396 | 0.8122295996674829 | 0.03569921875144979 | 0.006735048408784212 | 0.03791848223189004 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004493954729494337 | 0.7988906517415444 | 0.030493152372895904 | 0.006757535324739982 | 0.03794330677931242 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.004545624488066476 | 0.805446050750913 | 0.03163749875413353 | 0.006750191510098843 | 0.037641353450211475 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004277648882226618 | 0.8005298838146359 | 0.035097718694518346 | 0.0067569764615608395 | 0.03851573805271821 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.00433761676289362 | 0.8072614052492256 | 0.03776168959841916 | 0.006755474186770054 | 0.0381658466128556 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004214122196075013 | 0.7678861275489564 | 0.02591146242342117 | 0.006849949208031789 | 0.04074363184413232 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.004246482423071322 | 0.7705662915823994 | 0.02576991518955452 | 0.006838212865228302 | 0.04050521315151692 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.0017309252480096655 | 0.7591855586747972 | 0.02674619410306696 | 0.0068538991851127065 | 0.04941912836528593 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.0018357219205986475 | 0.7604268904279565 | 0.026285417183263802 | 0.006827713929330274 | 0.048660665389609205 | 266 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.008541646860223358 | 0.8257293955003909 | 0.044441597613964205 | 0.007977865448701764 | 0.04169807808871722 | 263 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.008526417166868444 | 0.8259247198929608 | 0.04437372410233007 | 0.007977745210952378 | 0.04169474117102583 | 200 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.008552106742093658 | 0.8259269437000868 | 0.044854503553230854 | 0.007976690633846937 | 0.041679624039218245 | 443 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.00855115278760291 | 0.8259419808720818 | 0.04480885623457723 | 0.007976839115445025 | 0.04168078890644415 | 347 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.007752216258508998 | 0.8434374657493993 | 0.05406332718368038 | 0.007940211910137053 | 0.040489938615104866 | 263 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.007727968427866604 | 0.843830656028397 | 0.05436711647258543 | 0.00793881115341844 | 0.04047700926272815 | 200 |
| logistic | lr_c_0p1_freq_25 | 2 | 0.007761534349144307 | 0.843375993366701 | 0.054444184693283264 | 0.007938099510097198 | 0.04047348087486879 | 443 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.007760293435556034 | 0.8432496069950383 | 0.054548702551625444 | 0.007937357081965581 | 0.040468900288533796 | 347 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.007590238495511031 | 0.812697303316173 | 0.04958293755193435 | 0.007981465824690198 | 0.04181533637423297 | 263 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.007525645964775235 | 0.8233542106464236 | 0.05163718372068303 | 0.00796031759628052 | 0.04120949156876404 | 200 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.007680400735557327 | 0.8025655851021867 | 0.04495001354877128 | 0.008019942030802711 | 0.042498362527095525 | 443 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.007589021732616608 | 0.8060138627900413 | 0.04892378343546423 | 0.007987490823135202 | 0.04214773176239864 | 347 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.007601887083067827 | 0.8351213272104511 | 0.051528196648630765 | 0.007958640386586393 | 0.04084155732454498 | 263 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.007555016820173333 | 0.8375721215067033 | 0.054639323328530456 | 0.007949155537160668 | 0.04068180171169375 | 200 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.007627279239087544 | 0.8336018315699072 | 0.051622538467743374 | 0.007958230829766709 | 0.040911892130428756 | 443 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.00760436662645057 | 0.8338737714127477 | 0.05200008046756411 | 0.00795506611606771 | 0.04086590189674104 | 347 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007426658132720959 | 0.8349437932748897 | 0.04931209475182079 | 0.00794680008700356 | 0.04075513005633036 | 200 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007428669226929015 | 0.8334447354807897 | 0.048434747569188044 | 0.007949968348641473 | 0.04084427643324801 | 347 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.007230382985547111 | 0.8349023351563256 | 0.05207436506861534 | 0.007994868830305068 | 0.04171633643287963 | 200 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.007058431376285937 | 0.8311998551348501 | 0.0514291564989879 | 0.008011044558008872 | 0.042198235999787635 | 347 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.007749747149936285 | 0.8258240131988247 | 0.04256872433913841 | 0.007991257300578444 | 0.041463967496245195 | 200 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.00762138938843038 | 0.8252330099811875 | 0.04266083487710033 | 0.007991572355173567 | 0.04150185781625696 | 347 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007390281026137413 | 0.832662273059159 | 0.04952473586351426 | 0.007977363920615158 | 0.04147109060595399 | 200 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007159117520792536 | 0.831301673732549 | 0.04857100833290523 | 0.007995570693591386 | 0.04195982544731692 | 347 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007503964170228398 | 0.8306643411997863 | 0.04895269263628298 | 0.007949790889556208 | 0.041004958261818894 | 200 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.007505265410940176 | 0.8285142903434353 | 0.04835673589104983 | 0.00795161010580683 | 0.041145634986742195 | 347 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007436624530188304 | 0.836630392136618 | 0.051378019913501397 | 0.007984042521503212 | 0.04128408961666705 | 200 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.007313962359876437 | 0.8358929882831838 | 0.051722191701101525 | 0.007993742406683768 | 0.04152776118136992 | 347 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.008523777392118176 | 0.7910428755308678 | 0.03702234659937756 | 0.008072647480107208 | 0.057049647082353355 | 200 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.008518671440579597 | 0.7918487144012162 | 0.036216009517875083 | 0.008080732436965379 | 0.05933819002049952 | 347 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.00783879708955938 | 0.8190231344773233 | 0.044775467992597795 | 0.007969244324483982 | 0.041389186123143754 | 200 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.00774363980468563 | 0.8220521715741325 | 0.046327756141700496 | 0.007971358336257514 | 0.04130951549798697 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.0073014187785334595 | 0.8371245009009067 | 0.046775627439981746 | 0.007961278165567576 | 0.04081359502900109 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.00735927580422511 | 0.8371210592946402 | 0.04727081778085757 | 0.00795870193459265 | 0.040783040414401166 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.007301639305781362 | 0.83673909394685 | 0.05033463134053017 | 0.007949070830984322 | 0.04074619532557697 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.007305149442263132 | 0.8364229427004325 | 0.050087458943606264 | 0.007953535040939626 | 0.040767740079647785 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.007163516389432393 | 0.8257590992098602 | 0.039538063368915854 | 0.008022079562840427 | 0.04176739523698983 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.007201484859279873 | 0.8250550524633165 | 0.039855891412532214 | 0.008020869914491672 | 0.04179925520615564 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006850586701611529 | 0.8201782434359303 | 0.03741934831697905 | 0.008053087571362793 | 0.04224920143046266 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.006872420967902321 | 0.8214909250137533 | 0.036920204016110486 | 0.008062057140782342 | 0.04228970417494854 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.007152146290711166 | 0.8192025215854898 | 0.036527375995044235 | 0.00806440203809837 | 0.042360045543565795 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.007125351044654396 | 0.8194754674363158 | 0.03757264141859019 | 0.008046229137609929 | 0.04222327972698906 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.007098865251423799 | 0.8148241630410668 | 0.03934472670366617 | 0.008062332807192304 | 0.042584139096678006 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.007161128861744176 | 0.8183387313603929 | 0.038419138638651214 | 0.00807745339014752 | 0.04246714190807013 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.0066268641089153365 | 0.7971393103974103 | 0.0297228619115406 | 0.008208751894274577 | 0.044997692656996005 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006725080414509412 5 | 0.7931002148091789 | 0.029024572723811714 | 0.008227927572501242 | 0.04524916653428156 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.00487463298899367 | 0.7826063866681704 | 0.026554874130520642 | 0.008281933780785605 | 0.048520888066473124 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.004942544925391964 | 0.7736133900721838 | 0.025270254761714805 | 0.008302616283560741 | 0.048931253666971354 | 347 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.011172354360423515 | 0.7424575529799231 | 0.07801965072024701 | 0.025919688304903637 | 0.1218568221290443 | 311 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.011161674476753291 | 0.7425516307990796 | 0.0779684879455145 | 0.025920731015781347 | 0.12187079059228262 | 242 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.011175344251646556 | 0.7424407016969548 | 0.07802227067148901 | 0.025919473287077582 | 0.12185365447950919 | 489 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.011174461321591162 | 0.7424316462600465 | 0.0780003402194159 | 0.025919723676474854 | 0.12185702267136103 | 411 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.01085419086631657 | 0.7326544127314268 | 0.07374276213250816 | 0.02595209537614776 | 0.12526378734086344 | 311 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.010860345592481958 | 0.7326356890836844 | 0.07414106753615535 | 0.025950198606156067 | 0.1252534815369759 | 242 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 3 | 0.010854973991417289 | 0.732904407471784 | 0.07400386456909995 | 0.025948176910108887 | 0.12521957356076296 | 489 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.010851230758927145 | 0.7328426364557319 | 0.07385823721613988 | 0.025950404491290012 | 0.12524164601732105 | 411 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.010684634011872897 | 0.7085652143629578 | 0.06816138056782312 | 0.026029371299353746 | 0.1304217814552684 | 311 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.010717342642595749 | 0.7137685841565666 | 0.06949761633066837 | 0.026015301991283657 | 0.12947290477445106 | 242 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.010779426015127597 | 0.7067103817316024 | 0.06762077521958927 | 0.026039206938898853 | 0.13113514563930664 | 489 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.010745805104608342 | 0.7072244820341153 | 0.06791770909456324 | 0.026031567058093997 | 0.1308532816412593 | 411 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.010759161173226804 | 0.7239114564826001 | 0.07077378262948278 | 0.025997240422529225 | 0.12758838052496174 | 311 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.010780372228157495 | 0.7243194617884052 | 0.07114155402349656 | 0.025992784518207544 | 0.12745689535613577 | 242 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.010789479639585176 | 0.7241581490523553 | 0.07110227767888246 | 0.025991108896624267 | 0.1275847015963574 | 489 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.010773876020421395 | 0.7241346934281835 | 0.07094045134314093 | 0.025994716405099944 | 0.12759241003885619 | 411 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010389005507171376 | 0.7186768522402572 | 0.0678997119032748 | 0.026010478967338255 | 0.12805816590211072 | 242 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.0103761274143287 | 0.7181310919515618 | 0.06804453595521114 | 0.026011104729862098 | 0.12831053223983852 | 411 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.00999420952359543 | 0.707499974978398 | 0.06521090988224423 | 0.026181326097932816 | 0.1262938513809584 | 242 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009518723529980531 | 0.7006098598456667 | 0.06270365938321483 | 0.026242013693034565 | 0.1277237251458702 | 411 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011357188658972563 | 0.708017641284731 | 0.06530879440826703 | 0.026043028653276182 | 0.13018306983499212 | 242 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.010739755968777295 | 0.7060112066819049 | 0.06579266474139456 | 0.026059207125007786 | 0.13108040482118596 | 411 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.010345876161405093 | 0.70789540139722 | 0.06544829042237595 | 0.026131389124839196 | 0.12569296253653825 | 242 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.00977999758292569 | 0.7030648024086644 | 0.06419252311516092 | 0.026198956366202678 | 0.12709215177672137 | 411 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.010719198692142825 | 0.7160771588587126 | 0.06813435267921972 | 0.025989257129336273 | 0.1297799612872083 | 242 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.010680087461386064 | 0.7166670729130844 | 0.06865032476364125 | 0.025985809103601463 | 0.12924556051836714 | 411 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.01095241749944234 | 0.7100033437030568 | 0.0665875173290009 | 0.02611119151980438 | 0.12443700635948725 | 242 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010413081906731632 | 0.7087488422828171 | 0.06556703414649984 | 0.026153903341146013 | 0.12538477316444338 | 411 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.01388521658428618 | 0.6987875127431444 | 0.06617284369051726 | 0.02594838008895569 | 0.17615828653794527 | 242 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.013181139322458114 | 0.6966224462901951 | 0.0668880168328217 | 0.02595953875323261 | 0.19598261358762595 | 411 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.012397290817259871 | 0.708354760562333 | 0.0672727804479448 | 0.02595861553025718 | 0.1236477748701359 | 242 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.011979741521344994 | 0.7106339221160255 | 0.06870132155650961 | 0.025979828832463917 | 0.12361293074796809 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.010095747494739203 | 0.7206561120216101 | 0.06944002067256941 | 0.02601797998319411 | 0.12909510849690908 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.01006793253574226 | 0.722849229903177 | 0.06963589256621049 | 0.026015696947997338 | 0.12880712964387458 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.00983147908051303 | 0.7219516607841504 | 0.06996970567295074 | 0.026023052416788234 | 0.1292067783343448 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.009830496942447456 | 0.72305472193715 | 0.0701529279182438 | 0.026016227251586876 | 0.12910133584556333 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.010578607858939812 | 0.714969289132543 | 0.06726828803294246 | 0.026030286593822453 | 0.13036666307211342 | 242 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.010515217341560672 | 0.7169375019617277 | 0.06857947604488591 | 0.026022194855609748 | 0.13013408462007145 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.010108929336020617 | 0.713980476273768 | 0.06634117978091891 | 0.026081893534673918 | 0.13202230067336176 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.009858979006407658 | 0.7155131855672131 | 0.06752050187791273 | 0.026066958148275688 | 0.13214859876116689 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.010740310100855062 | 0.7128253123155507 | 0.06720270024638765 | 0.02603883437420606 | 0.13053692692698932 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.010885441084022627 | 0.714596663010222 | 0.06809142040952168 | 0.026036424511883805 | 0.13009956703137138 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.010351918062906626 | 0.7078827374027469 | 0.06581402709800499 | 0.026081850829123735 | 0.13248618459696218 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.01034632237268203 | 0.708055335391936 | 0.06602163248272491 | 0.026091930326018236 | 0.13253300383803945 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.010329194473303428 | 0.6982502376711759 | 0.06069588210162519 | 0.02632749218406981 | 0.13965244922482625 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.010193312157924113 | 0.6945927518104236 | 0.06115032109330511 | 0.026271858706022753 | 0.13998098948830542 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008474627061110305 | 0.6841140283083853 | 0.058311919000162846 | 0.02642315121169652 | 0.14945950673433742 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.008409167000593906 | 0.6866542911298454 | 0.05895382671899929 | 0.026435766666782053 | 0.1491604450003295 | 411 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.013555417796543815 | 0.6962373869541657 | 0.09922067418572565 | 0.04516958712489122 | 0.21021930626878058 | 349 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.013551324919275335 | 0.6962809645311631 | 0.09923064721018066 | 0.04516994407551032 | 0.21022198936188297 | 268 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.013561011540442383 | 0.6963350699664625 | 0.09927382483623504 | 0.04516830820041131 | 0.2101894804690456 | 512 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.013560965349213746 | 0.6963109211594136 | 0.09926522542009635 | 0.04516860799874741 | 0.2101936505485279 | 449 |
| logistic | lr_c_0p1_freq_100 | 4 | 0.01318334998866265 | 0.6955645499220446 | 0.09901926071935518 | 0.045220514124876376 | 0.21399592278466018 | 349 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.013183712370453798 | 0.6956154487589934 | 0.0990151048236858 | 0.04522159920510632 | 0.21397786300187968 | 268 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.013194493990247465 | 0.6959482823170157 | 0.09930364846485151 | 0.04521446747058598 | 0.21389333665973195 | 512 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.013192549760926948 | 0.6957751354650649 | 0.09919322880828982 | 0.045216558624638206 | 0.2139245424283304 | 449 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.013167117805494512 | 0.6880892219059513 | 0.0971049991266646 | 0.0452748322774969 | 0.21763416599621774 | 349 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.013137789045397153 | 0.6902039120348535 | 0.09721493712754868 | 0.045270448824205896 | 0.21678581852062295 | 268 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.013219072698144535 | 0.6877856260927737 | 0.09765334132809833 | 0.045268409358154495 | 0.21818161969707228 | 512 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.01319601807388251 | 0.6874193549973766 | 0.09755182220823862 | 0.045265548392832124 | 0.21803220759701333 | 449 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.013151004659612366 | 0.6915889733425548 | 0.09790655519288405 | 0.04525827702225823 | 0.2161514297959501 | 349 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.01313617860778566 | 0.6921237469247397 | 0.09780050930597306 | 0.04525842213716465 | 0.2159155472680624 | 268 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.013189270047932222 | 0.6926358662206411 | 0.098674672888803 | 0.04524241079545182 | 0.21595990480810054 | 512 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.01317516731569606 | 0.6920391173216585 | 0.09843593508498262 | 0.04524557278795316 | 0.21603857258934678 | 449 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.012894826493529073 | 0.6731828434161838 | 0.09291809295514337 | 0.04527531889645956 | 0.2217770224320034 | 268 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012872267132474705 | 0.6733582036723063 | 0.09277142894178825 | 0.04527511861469358 | 0.22186510749470495 | 449 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.012802930384718674 | 0.6784622285605036 | 0.09267560124163446 | 0.04548326155527732 | 0.20904045526633339 | 268 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.012669178694741318 | 0.6792929077952198 | 0.09285501069162179 | 0.045538818511866086 | 0.20890239788107154 | 449 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.012892338798771054 | 0.6684572917233024 | 0.09183512430001908 | 0.04530843329061235 | 0.22189981747572496 | 268 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012818923001633984 | 0.6659621089042526 | 0.09083828462069597 | 0.0453376037240326 | 0.22181976000089143 | 449 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012832453319472083 | 0.6788892831396831 | 0.09362985748240796 | 0.045457791881471 | 0.20892415905003578 | 268 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012688709425495853 | 0.6799212970230623 | 0.09361328769196288 | 0.04551944235213808 | 0.20889090333361227 | 449 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.012969357026593901 | 0.6732058949801393 | 0.0931075721043314 | 0.04525352352146675 | 0.23538305808605542 | 268 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.01294784325874989 | 0.6727614216668507 | 0.09290266211841985 | 0.04525948811429222 | 0.23500025622422233 | 449 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.012939347602062652 | 0.6844748085660785 | 0.09404596966876386 | 0.04537164659067807 | 0.20945729771497826 | 268 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012866194183011326 | 0.6843252713586219 | 0.09402070017401677 | 0.04541765577296409 | 0.2086719206722237 | 449 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.01379945436635792 | 0.6583214566227455 | 0.08905778831490066 | 0.04527152615304406 | 0.38384895489416276 | 268 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.013740868348470062 | 0.6552008452404072 | 0.08904010259112287 | 0.04526942721463879 | 0.4029952815108542 | 449 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.013213460667152623 | 0.6772909215582047 | 0.09388083623989338 | 0.04525979787606604 | 0.21499135162976696 | 268 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.01314568063957884 | 0.6789154372530755 | 0.09509438946657964 | 0.0452744455550302 | 0.21301562253260534 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012879144599806413 | 0.6881057183883225 | 0.09966387618504767 | 0.04518237511385417 | 0.21810251517841783 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.012924536446495338 | 0.6879529272875542 | 0.099420360298149 | 0.04518016459132902 | 0.21787263697019657 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.012873087172674528 | 0.6867830722405556 | 0.09894915489441858 | 0.045194871963967584 | 0.2181310934256937 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.01288777131558909 | 0.6864579241136839 | 0.09909595477806066 | 0.045190130330786095 | 0.2180879082092989 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.013021008263565982 | 0.6853781044838007 | 0.09930098052647837 | 0.045167707908217246 | 0.22146206592851267 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.01302703636235676 | 0.6858083751203231 | 0.09829613671161468 | 0.04518666354004961 | 0.22149688947766316 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.01284141740424892 | 0.6849208000476128 | 0.09761573611447096 | 0.04522104978834608 | 0.22223334395981093 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.012826338699794741 | 0.6844533981372672 | 0.09922392110284198 | 0.045199576526294576 | 0.22212310656950796 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.013068403740399089 | 0.6867598362262526 | 0.0993227488841091 | 0.04516804122137734 | 0.22125256546964478 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.01305760116358725 | 0.6872882722836735 | 0.09967017347360105 | 0.04516035998416187 | 0.22093843610495326 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.012977501858756398 | 0.6855846510371577 | 0.09920214170267719 | 0.045186607671208696 | 0.2215488493837891 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.013045851623504669 | 0.6864717957257236 | 0.09971116212508477 | 0.045167702667823835 | 0.22125829889010087 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.013040391908323755 | 0.6768885360306445 | 0.095729276572521 | 0.04528610568267213 | 0.2304468799166957 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.013041583790809056 | 0.6752365230454912 | 0.0931154986872026 | 0.04535260417161984 | 0.2309275239204826 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.012309838582318924 | 0.6730972726433799 | 0.0922097975157319 | 0.04545166683706246 | 0.236047478002364 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.01244426835185546 | 0.6760446473133417 | 0.09453853481762231 | 0.04538873590380431 | 0.2342026310953382 | 449 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.2917478749877773 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.31410995799524244 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.28337487499811687 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.28031037499022204 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.32035529099812265 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.3361486670037266 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.3105165410088375 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.31069108299561776 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 18 | 0.43435629199666437 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 23 | 0.4602152500010561 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 18 | 0.43143108300864697 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 19 | 0.4516750829934608 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.36026845799642615 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 12 | 0.3579271670023445 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 23 | 0.4066334169910988 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 20 | 0.39537749999726657 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.3693863330117892 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.2738767080008984 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9174307499924907 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.8783876250090543 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.7760512079985347 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.5933987089956645 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.0199727909930516 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.992777625011513 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.462960666991421 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.6131750409986125 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 1.0249120840016985 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.9721253339957912 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 3.1399113340012264 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.9328649999952177 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.19640170900675 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.1893425000016578 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1295096249959897 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.1187317500007339 | 3.0 | NA | NA | 0.03 | 10 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.1256887089984957 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.1117587920016376 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.7398209999955725 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.741302666006959 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.7285779999947408 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7427141670050332 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.1146280829998432 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.108149040999706 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.2125638330035144 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.1150627089955378 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.7180261670000618 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.7237809590005782 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7436524579970865 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.7397236249962589 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.4494887920009205 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.4939576250035316 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.47124991699820384 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.44501020800089464 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.5085905419982737 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.5313347089977469 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.5120077920000767 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.5091502080031205 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.7795020839985227 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7378575409966288 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 21 | 0.8575232089933706 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 18 | 0.7981278340012068 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 13 | 0.6060364580043824 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.6221610419888748 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.6144076670025242 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.6271295829938026 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.5943354579940205 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.426982666002004 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.5603026669996325 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.3829240829945775 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 6.169783207995351 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 6.071031834013411 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.7613988750090357 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.5437738749897107 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 7.432494167005643 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 7.1823365410091355 | NA | 50 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.204253999996581 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.8547890829941025 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 10.308232333001797 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 11.021373750001658 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.913093665993074 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.7003955420077546 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.370077166997362 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.3540405419917079 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.3231283339991933 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.3320096249954076 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 2.0095261669921456 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 2.015161333998549 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 2.0444779160025064 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.9888677499984624 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3550092920049792 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.3440870839986019 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.3254793750093086 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.323746709007537 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 2.084143874992151 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 2.0053739580034744 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.9967150409938768 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.9911830000055488 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.6263371669920161 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.6502448749961331 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6443997909955215 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6180429999949411 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 11 | 0.7311520419898443 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 10 | 0.7145395830011694 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.7721124999952735 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.7751825830055168 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 18 | 1.1022333330038236 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.1202709580102237 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 20 | 1.2722047080023913 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 19 | 1.2638896249991376 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 14 | 0.8878088329947786 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 13 | 0.8574379579949891 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.9286617080069846 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 15 | 0.8854222919908352 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 10.568554333003704 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 10.77836437500082 | 12.0 | 50 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.366634249992785 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.0544179169955896 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 10.935528041009093 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 10.466028333001304 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.765359334007371 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.2333682919997955 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 15.511981083007413 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 15.71706254100718 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.5115543339925352 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 3.1296655000041937 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 22.016628375000437 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 24.084160290993168 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.504803499992704 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 5.092505708002136 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.566580749989953 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.573827333006193 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.5635629589960445 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.6586861249961657 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.2539407079893863 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.2930379169993103 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.2022616249887506 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.2203723330021603 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.5838174170057755 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.5497601249953732 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.548074040998472 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.5215073340077652 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.268760625011055 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.2453071249910863 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.310606832994381 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.2314740420115413 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 9 | 0.8877373329887632 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 9 | 0.8431981670000823 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 9 | 0.832086916998378 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 9 | 0.8493359589920146 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 11 | 1.0155310409900267 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 0.9536309590039309 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 1.006423665996408 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 11 | 1.0311150830093538 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 18 | 1.574763750002603 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 16 | 1.5637695839977823 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 22 | 1.8683298340038164 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 22 | 1.80725933299982 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 15 | 1.3807695830037119 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 14 | 1.292744791993755 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.3226894160034135 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 16 | 1.340706416987814 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 18.437358000010136 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 19.029609707999043 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.4513000000006286 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.8058391670056153 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 17.606644582992885 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 18.233546791001572 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 3.767314124997938 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 3.046822250005789 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 29.36472449998837 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 28.916927290993044 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.816379792013322 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 4.971817291996558 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 48.578570707992185 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 53.61774424999021 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 10.879047584006912 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.83719266699336 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.7860894999903394 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.8061361250001937 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.7838845829974161 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.8313988750014687 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.526094542001374 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.6468491249979706 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.5121733749983832 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.4722142920072656 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.7832749160006642 | 3.0 | NA | NA | 0.1 | 10 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7976948750001611 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.758106334003969 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.7587987500010058 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.5147304580023047 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.522370792008587 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.487024541987921 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.479599499987671 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45349 | 756 | logistic | raw | 45349 | 756 | 0.016670709387197072 | 0.009255052589132416 | 0.7730351775381265 |
| logistic | validation | calibrated | 45349 | 756 | logistic | calibrated | 45349 | 756 | 0.016670709387197072 | 0.021534254347790594 | 0.772928584420859 |
| rf | validation | raw | 45349 | 756 | rf | raw | 45349 | 756 | 0.016670709387197072 | 0.014636745985956887 | 0.7338040160288047 |
| rf | validation | calibrated | 45349 | 756 | rf | calibrated | 45349 | 756 | 0.016670709387197072 | 0.03646847703800124 | 0.733541100182165 |
| xgb | validation | raw | 45349 | 756 | xgb | raw | 45349 | 756 | 0.016670709387197072 | 0.011939073994976655 | 0.7584315051938895 |
| xgb | validation | calibrated | 45349 | 756 | xgb | calibrated | 45349 | 756 | 0.016670709387197072 | 0.030417563362989737 | 0.7582362945900946 |
| logistic | test | raw | 49308 | 856 | logistic | raw | 49308 | 856 | 0.017360266082582947 | 0.11035294697776564 | 0.7583232002999789 |
| logistic | test | calibrated | 49308 | 856 | logistic | calibrated | 49308 | 856 | 0.017360266082582947 | 0.1361782344688166 | 0.7542517751454181 |
| rf | test | raw | 49308 | 856 | rf | raw | 49308 | 856 | 0.017360266082582947 | 0.04684446554191703 | 0.7833790943305678 |
| rf | test | calibrated | 49308 | 856 | rf | calibrated | 49308 | 856 | 0.017360266082582947 | 0.1252096684004284 | 0.7784328752765044 |
| xgb | test | raw | 49308 | 856 | xgb | raw | 49308 | 856 | 0.017360266082582947 | 0.07855418817361254 | 0.7724029528983689 |
| xgb | test | calibrated | 49308 | 856 | xgb | calibrated | 49308 | 856 | 0.017360266082582947 | 0.10542460199126878 | 0.7703513994194853 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.060568151438870024 | 0.016148722724572902 | 0.07889594722767632 |
| logistic | 0.05812797625624569 | 0.016050593185993833 | 0.07697904026686594 |
| rf | 0.04852228236543319 | 0.016176093719277897 | 0.07907687876237712 |
| rf | 0.04541980257637707 | 0.016712439447772925 | 0.0859607741360717 |
| xgb | 0.049181219991598746 | 0.016168623944659614 | 0.07879475135125434 |
| xgb | 0.04787888319562756 | 0.01637165019890272 | 0.08194776834610179 |
| logistic | 0.05416177263184878 | 0.03168888165234993 | 0.15057525878102712 |
| logistic | 0.04564896558259901 | 0.03406563705219769 | 0.17507628527394145 |
| rf | 0.05699281028512053 | 0.017598787023911523 | 0.09505269842682544 |
| rf | 0.04910156723534144 | 0.03513704763014563 | 0.43272852248671284 |

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| xgb | 0.052128994452963484 | 0.023526371258256743 | 0.11861272271901935 |
| xgb | 0.04833147634689907 | 0.031141068315321152 | 0.35882121406354117 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45349 | 756 | logistic | raw | 0.7493955667172744 | 1.03358370334762 | 0.007415656798064 6565 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 45349 | 756 | logistic | calibrated | 0.05074054358576926 | 1.0946544039782657 | 0.004877023448654 038 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 45349 | 756 | rf | raw | 0.8657989530298572 | 1.1872921924067534 | 0.002033963401240 185 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 45349 | 756 | rf | calibrated | -0.46550166582744046 | 1.119614123108489 | 0.019807121590333 068 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 45349 | 756 | xgb | raw | -0.3196114660532633 | 0.8307182201192617 | 0.005240333709956 32 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 45349 | 756 | xgb | calibrated | 0.21820046011154762 | 1.2703803089141446 | 0.013746853975792 669 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 49308 | 856 | logistic | raw | -2.1135630734147046 | 0.9662319535608062 | 0.0929926808951827 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 49308 | 856 | logistic | calibrated | -1.3874006694147356 | 1.5575464215991475 | 0.11881796838623365 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 49308 | 856 | rf | raw | 4.332579608970688 | 2.9022437618520645 | 0.02948419945933408 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 49308 | 856 | rf | calibrated | -3.811996362566195 | 0.13403959544036123 | 0.10784940231784547 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 49308 | 856 | xgb | raw | -1.5932539606900815 | 1.0272160538324513 | 0.061193922091029584 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 49308 | 856 | xgb | calibrated | -3.7393096430538564 | 0.15575048689437193 | 0.08806433590868581 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R4_code96/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 303 | 0.006145047456802142 | 31 | 272 | 48180 | 825 |
| logistic | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 0 | 0.0 | 0 | 0 | 48452 | 856 |
| logistic | 1:1 | 1 | 1 | 0.24719101123595508 | 49308 | 856 | 4188 | 0.0849355074227306 | 271 | 3917 | 44535 | 585 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 1897 | 0.03847245883021011 | 147 | 1750 | 46702 | 709 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 0 | 0.0 | 0 | 0 | 48452 | 856 |
| logistic | 2:1 | 1 | 2 | 0.24719101123595505 | 49308 | 856 | 8573 | 0.17386630972661637 | 420 | 8153 | 40299 | 436 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 10416 | 0.21124361158432708 | 480 | 9936 | 38516 | 376 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 9035 | 0.18323598604688895 | 435 | 8600 | 39852 | 421 |
| logistic | 5:1 | 1 | 5 | 0.16326530612244897 | 49308 | 856 | 12218 | 0.2477894053703253 | 532 | 11686 | 36766 | 324 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 22082 | 0.4478380790135475 | 693 | 21389 | 27063 | 163 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 35069 | 0.7112233308996512 | 810 | 34259 | 14193 | 46 |
| logistic | 10:1 | 1 | 10 | 0.09322033898305085 | 49308 | 856 | 35069 | 0.7112233308996512 | 810 | 34259 | 14193 | 46 |
| rf | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 0 | 0.0 | 0 | 0 | 48452 | 856 |
| rf | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 499 | 0.010120061653281415 | 40 | 459 | 47993 | 816 |
| rf | 1:1 | 1 | 1 | 0.7879040981520614 | 49308 | 856 | 440 | 0.00892350125740245 | 34 | 406 | 48046 | 822 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 0 | 0.0 | 0 | 0 | 48452 | 856 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 523 | 0.010606798085503367 | 40 | 483 | 47969 | 816 |
| rf | 2:1 | 1 | 2 | 0.7879040981520614 | 49308 | 856 | 440 | 0.00892350125740245 | 34 | 406 | 48046 | 822 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 0 | 0.0 | 0 | 0 | 48452 | 856 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 8101 | 0.16429382655958466 | 450 | 7651 | 40801 | 406 |
| rf | 5:1 | 1 | 5 | 0.7879040981520614 | 49308 | 856 | 440 | 0.00892350125740245 | 34 | 406 | 48046 | 822 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 789 | 0.016001460209296665 | 56 | 733 | 47719 | 800 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 29968 | 0.6077715583678105 | 799 | 29169 | 19283 | 57 |
| rf | 10:1 | 1 | 10 | 0.1743119266055046 | 49308 | 856 | 8084 | 0.1639490549200941 | 450 | 7634 | 40818 | 406 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 68 | 0.0013790865579621967 | 6 | 62 | 48390 | 850 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49308 | 856 | 430 | 0.008720694410643303 | 37 | 393 | 48059 | 819 |
| xgb | 1:1 | 1 | 1 | 0.26530612244897966 | 49308 | 856 | 478 | 0.009694167275087207 | 42 | 436 | 48016 | 814 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 601 | 0.012188691490224 71 | 52 | 549 | 47903 | 804 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49308 | 856 | 463 | 0.009389570049484847 | 39 | 424 | 48028 | 817 |
| xgb | 2:1 | 1 | 2 | 0.26530612244897966 | 49308 | 856 | 478 | 0.009694167275087207 | 42 | 436 | 48016 | 814 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 5295 | 0.10738622535896812 | 308 | 4987 | 43465 | 548 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49308 | 856 | 6609 | 0.13403504502311997 | 366 | 6243 | 42209 | 490 |
| xgb | 5:1 | 1 | 5 | 0.26530612244897966 | 49308 | 856 | 478 | 0.009694167275087207 | 42 | 436 | 48016 | 814 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 15014 | 0.3044941997241827 | 603 | 14411 | 34041 | 253 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49308 | 856 | 28269 | 0.5733146751034315 | 784 | 27485 | 20967 | 72 |
| xgb | 10:1 | 1 | 10 | 0.16161616161616163 | 49308 | 856 | 7605 | 0.15423460696033098 | 393 | 7212 | 41240 | 463 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.036214953271028034 | 0.9943861966482291 | 0.10231023102310231 | 0.9831649831649831 | 0.005613803351770825 | 0.9637850467289719 | 1097 | 0.02224791108947838 | 856 |
| logistic | 0.0 | 1.0 | NA | 0.9826397339174171 | 0.0 | 1.0 | 856 | 0.017360266082582947 | 856 |
| logistic | 0.316588785046729 | 0.9191571039379179 | 0.06470869149952245 | 0.9870345744680851 | 0.08084289606208206 | 0.6834112149532711 | 4502 | 0.09130364241096779 | 856 |
| logistic | 0.17172897196261683 | 0.9638817799058862 | 0.07749077490774908 | 0.9850456645082365 | 0.036118220094113765 | 0.8282710280373832 | 3168 | 0.06424920905329765 | 1712 |
| logistic | 0.0 | 1.0 | NA | 0.9826397339174171 | 0.0 | 1.0 | 1712 | 0.03472053216516589 | 1712 |
| logistic | 0.49065420560747663 | 0.8317303723272517 | 0.04899101831330923 | 0.9892966736221922 | 0.16826962767274828 | 0.5093457943925234 | 9025 | 0.1830331792001298 | 1712 |
| logistic | 0.5607476635514018 | 0.7949310657970775 | 0.04608294930875576 | 0.990332201995269 | 0.2050689342029225 | 0.4392523364485981 | 11816 | 0.2396365701306076 | 4280 |
| logistic | 0.508177570093458 | 0.8225047469660696 | 0.04814609850581074 | 0.9895463461872719 | 0.1774952530339305 | 0.49182242990654207 | 10705 | 0.21710472945566642 | 4280 |
| logistic | 0.6214953271028038 | 0.7588128457029638 | 0.04354231461777705 | 0.9912644917767592 | 0.24118715429703624 | 0.37850467289719625 | 13306 | 0.26985479029772047 | 4280 |
| logistic | 0.8095794392523364 | 0.5585527945182861 | 0.03138302689973734 | 0.9940130757364284 | 0.44144720548171384 | 0.19042056074766356 | 23019 | 0.46684108055487955 | 8560 |
| logistic | 0.9462616822429907 | 0.29292908445471805 | 0.023093732242151187 | 0.9967694360559028 | 0.7070709155452819 | 0.05373831775700934 | 34719 | 0.7041250912630811 | 8560 |
| logistic | 0.9462616822429907 | 0.29292908445471805 | 0.023093732242151187 | 0.9967694360559028 | 0.7070709155452819 | 0.05373831775700934 | 34719 | 0.7041250912630811 | 8560 |
| rf | 0.0 | 1.0 | NA | 0.9826397339174171 | 0.0 | 1.0 | 856 | 0.017360266082582947 | 856 |
| rf | 0.04672897196261682 | 0.9905267068438868 | 0.08016032064128256 | 0.9832817718043804 | 0.009473293156113267 | 0.9532710280373832 | 1275 | 0.02585787296179119 | 856 |
| rf | 0.0397196261682243 | 0.9916205729381656 | 0.07727272727272727 | 0.9831791765572563 | 0.008379427061834394 | 0.9602803738317757 | 1228 | 0.0249046807820232 | 856 |
| rf | 0.0 | 1.0 | NA | 0.9826397339174171 | 0.0 | 1.0 | 1712 | 0.03472053216516589 | 1712 |
| rf | 0.04672897196261682 | 0.9900313712540246 | 0.07648183556405354 | 0.9832735471968843 | 0.009968628745975398 | 0.9532710280373832 | 2115 | 0.042893648089559505 | 1712 |
| rf | 0.0397196261682243 | 0.9916205729381656 | 0.07727272727272727 | 0.9831791765572563 | 0.008379427061834394 | 0.9602803738317757 | 2050 | 0.04157540358562505 | 1712 |
| rf | 0.0 | 1.0 | NA | 0.9826397339174171 | 0.0 | 1.0 | 4280 | 0.08680133041291474 | 4280 |
| rf | 0.5257009345794392 | 0.8420911417485346 | 0.05554869769164301 | 0.990147305069527 | 0.15790885825146536 | 0.4742990654205608 | 9681 | 0.19633730834752983 | 4280 |
| rf | 0.0397196261682243 | 0.9916205729381656 | 0.07727272727272727 | 0.9831791765572563 | 0.008379427061834394 | 0.9602803738317757 | 4516 | 0.0915875719964306 | 4280 |
| rf | 0.06542056074766354 | 0.984871625526294 | 0.07097591888466413 | 0.9835116140068839 | 0.015128374473705936 | 0.9345794392523364 | 8733 | 0.17711121927476273 | 8560 |
| rf | 0.9334112149532711 | 0.39798150747131184 | 0.026661772557394554 | 0.997052740434333 | 0.6020184925286882 | 0.06658878504672897 | 29739 | 0.603127281577026 | 8560 |
| rf | 0.5257009345794392 | 0.8424420044580203 | 0.05566551212271153 | 0.990151368135067 | 0.1575579955419797 | 0.4742990654205608 | 11694 | 0.23716232660014602 | 8560 |
| xgb | 0.007009345794392523 | 0.9987203830595228 | 0.08823529411764706 | 0.9827376116978067 | 0.001279616940477173 3 | 0.9929906542056075 | 912 | 0.01849598442443417 | 856 |
| xgb | 0.04322429906542056 | 0.9918888797160076 | 0.08604651162790698 | 0.9832439952534883 | 0.008111120283992405 | 0.9567757009345794 | 1212 | 0.024580189827208568 | 856 |
| xgb | 0.04906542056074766 | 0.991001403450838 | 0.08786610878661087 | 0.983329920131067 | 0.008998596549162057 | 0.9509345794392523 | 1250 | 0.025350855844893324 | 856 |
| xgb | 0.06074766355140187 | 0.9886691983819037 | 0.08652246256239601 | 0.9834931324039666 | 0.01133080161809626 | 0.9392523364485982 | 2157 | 0.04374543684594792 | 1712 |
| xgb | 0.0455607476635514 | 0.991249071245769 | 0.08423326133909287 | 0.983273620636708 | 0.008750928754230992 | 0.9544392523364486 | 2058 | 0.041737649063032366 | 1712 |
| xgb | 0.04906542056074766 | 0.991001403450838 | 0.08786610878661087 | 0.983329920131067 | 0.008998596549162057 | 0.9509345794392523 | 2064 | 0.04185933317108786 | 1712 |
| xgb | 0.3598130841121495 | 0.8970733922232312 | 0.05816808309726157 | 0.9875491332106423 | 0.10292660777676876 | 0.6401869158878505 | 7727 | 0.15670885049079256 | 4280 |
| xgb | 0.42757009345794394 | 0.871150829687113 | 0.05537902859736723 | 0.9885243214126794 | 0.12884917031288698 | 0.572429906542056 | 8693 | 0.17629999188772613 | 4280 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| xgb | 0.04906542056074766 | 0.991001403450838 | 0.08786610878661087 | 0.983329920131067 | 0.008998596549162057 | 0.9509345794392523 | 4506 | 0.09138476514967145 | 4280 |
| xgb | 0.7044392523364486 | 0.7025716172707009 | 0.04016251498601305 | 0.9926226162010847 | 0.2974283827292991 | 0.2955607476635514 | 16941 | 0.3435750790946702 | 8560 |
| xgb | 0.9158878504672897 | 0.4327375546933047 | 0.02773355972973929 | 0.9965777841152146 | 0.5672624453066953 | 0.08411214953271028 | 28205 | 0.572016711284173 | 8560 |
| xgb | 0.4591121495327103 | 0.8511516552464294 | 0.05167652859960552 4 | 0.9888976812219744 | 0.14884834475357053 | 0.5408878504672897 | 11842 | 0.2401638679321814 | 8560 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 1.2815420560747663 |
| logistic | 1.0 |
| logistic | 5.259345794392523 |
| logistic | 1.8504672897196262 |
| logistic | 1.0 |
| logistic | 5.27161214953271 |
| logistic | 2.7607476635514017 |
| logistic | 2.5011682242990654 |
| logistic | 3.108878504672897 |
| logistic | 2.6891355140186914 |
| logistic | 4.055957943925233 |
| logistic | 4.055957943925233 |
| rf | 1.0 |
| rf | 1.4894859813084111 |
| rf | 1.4345794392523366 |
| rf | 1.0 |
| rf | 1.2353971962616823 |
| rf | 1.1974299065420562 |
| rf | 1.0 |
| rf | 2.2619158878504675 |
| rf | 1.0551401869158878 |
| rf | 1.0202102803738318 |
| rf | 3.474182242990654 |
| rf | 1.3661214953271028 |
| xgb | 1.0654205607476634 |
| xgb | 1.4158878504672898 |
| xgb | 1.4602803738317758 |
| xgb | 1.2599299065420562 |
| xgb | 1.2021028037383177 |
| xgb | 1.205607476635514 |
| xgb | 1.8053738317757009 |
| xgb | 2.03107476635514 |
| xgb | 1.052803738317757 |
| xgb | 1.979088785046729 |
| xgb | 3.294976635514019 |
| xgb | 1.3834112149532711 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 49308 | 856 | 0 | 0 | 48452 | 856 | 856 | 0.017360266082582947 |
| test | all_non_risky | 2:1 | 2:1 | 49308 | 856 | 0 | 0 | 48452 | 856 | 1712 | 0.03472053216516589 |
| test | all_non_risky | 5:1 | 5:1 | 49308 | 856 | 0 | 0 | 48452 | 856 | 4280 | 0.08680133041291474 |
| test | all_non_risky | 10:1 | 10:1 | 49308 | 856 | 0 | 0 | 48452 | 856 | 8560 | 0.17360266082582948 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 38610 | 362 | 0.009375809375809376 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 4946 | 307 | 0.06207035988677719 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 1663 | 59 | 0.035478051713770296 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 4089 | 128 | 0.031303497187576426 |

## R5

Exclude First Payment Dates at least four months after the end of the Loan Identifier quarter, using the frozen unusual-FPD flag.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.019022034763542636 | 238986 | 4546 |
| validation | 0.017000727657604024 | 45351 | 771 |
| test | 0.017963950447071227 | 49321 | 886 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_200","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47132 | 252 | 47946 | 328 | 47946 | 328 | 0.006841029491511283 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.008229768485805727 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95078 | 580 | 48118 | 396 | 48118 | 396 | 0.00822976848580572 7 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143196 | 976 | 47293 | 1279 | 47293 | 1279 | 0.02704417144186243 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190489 | 2255 | 48497 | 2291 | 48497 | 2291 | 0.04724003546611131 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.006308882919173336 | 0.8186580653554866 | 0.029114637091527392 | 0.006727805331005335 | 0.036855211431392254 | 199 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.006300788386453577 | 0.8187366250106283 | 0.029219626866692557 | 0.006727620759836755 | 0.03684756089044953 | 137 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.00631294808388942 | 0.8174393342751102 | 0.028776917756002572 | 0.0067290325901340235 | 0.03688680793838019 | 340 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.006312113088506149 | 0.8176501072047974 | 0.02882126352431755 | 0.006728923065497951 | 0.03688267989627994 | 266 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.004888103137662224 | 0.8465738258436808 | 0.042193889978449285 | 0.006695588234725159 | 0.03540933472987566 | 199 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.004885177364905702 | 0.8461778903038306 | 0.041609936032362516 | 0.006698077520489612 5 | 0.035440903450617844 | 137 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.004887490598216257 | 0.8460582260858519 | 0.04192451098232818 | 0.006697319176947376 | 0.035430985433694974 | 340 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.004888491028336963 | 0.846098626364902 | 0.04205039000331207 | 0.006696955300208681 | 0.03542721901984535 | 266 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.004701167439610086 | 0.7985391105433588 | 0.038502221636242714 | 0.006713689732973712 | 0.03702443536765124 | 199 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.004627617592804201 | 0.8117925149231333 | 0.03990993077998426 | 0.00669983393846576 | 0.03649442242998946 | 137 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.004662071364338405 | 0.7920259581076637 | 0.03710174056735375 | 0.00672828058629957 | 0.03746066911618615 | 340 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.004686753648468882 | 0.7972515517292601 | 0.03886105560369009 | 0.006722056157628985 | 0.03716967116835017 | 266 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.004667583006538635 | 0.8360253193862948 | 0.0429777203425852 | 0.006690028340594557 | 0.03563565539637752 | 199 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.004642034556113798 | 0.8378291182162105 | 0.04338742338101584 | 0.006689469564685805 | 0.03557711959411866 | 137 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_25 | 1 | 0.004657771811049142 | 0.8351732640557117 | 0.0429609732055143 | 0.006692562292135554 5 | 0.03567448731491392 | 340 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.004663546705913502 | 0.8356987237865574 | 0.043153801161200836 | 0.006691632236853995 | 0.035641634642464615 | 266 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004897702773861737 | 0.8408412119212965 | 0.034129612505686656 | 0.006701167530868714 5 | 0.03560194847738843 | 137 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.00488427294161219 | 0.8382346576258825 | 0.03397412462297109 | 0.006702731306562007 5 | 0.03569839584423613 | 266 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004841276524004202 | 0.8393803992956138 | 0.03603995475225884 | 0.006716442892514047 5 | 0.03618557665433337 | 137 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004876918853918622 5 | 0.8349218987695778 | 0.03727805580083855 | 0.006725042171214951 | 0.03660779367282581 | 266 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.005065713044175753 | 0.8244092467595263 | 0.03089563676625869 | 0.006717610177154592 | 0.03633231358708842 | 137 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.004982711018252465 | 0.8242596824935027 | 0.028936567283016242 | 0.006729917659556392 | 0.036552019700977566 | 266 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.004855194559259276 | 0.8263389843357042 | 0.03493041806613683 | 0.006711416302624367 | 0.03627150245315005 | 137 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.004901972826051026 | 0.8250455991739135 | 0.03788580359240311 | 0.006718572213841819 | 0.036565542268566606 | 266 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.004904861754682331 | 0.8375963844375307 | 0.0338037189880038 | 0.006701007933222946 | 0.03582471756104807 | 137 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004900650660738736 | 0.8312202792241916 | 0.03336414023625714 | 0.006703081323026380 4 | 0.03611368573277551 | 266 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.004812170031468604 | 0.8404512948065346 | 0.03594039944765497 | 0.006711284796038246 | 0.0359397449532758 | 137 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004844793684144809 | 0.8386426940417079 | 0.036560585705050726 | 0.006717352755442797 | 0.0361857435088735 | 266 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.005408040202146899 | 0.7958756373128014 | 0.028129955270237584 | 0.006739648622163646 | 0.04950374436492175 | 137 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.00528124489086597 | 0.7739208707713521 | 0.024193330256445975 | 0.006769927602281972 | 0.06022209003984057 | 266 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.004958081348426083 | 0.8170752835830681 | 0.0324376195075863 | 0.006706406526381139 | 0.03635196966046495 | 137 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.004900470401593285 | 0.8119477134594523 | 0.03333337786264303 | 0.006709331328906136 5 | 0.03651074650786221 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.004662897233922012 | 0.8326566660076278 | 0.03838127452914442 | 0.006690887555951783 | 0.035829461044386224 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.00464223726191684 | 0.8348052437641432 | 0.038983296682422154 | 0.006689834076999131 | 0.035747291478346856 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.00460532030111051 | 0.8350191539579724 | 0.04112762664239875 | 0.006685288981572326 | 0.03580223131756479 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004602308648944908 | 0.8367583827697869 | 0.043959705823708874 | 0.00668178802107335 | 0.035731382320184205 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.004608376300880976 | 0.8146595901939111 | 0.03524809384421805 | 0.006712110725072927 | 0.03677426767839137 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.004605040798423743 | 0.8181222974710322 | 0.034706759057831105 | 0.006712935678165247 | 0.036653903914082755 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.003828419278112116 2 | 0.8062542833259406 | 0.03332535756518824 | 0.00674027724578054 | 0.038281690501988874 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.003941321186543364 | 0.8084257823184305 | 0.034979406355246725 | 0.006727913027119783 | 0.03805590890558558 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004623005538508704 | 0.798271738807522 | 0.03197048378175017 | 0.006739435258160048 | 0.03788216096685058 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.004608069926838536 | 0.8018837542474716 | 0.0323267827867914 | 0.006735962859585237 | 0.037672711330588575 | 266 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004312966737282621 | 0.8022342954959644 | 0.035717809261841406 | 0.006741182659342387 | 0.03833964308356951 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.004278443598569051 | 0.807173277629181 | 0.03781997624072914 | 0.006729184772150704 5 | 0.03808038634630503 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004251148494825747 | 0.7628885533652472 | 0.024807980820525433 | 0.006848081679992826 | 0.04102750312328588 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.004303061956206266 5 | 0.7649066145308855 | 0.02533043346278853 | 0.006831931566856054 | 0.040758805410161844 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.001705585221635051 4 | 0.7536365373208943 | 0.026952645431714967 | 0.006848488885649978 5 | 0.04962267949318494 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.001771249015419410 2 | 0.7579321242018544 | 0.02403263503231174 | 0.006841473068485104 6 | 0.04919909834501032 | 266 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.008579274940399402 | 0.82739759821085 | 0.04477780481996976 | 0.00801184324895287 | 0.04179842249381385 | 263 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.008564371757081175 | 0.8276209562199253 | 0.044750762455125256 | 0.008011726389178042 | 0.041794052083719196 | 200 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.008589253729495906 | 0.8275371374361358 | 0.04520449101418052 | 0.008010706078772427 | 0.0417821242508319 | 442 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.008588092438104815 | 0.8275544409350621 | 0.04513502418480045 | 0.00801089164233193 | 0.04178365545834803 | 345 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.00779937765421259 | 0.8447282429931944 | 0.05427017891013081 | 0.00797409438952673 | 0.040589863978166184 | 263 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.007777386492161357 | 0.8451909925286983 | 0.054493945825141506 | 0.00797301311505691 | 0.04057777968000204 | 200 |
| logistic | lr_c_0p1_freq_25 | 2 | 0.007808764936001498 | 0.8446722579722035 | 0.05472301014082035 | 0.00797194493769961 | 0.04057369861361709 | 442 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.007805793892221627 | 0.8445686486422417 | 0.054809926884114846 | 0.007971400719955963 | 0.04057127414546204 | 345 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.007633961867815773 | 0.8138359412404927 | 0.04995858003425749 | 0.008013910075109086 | 0.04191724816677282 | 263 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.007577339541719759 | 0.8245007702438238 | 0.05216373382592143 | 0.00799533916216105 | 0.041325838596654396 | 200 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.007727998202426692 | 0.8042513903123265 | 0.045324039686120685 | 0.008053477989681418 | 0.042572385867500666 | 442 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.007629355778972563 5 | 0.8075921826707628 | 0.049237023470403145 | 0.008020121245010153 | 0.04221875811019995 | 345 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.007648391825540977 | 0.8362210068498573 | 0.052232094271198126 | 0.00799191988870603 | 0.04094516614782127 | 263 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.007607356017046278 5 | 0.838707736600742 | 0.05469637462608831 | 0.007983963175793148 | 0.04079230375191986 | 200 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.007673785388981641 | 0.8348870499555718 | 0.05230185891549448 | 0.007991644886260968 | 0.04100912369965281 | 442 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.007647740225658912 | 0.835191475121696 | 0.05266069249959219 | 0.00798856766583308 | 0.04096331777071498 | 345 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007451845430212616 | 0.8349382725456655 | 0.05135568250440767 | 0.007975910847834956 | 0.040842368000156934 | 200 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007450925964961657 | 0.8351841198117549 | 0.05063423160750215 | 0.007978578196317464 | 0.040885684030026216 | 345 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.007265973511627482 | 0.835633217045354 | 0.050837653918220746 | 0.008028706052456314 | 0.04182983737789444 | 200 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.007085959684615398 | 0.8342284586783979 | 0.050720453699956834 | 0.008044681339558846 | 0.04227525577052722 | 345 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.007832982024848327 | 0.8289643850601062 | 0.04346959530272699 | 0.008021776568162274 | 0.041435579046325804 | 200 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.007691398128994341 | 0.8251712675982403 | 0.042461003714607704 | 0.008025009403920228 | 0.0416690202334581 | 345 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007409906237420634 | 0.8310483189888915 | 0.047631156680079036 | 0.008015852210445373 | 0.04171388670164684 | 200 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007233805552879555 | 0.8325944686375933 | 0.04790673364344018 | 0.008031904606785583 | 0.042090386990139256 | 345 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007545347631411085 | 0.8319241829467722 | 0.05030325770740932 | 0.007979843879316374 | 0.041030256095967194 | 200 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.007530270296440599 | 0.8296985931567467 | 0.04908615739756307 | 0.007982886033895374 | 0.04117613613183003 | 345 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007467546318529855 | 0.8380897318179913 | 0.051844189790712786 | 0.008016204434392596 | 0.041363745895998096 | 200 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.007382744994780807 | 0.8369886577945753 | 0.05098467560271924 | 0.008028437573063199 | 0.04166574089607944 | 345 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.00859973029749081 | 0.7986062693063658 | 0.037883266471658905 | 0.00810101993058857 | 0.05725754517330178 | 200 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.008508211625415414 | 0.7900554304623707 | 0.03710074233761921 | 0.008107851314260239 | 0.061832136959325795 | 345 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.00789778027479875 | 0.8198230047848671 | 0.04655796372596695 | 0.008001764106290216 | 0.04151576221917983 | 200 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.007791265474679778 | 0.8236469722157664 | 0.04641488149982149 | 0.008004077072754206 | 0.04141277668079422 | 345 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.007380656926699114 | 0.839035921005453 | 0.045312273608640326 | 0.008004825873927836 | 0.040930098998037484 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.007337200449926398 5 | 0.8381245240214898 | 0.047274328501696085 | 0.007993757327403178 | 0.04091065450898271 | 345 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.00732923172640929 | 0.8380111517081887 | 0.050673861657744255 | 0.00798714863781587 | 0.040841286474575036 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.007309713281867131 | 0.8377184738716108 | 0.048990973051881055 | 0.007993480207373081 | 0.04089552721873369 | 345 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.007247958825541931 | 0.8282191175406045 | 0.038894595973523766 | 0.008063350954463365 | 0.041862381229711684 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.007241608447802588 5 | 0.8251926720793282 | 0.03892239213894077 | 0.008064756041079339 | 0.04202137998556516 | 345 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006740912088060037 5 | 0.8196762425393874 | 0.03621607056118252 | 0.008100657598004242 | 0.0426040789979532 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.006796612607016062 | 0.8200087131319057 | 0.035822201099147584 | 0.008105496438313453 | 0.042624472997345654 | 345 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.007341125905309261 | 0.8231315184450007 | 0.03730607037715059 | 0.008098818025473555 | 0.04232568468405744 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.007331322363583928 | 0.8233698251955032 | 0.03978351797349287 | 0.008074472300853612 | 0.04215354749892086 | 345 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.007165691712637592 | 0.8187565377593037 | 0.03911490560854123 | 0.008105500133677189 | 0.04255237669391434 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.007204653548044351 | 0.817343207016733 | 0.03851863139256848 | 0.00811702974975424 | 0.042736375102344994 | 345 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.006736554359874218 | 0.7949481932183832 | 0.029558723443045595 | 0.008244797092026244 | 0.045254413087223895 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006703501189954508 5 | 0.7929873681282884 | 0.02880934871483072 | 0.008271029264390726 | 0.0455562021716937 | 345 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.004780798251139495 | 0.7730840581753159 | 0.025317159856527997 | 0.00830495278575105 | 0.049333225694686525 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.00495620690571614 | 0.7718971810219033 | 0.025068830132338817 | 0.008346191779421945 | 0.049333009010885304 | 345 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.011234290662966778 | 0.742599483523949 | 0.07833639762600968 | 0.02606855803771924 | 0.12244193798197892 | 311 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.01122322634729746 | 0.7426851561952812 | 0.07830405508192147 | 0.02606965793873724 | 0.12245654154850126 | 242 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.01123793328744955 | 0.7425842928519595 | 0.07834015135847677 | 0.026068314826862095 | 0.12243835527598376 | 489 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.011237731804387464 | 0.7425682865734204 | 0.07831587726263547 | 0.02606861903303298 | 0.1224419334235272 | 410 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.010911020655719103 | 0.7329491248762614 | 0.07418636718107574 | 0.02610111488462181 | 0.12585533809670624 | 311 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.010917535291119714 | 0.7329428718927131 | 0.07458454443070989 | 0.02609922067348607 | 0.1258431022385853 | 242 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.010914929288260242 | 0.7331685400299524 | 0.07446032086796275 | 0.026096940190458707 | 0.12581262226215797 | 489 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.010912363998872798 | 0.7330890013995468 | 0.0742631882351027 | 0.026099647676046903 | 0.12583919243103742 | 410 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.010736426693655118 | 0.7086441686357619 | 0.06857971489780262 | 0.026181039347361584 | 0.13110623043690336 | 311 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.010768514354450713 | 0.7137996686122621 | 0.06983978559217906 | 0.02616630138346035 | 0.1301507714362132 | 242 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.010839430557620726 | 0.7064493374267267 | 0.06789546888421133 | 0.02619276056177408 | 0.13188760102304273 | 489 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_50 | 3 | 0.010797169722961544 | 0.7070213155033586 | 0.06828255384167398 | 0.026184426301851364 | 0.13160150204331542 | 410 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.010813507350362532 | 0.7241481524829461 | 0.07125742261367186 | 0.02614697146156167 | 0.1282066691482675 | 311 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.010833298326836324 | 0.7245613251676166 | 0.07162408120965338 | 0.026142362895562696 | 0.12807529933076775 | 242 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.010850008650049116 | 0.7242848175554416 | 0.07155032338975197 | 0.02614110819590159 | 0.12822336742713877 | 489 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.010831801737879471 | 0.7242643084490756 | 0.07133616165329384 | 0.02614517532460412 | 0.12823277346495515 | 410 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010410210161629975 | 0.7180165243926 | 0.06922143355040078 | 0.02615676886509801 | 0.12895170192352626 | 242 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.010413203927419793 | 0.7191234044314554 | 0.06931856171841738 | 0.026153056982901112 | 0.1288700862529371 | 410 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.010084996954397294 | 0.7058163927604995 | 0.0657113170778617 | 0.026332141639704962 | 0.12687560384424393 | 242 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009558034823103218 | 0.7007251557833998 | 0.06470046806856199 | 0.026391645462216274 | 0.12830208828486783 | 410 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011338774033561493 | 0.7081012465424654 | 0.06558627112096999 | 0.026195501980752038 | 0.1311704637150738 | 242 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.010902972165255511 | 0.7091737351718057 | 0.06601115993956887 | 0.026210396924552223 | 0.13149682841859014 | 410 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.01039323464444186 | 0.7083283249993636 | 0.06623478605663927 | 0.026278877218408624 | 0.12630453283296125 | 242 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.00982943809704925 | 0.7003598320163157 | 0.0653239756059753 | 0.026348824403273265 | 0.12780961792871195 | 410 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.01078766656272414 | 0.715300605557278 | 0.06954430466151497 | 0.026132772269375586 | 0.13073194491793763 | 242 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.010861236193624342 | 0.7162152403356317 | 0.069920862849681 | 0.02612310366889533 | 0.12966828913947523 | 410 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010940681326636517 | 0.7106082851420309 | 0.06711467298853599 | 0.026261528597328825 | 0.12507822372390287 | 242 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010703890350681474 | 0.7085639333414282 | 0.06761725818219017 | 0.02629983491790378 | 0.1256383832784453 | 410 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.013955039077110603 | 0.6967588781236753 | 0.06523233004789585 | 0.026125703236611418 | 0.17687811990043859 | 242 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.013468135675731161 | 0.6989792904923079 | 0.06622634520286942 | 0.02612435148699939 | 0.183600268720135 | 410 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.012683449123374694 | 0.707748734594934 | 0.06855183606404504 | 0.026101950553868797 | 0.12387139011653184 | 242 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.01208564784252277 | 0.7073535222461614 | 0.06859595044312274 | 0.026133239381096074 | 0.12434833494995928 | 410 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.010264991254493405 | 0.725854188987524 | 0.07194438701904635 | 0.026130594645797273 | 0.12883103751380198 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.010226078352200345 | 0.7252303859113756 | 0.07163755394048901 | 0.026136444324712356 | 0.12891339361199694 | 410 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.01001410140701335 | 0.7254046708359794 | 0.07155271554328971 | 0.02614850048626219 | 0.1292740540553907 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.010034889363752509 | 0.7242632294695774 | 0.07149203830275311 | 0.02615039548422859 | 0.1294421267117596 | 410 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.0108982270150221 | 0.7164385890917448 | 0.06992264925704161 | 0.026149368277303346 | 0.130525853755585 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.01075860414657217 | 0.7177043679774788 | 0.07078149771282163 | 0.026133100584958633 | 0.13040753283354073 | 410 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.010460254687146118 | 0.7176667481253707 | 0.06894462998360283 | 0.02619999501389898 | 0.13216730902485077 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.010311430896365372 | 0.7177866762717932 | 0.06933023879655366 | 0.026185480643654915 | 0.13223232228706805 | 410 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.011192496311010583 | 0.7129572150135631 | 0.06872565155494986 | 0.02617662375198188 | 0.13066315671623266 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.011083866379563176 | 0.7177491498746021 | 0.07079766973006586 | 0.026171531183607943 | 0.1309604754750415 | 410 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.01081106804747089 | 0.7109715104214297 | 0.06647060932577913 | 0.0262401836938319 | 0.1329314328077952 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.010622929865540759 | 0.7116423722963194 | 0.06860320902922792 | 0.02624773974570122 | 0.13344290054612873 | 410 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.010789680601889987 | 0.6990462976679125 | 0.061874269770510394 | 0.02644458146721033 | 0.1397417807434286 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.010937501406739658 | 0.6971640306772732 | 0.06270150610043677 | 0.02640814204116589 | 0.13909845319004255 | 410 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008984952316633424 | 0.690631489488208 | 0.059429158814349944 | 0.02659854404908549 | 0.14902952804579897 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.009236241362639171 | 0.6928313077914587 | 0.0587966136429051 | 0.026631201061775207 | 0.14872955753289932 | 410 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.013628399775601345 | 0.6966275824017971 | 0.1003415764571716 | 0.045355172302578156 | 0.21088758009736783 | 348 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.01362410084705614 | 0.6966761002523136 | 0.10035509936322796 | 0.04535551728551323 | 0.21088978205417064 | 267 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.013634325464511331 | 0.6967241174318648 | 0.10039994633467171 | 0.04535383745131977 | 0.21085636857930024 | 512 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.013634480726368992 | 0.6967002363714858 | 0.10038585075145298 | 0.04535413596856174 | 0.21086037739901373 | 448 |
| logistic | lr_c_0p1_freq_100 | 4 | 0.013252369660110336 | 0.6960213737757579 | 0.10004827919641551 | 0.04540672701157193 | 0.21465739985054594 | 348 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.01325429044021471 | 0.6960577338237792 | 0.10007691310208364 | 0.04540645966938406 | 0.2146396239142685 | 267 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.013264057790715515 | 0.69643973632362 | 0.10032428599696691 | 0.0454007630305605 | 0.21454893983710918 | 512 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.013262709045898146 | 0.6962402236672909 | 0.10021658439018649 | 0.04540288424291486 | 0.21458338026026855 | 448 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.013224271438456992 | 0.688706230895506 | 0.0978098066006121 | 0.04547080920743297 | 0.21827833530115168 | 348 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.013201360049247384 | 0.6909069253998185 | 0.09825671407602066 | 0.04545729252245067 | 0.21740296102043766 | 267 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.013278964552681597 | 0.6885607812567986 | 0.09830329276506282 | 0.0454679663810026 | 0.21882868878461315 | 512 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.013254400224916583 | 0.6880793814004288 | 0.09837338268663527 | 0.04546434620072068 | 0.21870338531367464 | 448 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.013211987315211395 | 0.692152651441017 | 0.09873933330312051 | 0.04545005461409503 | 0.21681747290693668 | 348 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.013202201721567343 | 0.692658933699696 | 0.09883167587281044 | 0.045444919732599356 | 0.21657324522885057 | 267 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.013252260693204803 | 0.6933177505635713 | 0.09944473254513242 | 0.04543596050766293 | 0.2166206987505283 | 512 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.013237695597841433 | 0.6926349487264754 | 0.09923479567439543 | 0.04543872342009614 | 0.21670901017423902 | 448 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.012957935095344385 | 0.6737186833381407 | 0.09346315542862355 | 0.04546669595764557 | 0.22238587484110492 | 267 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012958441793690458 | 0.6747095442414874 | 0.09410237779066691 | 0.0454620917232099 | 0.22225437586024954 | 448 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.012832112206815224 | 0.6810390124138626 | 0.09433674586446995 | 0.04567734946684657 | 0.20948257657580202 | 267 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.012761763706387722 | 0.6819137034833456 | 0.0940375313408289 | 0.04573079222287059 | 0.2092546954181451 | 448 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.012969738204325952 | 0.6691028087773402 | 0.0921692042752366 | 0.04550711076344779 | 0.22249620071696574 | 267 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012896333157242163 | 0.667329266902647 | 0.09252293456322695 | 0.045510107606770316 | 0.22216983869913776 | 448 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012848350820394249 | 0.6793394328660032 | 0.09474417130395878 | 0.04564740836763915 | 0.2098569569867876 | 267 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012775982212445742 | 0.6824662269566425 | 0.09545364170749071 | 0.04570246118479955 | 0.209217432438297 | 448 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.013036688700643949 | 0.6732335378961538 | 0.09329907627456982 | 0.04545089848548449 | 0.23702058899107972 | 267 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.013024822927849183 | 0.6733914287360157 | 0.09386540402017717 | 0.045446934032955126 | 0.2364368665228245 | 448 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.01298995723864984 | 0.6842350313504099 | 0.09535138904069407 | 0.04556904886237181 | 0.21023543065464992 | 267 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012918725432549684 | 0.6855902531870399 | 0.09509084045288743 | 0.04561410807328253 | 0.20918599914411395 | 448 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.013881686243322812 | 0.6566775960304387 | 0.09019985739001606 | 0.04545557147945455 | 0.39227612917529864 | 267 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.013790903533564781 | 0.6543085154892389 | 0.08987905789136605 | 0.0454674727757437 | 0.4061634613913123 | 448 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.013301040865210001 | 0.676959668195338 | 0.09535598372571366 | 0.045446559812133615 | 0.21535950205861626 | 267 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.013257400314408554 | 0.6795870855079693 | 0.09528039368972926 | 0.045470271841774566 | 0.21349144127697556 | 448 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.013006766423608526 | 0.6888611932825525 | 0.10065345445364615 | 0.04536194895193392 | 0.218619665985313 | 267 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.0129872332886076 | 0.6890639650234663 | 0.10049935754424975 | 0.045368592625728094 | 0.21856505398707438 | 448 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.013005446830123423 | 0.6880330693361462 | 0.10010582159283435 | 0.045368159227156285 | 0.2185082920734924 | 267 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.01293758638538585 | 0.6879450268192432 | 0.10036896668902835 | 0.04537599505419699 | 0.21864832544111465 | 448 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.013066444518591519 | 0.6859017791635594 | 0.09845523293711952 | 0.045396654580564695 | 0.22238689217800126 | 267 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.013110866805979409 | 0.6865451366305557 | 0.09880884600780204 | 0.045386090669099925 | 0.22217864458392508 | 448 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.012922954559098233 | 0.6847344270216617 | 0.09813255938296442 | 0.045413535567250626 | 0.22305602361576776 | 267 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.012927517463645977 | 0.6855526272916725 | 0.09857896691697735 | 0.04541230127622976 | 0.22293373365136537 | 448 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.013280604754026891 | 0.6870575639168364 | 0.09847690659522165 | 0.045378001791024955 | 0.22173611248374203 | 267 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.013280372835348053 | 0.6870655935455238 | 0.09925550561843566 | 0.045358708314721344 | 0.22149170027748016 | 448 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.013161238447747192 | 0.6861868120887212 | 0.09884760352297006 | 0.04539614119564193 | 0.2218811101446345 | 267 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.013191749337819365 | 0.6861676023829142 | 0.09867590535512503 | 0.0453981066408166 | 0.22180569005447534 | 448 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.01317210457932427 | 0.6777237818311721 | 0.0927028537739754 | 0.04559745139281788 | 0.2314245327305324 | 267 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.013333894181127473 | 0.6769767996443082 | 0.09388027389381529 | 0.045545507438953046 | 0.23141243109077683 | 448 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.012529531902363573 | 0.6750465619274343 | 0.09331763512459601 | 0.04562036021243393 | 0.23582188078879487 | 267 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.012540070866405174 | 0.677868263191126 | 0.09330902913952925 | 0.04563843226581866 | 0.23489449009497307 | 448 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.29268733400385827 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.29197104199556634 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.29787266699713655 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.29460137500427663 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.30942770799447317 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.31728116699378006 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.3107824999897275 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.3111157499952242 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 17 | 0.3955712500028312 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 24 | 0.43779129200265743 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 18 | 0.4237554999999702 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 36 | 0.509018582990393 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 21 | 0.38811637500475626 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 13 | 0.35798587500175927 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 14 | 0.3626603750017239 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 13 | 0.34331929200561717 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.264860166993458 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.269330875002197 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.936198958006571 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.8966754590073833 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.591836667008465 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.6080790409905603 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 0.9662954579980578 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.8846599589887774 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.362553459010087 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.5323039590002736 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 1.0307440420001512 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.891314208987751 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 3.103592208994087 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.751180999999633 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.24985912500415 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.14865958399605 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1025484999991022 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.1086491250025574 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.134574584008078 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.0941324579907814 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.7290585419978015 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.7116981250001118 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.784234209000715 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7462683329940774 | 6.0 | NA | NA | 0.03 | 1 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.1351745409920113 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.1342770839983132 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.0881251249957131 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.0987434580019908 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.8363233330019284 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.745274957997026 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.749119291009265 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.755125916999532 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.4627645419968758 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.467222084000241 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.4666252079914557 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.45830949999799486 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.4987661249906523 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.5033745000109775 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.4968405419931514 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.4945648330030963 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 2 | 17 | 0.7131361670035403 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7281014170002891 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 20 | 0.8139137079997454 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 19 | 0.7510516250040382 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 14 | 0.6210076250135899 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.5842312499880791 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.5774889159947634 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.5932837919972371 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.619814125006087 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.369950041000266 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.562228166993009 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.3922519589978037 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 5.983859499989194 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 5.815798166993773 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.7154482920013834 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.436018458000035 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 6.846859917000984 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 7.468162457997096 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.154290874997969 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.825014749992988 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 10.404657250008313 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 10.673188000000664 | NA | 5 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.7637477079988457 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.6187609159969725 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.3068381250050152 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.3292532500054222 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.337571249998291 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.4234621669893386 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 1.9430781249975553 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 1.988952915999107 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 1.985501415998442 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.995225874998141 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3698315840010764 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.3492451250058366 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.3142541669949424 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.3035446660069283 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 1.9992265409964602 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 2.0559817910107085 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 2.059554042003583 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 2.004547457996523 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.6329265419917647 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.6351658749917988 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6356133749941364 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6307525419979356 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 11 | 0.7330337089952081 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 11 | 0.7338658749940805 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.7272917499940377 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.7307413750095293 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 18 | 1.1137471249967348 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.091296833998058 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 20 | 1.2423177079908783 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 20 | 1.2400175829970976 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 14 | 0.8693514160113409 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 13 | 0.8236719999986235 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.9109234579955228 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 15 | 0.9570717500027968 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 10.62608020901098 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 10.317292416992132 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.2851601250004023 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.093544875009684 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 10.497041082999203 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 10.210053542003152 | 12.0 | 5 | 0.5 | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.6822454999928595 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.2636508330033394 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 14.447424124999088 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 14.999683457994252 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.7321198340068804 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 3.247404291993007 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 21.812671542007593 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 25.036213667000993 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.420978957990883 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 4.704456167004537 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.5596087089943467 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.565862624993315 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.5724307500058785 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.551545583002735 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.185987582997768 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.2440711250092136 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.2106977500079665 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.18033208300767 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.6050752919982187 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.5778758750093402 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.5122163339983672 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.5190725840075174 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.2780580000107875 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.317068208998535 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.206913333007833 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.2099949169933097 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 8 | 0.8135541660012677 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 8 | 0.8359147090086481 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 8 | 0.8094719590008026 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 8 | 0.8118961669970304 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 10 | 0.9280832919903332 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 0.918735832994571 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 0.9755383330048062 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 11 | 0.9883759160002228 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 18 | 1.5625809999910416 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 16 | 1.5719817920034984 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 21 | 1.6471149589924607 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.7081850420072442 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 15 | 1.2623090840061195 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_200 | 4 | 13 | 1.1076948750123847 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.287640750000719 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 15 | 1.22221679201175 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 17.819794209004613 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 18.34293300000718 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.336487667009351 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.826886042006663 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 18.197686667001108 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 17.52974087499024 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 3.9312226670008386 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 3.0337555830046767 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 26.88906537501316 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 27.091099917000975 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.482161292005912 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 5.053567333001411 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 45.351296875000116 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 48.77651212499768 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 10.577629625011468 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.914611166997929 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.7552999579929747 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.8825703340116888 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.7996215830062283 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.7905417080037296 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.4907202090107603 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.6640605829888955 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.510612875004881 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.4600216660037404 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.753462375010713 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7995204579929123 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.758503125005518 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.8182823329989333 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.518857500006561 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.557704874998308 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.52613004099112 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.4944587079953635 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45351 | 771 | logistic | raw | 45351 | 771 | 0.01700072765760402 4 | 0.00928605407682073 7 | 0.7754430892392987 |
| logistic | validation | calibrated | 45351 | 771 | logistic | calibrated | 45351 | 771 | 0.01700072765760402 4 | 0.02158166816603237 | 0.7753967713648471 |
| rf | validation | raw | 45351 | 771 | rf | raw | 45351 | 771 | 0.01700072765760402 4 | 0.01475447312576785 9 | 0.7363134754174864 |
| rf | validation | calibrated | 45351 | 771 | rf | calibrated | 45351 | 771 | 0.01700072765760402 4 | 0.03684663278877454 | 0.7361028338276429 |
| xgb | validation | raw | 45351 | 771 | xgb | raw | 45351 | 771 | 0.01700072765760402 4 | 0.01147425233198126 3 | 0.7642301049891218 |
| xgb | validation | calibrated | 45351 | 771 | xgb | calibrated | 45351 | 771 | 0.01700072765760402 4 | 0.02948975393862485 | 0.7639969299861105 |
| logistic | test | raw | 49321 | 886 | logistic | raw | 49321 | 886 | 0.01796395044707122 7 | 0.11098116365715045 | 0.759355828399561 |
| logistic | test | calibrated | 49321 | 886 | logistic | calibrated | 49321 | 886 | 0.01796395044707122 7 | 0.13857696820140034 | 0.7545030679221252 |
| rf | test | raw | 49321 | 886 | rf | raw | 49321 | 886 | 0.01796395044707122 7 | 0.0481873480647129 | 0.7739977783168479 |
| rf | test | calibrated | 49321 | 886 | rf | calibrated | 49321 | 886 | 0.01796395044707122 7 | 0.1322014062134619 | 0.7695677295279028 |
| xgb | test | raw | 49321 | 886 | xgb | raw | 49321 | 886 | 0.01796395044707122 7 | 0.07874869459399922 | 0.775199815162673 |
| xgb | test | calibrated | 49321 | 886 | xgb | calibrated | 49321 | 886 | 0.01796395044707122 7 | 0.1061812563503729 | 0.7703737363215835 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.06311555547369835 | 0.016454084944311435 | 0.0800668002129027 |
| logistic | 0.0601461481922147 | 0.01633847549625692 | 0.07790493540775052 |
| rf | 0.050902615633820585 | 0.016482181175496147 | 0.08021966988895234 |
| rf | 0.047022381124513596 | 0.017022183879821986 | 0.0870502111029424 |
| xgb | 0.051973442883668605 | 0.01647001750711729 | 0.07993809216333075 |
| xgb | 0.04812440889038187 | 0.016619343860103865 | 0.08224924168956999 |
| logistic | 0.055987147496832965 | 0.032257577862356494 | 0.15213641802694056 |
| logistic | 0.04735893583675304 | 0.03557925822889996 | 0.17867577527577824 |
| rf | 0.05778502065499956 | 0.018226524016288125 | 0.09773693704373183 |
| rf | 0.049519718822354654 | 0.03732546325033485 | 0.4451734621401487 |
| xgb | 0.05455648509187403 | 0.02395674193812942 | 0.11975614006688341 |
| xgb | 0.050358179783323895 | 0.03020204198061348 | 0.2003797567999535 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45351 | 771 | logistic | raw | 0.8051184486374356 | 1.0428072412605154 | 0.00771467358078328 5 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 45351 | 771 | logistic | calibrated | 0.11547798854749543 | 1.1088768236288533 | 0.00460273728792883 4 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 45351 | 771 | rf | raw | 0.9527479601494856 | 1.2070231951906196 | 0.00224625453183616 5 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 45351 | 771 | rf | calibrated | -0.3345393891551638 | 1.16037584438107 | 0.01984590513117052 3 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 45351 | 771 | xgb | raw | -0.16129559022797496 | 0.8554782091426517 | 0.00587518033945762 8 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 45351 | 771 | xgb | calibrated | 0.3136869791490168 | 1.2823564576995554 | 0.01248902628102082 7 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 49321 | 886 | logistic | raw | -2.0776672148788746 | 0.9713338819267442 | 0.09301721321007925 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 49321 | 886 | logistic | calibrated | -1.5581789221969211 | 1.451723319658995 | 0.12061301775432909 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 49321 | 886 | rf | raw | 3.8553408914197527 | 2.746073893099925 | 0.03022339761764167 6 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 49321 | 886 | rf | calibrated | -3.778146536259626 | 0.14103889643745243 | 0.11423745576639065 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 49321 | 886 | xgb | raw | -1.5314556040256273 | 1.0420871161507674 | 0.06078474414692798 5 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 49321 | 886 | xgb | calibrated | -3.4978346440833104 | 0.24093640810895572 | 0.0882173059033017 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |

| model | calibration_logit_clip |
|---|---|
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R5_unusual_fpd/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 313 | 0.006346181139879565 | 32 | 281 | 48154 | 854 |
| logistic | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 0 | 0.0 | 0 | 0 | 48435 | 886 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.2528735632183909 | 49321 | 886 | 4250 | 0.08617019119644775 | 287 | 3963 | 44472 | 599 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 1950 | 0.03953691125484074 | 156 | 1794 | 46641 | 730 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 0 | 0.0 | 0 | 0 | 48435 | 886 |
| logistic | 2:1 | 1 | 2 | 0.25287356321839083 | 49321 | 886 | 8456 | 0.17144826747227349 | 431 | 8025 | 40410 | 455 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 10544 | 0.21378317552361062 | 501 | 10043 | 38392 | 385 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 10078 | 0.20433486750065896 | 484 | 9594 | 38841 | 402 |
| logistic | 5:1 | 1 | 5 | 0.14153910559948704 | 49321 | 886 | 11726 | 0.23774862634577562 | 538 | 11188 | 37247 | 348 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 22204 | 0.4501936294884532 | 722 | 21482 | 26953 | 164 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 35277 | 0.7152531376087266 | 841 | 34436 | 13999 | 45 |
| logistic | 10:1 | 1 | 10 | 0.1034103410341034 | 49321 | 886 | 34157 | 0.6925447578110744 | 834 | 33323 | 15112 | 52 |
| rf | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 0 | 0.0 | 0 | 0 | 48435 | 886 |
| rf | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 486 | 0.009853814805052614 | 40 | 446 | 47989 | 846 |
| rf | 1:1 | 1 | 1 | 0.27272727272727276 | 49321 | 886 | 521 | 0.010563451673729242 | 42 | 479 | 47956 | 844 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 0 | 0.0 | 0 | 0 | 48435 | 886 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 513 | 0.010401248960888 87 | 42 | 471 | 47964 | 844 |
| rf | 2:1 | 1 | 2 | 0.2727272727272727 | 49321 | 886 | 1064 | 0.021572960807769 51 | 82 | 982 | 47453 | 804 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 0 | 0.0 | 0 | 0 | 48435 | 886 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 10901 | 0.22102147158411226 | 534 | 10367 | 38068 | 352 |
| rf | 5:1 | 1 | 5 | 0.2727272727272727 | 49321 | 886 | 1064 | 0.021572960807769 51 | 82 | 982 | 47453 | 804 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 1134 | 0.022992234545122767 | 89 | 1045 | 47390 | 797 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 31472 | 0.6381054723140245 | 834 | 30638 | 17797 | 52 |
| rf | 10:1 | 1 | 10 | 0.17404612598031555 | 49321 | 886 | 8852 | 0.1794773017578719 | 465 | 8387 | 40048 | 421 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 59 | 0.001196245007197745 3 | 8 | 51 | 48384 | 878 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49321 | 886 | 664 | 0.013462825165750897 | 56 | 608 | 47827 | 830 |
| xgb | 1:1 | 1 | 1 | 0.24271844660194178 | 49321 | 886 | 774 | 0.015693112467306015 | 61 | 713 | 47722 | 825 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 622 | 0.012611260923338943 | 52 | 570 | 47865 | 834 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49321 | 886 | 729 | 0.014780722207578922 | 59 | 670 | 47765 | 827 |
| xgb | 2:1 | 1 | 2 | 0.24271844660194178 | 49321 | 886 | 774 | 0.015693112467306015 | 61 | 713 | 47722 | 825 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 5373 | 0.10893939701141502 | 329 | 5044 | 43391 | 557 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49321 | 886 | 5139 | 0.10419496766083412 | 314 | 4825 | 43610 | 572 |
| xgb | 5:1 | 1 | 5 | 0.24271844660194178 | 49321 | 886 | 774 | 0.015693112467306015 | 61 | 713 | 47722 | 825 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 14982 | 0.30376513047180714 | 632 | 14350 | 34085 | 254 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49321 | 886 | 27141 | 0.5502929786500679 | 808 | 26333 | 22102 | 78 |
| xgb | 10:1 | 1 | 10 | 0.14772727272727273 | 49321 | 886 | 7134 | 0.14464426917540196 | 399 | 6735 | 41700 | 487 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.03611738148984198 | 0.9941984102405286 | 0.10223642172523961 | 0.9825742735879857 | 0.0058015897594714564 | 0.963882618510158 | 1135 | 0.023012509884227813 | 886 |
| logistic | 0.0 | 1.0 | NA | 0.9820360495529288 | 0.0 | 1.0 | 886 | 0.017963950447071227 | 886 |
| logistic | 0.3239277652370203 | 0.9181790027872406 | 0.06752941176470588 | 0.9867098577799472 | 0.08182099721275937 | 0.6760722347629797 | 4562 | 0.09249609699722228 | 886 |
| logistic | 0.17607223476297967 | 0.9629606689377517 | 0.08 | 0.9845897278925925 | 0.037039331062248375 | 0.8239277652370203 | 3254 | 0.06597595344782141 | 1772 |
| logistic | 0.0 | 1.0 | NA | 0.9820360495529288 | 0.0 | 1.0 | 1772 | 0.03592790089414245 | 1772 |
| logistic | 0.48645598194130923 | 0.8343140291111799 | 0.05096972563859981 | 0.9888657775602594 | 0.16568597088882006 | 0.5135440180586908 | 8935 | 0.18116015490359078 | 1772 |
| logistic | 0.5654627539503386 | 0.7926499432228761 | 0.04751517450682853 | 0.9900714340975321 | 0.20735005677712398 | 0.4345372460496614 | 11968 | 0.24265525840919688 | 4430 |
| logistic | 0.54627539503386 | 0.8019200991018891 | 0.04802540186544949 | 0.9897561348520755 | 0.19807990089811087 | 0.45372460496614 | 11604 | 0.23527503497495997 | 4430 |
| logistic | 0.6072234762979684 | 0.7690100134200475 | 0.04588094831997271 | 0.9907434499268519 | 0.2309899865799525 | 0.3927765237020316 | 12928 | 0.2621195839500416 | 4430 |
| logistic | 0.8148984198645598 | 0.5564777536905131 | 0.03251666366420465 | 0.9939521333480842 | 0.44352224630948694 | 0.18510158013544017 | 23122 | 0.4688063907868859 | 8860 |
| logistic | 0.9492099322799097 | 0.2890265304015691 | 0.023839895682739462 | 0.9967957846767302 | 0.7109734695984309 | 0.050790067720090294 | 34886 | 0.7073254800186534 | 8860 |
| logistic | 0.9413092550790068 | 0.3120057809435326 | 0.024416664226952017 | 0.9965708256396729 | 0.6879942190564674 | 0.05869074492099323 | 33843 | 0.6861783013320898 | 8860 |
| rf | 0.0 | 1.0 | NA | 0.9820360495529288 | 0.0 | 1.0 | 886 | 0.017963950447071227 | 886 |
| rf | 0.045146726862302484 | 0.990791782801693 | 0.0823045267489712 | 0.982676359168629 | 0.00920821719830701 | 0.9548532731376975 | 1292 | 0.02619573812372012 | 886 |
| rf | 0.04740406320541761 | 0.9901104573139259 | 0.08061420345489444 | 0.9827049180327869 | 0.00988954268607412 | 0.9525959367945824 | 1323 | 0.026824273635976562 | 886 |
| rf | 0.0 | 1.0 | NA | 0.9820360495529288 | 0.0 | 1.0 | 1772 | 0.03592790089414245 | 1772 |
| rf | 0.04740406320541761 | 0.9902756271291422 | 0.08187134502923976 | 0.9827077528274053 | 0.009724372870857851 | 0.9525959367945824 | 2159 | 0.043774457127795464 | 1772 |
| rf | 0.09255079006772009 | 0.979725405182203 | 0.07706766917293233 | 0.9833392046749694 | 0.02027459481779705 | 0.90744920993228 | 2590 | 0.05251312828207052 | 1772 |
| rf | 0.0 | 1.0 | NA | 0.9820360495529288 | 0.0 | 1.0 | 4430 | 0.08981975223535614 | 4430 |
| rf | 0.6027088036117382 | 0.7859605657066171 | 0.048986331529217504 | 0.9908381051535659 | 0.21403943429338287 | 0.3972911963882618 | 12127 | 0.24587903732689928 | 4430 |
| rf | 0.09255079006772009 | 0.979725405182203 | 0.07706766917293233 | 0.9833392046749694 | 0.02027459481779705 | 0.90744920993228 | 5002 | 0.10141724620344275 | 4430 |
| rf | 0.10045146726862303 | 0.9784246928873749 | 0.07848324514991181 | 0.9834602693672567 | 0.021575307112625168 | 0.899548532731377 | 9015 | 0.18278218203199448 | 8860 |
| rf | 0.9413092550790068 | 0.3674409001754929 | 0.026499745805795626 | 0.9970866715222141 | 0.6325590998245071 | 0.05869074492099323 | 31158 | 0.6317390158350399 | 8860 |
| rf | 0.5248306997742663 | 0.8268400949726438 | 0.052535050158156348 5 | 0.9895969754626999 | 0.17315990502735626 | 0.4751693002257336 | 12597 | 0.25540844670627116 | 8860 |
| xgb | 0.009029345372460496 | 0.9989470424279963 | 0.13559322033898305 | 0.9821769315090739 | 0.00105295757200371 63 | 0.9909706546275395 | 929 | 0.018835790028588227 | 886 |
| xgb | 0.06320541760722348 | 0.9874470940435636 | 0.08433734939759036 | 0.9829418172102677 | 0.012552905956436461 | 0.9367945823927766 | 1438 | 0.029155937633056912 | 886 |
| xgb | 0.06884875846501129 | 0.98527924021885 | 0.07881136950904392 | 0.9830061589799576 | 0.014720759781149995 | 0.9311512415349887 | 1538 | 0.031183471543561565 | 886 |
| xgb | 0.05869074492099323 | 0.9882316506658408 | 0.08360128617363344 | 0.9828743916712869 | 0.011768349334159182 | 0.9413092550790068 | 2238 | 0.04537620891709414 | 1772 |
| xgb | 0.06659142212189616 | 0.9861670279756375 | 0.08093278463648834 | 0.9829807375699704 | 0.013832972024362547 | 0.9334085778781038 | 2324 | 0.04711988808012814 | 1772 |
| xgb | 0.06884875846501129 | 0.98527924021885 | 0.07881136950904392 | 0.9830061589799576 | 0.014720759781149995 | 0.9311512415349887 | 2363 | 0.047910626305224954 | 1772 |
| xgb | 0.37133182844243795 | 0.8958604315061423 | 0.06123208635771449 5 | 0.987325930645308 | 0.10413956849385775 | 0.628668171557562 | 7829 | 0.1587356298534093 | 4430 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| xgb | 0.3544018058690745 | 0.9003819551976876 | 0.06110138159174937 | 0.9870535512199539 | 0.09961804480231237 | 0.6455981941309256 | 7685 | 0.15581598102228258 | 4430 |
| xgb | 0.06884875846501129 | 0.98527924021885 | 0.07881136950904392 | 0.9830061589799576 | 0.014720759781149995 | 0.9311512415349887 | 4838 | 0.09809209059021512 | 4430 |
| xgb | 0.7133182844243793 | 0.7037266439558171 | 0.042183954078227205 | 0.9926031625848161 | 0.2962733560441829 | 0.2866817155756208 | 16890 | 0.3424504774842359 | 8860 |
| xgb | 0.9119638826185101 | 0.45632290698874783 | 0.029770457978703806 | 0.9964833183047791 | 0.5436770930112522 | 0.08803611738148984 | 27113 | 0.5497252691551267 | 8860 |
| xgb | 0.4503386004514673 | 0.8609476618148033 | 0.05592935239697225 | 0.9884561594804087 | 0.13905233818519666 | 0.5496613995485328 | 11605 | 0.235295310314065 | 8860 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 1.2810383747178329 |
| logistic | 1.0 |
| logistic | 5.148984198645598 |
| logistic | 1.8363431151241536 |
| logistic | 1.0 |
| logistic | 5.042325056433409 |
| logistic | 2.7015801354401807 |
| logistic | 2.61941309255079 |
| logistic | 2.9182844243792325 |
| logistic | 2.6097065462753952 |
| logistic | 3.937471783295711 |
| logistic | 3.8197516930022575 |
| rf | 1.0 |
| rf | 1.4582392776523703 |
| rf | 1.4932279909706547 |
| rf | 1.0 |
| rf | 1.2183972911963883 |
| rf | 1.4616252821670428 |
| rf | 1.0 |
| rf | 2.737471783295711 |
| rf | 1.129119638826185 |
| rf | 1.0174943566591421 |
| rf | 3.516704288939052 |
| rf | 1.421783295711061 |
| xgb | 1.0485327313769752 |
| xgb | 1.6230248306997743 |
| xgb | 1.7358916478555304 |
| xgb | 1.262979683972912 |
| xgb | 1.3115124153498872 |
| xgb | 1.3335214446952597 |
| xgb | 1.7672686230248307 |
| xgb | 1.734762979683973 |
| xgb | 1.0920993227990972 |
| xgb | 1.9063205417607223 |
| xgb | 3.060158013544018 |

| model | decision_relative_cost_vs_all_negative |
|---|---|
| xgb | 1.309819413092551 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 49321 | 886 | 0 | 0 | 48435 | 886 | 886 | 0.017963950447071227 |
| test | all_non_risky | 2:1 | 2:1 | 49321 | 886 | 0 | 0 | 48435 | 886 | 1772 | 0.03592790089414245 |
| test | all_non_risky | 5:1 | 5:1 | 49321 | 886 | 0 | 0 | 48435 | 886 | 4430 | 0.08981975223535614 |
| test | all_non_risky | 10:1 | 10:1 | 49321 | 886 | 0 | 0 | 48435 | 886 | 8860 | 0.17963950447071228 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 38443 | 373 | 0.00970267669016466 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 4339 | 285 | 0.06568333717446416 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 800 | 29 | 0.03625 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 5739 | 199 | 0.034675030493117265 |

## R6

T1 temporal sensitivity: unchanged Candidate A/19 predictors; 2020 validation maturity through 2023-03; test 2023Q2–Q4.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.019003069726648152 | 239435 | 4550 |
| validation | 0.016292168857125554 | 49533 | 807 |
| test | 0.017232996596618227 | 37022 | 638 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_50_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":50,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_200","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 50 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.006298225141270881 | 0.8173273795797431 | 0.029020236094958712 | 0.006757135976808477 | 0.0370143599068096 | 199 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.006290040898461586 | 0.8172752238851483 | 0.029125782367668447 | 0.006756960937007503 | 0.03700855003397112 | 137 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_25 | 1 | 0.006302309029516599 | 0.8161895468807274 | 0.028713360275695658 | 0.006758334708575664 | 0.0370444224799937 | 341 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.006301414770143446 | 0.8164148772689475 | 0.02875671091082145 | 0.006758219954888032 | 0.03703999506745102 | 266 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.004880370553832738 | 0.8457886976513419 | 0.042038997037784284 | 0.006725071230806392 | 0.03557448319952674 | 199 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.0048774664498217445 | 0.8454533384409069 | 0.04147947840077324 | 0.006727523184392375 5 | 0.035603947565501164 | 137 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.00488004807717957 | 0.8452919162485 | 0.04180387035311162 | 0.006726822813738316 | 0.03559485055573452 | 341 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.0048880711841627926 5 | 0.8453367663342297 | 0.0418961895113253 | 0.006726411855162391 | 0.035591191837875964 | 266 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.004693930374234669 | 0.7982083470710544 | 0.03838550374019917 | 0.006743235887838273 | 0.03719433735841162 | 199 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.004620373069184014 | 0.8115642706175399 | 0.039825709907718845 | 0.006729342474485534 | 0.03665694299398842 | 137 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.004658924092660816 | 0.7919209495004551 | 0.03754413868210855 | 0.006757886163084157 | 0.03762693652541289 | 341 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.0046794320388635787 4 | 0.7971097740788105 | 0.03877640576410692 | 0.006751459562721311 | 0.03733347715455922 | 266 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.004660406753367086 | 0.8354066651293124 | 0.04282412821015176 | 0.006719583258781023 | 0.03580504953511243 | 199 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.0046344997042161797 6 | 0.8373984282141343 | 0.043272577546409924 | 0.006718967482371953 | 0.03573986116027871 | 137 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.0046518797687300625 | 0.8345655036772622 | 0.04255769553023286 | 0.006722204934511342 | 0.03584268938056482 | 341 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.004656408123270278 | 0.8350937944887757 | 0.0430346833475333 | 0.006721150356982867 | 0.035810010134546 | 266 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.004883758232971338 | 0.8397111298727618 | 0.03552224528774165 | 0.006730801716680958 | 0.03579628858950473 | 137 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.004896315437906641 | 0.8393253301976389 | 0.03389874062609589 | 0.0067312977032418725 | 0.03579793122426348 | 266 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.004813301013531823 | 0.8395403056793804 | 0.03646083735848914 | 0.006743476927915912 | 0.036272883785083976 | 137 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.004869468603723089 | 0.8372324320753075 | 0.03721177897439956 | 0.006753910108969777 | 0.03671172241810044 | 266 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.00505094272844287 | 0.8220645587126373 | 0.029942877832815594 | 0.006751775771088811 | 0.03666922904084888 | 137 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.00495692798846819 | 0.8217385697396599 | 0.028508748104861738 | 0.006761612374386131 | 0.03676871316408859 | 266 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.004869370020362729 | 0.8264332175235511 | 0.033957877533911524 | 0.00674304799859196 | 0.03647235941879936 | 137 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.004869786407794138 | 0.8182135816986238 | 0.03411758036962894 | 0.006753514996010268 | 0.036951870131287035 | 266 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.004905369477546337 | 0.8356975871797363 | 0.03544773709454351 | 0.006730109193053086 | 0.036032649265242606 | 137 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.004903949945948333 | 0.8338463141297464 | 0.0333784907835279 | 0.006731779916207558 | 0.03666905544388343 | 266 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.004830578460552669 | 0.8407280070997811 | 0.03639981049866015 | 0.00673890487652875 | 0.03604234186228811 | 137 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.004808235924459074 | 0.8377629463329973 | 0.03677017564967789 | 0.006747159158126422 | 0.036355170912059975 | 266 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.005387927632480863 | 0.7884484347891823 | 0.02732521470025067 | 0.006773235578321083 | 0.05341876978492354 | 137 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.0052706913991904724 | 0.7860221785579745 | 0.0250521570405602 | 0.006792747185564057 | 0.05636092214403732 | 266 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.00496772379950651 | 0.8164744338629581 | 0.03164620819665059 | 0.006738140609874207 | 0.03653747326257435 | 137 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.004884203915025282 | 0.8162208339447811 | 0.03215471981731032 | 0.006739231728404199 | 0.03656749061417764 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.0046556126475501575 | 0.8332574185281676 | 0.04030083074232704 | 0.006713737391293457 | 0.03586766585973448 | 137 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.0046252387667089214 | 0.8349788423214557 | 0.03980194167578516 | 0.006715235766297011 | 0.03585707017371277 | 266 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.004636392398109815 | 0.837003417119138 | 0.040513087056998826 | 0.0067178237694321755 | 0.03590932658187942 | 137 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.004652868214052439 | 0.8368806193631668 | 0.04126833008795586 | 0.006717883886295844 | 0.03590044545124607 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.004611786340279562 | 0.8189303572188626 | 0.0372632509400017 | 0.006740558877161628 | 0.03675711758162531 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.004605072347571724 | 0.8206393932405965 | 0.03531623479199687 | 0.006741520330341467 | 0.03671201866002536 | 266 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.003924637230555585 | 0.8100482614874346 | 0.03315854452444982 | 0.006769730084783904 | 0.038325456247529664 | 137 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.00402522939987914 | 0.8121216568104475 | 0.03650364072812492 | 0.006751677519931934 | 0.037977415896447694 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.004617859237344292 | 0.8048100764039632 | 0.032479287578531926 | 0.006766162455635355 | 0.03766518537999987 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.004588967946268083 | 0.8079080357250626 | 0.0331383278263891 | 0.006766742547054068 | 0.03752774719260256 | 266 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.004332290748423403 | 0.8026997401109054 | 0.03562876057810091 | 0.006766169798238659 | 0.03836836702592234 | 137 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.0043994889691234655 | 0.8054408998477257 | 0.035872470596354046 | 0.006781482049557501 | 0.038370936628367096 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.004256949010941296 | 0.7661906586038156 | 0.02578469325236084 | 0.006869308352235934 | 0.0408940116856225 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.004271786383346765 | 0.7616477451397055 | 0.024248508544919296 | 0.006875363027102823 | 0.04102147643904621 | 266 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.0017635166526573287 | 0.7577217744625135 | 0.024663627020614105 | 0.006871653557443423 | 0.049226439553074534 | 137 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.001904462205942283 | 0.7652901629023724 | 0.0280408191269328 | 0.00684003058958302 | 0.04822254475483483 | 266 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.008566846502098808 | 0.825735461689867 | 0.044767359503727504 | 0.008016856645739272 | 0.04186062721990486 | 263 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.008552337572713582 | 0.8259605151321573 | 0.04472243105857647 | 0.008016667659020975 | 0.04185614459257684 | 200 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.00857707638975317 | 0.825911574437427 | 0.045189995703829935 | 0.008015706332544097 | 0.041842813262378875 | 443 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.008576198681017652 | 0.8259198453621553 | 0.04514961690664096 | 0.00801582768813987 | 0.041843935490058055 | 347 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.007785029907250061 | 0.8430278053738673 | 0.05408983003439572 | 0.007979787428175542 | 0.04066668975754616 | 263 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.0077623175200626894 | 0.8434584729512841 | 0.054371368594793246 | 0.00797829332559861 | 0.0406515208530215 | 200 |
| logistic | lr_c_0p1_freq_25 | 2 | 0.00779486908141632 | 0.8429517339387861 | 0.05453055984817312 | 0.007977567388608602 | 0.04065052542084445 | 443 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.007792979820419179 | 0.8428104433647639 | 0.05471877531975221 | 0.00797678055683611 | 0.04064663951135589 | 347 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.007628867129252464 | 0.8121143022832282 | 0.049588887040177915 | 0.008020869889561964 | 0.042008643213356095 | 263 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.00756480746983472 | 0.8228233059697428 | 0.052078018807325724 | 0.008000097920093062 | 0.04140127393633611 | 200 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.0077228756898670425 | 0.8020961051952049 | 0.04505848028227023 | 0.00805990658889634 | 0.04269137109704126 | 443 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.007628529724677419 | 0.8055374420547968 | 0.04894197642110243 | 0.008026821013394999 | 0.04233845294201983 | 347 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.007640707250804669 | 0.8344957933655383 | 0.05148430119414243 | 0.007998522482994305 | 0.041031385527606636 | 263 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.007594193032741093 | 0.8370134207080144 | 0.054538214741421696 | 0.007988968058667912 | 0.04086831636984972 | 200 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.007666694938684552 | 0.8329288482373816 | 0.05201046639384085 | 0.007998149617311498 | 0.04110372425184773 | 443 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.007642409862379748 | 0.83320911141926 | 0.05198089015441204 | 0.007994726480888151 | 0.04105672697384555 | 347 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.007473811142344571 | 0.8361900159086233 | 0.05032636077159143 | 0.007982870240909197 | 0.040852651024121736 | 200 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.007471742662748061 | 0.8346255994708294 | 0.049746662073301647 | 0.007984616709625986 | 0.04091541304747388 | 347 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.007210774651812034 | 0.8377407352736184 | 0.05180974020182197 | 0.008032119240318828 | 0.04178199934882259 | 200 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.007086196727865568 | 0.8310992880788369 | 0.050636334587852175 | 0.008051722870390158 | 0.04239686274434906 | 347 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.007821382213659236 | 0.8270755095390102 | 0.04363829762729725 | 0.00802681583882523 | 0.04155695919416671 | 200 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.007664992296825264 | 0.8250054867312768 | 0.04258689315491025 | 0.00803102554195993 | 0.041671593010831724 | 347 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.007372596038643427 | 0.8320333757626505 | 0.04950607756949375 | 0.008016709290731746 | 0.041660448999976514 | 200 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.007232909925381737 | 0.8287550867504105 | 0.04588640200658971 | 0.00804058585267298 | 0.042244684175166955 | 347 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.007552111412888348 | 0.8313473894801634 | 0.05022745276762364 | 0.007985048994619628 | 0.04112217987144976 | 200 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.007562386031294974 | 0.8300060246047893 | 0.05046025028795235 | 0.007985220704522986 | 0.04112502706497617 | 347 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.007441555457873739 | 0.8369533643118976 | 0.052762868326629464 | 0.008019730834344142 | 0.04138148891503907 | 200 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.007356954481049943 | 0.8361842209932084 | 0.05077410995829807 | 0.0080327335502495 | 0.04168527403689396 | 347 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.008567593198189477 | 0.7995375183369562 | 0.038647941707235796 | 0.00810367576939308 | 0.05442014892976076 | 200 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.008518742472255994 | 0.7917280269187519 | 0.035830920292633114 | 0.00812320111361961 | 0.05900161395562077 | 347 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.007893536665995626 | 0.8214685074268163 | 0.04490462548405352 | 0.00800815324912326 | 0.04149415577173598 | 200 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.007787591123225894 | 0.8219945277033307 | 0.04560214433979277 | 0.008011730012445742 | 0.04149565542818059 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.007318955241087229 | 0.837522899135072 | 0.046179024536297714 | 0.008006484716173956 | 0.040995428603605055 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.007349071452439088 | 0.8370358628349843 | 0.04828976318055842 | 0.007995945324485719 | 0.04095526148267497 | 347 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.0073261931585090565 | 0.8374946620926874 | 0.047031218164936155 | 0.008000322829345262 | 0.04093409956426049 | 200 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.007309196865505592 | 0.8373987825830973 | 0.050055240458039184 | 0.0079932522578528 | 0.040915929463734776 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.007160219102279065 | 0.8279858380696232 | 0.03955292269164933 | 0.008061904354077658 | 0.04187261497954202 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.0071715444988456 | 0.8270640250702791 | 0.04037889543368633 | 0.008059804531512879 | 0.04191314438861903 | 347 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006773226216656993 | 0.8211621407850467 | 0.03703475273468293 | 0.008094023427715282 | 0.04248432464894268 | 200 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.0068382620158896415 | 0.8203456108625795 | 0.03754644164580798 | 0.008093895707050845 | 0.04246950107063251 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.007082319335268827 | 0.8214707463714084 | 0.036533918167438235 | 0.00810149327688733 | 0.04242817920693373 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.007163669112827945 | 0.8210967899434827 | 0.03766180098580184 | 0.008095023499490345 | 0.04238386241197888 | 347 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.006969360798570251 | 0.8191389777705991 | 0.03628506515946588 | 0.00813412125013689 | 0.042763913548206724 | 200 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.007166988619985396 | 0.8183062220849706 | 0.038040549821818856 | 0.008136788681543152 | 0.0427204482709084 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.006599021733829052 | 0.7954113278374356 | 0.02942963049416516 | 0.00826029916717422 | 0.04539171160809075 | 200 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006677742673742915 | 0.7937853272530975 | 0.03076832883427942 | 0.008252518935185211 | 0.045363786310433814 | 347 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.00494723097902312 | 0.7761710233367567 | 0.026262250123569735 | 0.008332904620564205 | 0.0491173204187187 | 200 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.0049121028692724125 | 0.7712429219059287 | 0.02549498857692945 | 0.008334985358899727 | 0.0492870131279642 | 347 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.011212410976200472 | 0.7426950505737505 | 0.07819751475921309 | 0.02601572736439355 | 0.1222247258492797 | 311 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.011201601935211514 | 0.7427851831184389 | 0.07816561303869263 | 0.02601670137462005 | 0.12223765428079157 | 242 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.011215276282528422 | 0.7426923716907445 | 0.07819829471657119 | 0.026015424769675843 | 0.12222044489958916 | 489 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.011214588676868472 | 0.7426781295279269 | 0.07817422788873749 | 0.026015730889963132 | 0.12222427923754305 | 411 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.010891652804710661 | 0.732838982210284 | 0.0739760207046796 | 0.02604905744461674 | 0.1256436466440567 | 311 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.010897211497068716 | 0.7328286227323297 | 0.0743616874169585 | 0.026047008285902874 | 0.12563178476275194 | 242 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.010894021366990699 | 0.7330862363297703 | 0.07424730322603718 | 0.026044536287797498 | 0.12559513234588623 | 489 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.010889554219439178 | 0.7330202306870931 | 0.07406769517723034 | 0.026047228666059522 | 0.12562081738944209 | 411 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.010720508054346382 | 0.7083271518390549 | 0.06832024541491172 | 0.026128380689268767 | 0.13087434516463511 | 311 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.01075400369935155 | 0.7135304582046269 | 0.06964215125261035 | 0.02611337807918746 | 0.12991384475577158 | 242 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.010821179509027569 | 0.7064764166908246 | 0.06779540256206382 | 0.026137372914309922 | 0.13158529006848937 | 489 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.010784003668800585 | 0.7069881172549127 | 0.0680935038029653 | 0.026130205111700642 | 0.13130568361060557 | 411 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.010797405697162385 | 0.7239288816921506 | 0.07098191338476914 | 0.026094723671767274 | 0.12798376570463574 | 311 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.010817363671624995 | 0.7243601988110983 | 0.07138114825003694 | 0.026089752748404117 | 0.12784932195547788 | 242 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.010831972550813137 | 0.7241491265671509 | 0.07136290523583479 | 0.026087588009737518 | 0.12797989450247446 | 489 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.010812739424230014 | 0.7241348674493775 | 0.07113606878717327 | 0.026091875839195287 | 0.12798809868288907 | 411 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.010397204484506673 | 0.7184356964483741 | 0.06895192159599234 | 0.026103398878162352 | 0.12866326356478033 | 242 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.010406043294216641 | 0.7195199658676396 | 0.06903321120337483 | 0.02609996145183583 | 0.1286075023897751 | 411 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.010038123862833821 | 0.7063873692633035 | 0.06669526053086149 | 0.026276700873730935 | 0.12670881392623784 | 242 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009698919558492385 | 0.7016678047398122 | 0.06472863762885028 | 0.026329143421651165 | 0.12780192487448597 | 411 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.011404911913950048 | 0.7092103185283452 | 0.06579435989637236 | 0.026141230636974128 | 0.13057387952098568 | 242 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.010903701729877226 | 0.7082410545738316 | 0.0661322831631501 | 0.026159547816396195 | 0.1317089124391802 | 411 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.010559800746017393 | 0.7060205318410169 | 0.06510356885790626 | 0.026222834011437345 | 0.1260124252230204 | 242 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.00985934893820875 | 0.6986508907811598 | 0.06347426479718779 | 0.026292075364201854 | 0.127663181716197 | 411 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.010826607040696538 | 0.7165769161736477 | 0.06992349126523084 | 0.026074346999320184 | 0.12892183026667875 | 242 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.010867405965686815 | 0.7169822159130195 | 0.06987043228473347 | 0.02606819417230029 | 0.12960010950354178 | 411 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.010829348891769851 | 0.7125919844497284 | 0.06704730264720829 | 0.02620704217359632 | 0.12492488089923726 | 242 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.010622399921085043 | 0.7097578449139015 | 0.06674338032214995 | 0.026243802377149048 | 0.12544017169690447 | 411 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.01404738035836603 | 0.6994192215552558 | 0.0662666801707327 | 0.026056078114020272 | 0.1715179117376807 | 242 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.013422003286425298 | 0.7000703681527877 | 0.06649201244226514 | 0.0260719142353332 | 0.1803146644229054 | 411 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.01242939093866895 | 0.7093206189928803 | 0.06889328223302982 | 0.02604505905964805 | 0.12446043453523173 | 242 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.012191144296709262 | 0.7100054550874582 | 0.0686582950263595 | 0.026068721098404773 | 0.12374518110906546 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.010314136459259054 | 0.7236771938517396 | 0.07083513974854984 | 0.026086923437668774 | 0.12864975131687822 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.010261073806939061 | 0.7231142638888978 | 0.07081027602142484 | 0.026092553156956067 | 0.12891260148522488 | 411 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.009989239090944173 | 0.7221474499254881 | 0.07145356557684987 | 0.026098905933042885 | 0.12932616382594395 | 242 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.010019875692353172 | 0.7227992747212495 | 0.07042842911597287 | 0.026105279073950195 | 0.12932511291482482 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.010811429945999346 | 0.714530419106499 | 0.06821220928934008 | 0.026112161469332384 | 0.13074688049094585 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.010924414264637425 | 0.7165074432425228 | 0.06962250867137613 | 0.02609455122548406 | 0.13016190531711877 | 411 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.010194135653086746 | 0.7151232745662133 | 0.0662678453874736 | 0.026181930466361057 | 0.13242517942824975 | 242 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.010179199760749 | 0.7161201411886636 | 0.06826282067936575 | 0.02615066062967715 | 0.13241852923614175 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.011180009157868832 | 0.7151162976019284 | 0.06894925117979625 | 0.026112789355848678 | 0.13040569971914875 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.011022807313307674 | 0.714227569687157 | 0.06898668348099925 | 0.02611512838676113 | 0.13028950144912646 | 411 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.01029791507743422 | 0.7083683439040135 | 0.06581267234164852 | 0.026216965212459527 | 0.13374320454163838 | 242 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.010592268400367595 | 0.7106979378670727 | 0.06685521836852976 | 0.026205599804902998 | 0.13292245551857532 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.01036971437359398 | 0.6968421021934186 | 0.0617462214972945 | 0.026369444593471448 | 0.14018099113711982 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.010571468697788942 | 0.6932892505614549 | 0.060901821758469914 | 0.02638571213883279 | 0.1398771468071607 | 411 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.008800491976048142 | 0.6871996662722153 | 0.05791562190939881 | 0.026599349993470762 | 0.1501118972834162 | 242 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.00902784942745175 | 0.6861871841355328 | 0.05976989257673336 | 0.026585546287195474 | 0.1491805793740167 | 411 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.013613864902385746 | 0.6968064433411527 | 0.10034978041646668 | 0.04529862835415747 | 0.21061304838813139 | 349 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.013609486347931923 | 0.6968496425914784 | 0.1003516014076626 | 0.045299008894753516 | 0.21061654761515514 | 268 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.013619456904082252 | 0.6969005051607032 | 0.100399186433299 | 0.045297332318148975 | 0.2105831085155579 | 512 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.01361951480407039 | 0.6968809179815244 | 0.10038946348572833 | 0.04529761182569092 | 0.210586654310025 | 449 |
| logistic | lr_c_0p1_freq_100 | 4 | 0.013240389022405506 | 0.6961657785700122 | 0.10003292562675038 | 0.04535005635261595 | 0.21436989156840133 | 349 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.013241290919699472 | 0.6962107216259916 | 0.10007210837067738 | 0.04534975497085705 | 0.21435173178093486 | 268 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.013251233882987223 | 0.6965724623263514 | 0.10030960951176994 | 0.04534431023513997 | 0.21426644460076522 | 512 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.013250033756513997 | 0.6963809641876613 | 0.10017117128758564 | 0.045346236150234004 | 0.21429788980125775 | 449 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.013214975845567863 | 0.6888435009268186 | 0.09774090852854891 | 0.04541434169799133 | 0.21800342963185693 | 349 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.013191070837122346 | 0.6909597457640698 | 0.09818380611550591 | 0.04540102261173525 | 0.21711639098428034 | 268 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.013269594499277068 | 0.6884917046316723 | 0.09819870063127223 | 0.04541197240426763 | 0.2185840708337649 | 512 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.013244064237622677 | 0.6880779796708834 | 0.09807950829463938 | 0.045407381243937804 | 0.21843662839723182 | 449 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.013202557790584955 | 0.6922471550611969 | 0.09863071427419665 | 0.04539351916771947 | 0.21652908600270884 | 349 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.013191352453504172 | 0.6927682136164565 | 0.09876143518913287 | 0.04538830030350409 | 0.21628078111650542 | 268 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.013242327966319214 | 0.6933540569082397 | 0.09934305747502266 | 0.04537968193167104 | 0.21634208831807317 | 512 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.013226990086650016 | 0.6926932111214005 | 0.09911207321338722 | 0.04538187125334194 | 0.21642442604499015 | 449 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.01294374683341252 | 0.6743092766954306 | 0.09370890124195733 | 0.04540690652352956 | 0.2221389084864037 | 268 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.012959757668006737 | 0.6754137843444409 | 0.09400567841743171 | 0.04540389338379717 | 0.22187315003447333 | 449 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.012835617580657401 | 0.6802369622626505 | 0.09462802343071144 | 0.04562238116710027 | 0.20923072811753649 | 268 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.01273888808612747 | 0.6794528718329992 | 0.09303359096889358 | 0.04568551939529583 | 0.2093828961626628 | 449 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.012945575920372489 | 0.669081677255243 | 0.09192658940333451 | 0.045459457441712085 | 0.22246088234997274 | 268 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.012860893849419288 | 0.6667458872862089 | 0.09227081980556243 | 0.045462748783680694 | 0.2222147819949983 | 449 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.012835566570718661 | 0.6805490469187784 | 0.09483804665151033 | 0.04559591410880352 | 0.20943918119742488 | 268 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.012781920756560313 | 0.678925818356682 | 0.09427792686767168 | 0.0456535278634094 | 0.20932872866427485 | 449 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.013036216313765335 | 0.673180233229576 | 0.0937594551438035 | 0.045385817900779946 | 0.23799991294002762 | 268 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.013032327214514883 | 0.6731729658016901 | 0.09386849663972416 | 0.04538680553159157 | 0.24029371402791275 | 449 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.012989613938273168 | 0.6845073820666121 | 0.09519682804863408 | 0.04551238304266364 | 0.20984763703907847 | 268 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.012962301813836641 | 0.684268094227038 | 0.09494034012373534 | 0.04555650109200142 | 0.20905407283179628 | 449 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.01386489552234178 | 0.6570000921106425 | 0.09033935562004747 | 0.04539623244480453 | 0.38806925431900624 | 268 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.013836087266587489 | 0.65439753758052 | 0.0895318617238718 | 0.04541118514415433 | 0.3994230807751628 | 449 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.013283893181886242 | 0.678161381088045 | 0.09531019014448162 | 0.045389477206279706 | 0.21494450352213762 | 268 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.01327580593246454 | 0.6779941925427632 | 0.09526529020864796 | 0.04541365044800027 | 0.21336224043755808 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.012954355757617171 | 0.6879870378495562 | 0.10004652405116389 | 0.04531946510294601 | 0.21859956776676648 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.012942949991846375 | 0.6888437931320902 | 0.10040856992679877 | 0.04531169538873583 | 0.21846966323946218 | 449 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.012940042213187635 | 0.6871092626394144 | 0.09981883982278371 | 0.045326618098423095 | 0.21850892279955889 | 268 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.012949402048934243 | 0.6882878773157126 | 0.10049215516139583 | 0.045312151813932985 | 0.21829887485126956 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.013125544902736496 | 0.6860383727356119 | 0.09951480473546617 | 0.045311325617582957 | 0.221902986209303 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.013126876047502055 | 0.6855360106047514 | 0.09942289053493136 | 0.04531338725920146 | 0.22200246265236417 | 449 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.012900876414834178 | 0.6848190991096562 | 0.09892614803849645 | 0.045343479191506345 | 0.22292259654148308 | 268 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.012887702656482274 | 0.686136959023885 | 0.09921855200905794 | 0.045338641828641627 | 0.22232883051790248 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.013283427242465068 | 0.6864404189115101 | 0.09959524093942089 | 0.04529488049767752 | 0.22151457376745406 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.013249117491596712 | 0.6862981008052385 | 0.09967763115001373 | 0.04529774086982183 | 0.22160450654816147 | 449 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.013149395420844007 | 0.6835262934409795 | 0.09960924612907143 | 0.04530627082722818 | 0.22204968633061795 | 268 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.013191373678862976 | 0.6873337045659921 | 0.10066496623995196 | 0.0452950586880526 | 0.2211762239623465 | 449 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.013425094860915325 | 0.6769291184664645 | 0.09264692367982039 | 0.0455222396659496 | 0.23081506875458002 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.013260129611306896 | 0.675472536456001 | 0.09386755374125236 | 0.045496887808654726 | 0.231640600426562 | 449 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.012501650304004361 | 0.6761611700215789 | 0.09368568137539884 | 0.04558457971757065 | 0.2355234359156027 | 268 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.012507797655789955 | 0.6775430134643288 | 0.09614299433852325 | 0.04549076775167126 | 0.23455092107154174 | 449 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 8 | 0.287194582997472 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 8 | 0.29763683400233276 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 8 | 0.2937532499927329 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 8 | 0.2914160840009572 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.3041819999925792 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.3089747080084635 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.3080969169968739 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.3059394160081865 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 18 | 0.40900462499121204 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 23 | 0.4239572920050705 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 32 | 0.49459083398687653 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 30 | 0.45049699999799486 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.33930570899974555 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 13 | 0.34879499999806285 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 14 | 0.3519288340030471 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 14 | 0.3655255420017056 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.381774207999115 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.1917992499948014 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9541906660015229 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.8266823340090923 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.521491249994142 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.480593874992337 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 0.9156225839979015 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.844922708987724 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.3720201670075767 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.55161287500232 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9406060829933267 | NA | 50 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.890752874998725 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 3.086528208004893 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.9767854579986306 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.1852937499934342 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.1124938330031 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.218421124998713 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.1025993330113124 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.0908836250018794 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.095088999994914 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.691228208001121 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.7451255420019152 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.7600537079997594 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.696636499997112 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.0939744579955004 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.0858771670027636 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.1006446249957662 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.0811121659935452 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.6972540410060901 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.6781807500083232 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7089284170069732 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.7690855419932632 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.45176095799251925 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.5020009159925394 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.46509504099958576 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.4569181669940008 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.48716024999157526 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.48945366599946283 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.4875159579969477 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.49209562499891035 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.7517177499976242 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7178848330077017 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 20 | 0.794208167004399 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 20 | 0.7790823749965057 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 14 | 0.602222833011183 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.5722404170082882 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 14 | 0.5857632080005715 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.583556249999674 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.865233541000634 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.185101541996119 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.5662635829939973 | 12.0 | 50 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.329166792012984 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 5.611652959007188 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 5.478999459010083 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.6100107909878716 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.4433134169958066 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 6.816684499994153 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 6.966024624998681 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.005396541004302 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.8161518330016406 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 9.32708800000546 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 10.956607959000394 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.881226790996152 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.726824166995357 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.335589375012205 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.305888999995659 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.2995968750037719 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.3114628339972114 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 2.0078121249971446 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 1.9751335000037216 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 1.9768411249970086 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.961432291995152 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3113512500131037 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.3045044999889797 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.2971005830040667 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.2958602500002598 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 2.0704994999978226 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 2.005122750008013 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.954718375010998 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.9609295830014162 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.619648000007146 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.6276070419989992 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6319537919916911 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6265630419948138 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 11 | 0.735052584001096 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 10 | 0.6868119590071728 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.7159681250050198 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.713666291005211 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_100 | 3 | 18 | 1.143053375009913 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.0821533749985974 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 20 | 1.1688022919988725 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 20 | 1.2978394579986343 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 15 | 0.8926097089861287 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 14 | 0.8886338750016876 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 0.9011748329940019 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 16 | 0.9187532499927329 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 10.906278666006983 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 9.995070207995013 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.2304524170031073 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.062886083003832 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 10.651322375008021 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 9.693021208004211 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.4313818749942584 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.0445846660004463 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 14.545700333008426 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 14.695490458005224 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.42102745900047 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.936587917007273 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 20.16563795899856 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 23.6963765410037 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.162440832995344 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 5.397140041997773 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.536733250002726 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.5328578330081655 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.5182470419968013 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.5636146670003654 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.2149059169896645 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.236514000003808 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.253031332991668 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.205419083999004 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.6596867080079392 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.5229087920015445 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.5239676669880282 | 3.0 | NA | NA | 0.1 | 1 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.520191709001665 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.2380471250071423 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.224075374993845 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.296964958004537 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.2007104579970473 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 8 | 0.795059083000524 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 8 | 0.7837462089955807 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 9 | 0.8466414170106873 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 8 | 0.7979528750001919 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 11 | 0.9771547080017626 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 0.9056027920014458 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 0.9781488750013523 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 11 | 0.9591483749973122 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 18 | 1.5114690829941537 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 16 | 1.5605496670032153 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 22 | 1.772978167005931 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.663714082998922 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 14 | 1.138516791004804 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 13 | 1.1022778329934226 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.273444375008694 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 16 | 1.3001127910101786 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 18.168112125000334 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 18.48248325000168 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.1218297090090346 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.974307249998674 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 18.283162708990858 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 16.07790654200653 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 3.4716477500041947 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 2.852862790998188 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 28.481357707991265 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 26.787284167003236 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.408756833989173 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 4.422260708000977 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 44.506090708004194 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 47.145883042001515 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 11.4648326250026 | NA | 5 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.777823625001474 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.763012333001825 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.8000250000040978 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.7583303330029594 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.7830009579920443 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.487365958004375 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.620298499998171 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.4801191669976106 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.4857018329930725 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.77084929200646 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7677880409901263 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.7506604999944102 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.7654374999983702 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.4891890410071937 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.5113821669947356 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.4654041250032606 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.4742257079924457 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 49533 | 807 | logistic | raw | 49533 | 807 | 0.016292168857125554 | 0.009054181972022077 | 0.7766196694247747 |
| logistic | validation | calibrated | 49533 | 807 | logistic | calibrated | 49533 | 807 | 0.016292168857125554 | 0.021104621499077724 | 0.776318374079857 |
| rf | validation | raw | 49533 | 807 | rf | raw | 49533 | 807 | 0.016292168857125554 | 0.014795153442495034 | 0.7354671630416876 |
| rf | validation | calibrated | 49533 | 807 | rf | calibrated | 49533 | 807 | 0.016292168857125554 | 0.037056469695139135 | 0.7346846598034144 |
| xgb | validation | raw | 49533 | 807 | xgb | raw | 49533 | 807 | 0.016292168857125554 | 0.011069971214091370 | 0.7666673100743245 |
| xgb | validation | calibrated | 49533 | 807 | xgb | calibrated | 49533 | 807 | 0.016292168857125554 | 0.028797665468744310 | 0.7659311296443034 |
| logistic | test | raw | 37022 | 638 | logistic | raw | 37022 | 638 | 0.017232996596618227 | 0.11760389741948876 | 0.7734521685097725 |
| logistic | test | calibrated | 37022 | 638 | logistic | calibrated | 37022 | 638 | 0.017232996596618227 | 0.1438094601820402 | 0.7688401607168951 |
| rf | test | raw | 37022 | 638 | rf | raw | 37022 | 638 | 0.017232996596618227 | 0.046445594700486360 | 0.796049557075624 |
| rf | test | calibrated | 37022 | 638 | rf | calibrated | 37022 | 638 | 0.017232996596618227 | 0.1227888382941743 | 0.7899249049842434 |
| xgb | test | raw | 37022 | 638 | xgb | raw | 37022 | 638 | 0.017232996596618227 | 0.075672821997728570 | 0.7902944394242672 |
| xgb | test | calibrated | 37022 | 638 | xgb | calibrated | 37022 | 638 | 0.017232996596618227 | 0.104204505050321220 | 0.788907952925672 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.061900304690808164 | 0.015777625054154573 | 0.07717497640044939 |
| logistic | 0.059318280817984334 | 0.015676972312680004 | 0.0753584899249814 |
| rf | 0.04796082788165338 | 0.015809341515726133 | 0.07751059858043519 |
| rf | 0.04538190102845495 | 0.016418330864122865 | 0.08524524950910237 |
| xgb | 0.05179793705054798 | 0.01578635563014624 | 0.07703500065267808 |
| xgb | 0.050068396430894294 | 0.015927556764526996 | 0.07952167406410683 |
| logistic | 0.05914546228221564 | 0.03362853223298878 | 0.1571404801859408 |
| logistic | 0.04908627224747175 | 0.03658934313810796 | 0.18272016656202764 |
| rf | 0.06326783916870828 | 0.017419730375058434 | 0.09405952448103891 |
| rf | 0.05237902839152505 | 0.029476232339109988 | 0.16050402607524264 |
| xgb | 0.05773209570962165 | 0.022826872117883056 | 0.11470727666053984 |
| xgb | 0.05321096600845386 | 0.031542384618544334 | 0.4154101525153789 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 49533 | 807 | logistic | raw | 0.8341674443302104 | 1.0538280499504602 | 0.007237986885103474 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 49533 | 807 | logistic | calibrated | 0.11552519391252272 | 1.1142787316741336 | 0.004832409150686277 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 49533 | 807 | rf | raw | 0.9122833569446295 | 1.2087183101638417 | 0.00149701541463052 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 49533 | 807 | rf | calibrated | -0.48371595007631646 | 1.127715339991219 | 0.020764300838013592 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 49533 | 807 | xgb | raw | -0.16095159242022 | 0.8592667877496902 | 0.005444796093127514 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 49533 | 807 | xgb | calibrated | 0.3803183052968254 | 1.307933820852546 | 0.012505496611618762 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | raw | 37022 | 638 | logistic | raw | -2.1014643597148535 | 1.0341900893939382 | 0.10037090082287053 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 37022 | 638 | logistic | calibrated | -1.52971133419676 | 1.5529475899317657 | 0.12657646358542193 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 37022 | 638 | rf | raw | 4.692565768332057 | 3.0308644885014933 | 0.029212598103868146 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 37022 | 638 | rf | calibrated | -1.1912730015022637 | 1.5464280339718026 | 0.1055558416975561 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 37022 | 638 | xgb | raw | -1.4539226450679361 | 1.0834288809874992 | 0.058439825401110326 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 37022 | 638 | xgb | calibrated | -3.7582709202711873 | 0.15596204589319754 | 0.08697150845370302 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R6_t1_temporal/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 301 | 0.00813030090216628 | 32 | 269 | 36115 | 606 |
| logistic | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 0 | 0.0 | 0 | 0 | 36384 | 638 |
| logistic | 1:1 | 1 | 1 | 0.2528735632183909 | 37022 | 638 | 3731 | 0.10077791583382853 | 248 | 3483 | 32901 | 390 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 1769 | 0.04778239965425963 | 139 | 1630 | 34754 | 499 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 0 | 0.0 | 0 | 0 | 36384 | 638 |
| logistic | 2:1 | 1 | 2 | 0.25287356321839083 | 37022 | 638 | 7119 | 0.19229107017449085 | 351 | 6768 | 29616 | 287 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 8700 | 0.2349954081357031 | 398 | 8302 | 28082 | 240 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 8368 | 0.22602776727351304 | 385 | 7983 | 28401 | 253 |
| logistic | 5:1 | 1 | 5 | 0.1402439024390244 | 37022 | 638 | 12948 | 0.3497379936254119 | 479 | 12469 | 23915 | 159 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 17637 | 0.47639241532062016 | 543 | 17094 | 19290 | 95 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 27057 | 0.7308357193020366 | 612 | 26445 | 9939 | 26 |
| logistic | 10:1 | 1 | 10 | 0.1 | 37022 | 638 | 26477 | 0.7151693587596564 | 607 | 25870 | 10514 | 31 |
| rf | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 0 | 0.0 | 0 | 0 | 36384 | 638 |
| rf | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 341 | 0.009210739560261466 | 30 | 311 | 36073 | 608 |
| rf | 1:1 | 1 | 1 | 0.3366475670173503 | 37022 | 638 | 372 | 0.010048079520285236 | 32 | 340 | 36044 | 606 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 0 | 0.0 | 0 | 0 | 36384 | 638 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 372 | 0.010048079520285236 | 32 | 340 | 36044 | 606 |
| rf | 2:1 | 1 | 2 | 0.3366475670173503 | 37022 | 638 | 372 | 0.010048079520285236 | 32 | 340 | 36044 | 606 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 0 | 0.0 | 0 | 0 | 36384 | 638 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 8517 | 0.2300524012749176 | 424 | 8093 | 28291 | 214 |
| rf | 5:1 | 1 | 5 | 0.2857142857142857 | 37022 | 638 | 649 | 0.017530117227594403 | 56 | 593 | 35791 | 582 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 671 | 0.018124358489546757 | 58 | 613 | 35771 | 580 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 22488 | 0.6074226135811139 | 601 | 21887 | 14497 | 37 |
| rf | 10:1 | 1 | 10 | 0.17105263157894737 | 37022 | 638 | 8505 | 0.22972826967748905 | 424 | 8081 | 28303 | 214 |
| xgb | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 57 | 0.001539625087785641 | 8 | 49 | 36335 | 630 |
| xgb | 1:1 | 1 | 1 | 0.5 | 37022 | 638 | 368 | 0.009940035654475717 | 38 | 330 | 36054 | 600 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | 1:1 | 1 | 1 | 0.26562500000000006 | 37022 | 638 | 379 | 0.010237156285451894 | 38 | 341 | 36043 | 600 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 452 | 0.01220895683647561 | 41 | 411 | 35973 | 597 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 37022 | 638 | 375 | 0.010129112419642376 | 38 | 337 | 36047 | 600 |
| xgb | 2:1 | 1 | 2 | 0.26562500000000006 | 37022 | 638 | 379 | 0.010237156285451894 | 38 | 341 | 36043 | 600 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 3739 | 0.10099400356544758 | 232 | 3507 | 32877 | 406 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 37022 | 638 | 5540 | 0.14964075414618336 | 313 | 5227 | 31157 | 325 |
| xgb | 5:1 | 1 | 5 | 0.265625 | 37022 | 638 | 1705 | 0.04605369780130733 | 122 | 1583 | 34801 | 516 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 10600 | 0.28631624439522446 | 448 | 10152 | 26232 | 190 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 37022 | 638 | 17414 | 0.47036896980173953 | 569 | 16845 | 19539 | 69 |
| xgb | 10:1 | 1 | 10 | 0.1065015479876161 | 37022 | 638 | 14304 | 0.3863648641348387 | 527 | 13777 | 22607 | 111 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.050156739811912224 | 0.9926066402814424 | 0.10631229235880399 | 0.983497181449307 | 0.007393359718557608 | 0.9498432601880877 | 875 | 0.02363459564583221 | 638 |
| logistic | 0.0 | 1.0 | NA | 0.9827670034033817 | 0.0 | 1.0 | 638 | 0.017232996596618227 | 638 |
| logistic | 0.3887147335423197 | 0.9042711081794196 | 0.06647011525060306 | 0.9882851221050735 | 0.09572889182058048 | 0.6112852664576802 | 3873 | 0.10461347307006645 | 638 |
| logistic | 0.21786833855799373 | 0.9552000879507476 | 0.07857546636517806 | 0.9858451762970527 | 0.04479991204925242 | 0.7821316614420063 | 2628 | 0.07098481983685376 | 1276 |
| logistic | 0.0 | 1.0 | NA | 0.9827670034033817 | 0.0 | 1.0 | 1276 | 0.034465993193236455 | 1276 |
| logistic | 0.5501567398119123 | 0.8139841688654353 | 0.0493046776232617 | 0.9904023007724977 | 0.18601583113456466 | 0.4498432601880878 | 7342 | 0.1983145156933715 | 1276 |
| logistic | 0.6238244514106583 | 0.7718227792436235 | 0.04574712643678161 | 0.9915260221735753 | 0.22817722075637642 | 0.3761755485893417 | 9502 | 0.2566582032305116 | 3190 |
| logistic | 0.603448275862069 | 0.7805903693931399 | 0.04600860420650096 | 0.9911705172052767 | 0.21940963060686017 | 0.39655172413793105 | 9248 | 0.24979741775160716 | 3190 |
| logistic | 0.7507836990595611 | 0.6572944151275286 | 0.03699413036762434 | 0.9933953642934286 | 0.34270558487247144 | 0.24921630094043887 | 13264 | 0.3582734590243639 | 3190 |
| logistic | 0.8510971786833855 | 0.5301781002638523 | 0.030787548902874638 | 0.9950993035852463 | 0.46982189973614774 | 0.14890282131661442 | 18044 | 0.4873858786667387 | 6380 |
| logistic | 0.9592476489028213 | 0.27316952506596304 | 0.022618915622574565 | 0.9973908680381335 | 0.726830474934037 | 0.04075235109717868 | 26705 | 0.721327859110799 | 6380 |
| logistic | 0.95141065830721 | 0.28897317502198766 | 0.022925558031499038 | 0.9970602181128497 | 0.7110268249780123 | 0.048589341692789965 | 26180 | 0.7071471017232996 | 6380 |
| rf | 0.0 | 1.0 | NA | 0.9827670034033817 | 0.0 | 1.0 | 638 | 0.017232996596618227 | 638 |
| rf | 0.047021943573667714 | 0.9914522867194371 | 0.08797653958944282 | 0.9834246612687767 | 0.008547713280562886 | 0.9529780564263323 | 919 | 0.024823078169736914 | 638 |
| rf | 0.050156739811912224 | 0.9906552330694811 | 0.08602150537634409 | 0.9834652114597544 | 0.009344766930518914 | 0.9498432601880877 | 946 | 0.025552374263951166 | 638 |
| rf | 0.0 | 1.0 | NA | 0.9827670034033817 | 0.0 | 1.0 | 1276 | 0.034465993193236455 | 1276 |
| rf | 0.050156739811912224 | 0.9906552330694811 | 0.08602150537634409 | 0.9834652114597544 | 0.009344766930518914 | 0.9498432601880877 | 1552 | 0.041921019934093244 | 1276 |
| rf | 0.050156739811912224 | 0.9906552330694811 | 0.08602150537634409 | 0.9834652114597544 | 0.009344766930518914 | 0.9498432601880877 | 1552 | 0.041921019934093244 | 1276 |
| rf | 0.0 | 1.0 | NA | 0.9827670034033817 | 0.0 | 1.0 | 3190 | 0.08616498298309114 | 3190 |
| rf | 0.664576802507837 | 0.7775670624450308 | 0.04978278736644358 | 0.9924925451675145 | 0.2224329375549692 | 0.335423197492163 | 9163 | 0.24750148560315488 | 3190 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| rf | 0.0877742946708464 | 0.9837016270888302 | 0.08628659476117104 | 0.9839991202265417 | 0.016298372911169746 | 0.9122257053291536 | 3503 | 0.09461941548268597 | 3190 |
| rf | 0.09090909090909091 | 0.9831519349164468 | 0.08643815201192251 | 0.9840444554482682 | 0.016848065083553212 | 0.9090909090909091 | 6413 | 0.1732213278591108 | 6380 |
| rf | 0.9420062695924765 | 0.3984443711521548 | 0.026725364638918534 | 0.9974542452181092 | 0.6015556288478452 | 0.05799373040752351 | 22257 | 0.6011830803306142 | 6380 |
| rf | 0.664576802507837 | 0.7778968777484608 | 0.04985302763080541 | 0.9924957043167234 | 0.22210312225153914 | 0.335423197492163 | 10221 | 0.27607908810977255 | 6380 |
| xgb | 0.012539184952978056 | 0.9986532541776605 | 0.14035087719298245 | 0.9829568510753416 | 0.0013467458223394898 | 0.987460815047022 | 679 | 0.018340446221165794 | 638 |
| xgb | 0.05956112852664577 | 0.9909300791556728 | 0.10326086956521739 | 0.9836307087903093 | 0.009069920844327176 | 0.9404388714733543 | 930 | 0.02512019880071309 | 638 |
| xgb | 0.05956112852664577 | 0.9906277484608619 | 0.10026385224274406 | 0.9836257948312093 | 0.009372251539138083 | 0.9404388714733543 | 941 | 0.025417319431689265 | 638 |
| xgb | 0.06426332288401254 | 0.9887038258575198 | 0.09070796460176991 | 0.9836751435602953 | 0.011296174142480 21 | 0.9357366771159875 | 1605 | 0.04335260115606936 | 1276 |
| xgb | 0.05956112852664577 | 0.9907376868953386 | 0.10133333333333333 | 0.9836275820667449 | 0.009262313104661 39 | 0.9404388714733543 | 1537 | 0.04151585543730755 | 1276 |
| xgb | 0.05956112852664577 | 0.9906277484608619 | 0.10026385224274406 | 0.9836257948312093 | 0.009372251539138083 | 0.9404388714733543 | 1541 | 0.041623899303117065 | 1276 |
| xgb | 0.36363636363636365 | 0.9036114775725593 | 0.06204867611660872 | 0.9878015803863834 | 0.09638852242744063 | 0.6363636363636364 | 5537 | 0.1495597212468262 | 3190 |
| xgb | 0.49059561128526646 | 0.8563379507475813 | 0.05649819494584837 | 0.9896766406200368 | 0.14366204925241866 | 0.5094043887147336 | 6852 | 0.18507914213170548 | 3190 |
| xgb | 0.19122257053291536 | 0.9564918645558487 | 0.07155425219941348 | 0.985389472492001 | 0.043508135444151275 | 0.8087774294670846 | 4163 | 0.11244665334125654 | 3190 |
| xgb | 0.7021943573667712 | 0.7209762532981531 | 0.04226415094339623 | 0.9928090227840436 | 0.279023746701847 | 0.29780564263322884 | 12052 | 0.3255361676840797 | 6380 |
| xgb | 0.8918495297805643 | 0.5370217678100264 | 0.032674859308602276 | 0.9964810281517748 | 0.46297823218997364 | 0.10815047021943573 | 17535 | 0.47363729674247745 | 6380 |
| xgb | 0.8260188087774295 | 0.6213445470536499 | 0.03684284116331096 | 0.995114006514658 | 0.37865545294635006 | 0.17398119122257052 | 14887 | 0.4021122575765761 | 6380 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 1.371473354231975 |
| logistic | 1.0 |
| logistic | 6.070532915360501 |
| logistic | 2.0595611285266457 |
| logistic | 1.0 |
| logistic | 5.753918495297806 |
| logistic | 2.9786833855799375 |
| logistic | 2.8990595611285266 |
| logistic | 4.157993730407523 |
| logistic | 2.8282131661442005 |
| logistic | 4.185736677115988 |
| logistic | 4.103448275862069 |
| rf | 1.0 |
| rf | 1.4404388714733543 |
| rf | 1.4827586206896552 |
| rf | 1.0 |
| rf | 1.2163009404388714 |
| rf | 1.2163009404388714 |
| rf | 1.0 |
| rf | 2.872413793103448 |

| model | decision_relative_cost_vs_all_negative |
|---|---|
| rf | 1.0981191222570532 |
| rf | 1.0051724137931035 |
| rf | 3.4885579937304074 |
| rf | 1.602037617554859 |
| xgb | 1.0642633228840126 |
| xgb | 1.457680250783699 |
| xgb | 1.474921630094044 |
| xgb | 1.2578369905956113 |
| xgb | 1.2045454545454546 |
| xgb | 1.207680250783699 |
| xgb | 1.7357366771159874 |
| xgb | 2.147962382445141 |
| xgb | 1.3050156739811911 |
| xgb | 1.8890282131661442 |
| xgb | 2.748432601880878 |
| xgb | 2.333385579937304 |

All Non Risky Benchmark (4 records)

| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 37022 | 638 | 0 | 0 | 36384 | 638 | 638 | 0.017232996596618227 |
| test | all_non_risky | 2:1 | 2:1 | 37022 | 638 | 0 | 0 | 36384 | 638 | 1276 | 0.034465993193236455 |
| test | all_non_risky | 5:1 | 5:1 | 37022 | 638 | 0 | 0 | 36384 | 638 | 3190 | 0.08616498298309114 |
| test | all_non_risky | 10:1 | 10:1 | 37022 | 638 | 0 | 0 | 36384 | 638 | 6380 | 0.17232996596618227 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 27701 | 224 | 0.008086350673260892 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 4587 | 284 | 0.06191410507957271 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 953 | 29 | 0.030430220356768102 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 3781 | 101 | 0.026712509918011107 |

R7

Add First Payment Date continuous month index relative to 2015-01 and calendar-quarter sine/cosine. Scheduled-payment-date proxy for origination timing, not exact origination month.

Role Count (3 records)

| role | role_event_rate | role_n | role_positive_n |
|---|---|---|---|
| train | 0.019003069726648152 | 239435 | 4550 |
| validation | 0.01699585574464333 | 45364 | 771 |
| test | 0.017957760752361263 | 49338 | 886 |

Selected Settings (3 records)

*Column panel 1 of 3 — Selected Settings (3 records)*

| model | config_id | role | setting_C | setting_config_id | setting_folds | setting_grid_size | setting_min_frequency | setting_selection_metric | setting_tie_tolerance_absolute_log_loss |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | train | 0.01 | lr_c_0p01_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | train | NA | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | train | NA | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 16 | 200 | unweighted mean forward-fold log loss | 0.0001 |

*Column panel 2 of 3 — Selected Settings (3 records)*

| model | config_id | setting_configuration_json | fixed_protocol_settings_json | fixed_settings_source | setting_max_depth | setting_max_features |
|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | {"C":0.01,"config_id":"lr_c_0p01_freq_200","folds":4,"grid_size":16,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "max_iter": 2000, "penalty": "l2", "random_state": 20260925, "resampling": false, "solver": "liblinear"} | 代码/07_robustness/robustness_protocol_freeze.md | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | {"config_id":"rf_depth_none_leaf_5_features_sqrt_freq_200","folds":4,"grid_size":16,"max_depth":"None","max_features":"sqrt","min_frequency":200,"min_samples_leaf":5,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"bootstrap": true, "class_weight": null, "criterion": "gini", "n_estimators": 400, "n_jobs": -1, "random_state": 20260925, "resampling": false} | 代码/07_robustness/robustness_protocol_freeze.md | None | sqrt |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | {"config_id":"xgb_lr_0p03_depth_3_child_10_freq_200","folds":4,"grid_size":16,"learning_rate":0.03,"max_depth":3,"min_child_weight":10,"min_frequency":200,"selection_metric":"unweighted mean forward-fold log loss","tie_tolerance_absolute_log_loss":0.0001} | {"class_weight": null, "colsample_bytree": 0.8, "n_estimators": 600, "n_jobs": -1, "random_state": 20260925, "reg_alpha": 0, "reg_lambda": 1, "resampling": false, "subsample": 0.8, "tree_method": "hist"} | 代码/07_robustness/robustness_protocol_freeze.md | 3 | NA |

*Column panel 3 of 3 — Selected Settings (3 records)*

| model | config_id | setting_min_samples_leaf | setting_learning_rate | setting_min_child_weight |
|---|---|---|---|---|
| logistic | lr_c_0p01_freq_200 | NA | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 5 | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 0.03 | 10 |

Development Fold (192 records)

*Column panel 1 of 4 — Development Fold (192 records)*

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 1 |
| logistic | lr_c_0p01_freq_200 | 1 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 1 |
| logistic | lr_c_0p01_freq_25 | 1 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 1 |
| logistic | lr_c_0p01_freq_50 | 1 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 1 |
| logistic | lr_c_0p1_freq_100 | 1 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 1 |
| logistic | lr_c_0p1_freq_200 | 1 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 1 |
| logistic | lr_c_0p1_freq_25 | 1 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 1 |
| logistic | lr_c_0p1_freq_50 | 1 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 1 |
| logistic | lr_c_10p0_freq_100 | 1 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 1 |
| logistic | lr_c_10p0_freq_200 | 1 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 1 |
| logistic | lr_c_10p0_freq_25 | 1 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 1 |
| logistic | lr_c_10p0_freq_50 | 1 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 1 |
| logistic | lr_c_1p0_freq_100 | 1 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 1 |
| logistic | lr_c_1p0_freq_200 | 1 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 1 |
| logistic | lr_c_1p0_freq_25 | 1 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 1 |
| logistic | lr_c_1p0_freq_50 | 1 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | development | True | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 1 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 1 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 2 |
| logistic | lr_c_0p01_freq_200 | 2 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 2 |
| logistic | lr_c_0p01_freq_25 | 2 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 2 |
| logistic | lr_c_0p01_freq_50 | 2 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 2 |
| logistic | lr_c_0p1_freq_100 | 2 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 2 |
| logistic | lr_c_0p1_freq_200 | 2 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 2 |
| logistic | lr_c_0p1_freq_25 | 2 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 2 |
| logistic | lr_c_0p1_freq_50 | 2 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 2 |
| logistic | lr_c_10p0_freq_100 | 2 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 2 |
| logistic | lr_c_10p0_freq_200 | 2 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 2 |
| logistic | lr_c_10p0_freq_25 | 2 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 2 |
| logistic | lr_c_10p0_freq_50 | 2 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 2 |
| logistic | lr_c_1p0_freq_100 | 2 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 2 |
| logistic | lr_c_1p0_freq_200 | 2 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 2 |
| logistic | lr_c_1p0_freq_25 | 2 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 2 |
| logistic | lr_c_1p0_freq_50 | 2 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | development | True | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 2 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 2 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 2 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 2 |
| logistic | lr_c_0p01_freq_100 | 3 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 3 |
| logistic | lr_c_0p01_freq_200 | 3 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 3 |
| logistic | lr_c_0p01_freq_25 | 3 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 3 |
| logistic | lr_c_0p01_freq_50 | 3 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 3 |
| logistic | lr_c_0p1_freq_100 | 3 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 3 |
| logistic | lr_c_0p1_freq_200 | 3 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 3 |
| logistic | lr_c_0p1_freq_25 | 3 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 3 |
| logistic | lr_c_0p1_freq_50 | 3 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 3 |
| logistic | lr_c_10p0_freq_100 | 3 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 3 |
| logistic | lr_c_10p0_freq_200 | 3 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 3 |
| logistic | lr_c_10p0_freq_25 | 3 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 3 |
| logistic | lr_c_10p0_freq_50 | 3 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 3 |
| logistic | lr_c_1p0_freq_100 | 3 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 3 |
| logistic | lr_c_1p0_freq_200 | 3 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 3 |
| logistic | lr_c_1p0_freq_25 | 3 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 3 |
| logistic | lr_c_1p0_freq_50 | 3 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | development | True | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 3 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 3 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 3 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 3 |
| logistic | lr_c_0p01_freq_100 | 4 | development | False | logistic | lr_c_0p01_freq_100 | 0.01 | 100 | 4 |
| logistic | lr_c_0p01_freq_200 | 4 | development | True | logistic | lr_c_0p01_freq_200 | 0.01 | 200 | 4 |
| logistic | lr_c_0p01_freq_25 | 4 | development | False | logistic | lr_c_0p01_freq_25 | 0.01 | 25 | 4 |
| logistic | lr_c_0p01_freq_50 | 4 | development | False | logistic | lr_c_0p01_freq_50 | 0.01 | 50 | 4 |
| logistic | lr_c_0p1_freq_100 | 4 | development | False | logistic | lr_c_0p1_freq_100 | 0.1 | 100 | 4 |
| logistic | lr_c_0p1_freq_200 | 4 | development | False | logistic | lr_c_0p1_freq_200 | 0.1 | 200 | 4 |
| logistic | lr_c_0p1_freq_25 | 4 | development | False | logistic | lr_c_0p1_freq_25 | 0.1 | 25 | 4 |
| logistic | lr_c_0p1_freq_50 | 4 | development | False | logistic | lr_c_0p1_freq_50 | 0.1 | 50 | 4 |
| logistic | lr_c_10p0_freq_100 | 4 | development | False | logistic | lr_c_10p0_freq_100 | 10.0 | 100 | 4 |
| logistic | lr_c_10p0_freq_200 | 4 | development | False | logistic | lr_c_10p0_freq_200 | 10.0 | 200 | 4 |
| logistic | lr_c_10p0_freq_25 | 4 | development | False | logistic | lr_c_10p0_freq_25 | 10.0 | 25 | 4 |
| logistic | lr_c_10p0_freq_50 | 4 | development | False | logistic | lr_c_10p0_freq_50 | 10.0 | 50 | 4 |
| logistic | lr_c_1p0_freq_100 | 4 | development | False | logistic | lr_c_1p0_freq_100 | 1.0 | 100 | 4 |
| logistic | lr_c_1p0_freq_200 | 4 | development | False | logistic | lr_c_1p0_freq_200 | 1.0 | 200 | 4 |
| logistic | lr_c_1p0_freq_25 | 4 | development | False | logistic | lr_c_1p0_freq_25 | 1.0 | 25 | 4 |
| logistic | lr_c_1p0_freq_50 | 4 | development | False | logistic | lr_c_1p0_freq_50 | 1.0 | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_0p5_freq_50 | NA | 50 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | development | True | rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | NA | 200 | 4 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | development | False | rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | development | True | xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | NA | 200 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | NA | 50 | 4 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | NA | 200 | 4 |

| model | config_id | fold | role | selected_configuration | fold_model | fold_config_id | fold_C | fold_min_frequency | fold_fold |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | development | False | xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | NA | 50 | 4 |

*Column panel 2 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_10p0_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_100 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_25 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_1p0_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 2015 | 2016 | 47221 | 252 | 48031 | 330 | 48031 | 330 | 0.006870562761549832 |
| logistic | lr_c_0p01_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_10p0_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_100 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_25 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_1p0_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 2015,2016 | 2017 | 95252 | 582 | 48211 | 397 | 48211 | 397 | 0.008234635249216984 |
| logistic | lr_c_0p01_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_10p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_100 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_25 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_1p0_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 2015,2016,2017 | 2018 | 143463 | 979 | 47393 | 1279 | 47393 | 1279 | 0.026987107800730065 |
| logistic | lr_c_0p01_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p01_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_0p1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_10p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_100 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| logistic | lr_c_1p0_freq_25 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_1p0_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

| model | config_id | fold | fold_fit_years | fold_validation_year | fold_fit_n | fold_fit_positive_n | fold_validation_n | fold_validation_positive_n | fold_n | fold_positive_n | fold_prevalence |
|---|---|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 2015,2016,2017,2018 | 2019 | 190856 | 2258 | 48579 | 2292 | 48579 | 2292 | 0.04718088062743161 |

*Column panel 3 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 0.010937970371420054 | 0.8136134621407467 | 0.028649197132095003 | 0.006771495080130922 5 | 0.037952484924665096 | 202 |
| logistic | lr_c_0p01_freq_200 | 1 | 0.01093426031315728 | 0.8135658803925715 | 0.028775993913756452 | 0.006771074222758517 | 0.03794514829621414 | 140 |
| logistic | lr_c_0p01_freq_25 | 1 | 0.01092425291310773 | 0.8123351076433819 | 0.028431000925215714 | 0.006773078281316821 | 0.037977611463216744 | 344 |
| logistic | lr_c_0p01_freq_50 | 1 | 0.010929525925465635 | 0.8125664730997951 | 0.028478304224520166 | 0.006772962421406947 | 0.03797515042576807 | 269 |
| logistic | lr_c_0p1_freq_100 | 1 | 0.01179893724604475 | 0.8431168141446751 | 0.04042225273937665 | 0.006784105001840768 | 0.03659462204628081 | 202 |
| logistic | lr_c_0p1_freq_200 | 1 | 0.011773512670533086 | 0.8427421317004344 | 0.03994014438381027 | 0.006787022859830852 | 0.03661053825318829 | 140 |
| logistic | lr_c_0p1_freq_25 | 1 | 0.011741049181260479 | 0.842393749448109 | 0.04025223711791156 | 0.006785276350579007 5 | 0.036590009875687404 | 344 |
| logistic | lr_c_0p1_freq_50 | 1 | 0.011763982502664192 | 0.8424751275781652 | 0.04028720859412929 | 0.006785119136428535 | 0.03659710495716702 | 269 |
| logistic | lr_c_10p0_freq_100 | 1 | 0.012014574885529745 | 0.7951438664966685 | 0.03842341354245253 | 0.006996276752105093 | 0.038213423118370546 | 202 |
| logistic | lr_c_10p0_freq_200 | 1 | 0.011999569667932506 | 0.8089493073329889 | 0.039574031719966325 | 0.006916276236589269 | 0.037650609784220176 | 140 |
| logistic | lr_c_10p0_freq_25 | 1 | 0.011948623028512906 | 0.7886240870371182 | 0.03728814691049119 | 0.007042155648984293 | 0.03860572041502864 | 344 |
| logistic | lr_c_10p0_freq_50 | 1 | 0.012193160450737876 | 0.7935821179023629 | 0.03911633803795314 | 0.007031648606528365 4 | 0.038451242068217324 | 269 |
| logistic | lr_c_1p0_freq_100 | 1 | 0.011908469424620843 | 0.8324009470610171 | 0.04219132519634189 | 0.006867047018177447 | 0.03677079532022307 | 202 |
| logistic | lr_c_1p0_freq_200 | 1 | 0.011929286706614382 | 0.8348282514882797 | 0.042458271690052285 | 0.006845579790561131 | 0.0367069877365547 | 140 |
| logistic | lr_c_1p0_freq_25 | 1 | 0.011836762097304361 | 0.8312899227701851 | 0.04234472406353977 | 0.006863543713022519 | 0.03676839823515438 | 344 |
| logistic | lr_c_1p0_freq_50 | 1 | 0.011903669652414938 | 0.831861475491588 | 0.04247943159424028 | 0.006863723130428825 | 0.03677021518961608 | 269 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | 0.008229067623178765 | 0.8504878558546197 | 0.03770260575955906 | 0.006704587228849308 | 0.035279169895704524 | 140 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | 0.007934368931585676 | 0.8469757002743733 | 0.03723767418952374 | 0.006706371632578499 | 0.03524239179558827 | 269 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | 0.006608094033597252 | 0.8452578657584842 | 0.03760061807034768 | 0.006730920219286495 | 0.03621744392004315 | 140 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | 0.006499289777786076 | 0.8421359567457133 | 0.03633508766356185 | 0.006744369704518722 | 0.036694695036385054 | 269 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | 0.01009293089877339 | 0.8200862951224579 | 0.036131239436625655 | 0.00674042491693201 | 0.03643327496282612 | 140 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | 0.008303255413319553 | 0.8244052757930874 | 0.033308573946748694 | 0.00673537728476169 | 0.03604352892435908 | 269 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | 0.007825288325692131 | 0.830088880672726 | 0.03889590635503128 | 0.006713551413695244 | 0.0361724725960129 | 140 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | 0.007169736334681731 | 0.8332748884624106 | 0.037790410000686066 | 0.006724391520065689 | 0.036312747368845925 | 269 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | 0.008330982474684601 | 0.8458531775904577 | 0.037244960417827065 | 0.006707524521722795 | 0.03543000344225826 | 140 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | 0.008132141488484602 | 0.8450948553902371 | 0.03722484059669917 | 0.006706794234038853 | 0.03534052753296647 | 269 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | 0.006730142686050764 | 0.8477201735812666 | 0.03770753077011961 | 0.006728856649904061 | 0.036110995906251706 | 140 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | 0.006492572578328444 | 0.838879561002787 | 0.03569322548718899 | 0.006740666652253513 | 0.03652652944021722 | 269 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | 0.011305903581362823 | 0.8134578526719153 | 0.034687859542525526 | 0.006783619463372568 | 0.040204330521949876 | 140 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | 0.01093688406154841 | 0.7974321102473552 | 0.031302216732710594 | 0.006804527787226238 | 0.040095414741808354 | 269 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | 0.00862116565466406 | 0.8249942031581828 | 0.03525203726233232 | 0.006716678025529667 | 0.03622915492305976 | 140 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | 0.007728733433298841 | 0.8283954405377436 | 0.034845780021179974 | 0.006718638925035571 4 | 0.0360387054400713 | 269 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | 0.007245960984973122 | 0.8371147164820253 | 0.040292172689715765 | 0.006702783935760467 | 0.0352514454753214 | 140 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | 0.007506731570397892 | 0.838192706715379 | 0.04112052160525384 | 0.006703136156473771 | 0.0352447623209785 | 269 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | 0.007484546181950726 | 0.8471240358978561 | 0.046129382059700644 | 0.006702408241203045 | 0.03490279657442868 | 140 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | 0.007478386024343277 | 0.8467285483501075 | 0.047076208191054354 | 0.006714345535354781 | 0.034935683481270456 | 269 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | 0.006950764391020781 | 0.8215478298212413 | 0.03853682060490279 | 0.006735808879708879 | 0.03600892716858657 | 140 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | 0.006813654011437966 | 0.8237184850327133 | 0.038958796059999246 | 0.006739596505436217 | 0.035989794521106545 | 269 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | 0.005503560075569835 | 0.8222248056549224 | 0.03631398289514122 | 0.006776908689951058 | 0.03690026784015731 | 140 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | 0.005618031841197781 5 | 0.8234923287930562 | 0.03933904429844104 | 0.00677043365618203 | 0.03675224084237356 | 269 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | 0.007044947842303404 | 0.8124525691285297 | 0.03390824795851356 | 0.006801203808645446 | 0.03679541146477767 | 140 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | 0.006998443925170269 | 0.8096691956778748 | 0.035883511508332314 | 0.006800314803652874 | 0.0368977588494267 | 269 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | 0.006637959611490484 | 0.817166370313055 | 0.038668128386335064 | 0.006860187383741965 5 | 0.03708335953240143 | 140 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | 0.006578918349283256 | 0.8117553281711267 | 0.03727553281799936 | 0.006870821520169684 | 0.03741313851867943 | 269 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | 0.005454410228223608 | 0.757039970574278 | 0.02750919965135956 | 0.006914868889715851 | 0.040663212468698015 | 140 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | 0.005786989538341933 | 0.7711804847493826 | 0.029736169521440023 | 0.006914677885833632 | 0.039875599492582306 | 269 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | 0.002051933007399108 | 0.7776614174278792 | 0.029932981288629354 | 0.006872342510854629 5 | 0.04752975007071323 | 140 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | 0.002325768671405065 | 0.7797825850801678 | 0.03061394602047675 | 0.006917538878251105 5 | 0.04739407919496841 | 269 |
| logistic | lr_c_0p01_freq_100 | 2 | 0.013991883546692424 | 0.8208622539123318 | 0.042346051959531276 | 0.008072449233541371 | 0.04345711619577892 | 266 |
| logistic | lr_c_0p01_freq_200 | 2 | 0.01396899462095669 | 0.8210541709746595 | 0.042334392587939876 | 0.008072049241110033 | 0.043442813553451025 | 203 |
| logistic | lr_c_0p01_freq_25 | 2 | 0.014011039778913131 | 0.8210597551658774 | 0.0424811392082453 | 0.008071081281898968 | 0.043447687490812806 | 446 |
| logistic | lr_c_0p01_freq_50 | 2 | 0.014009215652987067 | 0.8210671305127689 | 0.04243784581000202 | 0.008071218427199763 | 0.04344810618732924 | 350 |
| logistic | lr_c_0p1_freq_100 | 2 | 0.014858168146009744 | 0.8403156269166024 | 0.05121276740543513 | 0.008180550001711907 | 0.042708083989790986 | 266 |
| logistic | lr_c_0p1_freq_200 | 2 | 0.014816585674322435 | 0.8407326501022697 | 0.05141394568183096 | 0.008175776805591178 | 0.042671823514397515 | 203 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_25 | 2 | 0.01488973606005993 | 0.8402680032481027 | 0.05154945880309654 | 0.00817850042063128 | 0.0427073919654708 | 446 |
| logistic | lr_c_0p1_freq_50 | 2 | 0.01488129619279946 1 | 0.8400777192983011 | 0.05167332452669948 | 0.00817610384534751 | 0.04270000030483828 | 350 |
| logistic | lr_c_10p0_freq_100 | 2 | 0.01522970627534661 8 | 0.8097807952077947 | 0.048229511098307455 | 0.00842359014639901 | 0.044275182610344584 | 266 |
| logistic | lr_c_10p0_freq_200 | 2 | 0.01497254470219552 | 0.8205569672320713 | 0.049102512288406786 | 0.00833725213772881 | 0.04352558289240022 | 203 |
| logistic | lr_c_10p0_freq_25 | 2 | 0.01542202680814434 7 | 0.8000693598694099 | 0.04309138867334035 | 0.00845566423664098 | 0.04507028205654638 | 446 |
| logistic | lr_c_10p0_freq_50 | 2 | 0.01523530369087466 9 | 0.8031460385062646 | 0.047523144537176626 | 0.00845076203731702 2 | 0.04461175884754569 | 350 |
| logistic | lr_c_1p0_freq_100 | 2 | 0.01510605962861949 7 | 0.8319494021701853 | 0.049235542531891295 | 0.00831833303125630 9 | 0.043215701264969174 | 266 |
| logistic | lr_c_1p0_freq_200 | 2 | 0.01496663127291567 | 0.8344922637352402 | 0.051819763732448745 | 0.00828541235784441 1 | 0.04298070528809483 | 203 |
| logistic | lr_c_1p0_freq_25 | 2 | 0.01519474220643006 1 | 0.8304714353341701 | 0.049244414991931704 | 0.00833016633603714 6 | 0.04333300735121623 | 446 |
| logistic | lr_c_1p0_freq_50 | 2 | 0.01513878943331304 5 | 0.8305935500062743 | 0.04965762412455448 | 0.00831651350633792 8 | 0.04325795147937848 | 350 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | 0.01039759565058352 2 | 0.8280493187339395 | 0.05045338051242494 | 0.00799449651083064 4 | 0.04142957086848674 | 203 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | 0.01037201859346550 1 | 0.8290655361734952 | 0.04987173511357592 | 0.00799632440935092 6 | 0.04145501378463545 | 350 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | 0.00829998064061701 6 | 0.8375208445741522 | 0.053733359394561926 | 0.00802322443621309 4 | 0.04186701638701852 | 203 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | 0.00790723116168802 6 | 0.8317948359717584 | 0.05142479854052507 | 0.00804469612090878 9 | 0.0424164581551413 | 350 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | 0.01489929031043229 5 | 0.8010393496882704 | 0.041041521923972916 | 0.00815506865563111 5 | 0.04408138569207734 | 203 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | 0.01368314966025053 9 | 0.8079223658342745 | 0.041899186777611855 | 0.00811415774176248 9 | 0.04335818265788054 | 350 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | 0.00947503718017499 2 | 0.8267215982503148 | 0.04939537790969275 | 0.00800953666726951 9 | 0.042112398496027194 | 203 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | 0.00876172437087749 7 | 0.8278886415338024 | 0.04904746418782282 | 0.00802628009163152 9 | 0.04232018624224171 | 350 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | 0.01098184760665072 9 | 0.8253966698622992 | 0.050380234919888627 | 0.00799890368571295 5 | 0.04167182066703604 | 203 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | 0.01065355446509992 3 | 0.8258603157765307 | 0.05064318137410608 | 0.00799801534638068 9 | 0.04153628126292638 | 350 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | 0.00885361324224060 6 | 0.8367690333206582 | 0.05262459189725955 | 0.00801476932223539 | 0.04166927413842042 | 203 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | 0.00857631215570438 8 | 0.8347273792579327 | 0.05220308519587398 | 0.00802994250517960 5 | 0.04203414617433081 | 350 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | 0.01954109030219904 4 | 0.7770396021358583 | 0.03795447716067282 | 0.00838549100387242 | 0.04759975736613913 | 203 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | 0.01888397440227195 3 | 0.773590126054161 | 0.036632248413382945 | 0.00836685152367000 6 | 0.04733195710707715 | 350 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | 0.01278645996403472 5 | 0.8175874945303903 | 0.048933487744556 | 0.00802355595767946 | 0.042907233596346135 | 203 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | 0.01179427657878255 8 | 0.8132744970303165 | 0.049280315110359546 | 0.00802296160523526 3 | 0.0428380629163129 | 350 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | 0.00794574957097551 7 | 0.8347070443729316 | 0.045547159284303966 | 0.00802484228039536 4 | 0.04117997750840866 | 203 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | 0.00811004881552247 4 | 0.8349754279782098 | 0.045732005354507874 | 0.00802547022304352 | 0.04114141730438595 | 350 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | 0.00836983281972908 7 | 0.8374448258201201 | 0.04658464749335628 | 0.00802451489986650 6 | 0.04096502381976541 | 203 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | 0.00853234933128667 2 | 0.8358123454667272 | 0.047511422110186934 | 0.00802661223095569 1 | 0.04102324423300787 | 350 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | 0.00743018186692367 5 | 0.8238906503675715 | 0.03941062941331661 | 0.00808483424920466 3 | 0.04214204334644366 | 203 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | 0.007500960142845343 | 0.8201395752790595 | 0.03966447692436781 | 0.008094053603049706 | 0.04234696776278764 | 350 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | 0.006822304351400347 | 0.8170915287924586 | 0.036633112930626634 | 0.008114647249964812 | 0.04271896415782953 | 203 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | 0.0069541593728768055 | 0.8162508446089218 | 0.035954636335078116 | 0.008117824118708612 | 0.042778166921867825 | 350 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | 0.00675064250234409 | 0.816448714629812 | 0.0368247538920876 | 0.008105749280003132 | 0.04286021044259399 | 203 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | 0.006944307112507819 | 0.8159821975983975 | 0.03655340178303761 | 0.008137182130532243 | 0.04303956570305958 | 350 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | 0.00700154917001785 | 0.8152629432333247 | 0.03813001750052926 | 0.008118084816578204 | 0.04297298117450893 | 203 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | 0.007266325399047901 | 0.8088863763540479 | 0.03590112863208511 | 0.00815923409135512 | 0.043276183836975936 | 350 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | 0.00645529041766494 | 0.7772600459863415 | 0.026438524233646312 | 0.00833459972109109 | 0.04670110158363384 | 203 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | 0.006065080740932869 | 0.7766311132801655 | 0.02587024411016906 | 0.008334465738652623 | 0.04717340768074935 | 350 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | 0.003617291441946546 | 0.7817552408951607 | 0.026328316059196475 | 0.00825429630448655 | 0.05022433899603056 | 203 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | 0.003987630795342951 | 0.7795648945709966 | 0.026005185534826343 | 0.008288389168546849 | 0.04998165315546123 | 350 |
| logistic | lr_c_0p01_freq_100 | 3 | 0.012997741116901983 | 0.7454336658889653 | 0.0793362341812003 | 0.02591001780415207 | 0.11963914724029864 | 314 |
| logistic | lr_c_0p01_freq_200 | 3 | 0.012990016351419746 | 0.7455445852093852 | 0.07928191281243166 | 0.025910944338142405 | 0.11964551980117794 | 245 |
| logistic | lr_c_0p01_freq_25 | 3 | 0.013000547885292818 | 0.7454236624650816 | 0.07933701714962474 | 0.025909740301719816 | 0.11963623294199294 | 492 |
| logistic | lr_c_0p01_freq_50 | 3 | 0.012999675854228638 | 0.7454092507527068 | 0.07931813428895423 | 0.025910092641466334 | 0.11964017873916447 | 414 |
| logistic | lr_c_0p1_freq_100 | 3 | 0.012868570912617389 | 0.7355901102828314 | 0.07477707297924037 | 0.025960347241812683 | 0.12259157348916636 | 314 |
| logistic | lr_c_0p1_freq_200 | 3 | 0.012874943547080221 | 0.7356147797434262 | 0.07498482627835017 | 0.02595817399665963 | 0.12257732175429717 | 245 |
| logistic | lr_c_0p1_freq_25 | 3 | 0.012879858640752723 | 0.7358318370867479 | 0.07507467467906331 | 0.025955226666891165 | 0.1225361288567146 | 492 |
| logistic | lr_c_0p1_freq_50 | 3 | 0.012874904989142554 | 0.7357574896058491 | 0.07489031542761451 | 0.025958306841056003 | 0.12256180190927561 | 414 |
| logistic | lr_c_10p0_freq_100 | 3 | 0.01274060344959874 | 0.7105076608763345 | 0.06894013890281091 | 0.026071214314024888 | 0.12767172330814042 | 314 |
| logistic | lr_c_10p0_freq_200 | 3 | 0.012791769860552126 | 0.7158587127261828 | 0.07027148750644939 | 0.02605155080180388 | 0.12670065251673768 | 245 |
| logistic | lr_c_10p0_freq_25 | 3 | 0.012790230512871101 | 0.7087124192982256 | 0.06842304333768609 | 0.026088887692616948 | 0.12845686776307086 | 492 |
| logistic | lr_c_10p0_freq_50 | 3 | 0.012806797016716874 | 0.7092119122941842 | 0.06868456550871312 | 0.02607787058742125 | 0.1281130903264364 | 414 |
| logistic | lr_c_1p0_freq_100 | 3 | 0.012821679071711452 | 0.7265040851439899 | 0.07167830762965403 | 0.026025494582423155 | 0.12480654816523298 | 314 |
| logistic | lr_c_1p0_freq_200 | 3 | 0.01285022603716422 | 0.7269912179772176 | 0.07211260694866414 | 0.026019257131653938 | 0.12466491632557213 | 245 |
| logistic | lr_c_1p0_freq_25 | 3 | 0.012857500609487017 | 0.7267660900749656 | 0.07207742153338188 | 0.026019115738232977 | 0.12480997441013006 | 492 |
| logistic | lr_c_1p0_freq_50 | 3 | 0.012853224495018198 | 0.7267353846501292 | 0.07186427315312013 | 0.02602317492228757 | 0.12479715702418752 | 414 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | 0.011486786536400902 | 0.7197222045796487 | 0.07137966988062493 | 0.026032987355649725 | 0.12613847797004654 | 245 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | 0.011530797133406245 | 0.72030604848039 | 0.07071317980501689 | 0.026034964598472335 | 0.12620568288666806 | 414 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | 0.010599720424135933 | 0.7096396688724272 | 0.06717028213250413 | 0.02625364692593907 | 0.12559925248402698 | 245 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | 0.009998735806501511 | 0.7017183135529472 | 0.06396253529814948 | 0.02631660524397955 | 0.12728494221720052 | 414 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | 0.01633019008117351 | 0.6954328232276653 | 0.060713892144882015 | 0.026081981976892175 | 0.1238279030100992 | 245 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | 0.014471437103018783 | 0.6971026066108118 | 0.061737513858695685 | 0.026126176985018547 | 0.12619876831764643 | 414 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | 0.014773942193169161 | 0.7041137944739932 | 0.0633008844673639 | 0.026085925470078385 | 0.12097248553189535 | 245 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | 0.013009184255487916 | 0.6992871085401672 | 0.06229118627792167 | 0.02617491411080054 | 0.12316705680312309 | 414 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | 0.01212854146932699 | 0.7202743003257759 | 0.07200831437896979 | 0.025997266127746003 | 0.12584427651022354 | 245 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | 0.012195156415282574 | 0.7206380943335081 | 0.07159801345415916 | 0.025997998852652995 | 0.12566869409866183 | 414 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | 0.01167000998526794 | 0.7114957109218025 | 0.06713980494136286 | 0.02618431201099776 | 0.12379011755805269 | 245 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | 0.011187543767532036 | 0.7122366085775189 | 0.06716532926710908 | 0.026220093575521814 | 0.12448944631191668 | 414 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | 0.0196854370581195 | 0.7007467098145423 | 0.06511698757642195 | 0.025961052953761353 | 0.12865292570307174 | 245 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | 0.019444021888068593 | 0.7049745467118016 | 0.06621869606003669 | 0.025945527696134296 | 0.12827670597059254 | 414 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | 0.019226961140272964 | 0.7100907215598504 | 0.06933039049478408 | 0.025878862017351966 | 0.11744864027101694 | 245 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | 0.01712354441403844 | 0.7145784609735746 | 0.06913086094508666 | 0.025928482030513116 | 0.11827084411557362 | 414 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | 0.009124351143594445 | 0.7336842715284617 | 0.07636688892811452 | 0.026087484353953277 | 0.12888822862842744 | 245 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | 0.008796851547215616 | 0.7297305538780511 | 0.0749874553746544 | 0.02612133118789721 | 0.13003567145610767 | 414 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | 0.00844886319830538 | 0.73014973972617 | 0.07479663649198051 | 0.02614414482280141 | 0.13063158921113127 | 245 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | 0.008430829299317207 | 0.7289135369485618 | 0.07377833303609109 | 0.026155739168430486 | 0.13110900303989464 | 414 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | 0.010755720241165594 | 0.7268787133684367 | 0.07363571353343107 | 0.026023929047357914 | 0.12862115921680564 | 245 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | 0.010803930083583269 | 0.7252913904125083 | 0.07217564406983554 | 0.02603809555957109 | 0.12874181368187207 | 414 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | 0.009356722863681756 | 0.7241223207821335 | 0.06958281279207809 | 0.02614184022026648 | 0.13176589280906884 | 245 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | 0.009695145474174249 | 0.725627997148719 | 0.07087700720882689 | 0.02611303574865398 | 0.1309124776861942 | 414 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | 0.010767402004973673 | 0.7164785350429943 | 0.07286913359862641 | 0.026049710004006132 | 0.13004577370158188 | 245 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | 0.011416454062002061 | 0.7215602319885556 | 0.07336013085608935 | 0.026031038529373527 | 0.12899398659421504 | 414 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | 0.010403694777659908 | 0.7159134857106854 | 0.06898367030692482 | 0.026158319479800476 | 0.13205076340098995 | 245 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | 0.010033873500350681 | 0.7150509023376577 | 0.0686877027658625 | 0.026174154471136794 | 0.13299814886402894 | 414 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | 0.011165770629172062 | 0.7025369920680987 | 0.0655171817695723 | 0.026305309763532514 | 0.137971281230783 | 245 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | 0.011546467604763907 | 0.7065259082066158 | 0.06475162789130082 | 0.02637090458190341 | 0.13741573153706602 | 414 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | 0.007711931596050403 | 0.6944185269107193 | 0.06036380619647214 | 0.026529137966028557 | 0.15080395440731362 | 245 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | 0.007708145977369976 | 0.6978418850682553 | 0.06021075953593196 | 0.02657869305840814 | 0.15297420312570886 | 414 |
| logistic | lr_c_0p01_freq_100 | 4 | 0.022049852088210847 | 0.7217042553872568 | 0.11224738176097718 | 0.04440942949416063 | 0.18845346884523498 | 352 |
| logistic | lr_c_0p01_freq_200 | 4 | 0.022048602631395454 | 0.7217558154787428 | 0.11222728340873636 | 0.044409408437950985 | 0.18844533700114888 | 271 |
| logistic | lr_c_0p01_freq_25 | 4 | 0.022053135785006624 | 0.7218028511015063 | 0.11228739891806397 | 0.04440808802205282 | 0.18843437021398704 | 515 |
| logistic | lr_c_0p01_freq_50 | 4 | 0.022052451288244956 | 0.7217702655007261 | 0.11226611749395582 | 0.04440853280506416 | 0.18844005943005154 | 452 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p1_freq_100 | 4 | 0.021670009184507105 | 0.7215801624065589 | 0.11179923678468999 | 0.04444109868922813 | 0.190153537849006 | 352 |
| logistic | lr_c_0p1_freq_200 | 4 | 0.021658197194337675 | 0.7216686440480181 | 0.1117520442238884 | 0.04444257159791916 | 0.19015408956813654 | 271 |
| logistic | lr_c_0p1_freq_25 | 4 | 0.02168255865341839 | 0.7220188567791115 | 0.11207548182569058 | 0.04443374035542509 | 0.19006188446520525 | 515 |
| logistic | lr_c_0p1_freq_50 | 4 | 0.02167480321034218 | 0.7217705859839273 | 0.11194480773899085 | 0.04443697943016637 | 0.19010624735727363 | 452 |
| logistic | lr_c_10p0_freq_100 | 4 | 0.021674904306580492 | 0.7105589524889688 | 0.10840910067510981 | 0.04454276066703226 | 0.1932983342847692 | 352 |
| logistic | lr_c_10p0_freq_200 | 4 | 0.02160052762778913 | 0.7139272309335212 | 0.1090758393397116 | 0.04452365554811544 | 0.19247474350714103 | 271 |
| logistic | lr_c_10p0_freq_25 | 4 | 0.021715253487816794 | 0.7093688663992631 | 0.10804701091895161 | 0.04455795751962706 | 0.19394755499823166 | 515 |
| logistic | lr_c_10p0_freq_50 | 4 | 0.021662146422235823 | 0.7092175511984168 | 0.10839512875516967 | 0.04454509983958473 | 0.19381845799312455 | 452 |
| logistic | lr_c_1p0_freq_100 | 4 | 0.02165006563314452 | 0.7156949597154502 | 0.10972948843291948 | 0.044506466921228956 | 0.1919119934259627 | 352 |
| logistic | lr_c_1p0_freq_200 | 4 | 0.021593575111551685 | 0.716695923012545 | 0.1099211128803369 | 0.04450233632185413 | 0.1917156269488396 | 271 |
| logistic | lr_c_1p0_freq_25 | 4 | 0.021685959804954955 | 0.716766561280479 | 0.11013706446110381 | 0.04449535473555892 | 0.1917825263967915 | 515 |
| logistic | lr_c_1p0_freq_50 | 4 | 0.02164796055836572 | 0.7160204763880984 | 0.11013679814888025 | 0.04449742081211918 | 0.19188474598056096 | 452 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | 0.03478261923728115 | 0.7171330526729977 | 0.10910754011291784 | 0.043816761109214016 | 0.17854332061909964 | 271 |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | 0.0349765327397572 | 0.7163604619346832 | 0.10845049999657351 | 0.043818456374909234 | 0.17857628052759925 | 452 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | 0.023359511898723987 | 0.715033482388185 | 0.10436822033000048 | 0.04484990089098388 | 0.18896953167611286 | 271 |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | 0.02163551539584283 | 0.7078388513188317 | 0.10066412819565312 | 0.045059118880085236 | 0.19212597544157303 | 452 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | 0.040927127011986954 | 0.6829326407276612 | 0.10206136669500548 | 0.04394202692050462 | 0.1805136991881145 | 271 |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | 0.03875743017860188 | 0.6926526888484024 | 0.10356160143224774 | 0.04391493494547013 | 0.18016027365428872 | 452 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | 0.024582620558228085 | 0.7083289172633405 | 0.10441184185955879 | 0.04474859497034871 | 0.18799411622453177 | 271 |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | 0.02190301525994084 | 0.7033934005571355 | 0.10259099697483005 | 0.04499050181216565 | 0.19160931195356817 | 452 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | 0.03546296176146024 | 0.7165714906966932 | 0.10910986735580365 | 0.04379360765491783 | 0.17831950598319368 | 271 |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | 0.03521392265244519 | 0.7160670784159427 | 0.10937251175453257 | 0.04379920918221197 | 0.17845229334916604 | 452 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | 0.026096968683694644 | 0.7210164607335877 | 0.10684314406323538 | 0.04463250854033956 | 0.18567153192726638 | 271 |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | 0.024104391132067273 | 0.7157943754896559 | 0.1037840624562191 | 0.04482529703654627 | 0.18828599158970866 | 452 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | 0.05274505990051458 | 0.6620323099098195 | 0.0978487651132118 | 0.04413309637598919 | 0.18271246648199688 | 271 |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | 0.05049841099454696 | 0.6719130850689479 | 0.0990318249735126 | 0.04404642769846847 | 0.18191983159951217 | 452 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | 0.03274062662275171 | 0.7139609853553882 | 0.10733929181387994 | 0.04413989796371802 | 0.18092545361029969 | 271 |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | 0.029754559723145012 | 0.7170667597802329 | 0.10814471995851155 | 0.04428499255554874 | 0.1822327900292966 | 452 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | 0.02956620473383698 | 0.7231906046315252 | 0.11416663453735887 | 0.04388030067021974 | 0.18150921049279967 | 271 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | 0.02926067277103234 | 0.7234729503317774 | 0.11543031610233326 | 0.04386022258070923 | 0.1817177946198793 | 452 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | 0.029287907946069044 | 0.7229389734757168 | 0.11419209755417581 | 0.04389582582412464 | 0.18181977912999178 | 271 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | 0.02937834279792957 | 0.7230366737221986 | 0.11505772984533333 | 0.04387483103459335 | 0.1817654645097791 | 452 |

| model | config_id | fold | fold_mean_probability | fold_roc_auc | fold_pr_auc | fold_brier_score | fold_log_loss | fold_transformed_feature_count |
|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | 0.03236715091453178 | 0.719493185226358 | 0.10961638661362083 | 0.043931401380276874 | 0.18124635575195033 | 271 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | 0.030679187414991965 | 0.7187163198076979 | 0.11044244646888698 | 0.04398384637870936 | 0.18269689208541828 | 452 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | 0.02906918275466903 | 0.7146461548746004 | 0.10847452552259516 | 0.044092712351831286 | 0.18386015748469514 | 271 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | 0.03006670240449971 | 0.7182082172571457 | 0.11141277910249653 | 0.04399049373375279 | 0.18286512018714612 | 452 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | 0.034793195959570984 | 0.7204447705455277 | 0.11028411416120168 | 0.04391939848267377 | 0.1807506075288329 | 271 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | 0.033967205160644574 | 0.7201705547500116 | 0.1113710396994556 | 0.04391900942184682 | 0.18108193671127013 | 452 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | 0.027161667881523594 | 0.7151261491632126 | 0.10899948086862996 | 0.04416500348642565 | 0.18571218989050967 | 271 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | 0.026538204066139683 | 0.7168404609362836 | 0.10967736458639081 | 0.044212402282361896 | 0.1867921118371637 | 452 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | 0.03629257654552063 | 0.6985418693015966 | 0.09787386994954365 | 0.044994819692032686 | 0.18913033085230185 | 271 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | 0.03576175680245201 | 0.69674559394982 | 0.0963481210534829 | 0.04506992717726164 | 0.19016434577801886 | 452 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | 0.0237099865469718 | 0.6896950577833098 | 0.09439723647447688 | 0.04502392238865134 | 0.19927997440387796 | 271 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | 0.023667072566018334 | 0.69481561112131 | 0.09635336793385899 | 0.04500248582510568 | 0.19906696377949415 | 452 |

*Column panel 4 of 4 — Development Fold (192 records)*

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_0p01_freq_100 | 1 | 9 | 0.3131924170011189 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 1 | 9 | 0.3143733750039246 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 1 | 9 | 0.3171328329917742 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 1 | 9 | 0.3124661250039935 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 1 | 10 | 0.3311766249971697 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 1 | 10 | 0.336397749997559 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 1 | 10 | 0.33461599999282043 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 1 | 10 | 0.33471799999824725 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 1 | 16 | 0.4402620839973679 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 1 | 15 | 0.449491167004453 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 1 | 19 | 0.4726519170071697 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 1 | 18 | 0.47306745799141936 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 1 | 13 | 0.37861045901081525 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 1 | 12 | 0.37203633300669026 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 1 | 13 | 0.37455104100808967 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 1 | 13 | 0.3808031249936903 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 1 | NA | 2.3705696660035755 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 1 | NA | 2.418668833008269 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 1 | NA | 0.9255515830009244 | 12.0 | 50 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.8332516669906909 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 1 | NA | 2.743582167007844 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 1 | NA | 2.8845040000014706 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.0788297079998301 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 1 | NA | 0.9387061669985997 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 1 | NA | 2.5571653340011835 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 1 | NA | 2.7348860839992994 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 1 | NA | 1.0043489170057 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 1 | NA | 0.9040940840059193 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 1 | NA | 3.215423166999244 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 1 | NA | 3.9970923340006266 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 1 | NA | 1.2226561249990482 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 1 | NA | 1.1517517500033136 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 1 | NA | 1.1947981660050573 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 1 | NA | 1.1043682500021532 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 1 | NA | 1.100260833001812 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 1 | NA | 1.1039745830057655 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 1 | NA | 1.7097240839939332 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 1 | NA | 1.7095064579916652 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 1 | NA | 1.7664646670018556 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 1 | NA | 1.7111287910083774 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 1 | NA | 1.1061493750021327 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 1 | NA | 1.1045960829942487 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 1 | NA | 1.1522714169987012 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 1 | NA | 1.1023339999956079 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 1 | NA | 1.702376125002047 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 1 | NA | 1.700454792007804 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 1 | NA | 1.7188430420064833 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 1 | NA | 1.7201974170020549 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 2 | 9 | 0.4908512090041768 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 2 | 9 | 0.4976724580046721 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 2 | 9 | 0.4937538329977542 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 2 | 9 | 0.4877702919911826 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 2 | 10 | 0.5362783749878872 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 2 | 10 | 0.5416608749947045 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 2 | 10 | 0.5344942079973407 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 2 | 10 | 0.5357551670022076 | NA | NA | NA | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| logistic | lr_c_10p0_freq_100 | 2 | 18 | 0.8950084169919137 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 2 | 16 | 0.7982442909997189 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 2 | 20 | 0.8840877500042552 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 2 | 20 | 0.9073727919894736 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 2 | 14 | 0.6695284169982187 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 2 | 13 | 0.6317051660007564 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 2 | 15 | 0.6761112079984741 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 2 | 14 | 0.6495580420014448 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 2 | NA | 5.760092707991134 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 2 | NA | 5.8201264170056675 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 2 | NA | 1.6132645000034245 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.350639708005474 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 2 | NA | 6.133171749999747 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 2 | NA | 7.044903791000252 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 2 | NA | 1.9322971660003532 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 2 | NA | 1.561997041993891 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 2 | NA | 7.5671650000003865 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 2 | NA | 7.779776833995129 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 2 | NA | 2.1160196249984438 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 2 | NA | 1.8163555829960387 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 2 | NA | 10.971002790989587 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 2 | NA | 11.911807707991102 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 2 | NA | 2.91761679200863 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 2 | NA | 2.7321155419922434 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 2 | NA | 1.3339541669993196 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 2 | NA | 1.3280041660036659 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 2 | NA | 1.323773208001512 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 2 | NA | 1.3346964999946067 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 2 | NA | 1.976263292002841 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 2 | NA | 1.9895307090046117 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 2 | NA | 1.970503207994625 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 2 | NA | 1.964858124993043 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 2 | NA | 1.3333386250014883 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 2 | NA | 1.326946291999775 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 2 | NA | 1.3252354170108447 | 3.0 | NA | NA | 0.1 | 1 |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 2 | NA | 1.3251211669994518 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 2 | NA | 2.1011942500044825 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 2 | NA | 1.98539112501021 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 2 | NA | 1.9660272500041174 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 2 | NA | 1.9813969579990953 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 3 | 9 | 0.6780093749985099 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 3 | 9 | 0.7311657079990255 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 3 | 9 | 0.6863537919998635 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 3 | 9 | 0.6815700830047717 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 3 | 11 | 0.8000446250080131 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 3 | 11 | 0.802278415998444 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 3 | 11 | 0.7956212499993853 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 3 | 11 | 0.7991864579962566 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 3 | 19 | 1.3694663749920437 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 3 | 17 | 1.3279072919976898 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 3 | 21 | 1.4182294999918668 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 3 | 20 | 1.4020091669954127 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 3 | 14 | 0.9777985000109766 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 3 | 13 | 0.9351792920060689 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 3 | 15 | 1.0201452500041341 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 3 | 15 | 1.0388574999960838 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 3 | NA | 11.70144774999062 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 3 | NA | 11.51657587500813 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 3 | NA | 2.3925375420076307 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 3 | NA | 2.2012933749938384 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 3 | NA | 11.514515999995638 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 3 | NA | 11.278543499996886 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 3 | NA | 2.9502907090063673 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 3 | NA | 2.3280480840039672 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 3 | NA | 16.560073582993937 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 3 | NA | 16.45757191699522 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 3 | NA | 3.61647979200643 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 3 | NA | 3.099515458001406 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 3 | NA | 23.98587674999726 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 3 | NA | 26.861935625012848 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 3 | NA | 5.408917208987987 | NA | 5 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 3 | NA | 5.16776295901218 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 3 | NA | 1.5623690420034109 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 3 | NA | 1.560301999998046 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 3 | NA | 1.5489193749963306 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 3 | NA | 1.656156707991613 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 3 | NA | 2.2522191669995664 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 3 | NA | 2.2570583750057267 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 3 | NA | 2.225052333000349 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 3 | NA | 2.226129292001133 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 3 | NA | 1.6451356250036042 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 3 | NA | 1.557488917009323 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 3 | NA | 1.539110375000746 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 3 | NA | 1.5385084169975016 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 3 | NA | 2.269374292009161 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 3 | NA | 2.253126040988718 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 3 | NA | 2.3355088329990394 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 3 | NA | 2.2303070410125656 | 6.0 | NA | NA | 0.1 | 1 |
| logistic | lr_c_0p01_freq_100 | 4 | 9 | 0.9027116670040414 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_200 | 4 | 9 | 0.9041572499991162 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_25 | 4 | 9 | 0.914727417009999 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p01_freq_50 | 4 | 9 | 0.9085878329933621 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_100 | 4 | 10 | 1.0348698750021867 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_200 | 4 | 10 | 1.0297209169948474 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_25 | 4 | 11 | 1.1012927499978105 | NA | NA | NA | NA | NA |
| logistic | lr_c_0p1_freq_50 | 4 | 10 | 1.0136219580017496 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_100 | 4 | 19 | 1.9178338330093538 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_200 | 4 | 16 | 1.876585582998814 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_25 | 4 | 21 | 2.0104859590064734 | NA | NA | NA | NA | NA |
| logistic | lr_c_10p0_freq_50 | 4 | 20 | 1.966217166002025 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_100 | 4 | 14 | 1.2948693329963135 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_200 | 4 | 13 | 1.2972651249874616 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_25 | 4 | 16 | 1.537381624992122 | NA | NA | NA | NA | NA |
| logistic | lr_c_1p0_freq_50 | 4 | 15 | 1.3482389999990119 | NA | NA | NA | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_200 | 4 | NA | 19.732329666003352 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_0p5_freq_50 | 4 | NA | 19.559802833988215 | 12.0 | 50 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_200 | 4 | NA | 3.3701428750064224 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_50_features_sqrt_freq_50 | 4 | NA | 2.913740374991903 | 12.0 | 50 | sqrt | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_200 | 4 | NA | 20.002198582995334 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_0p5_freq_50 | 4 | NA | 19.40565683299792 | 12.0 | 5 | 0.5 | NA | NA |
| rf | rf_depth_12_leaf_5_features_sqrt_freq_200 | 4 | NA | 4.188632541001425 | 12.0 | 5 | sqrt | NA | NA |

| model | config_id | fold | fold_iterations | fold_runtime_seconds | fold_max_depth | fold_min_samples_leaf | fold_max_features | fold_learning_rate | fold_min_child_weight |
|---|---|---|---|---|---|---|---|---|---|
| rf | rf_depth_12_leaf_5_features_sqrt_freq_50 | 4 | NA | 3.3620825419930043 | 12.0 | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_200 | 4 | NA | 29.01848491599958 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_0p5_freq_50 | 4 | NA | 30.12743366700306 | NA | 50 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_200 | 4 | NA | 5.646645332992193 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_50_features_sqrt_freq_50 | 4 | NA | 4.652386875008233 | NA | 50 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_200 | 4 | NA | 51.65452199999709 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_0p5_freq_50 | 4 | NA | 55.062459166991175 | NA | 5 | 0.5 | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_200 | 4 | NA | 10.644144000005326 | NA | 5 | sqrt | NA | NA |
| rf | rf_depth_none_leaf_5_features_sqrt_freq_50 | 4 | NA | 9.585657667004853 | NA | 5 | sqrt | NA | NA |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_200 | 4 | NA | 1.7811102499981644 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_10_freq_50 | 4 | NA | 1.799468875004095 | 3.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_200 | 4 | NA | 1.7822993330046302 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_3_child_1_freq_50 | 4 | NA | 1.891248458006885 | 3.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_200 | 4 | NA | 2.5016177500074264 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_10_freq_50 | 4 | NA | 2.6171712080104044 | 6.0 | NA | NA | 0.03 | 10 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_200 | 4 | NA | 2.4884513750002952 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p03_depth_6_child_1_freq_50 | 4 | NA | 2.498635458992794 | 6.0 | NA | NA | 0.03 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_200 | 4 | NA | 1.7775732919981238 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_10_freq_50 | 4 | NA | 1.7859797090059146 | 3.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_200 | 4 | NA | 1.7645422089990461 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_3_child_1_freq_50 | 4 | NA | 1.7700494169985177 | 3.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_200 | 4 | NA | 2.5383118339959765 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_10_freq_50 | 4 | NA | 2.519634832991869 | 6.0 | NA | NA | 0.1 | 10 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_200 | 4 | NA | 2.4773671250004554 | 6.0 | NA | NA | 0.1 | 1 |
| xgb | xgb_lr_0p1_depth_6_child_1_freq_50 | 4 | NA | 2.4785815000068396 | 6.0 | NA | NA | 0.1 | 1 |

Prediction Metric (12 records)

*Column panel 1 of 2 — Prediction Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45364 | 771 | logistic | raw | 45364 | 771 | 0.01699585574464333 | 0.044960377591741776 | 0.7286163896010271 |
| logistic | validation | calibrated | 45364 | 771 | logistic | calibrated | 45364 | 771 | 0.01699585574464333 | 0.06303500306312243 | 0.727487516943488 |
| rf | validation | raw | 45364 | 771 | rf | raw | 45364 | 771 | 0.01699585574464333 | 0.03694632043833322 | 0.7283688415440261 |
| rf | validation | calibrated | 45364 | 771 | rf | calibrated | 45364 | 771 | 0.01699585574464333 | 0.0520710869698955 | 0.7284918448025219 |
| xgb | validation | raw | 45364 | 771 | xgb | raw | 45364 | 771 | 0.01699585574464333 | 0.03437418229621991 | 0.7588928752725728 |
| xgb | validation | calibrated | 45364 | 771 | xgb | calibrated | 45364 | 771 | 0.01699585574464333 | 0.05070255452879132 | 0.7589189941957528 |
| logistic | test | raw | 49338 | 886 | logistic | raw | 49338 | 886 | 0.017957760752361263 | 0.46823979640791763 | 0.768731670673021 |

| model | role | probability_type | context_role_n | context_role_positive_n | prediction_model | prediction_probability_type | prediction_n | prediction_positive_n | prediction_prevalence | prediction_mean_probability | prediction_roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 49338 | 886 | logistic | calibrated | 49338 | 886 | 0.017957760752361263 | 0.30372133314515043 | 0.6303155514130576 |
| rf | test | raw | 49338 | 886 | rf | raw | 49338 | 886 | 0.017957760752361263 | 0.09426678063303633 | 0.7462892925702085 |
| rf | test | calibrated | 49338 | 886 | rf | calibrated | 49338 | 886 | 0.017957760752361263 | 0.15085788176380488 | 0.7370969434924214 |
| xgb | test | raw | 49338 | 886 | xgb | raw | 49338 | 886 | 0.017957760752361263 | 0.08178461203685729 | 0.7762426997168685 |
| xgb | test | calibrated | 49338 | 886 | xgb | calibrated | 49338 | 886 | 0.017957760752361263 | 0.09657108775910662 | 0.7749095285758134 |

*Column panel 2 of 2 — Prediction Metric (12 records)*

| model | prediction_pr_auc | prediction_brier_score | prediction_log_loss |
|---|---|---|---|
| logistic | 0.043379437227034356 | 0.0180658506639475 | 0.0927989406651624 |
| logistic | 0.041149623126218046 | 0.019733108884892058 | 0.10629064647015775 |
| rf | 0.04721582699895218 | 0.016955805378059734 | 0.0877072653576941 |
| rf | 0.04493044406604759 | 0.01839683721682325 | 0.09748774838362906 |
| xgb | 0.05415071314138024 | 0.016981710720008376 | 0.08431378835375367 |
| xgb | 0.05134625118055551 | 0.017911796406784187 | 0.09508946770992628 |
| logistic | 0.05966222122343541 | 0.2573222746838045 | 0.7161708162756691 |
| logistic | 0.024268052651752442 | 0.10236878558353313 | 0.37845072938582597 |
| rf | 0.053865792042572705 | 0.023481630172721604 | 0.1353573011262802 |
| rf | 0.04392782431184389 | 0.038359247673371795 | 0.2244971513839899 |
| xgb | 0.05483208412521002 | 0.024157224295732152 | 0.12224394206505904 |
| xgb | 0.05098063021207939 | 0.025295797206078587 | 0.13577762897412873 |

Calibration Metric (12 records)

*Column panel 1 of 2 — Calibration Metric (12 records)*

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | validation | raw | 45364 | 771 | logistic | raw | -1.302043770010909 | 0.8998211969799992 | 0.02796452184709845 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | validation | calibrated | 45364 | 771 | logistic | calibrated | -1.3069897766101535 | 1.0361517725106284 | 0.04603914731847911 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | raw | 45364 | 771 | rf | raw | 0.18048828153885826 | 1.3296835440285277 | 0.01995046469368989 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | validation | calibrated | 45364 | 771 | rf | calibrated | -1.1248660684067355 | 1.0223554220253095 | 0.03509232679400118 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | raw | 45364 | 771 | xgb | raw | -0.7301302670957726 | 1.0045349707050066 | 0.017378326551576586 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | validation | calibrated | 45364 | 771 | xgb | calibrated | -0.2583167290353169 | 1.3462521679833146 | 0.03370669878414798 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

| model | role | probability_type | context_role_n | context_role_positive_n | calibration_model | calibration_probability_type | calibration_calibration_intercept | calibration_calibration_slope | calibration_ece | calibration_ece_rule |
|---|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 49338 | 886 | logistic | raw | -4.33780077590146 | 1.0477099505020997 | 0.4502820356555563 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| logistic | test | calibrated | 49338 | 886 | logistic | calibrated | -0.8543352123560405 | 4.21175084965155 | 0.28576357239278916 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | raw | 49338 | 886 | rf | raw | 1.3365947793332367 | 2.4837770288676624 | 0.07630901988067505 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| rf | test | calibrated | 49338 | 886 | rf | calibrated | -3.5620521278182706 | 0.2635510037118555 | 0.1329001210114436 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | raw | 49338 | 886 | xgb | raw | -1.4769088252441482 | 1.0903075229193981 | 0.06382685128449603 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |
| xgb | test | calibrated | 49338 | 886 | xgb | calibrated | -0.9586237977623691 | 1.451980771630372 | 0.07861332700674536 | 10 fixed equal-width probability bins: [0.0,0.1), ..., [0.9,1.0] |

*Column panel 2 of 2 — Calibration Metric (12 records)*

| model | calibration_logit_clip |
|---|---|
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |
| logistic | 1e-06 |
| logistic | 1e-06 |
| rf | 1e-06 |
| rf | 1e-06 |
| xgb | 1e-06 |
| xgb | 1e-06 |

Decision (36 records)

*Column panel 1 of 4 — Decision (36 records)*

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| logistic | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| logistic | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | logistic | primary_benchmark | raw |
| logistic | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | logistic | primary_benchmark | calibrated |
| logistic | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | logistic | primary_benchmark | calibrated |
| rf | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| rf | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | rf | primary_benchmark | raw |
| rf | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | rf | primary_benchmark | calibrated |
| rf | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | rf | primary_benchmark | calibrated |
| xgb | test | raw | 1:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 1:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |

| model | role | probability_type | cost_ratio | regime | threshold_source | decision_analysis | decision_model | decision_model_role | decision_probability_type |
|---|---|---|---|---|---|---|---|---|---|
| xgb | test | calibrated | 1:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 2:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 2:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 2:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 5:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 5:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 5:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |
| xgb | test | raw | 10:1 | raw_common | theoretical_common_rule | robustness_raw_common_secondary | xgb | primary_benchmark | raw |
| xgb | test | calibrated | 10:1 | calibrated_common_reference | theoretical_common_rule | robustness_primary_common_theoretical | xgb | primary_benchmark | calibrated |
| xgb | test | calibrated | 10:1 | validation_2020_optimal_transported | 代码/07_robustness/R7_time_features/validation/validation_optimal_thresholds.json | frozen_robustness_validation_optimal_secondary | xgb | primary_benchmark | calibrated |

*Column panel 2 of 4 — Decision (36 records)*

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| logistic | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 22029 | 0.44649154809680164 | 727 | 21302 | 27150 | 159 |
| logistic | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 1:1 | 1 | 1 | 0.33333333333333337 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 34848 | 0.7063115651222182 | 848 | 34000 | 14452 | 38 |
| logistic | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 34338 | 0.695974705095464 | 840 | 33498 | 14954 | 46 |
| logistic | 2:1 | 1 | 2 | 0.33333333333333337 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 45293 | 0.9180145121407435 | 881 | 44412 | 4040 | 5 |
| logistic | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 44889 | 0.9098260975313146 | 880 | 44009 | 4443 | 6 |
| logistic | 5:1 | 1 | 5 | 0.33333333333333337 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 48162 | 0.9761644168794844 | 884 | 47278 | 1174 | 2 |
| logistic | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 48929 | 0.991710243625603 | 886 | 48043 | 409 | 0 |
| logistic | 10:1 | 1 | 10 | 0.3333333333333333 | 49338 | 886 | 34338 | 0.695974705095464 | 840 | 33498 | 14954 | 46 |
| rf | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| rf | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 306 | 0.006202116016052536 | 32 | 274 | 48178 | 854 |
| rf | 1:1 | 1 | 1 | 0.6122383096347276 | 49338 | 886 | 204 | 0.004134744010701690 5 | 24 | 180 | 48272 | 862 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| rf | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 1251 | 0.025355709595038307 | 95 | 1156 | 47296 | 791 |

| model | decision_cost_ratio_fn_to_fp | decision_false_positive_cost | decision_false_negative_cost | decision_threshold | decision_n | decision_positive_n | decision_predicted_positive_n | decision_predicted_positive_rate | decision_tp | decision_fp | decision_tn | decision_fn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| rf | 2:1 | 1 | 2 | 0.6122383096347276 | 49338 | 886 | 204 | 0.0041347440107016905 | 24 | 180 | 48272 | 862 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 1334 | 0.027037982893510074 | 103 | 1231 | 47221 | 783 |
| rf | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 11023 | 0.22341805504884674 | 498 | 10525 | 37927 | 388 |
| rf | 5:1 | 1 | 5 | 0.6122383096347276 | 49338 | 886 | 204 | 0.0041347440107016905 | 24 | 180 | 48272 | 862 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 24859 | 0.5038509870687908 | 730 | 24129 | 24323 | 156 |
| rf | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 42706 | 0.8655802829462078 | 873 | 41833 | 6619 | 13 |
| rf | 10:1 | 1 | 10 | 0.6122383096347276 | 49338 | 886 | 204 | 0.0041347440107016905 | 24 | 180 | 48272 | 862 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 52 | 0.0010539543556690584 | 5 | 47 | 48405 | 881 |
| xgb | 1:1 | 1 | 1 | 0.5 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| xgb | 1:1 | 1 | 1 | 0.37500000000000006 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 566 | 0.011471887794397827 | 46 | 520 | 47932 | 840 |
| xgb | 2:1 | 1 | 2 | 0.3333333333333333 | 49338 | 886 | 487 | 0.009870687907900603 | 43 | 444 | 48008 | 843 |
| xgb | 2:1 | 1 | 2 | 0.37500000000000006 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 5523 | 0.11194211358385017 | 332 | 5191 | 43261 | 554 |
| xgb | 5:1 | 1 | 5 | 0.16666666666666666 | 49338 | 886 | 3671 | 0.0744051238396368 | 247 | 3424 | 45028 | 639 |
| xgb | 5:1 | 1 | 5 | 0.37500000000000006 | 49338 | 886 | 0 | 0.0 | 0 | 0 | 48452 | 886 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 16147 | 0.32727309578823627 | 648 | 15499 | 32953 | 238 |
| xgb | 10:1 | 1 | 10 | 0.09090909090909091 | 49338 | 886 | 22503 | 0.456098747415785 | 766 | 21737 | 26715 | 120 |
| xgb | 10:1 | 1 | 10 | 0.1582089552238806 | 49338 | 886 | 5711 | 0.11575256394665369 | 342 | 5369 | 43083 | 544 |

*Column panel 3 of 4 — Decision (36 records)*

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.8205417607223476 | 0.5603483860315364 | 0.033001951972400016 | 0.9941777436010106 | 0.43965161396846364 | 0.17945823927765236 | 21461 | 0.43497912359641655 | 886 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| logistic | 0.9571106094808126 | 0.2982745810286469 | 0.024334251606978878 | 0.9973775017253278 | 0.7017254189713531 | 0.04288939051918736 | 34076 | 0.6906643966111313 | 1772 |
| logistic | 0.9480812641083521 | 0.30863535044992985 | 0.024462694391053644 | 0.9969333333333333 | 0.6913646495500702 | 0.05191873589164785 | 33590 | 0.6808139770562244 | 1772 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 1772 | 0.035915521504722527 | 1772 |
| logistic | 0.9943566591422122 | 0.08338149096012548 | 0.019451129313580465 | 0.9987639060568603 | 0.9166185090398745 | 0.00564334085778178106 | 44437 | 0.9006648019781912 | 4430 |
| logistic | 0.9932279909706546 | 0.09169900107322711 | 0.01960391187150527 | 0.9986513823331086 | 0.9083009989267729 | 0.006772009029345372 | 44039 | 0.8925979974867242 | 4430 |
| logistic | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 4430 | 0.08978880376180631 | 4430 |
| logistic | 0.9977426636568849 | 0.024230165937422603 | 0.018354719488393338 | 0.9982993197278912 | 0.9757698340625774 | 0.002257336343115124 | 47298 | 0.9586525598929831 | 8860 |
| logistic | 1.0 | 0.008441344010567 16 | 0.018107870587994 85 | 1.0 | 0.9915586559894328 | 0.0 | 48043 | 0.9737524828732417 | 8860 |

| model | decision_sensitivity_recall | decision_specificity | decision_precision_ppv | decision_npv | decision_false_positive_rate | decision_false_negative_rate | decision_total_cost | decision_cost_per_loan | decision_all_negative_cost |
|---|---|---|---|---|---|---|---|---|---|
| logistic | 0.9480812641083521 | 0.30863535044992985 | 0.024462694391053644 | 0.9969333333333333 | 0.6913646495500702 | 0.05191873589164785 | 33958 | 0.6882727309578823 | 8860 |
| rf | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| rf | 0.03611738148984198 | 0.9943449186824074 | 0.10457516339869281 | 0.9825828030673845 | 0.005655081317592669 | 0.963882618510158 | 1128 | 0.022862702176821112 | 886 |
| rf | 0.02708803611738149 | 0.996284983076034 | 0.11764705882352941 | 0.9824561403508771 | 0.003715016923965987 | 0.9729119638826185 | 1042 | 0.021119623819368438 | 886 |
| rf | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 1772 | 0.035915521504722527 | 1772 |
| rf | 0.1072234762979684 | 0.976141335754974 | 0.0759392486011191 | 0.9835506477842244 | 0.023858664245026005 | 0.8927765237020316 | 2738 | 0.055494750496574645 | 1772 |
| rf | 0.02708803611738149 | 0.996284983076034 | 0.11764705882352941 | 0.9824561403508771 | 0.003715016923965987 | 0.9729119638826185 | 1904 | 0.038590944099882445 | 1772 |
| rf | 0.1162528216704289 | 0.9745934120366548 | 0.07721139430284858 | 0.9836888592617282 | 0.025406587963345165 | 0.8837471783295711 | 5146 | 0.1043009445052495 | 4430 |
| rf | 0.5620767494356659 | 0.7827747048625444 | 0.04517826363059058 | 0.9898734177215189 | 0.21722529513745562 | 0.43792325056433407 | 12465 | 0.25264502006566947 | 4430 |
| rf | 0.02708803611738149 | 0.996284983076034 | 0.11764705882352941 | 0.9824561403508771 | 0.003715016923965987 | 0.9729119638826185 | 4490 | 0.09100490494142446 | 4430 |
| rf | 0.8239277652370203 | 0.5020019813423594 | 0.029365622108693028 | 0.9936271906532129 | 0.49799801865764054 | 0.17607223476297967 | 25689 | 0.5206737200535084 | 8860 |
| rf | 0.9853273137697517 | 0.13660942788739372 | 0.02044209244602632 | 0.9980398069963812 | 0.8633905721126063 | 0.014672686230248307 | 41963 | 0.8505208966719364 | 8860 |
| rf | 0.02708803611738149 | 0.996284983076034 | 0.11764705882352941 | 0.9824561403508771 | 0.003715016923965987 | 0.9729119638826185 | 8800 | 0.17836150634399447 | 8860 |
| xgb | 0.005643340857787810 6 | 0.9990299678031866 | 0.09615384615384616 | 0.9821247413058475 | 0.000970032196813341 | 0.9943566591422122 | 928 | 0.018809031578093965 | 886 |
| xgb | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| xgb | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 886 | 0.017957760752361263 | 886 |
| xgb | 0.05191873589164785 | 0.9892677288863205 | 0.0812720848056537 | 0.9827770031985565 | 0.010732271113679517 | 0.9480812641083521 | 2200 | 0.04459037658599862 | 1772 |
| xgb | 0.04853273137697517 | 0.9908362915875506 | 0.08829568788501027 | 0.9827434443511903 | 0.009163708412449435 | 0.9514672686230248 | 2130 | 0.043171591876444 12 | 1772 |
| xgb | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 1772 | 0.035915521504722527 | 1772 |
| xgb | 0.3747178329571106 | 0.8928630397094032 | 0.06011225783088901 | 0.987355928335045 | 0.10713696029059688 | 0.6252821670428894 | 7961 | 0.1613563581823341 | 4430 |
| xgb | 0.27878103837471785 | 0.9293321225130026 | 0.06728411876872786 | 0.9860074014058292 | 0.07066787748699745 | 0.7212189616252822 | 6619 | 0.13415622846487493 | 4430 |
| xgb | 0.0 | 1.0 | NA | 0.9820422392476388 | 0.0 | 1.0 | 4430 | 0.08978880376180631 | 4430 |
| xgb | 0.7313769751693002 | 0.6801164038636176 | 0.040131293738775005 | 0.9928293814588292 | 0.3198835961363824 | 0.2686230248306998 | 17879 | 0.36237788317321334 | 8860 |
| xgb | 0.8645598194130926 | 0.5513704284652853 | 0.03403990579033907 | 0.9955282280603689 | 0.4486295715347148 | 0.13544018058690746 | 22937 | 0.4648952126150229 | 8860 |
| xgb | 0.3860045146726862 | 0.889189300751259 | 0.05988443354929084 | 0.987530657620281 | 0.11081069924874103 | 0.6139954853273137 | 10809 | 0.21908062750820867 | 8860 |

*Column panel 4 of 4 — Decision (36 records)*

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 24.22234762979684 |
| logistic | 1.0 |
| logistic | 1.0 |
| logistic | 19.230248306997744 |

| model | decision_relative_cost_vs_all_negative |
|---|---|
| logistic | 18.955981941309254 |
| logistic | 1.0 |
| logistic | 10.030925507900678 |
| logistic | 9.941083521444694 |
| logistic | 1.0 |
| logistic | 5.3383747178329575 |
| logistic | 5.422460496613995 |
| logistic | 3.832731376975169 |
| rf | 1.0 |
| rf | 1.27313769751693 |
| rf | 1.1760722347629797 |
| rf | 1.0 |
| rf | 1.5451467268623025 |
| rf | 1.074492099322799 |
| rf | 1.161625282167043 |
| rf | 2.8137697516930023 |
| rf | 1.0135440180586908 |
| rf | 2.899435665914221 |
| rf | 4.736230248306998 |
| rf | 0.9932279909706546 |
| xgb | 1.0474040632054176 |
| xgb | 1.0 |
| xgb | 1.0 |
| xgb | 1.2415349887133182 |
| xgb | 1.2020316027088036 |
| xgb | 1.0 |
| xgb | 1.7970654627539504 |
| xgb | 1.4941309255079007 |
| xgb | 1.0 |
| xgb | 2.0179458239277652 |
| xgb | 2.5888261851015804 |
| xgb | 1.2199774266365688 |

All Non Risky Benchmark (4 records)

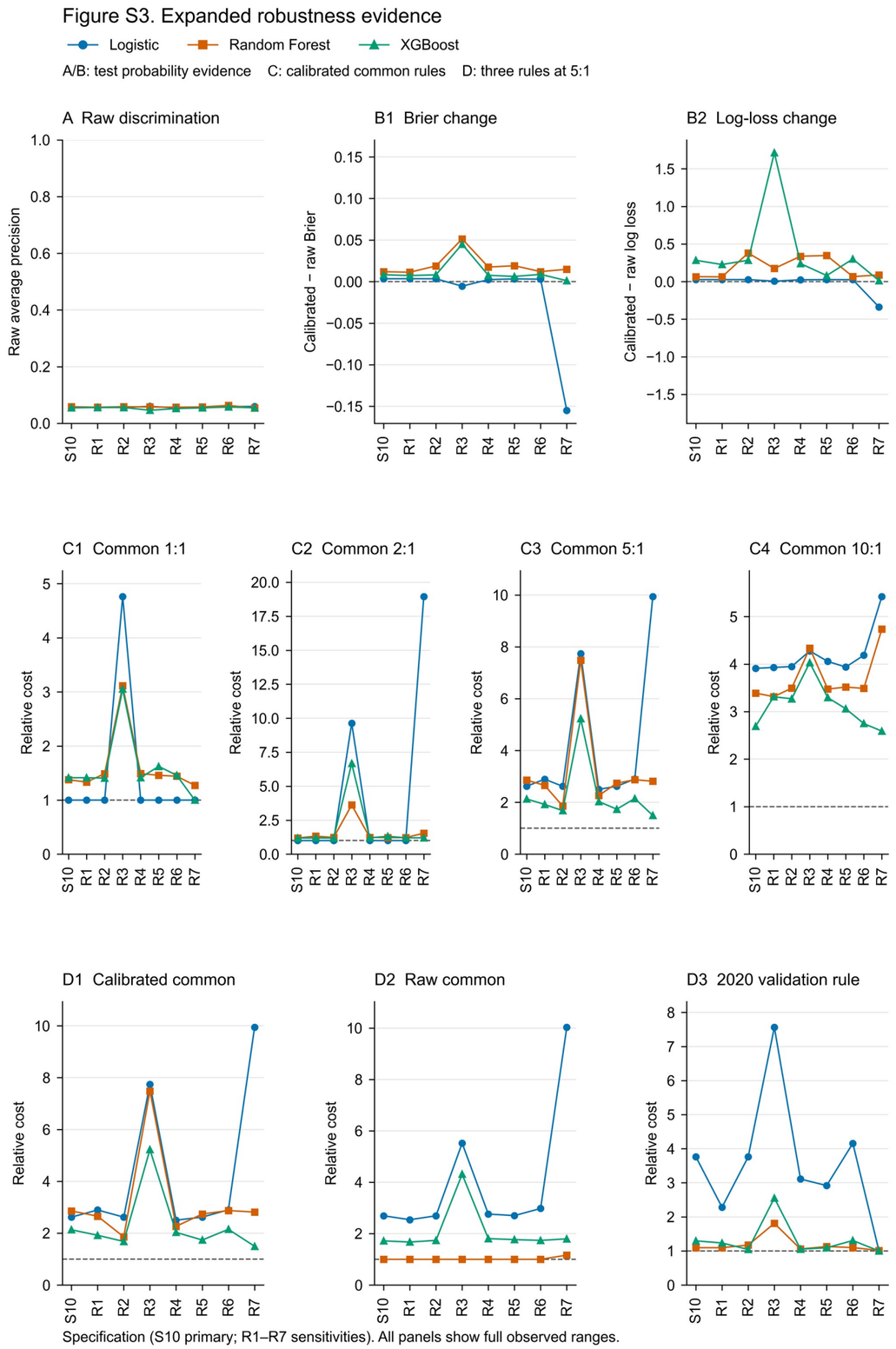
| role | model | cost_ratio | benchmark_cost_ratio_fn_to_fp | benchmark_n | benchmark_positive_n | benchmark_tp | benchmark_fp | benchmark_tn | benchmark_fn | benchmark_total_cost | benchmark_cost_per_loan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| test | all_non_risky | 1:1 | 1:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 886 | 0.017957760752361263 |
| test | all_non_risky | 2:1 | 2:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 1772 | 0.035915521504722527 |
| test | all_non_risky | 5:1 | 5:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 4430 | 0.08978880376180631 |
| test | all_non_risky | 10:1 | 10:1 | 49338 | 886 | 0 | 0 | 48452 | 886 | 8860 | 0.17957760752361263 |

Rq3 Cell (4 records)

| role | probability_type | cost_ratio | regime | classification_cell | model_comparison | rq3_classification_cell | rq3_n | rq3_positive_n | rq3_event_rate |
|---|---|---|---|---|---|---|---|---|---|
| test | calibrated | 5:1 | calibrated_common_reference | agreement_non_risky | Logistic versus XGBoost | agreement_non_risky | 4449 | 6 | 0.0013486176668914363 |
| test | calibrated | 5:1 | calibrated_common_reference | agreement_risky | Logistic versus XGBoost | agreement_risky | 3671 | 247 | 0.06728411876872786 |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_non_risky_xgb_risky | Logistic versus XGBoost | logistic_non_risky_xgb_risky | 0 | 0 | NA |
| test | calibrated | 5:1 | calibrated_common_reference | logistic_risky_xgb_non_risky | Logistic versus XGBoost | logistic_risky_xgb_non_risky | 41218 | 633 | 0.015357368140132952 |

Figure S3. Expanded discrimination, calibration and decision sensitivities.

The primary analysis and R1–R7 are compared in their own purchased/eligible cohorts using the three primary models. Panels report raw average precision (AP), calibrated-minus-raw Brier/log-loss changes, costs under the four common calibrated illustrative loss ratios, and common-5:1 raw-probability/unchanged 2020-validation-optimal sensitivities alongside the calibrated-common reference. AP is average precision, not trapezoidal precision–recall area. Positive probability-loss changes indicate worsening; costs use each specification's own ratio-specific all-non-risky benchmark. R2 changes the 90+ day delinquency during the first 24 scheduled-payment months outcome to distress composite; R3 conditions on remaining on book and changes the estimand; R7 uses First Payment Date as a timing proxy with out-of-support extrapolation. R7 Logistic's probability-quality improvement remains visible, so the evidence is not a blanket claim that calibration worsens. Secondary rules do not replace the primary illustrative calibrated 5:1 rule. Loss is not measured lender economics; different panel ranges are labelled and no uncertainty bands are shown.




*Typesetting note: Figure S3 must be placed at native readable scale, likely as a dedicated full page or equivalent; do not shrink below the established 8 pt floor.*

## S.F Numerical estimation and provenance

The final directional association analysis retained the frozen 5,950 disagreement loans and the identical design: 27 terms, including the intercept and the Classic FICO missingness indicator. Ordinary maximum-likelihood logistic regression was attempted first. Its frozen estimability guard failed because the Classic FICO missingness indicator produced an extreme coefficient (absolute magnitude approximately 21.44). The identical covariates, references, outcome and sample were then estimated using the approved Firth bias-reduced logistic fallback. No numerical preconditioning was adopted. Firth was not selected for preferred coefficient signs or substantive results.

The accepted Firth fit converged after 47 iterations under the approved plateau stopping amendment. The final adjusted-score infinity norm was approximately $1.76\times10^{-7}$, the proposed Newton-step infinity norm approximately $1.83\times10^{-7}$, and the consecutive plateau streak five. This amendment requires both norms to be at most $10^{-6}$; at least five consecutive iterations with absolute penalized-log-likelihood change at most $10^{-10}$ and accepted parameter-change infinity norm at most $10^{-12}$; a minimum of 20 iterations; and all existing finite-value, positive-definite information, covariance, coefficient and identification guards. The original strict-step stopping criterion remains an alternative; failure of both criteria fails closed. The final expected-information condition number was approximately 3,454. This was an approved numerical stopping amendment, not a change to the Firth objective, adjusted score, scientific design or interpretation rules, and it preceded acceptance of the association estimates.

Table S8 preserves the complete term archive and reference/scaling information. Its 95% confidence intervals are approximate Wald intervals based on expected information, not profile-likelihood intervals. Eleven sparse terms remain in the fitted design but are suppressed from substantive interpretation under the frozen fewer-than-100 subgroup-direction rule. Associations describe characteristics associated with disagreement direction between mortgages classified as risky/non-risky; they are not delinquency effects or causal estimates.

The final PASS manifest and its approved protocol hash govern this completed analysis. Earlier diagnostic wording stating that the RQ3 result directory was absent described an earlier stage and is superseded by the authoritative completed manifest. Source, protocol and output hashes are recorded in the accompanying display provenance; this supplementary note reports saved diagnostics only and performs no new estimation.

## S.G Step 12 source-reproduction implementation correction

Before accepted robustness model fitting, an implementation mismatch was found between annual source reproduction and role-specific calendar-maturity eligibility. Step 6 reproduced annual outcomes using the full Release 47 source endpoint through March 2026 and then applied role eligibility separately. The initial Step 12 helper instead used the December 2022 development-role cutoff while attempting to reproduce the earlier annual records. These are distinct operations: a record may have a valid annual-source label under the later endpoint while remaining ineligible for a role whose 24-month window must already have matured.

The mismatch stopped the first R1 attempt during development-cohort construction. The preserved log and correction audit show that this attempt reached no hyperparameter tuning, model fit, calibration, accepted robustness result or 2023 scientific evaluation. The helper was corrected to use the annual March 2026 source endpoint for source reproduction, while all role-specific maturity cutoffs remained unchanged. The corrected pre-run manifest is bound by hash to all seven accepted R1–R7 manifests, documenting that the implementation correction preceded their accepted model fitting. The failed-attempt evidence and stale checkpoints were retained, not deleted. Historical “not started” statements in the correction record describe that earlier stage, not the subsequently completed PASS robustness suite.

This was a source-reproduction versus role-eligibility cutoff correction, not an outcome-rule revision, post-hoc robustness redefinition or scientific specification change. Candidate A/B/C, the predictor set, temporal splits, calibration, decision rules, RQ3 rules and the scientific robustness protocol were unchanged. The primary development counts remained 239,435 loans / 4,550 events for training and 45,364 / 771 for validation; Step 10 and Step 11 remained untouched.

Table S9 records the role-specific flows and their calendar/quarter-pool denominators. In the 2020 audit, 4,255 loans were calendar-immature for T2 validation; 4,202 nevertheless had valid annual-source labels under the later source endpoint and 53 remained unlabeled. Those later annual labels did not admit these loans to validation. This distinction explains why annual-source reproduction must not substitute for role-specific maturity eligibility. The note summarizes the saved correction audit and preservation hashes only; no cohort is reconstructed here.